

Image Colorization

October 18, 2025

1 Image Colorization

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1.1 Project Overview

Objective

The goal of our project is to perform automatic colorization of black and white images. Specifically, we aim to train a neural network that learns to map grayscale input images to their corresponding colored versions. In this setup, the predictors are the black and white images, and the labels are the fully colored versions of those same images.

Motivation

Colorizing black and white images is an important prediction task with both historical and practical significance. We believe this project could benefit a range of industries, particularly education assistance, historical archives, museums, and the media & film industry. For educational use in the medical field, colorizing images can enhance visualization and offer more accurate diagnostic information for doctors during clinical procedures compared to grayscale images.

In the context of museums and archives, AI powered colorization offers a compelling way to restore and reimagine old photographs, making them more engaging and accessible for modern audiences. Adding color not only enhances visual storytelling but also helps preserve cultural memory in a more immersive and relatable format.

In the media and film industry, this technology is often used to remaster classic films and footage for modern distribution. Whether it's archivists preparing exhibitions or editors bringing old scenes back to life, being able to generate realistic color from grayscale images opens up space for both creative exploration and historical impact.

While image colorization is not a new concept, we believe that there is still a lot to uncover in this domain. Our approach brings a fresh angle in terms of both data and model design. Instead of relying on public datasets, we're planning to use our own set of images and build a full deep learning pipeline from scratch. This allows us to take on a classic challenge in a customized approach.

Data

We began our project by compiling a dataset of 265 images from full scene portraits from our own camera rolls. Later in the process, we introduced another category, the MSBA student headshots (186 images), to explore whether the model could generalize to clean studio-like portraits. The

dataset was randomly split into 90/10 for training and validation sets during training. A small number of additional images for each dataset were reserved as a test set to evaluate generalization on unseen photos.

Dataset: [Google Drive](#)

The results of our work on this project is also available to read as a Medium article, please [click here](#) to view. The slide deck used to present our work can be found [here](#).

2 Initializations and Importations

```
[ ]: !pip install pillow-heif
```

```
Collecting pillow-heif
  Downloading pillow_heif-0.22.0-cp311-cp311-
manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (9.6 kB)
Requirement already satisfied: pillow>=10.1.0 in /usr/local/lib/python3.11/dist-
packages (from pillow-heif) (11.2.1)
Downloading
pillow_heif-0.22.0-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl
(7.8 MB)
7.8/7.8 MB
42.8 MB/s eta 0:00:00
Installing collected packages: pillow-heif
Successfully installed pillow-heif-0.22.0
```

```
[ ]: import os
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from PIL import Image
from pillow_heif import register_heif_opener

from google.colab import drive

import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers, models
from tensorflow.keras.applications import VGG16
from tensorflow.keras.applications.vgg16 import preprocess_input
from tensorflow.keras.models import Model, Sequential
from tensorflow.keras.layers import Input, Conv2D, UpSampling2D,
↳BatchNormalization, Concatenate, Lambda, LayerNormalization, Dense,
↳MultiHeadAttention, Add, Flatten, Reshape, Conv2D, UpSampling2D, Activation
from tensorflow.keras.optimizers import Adam
from tensorflow.keras.preprocessing.image import ImageDataGenerator,
↳img_to_array, load_img
```

```
from tensorflow.keras.callbacks import EarlyStopping, ModelCheckpoint

from skimage.color import rgb2lab, lab2rgb
```

```
[ ]: # Allows for heic/heif (iPhone image formats) to be handled, and declaring
      ↪ acceptable image extensions
register_heif_opener()
image_extensions = ('.png', '.jpg', '.jpeg', '.heic', '.heif')

[ ]: drive.mount('/content/drive', force_remount= True)
```

Mounted at /content/drive

3 Image Preprocessing and Visualization Functions

Preprocessing Methodology

We first standardized color images, converting them to RGB format to ensure compatibility with JPEG, rotating portrait images to landscape orientation when necessary, and then resized and center-cropped photos to fix the resolution of 176x120 pixels preserving the ratio. This ensured that all images fed into the model had consistent sizing and formatting. After saving the resized images as JPG files, we converted them to grayscale.

The standardized color images were normalized as NumPy arrays, scaling pixel values in the [0,1] range for model training stability. To enhance the diversity of the dataset and reduce the risk of overfitting, data augmentation was applied using TensorFlow's ImageDataGenerator. Augmentation techniques included random rotations, width and height shifts, shearing, zooming, and horizontal flipping. The augmented images were then incorporated into the main dataset getting both datasets to the 400-500 range.

To prepare for model training, we created two types of input/output pairs and a visualization function that displays the grayscale input, the model's predicted color output, and the ground truth color image side by side to easily evaluate the model's performance. * In the RGB setup, each color image was converted to a grayscale version as the model input and the RGB image as the output target. The model directly predicts an RGB image, so the image visualization was able to display the prediction directly. * For LAB colorization, images were transformed into the LAB color space, with the lightness channel being the input and the normalized A and B color channels used as output. Grayscale inputs were reshaped appropriately to match the input dimensions. To visualize the predicted colorization, we needed to reconstruct the images and convert from LAB to RGB to show the prediction.

The preprocessing pipeline standardizes, augments, and structures the dataset to provide the best possible foundation for training and evaluating high-quality image colorization models. This section contains the functions used in the preprocessing flow.

```
[ ]: # This function will a) convert an incoming image to RGB to allow conversion to
      ↪ the JPEG format, b) rotate to landscape if necessary, c) crop/resize the
      ↪ image to a target resolution (default to 720x480px, ie a 3:2 aspect ratio),
      ↪ and d) save the image to the output folder with a .jpg extension
```

```

def image_resizer(input_path, output_path, target_size=(720, 480)):
    try:
        img = Image.open(input_path)

        # Converting images to RGB in case they have transparency in them, also
        ↪ makes them compatible with the JPEG format
        img = img.convert('RGB')

        if img.height > img.width:
            img = img.rotate(90, expand=True)

        # Resize image without distortion or black bars (ideally)
        target_width, target_height = target_size
        img_width, img_height = img.size

        img_ratio = img_width / img_height
        target_ratio = target_width / target_height

        if img_ratio > target_ratio:
            # If the image is wider than our target resolution, scale height to
            ↪ match target
            new_height = target_height
            new_width = int(new_height * img_ratio)
        else:
            # If the image is taller than our target resolution, scale width to
            ↪ match target
            new_width = target_width
            new_height = int(new_width / img_ratio)

        img = img.resize((new_width, new_height), Image.Resampling.LANCZOS)

        left = (new_width - target_width) // 2
        top = (new_height - target_height) // 2
        right = left + target_width
        bottom = top + target_height

        img = img.crop((left, top, right, bottom))

        # Changing final image file name to have a .jpg extension
        base, _ = os.path.splitext(output_path)
        output_path = base + '.jpg'

        img.save(output_path, 'JPEG', quality=100)
        # print(f"Processed: {os.path.basename(input_path)}")
        return True

    except Exception as e:

```

```

print(f"Failed to process {os.path.basename(input_path)}: {e}")
return False

```

```

[ ]: # This function will help us show the image predictions for the models that
      ↪process images in the RGB colour space

```

```

def show_predictions(model, X, Y, num_samples=5):
    preds = model.predict(X[:num_samples])

    plt.figure(figsize=(15, 5 * num_samples))

    for i in range(num_samples):
        # Grayscale Input
        ax = plt.subplot(num_samples, 3, i * 3 + 1)
        plt.imshow(tf.squeeze(X[i]), cmap='gray')
        ax.set_title("Input (Grayscale)")
        ax.axis("off")

        # Model Prediction
        ax = plt.subplot(num_samples, 3, i * 3 + 2)
        pred_img = preds[i]
        plt.imshow(np.clip(pred_img, 0, 1))
        ax.set_title("Predicted (RGB)")
        ax.axis("off")

        # Ground Truth
        ax = plt.subplot(num_samples, 3, i * 3 + 3)
        plt.imshow(np.clip(Y[i], 0, 1))
        ax.set_title("Ground Truth (RGB)")
        ax.axis("off")

    plt.tight_layout()
    plt.show()

```

```

[ ]: # This function will help us show the image predictions for the models that
      ↪process images in the LAB colour space

```

```

def show_predictions_lab(model, X, Y, num_samples=5):
    preds = model.predict(X[:num_samples])

    plt.figure(figsize=(15, 5 * num_samples))

    for i in range(num_samples):
        # Grayscale Input
        ax = plt.subplot(num_samples, 3, i * 3 + 1)
        plt.imshow(tf.squeeze(X[i]), cmap='gray')
        ax.set_title("Input (Grayscale)")
        ax.axis("off")

```

```

# Model Prediction
ax = plt.subplot(num_samples, 3, i * 3 + 2)
pred_img = preds[i].reshape(120, 176, 2)
L_channel = X[i].reshape(120, 176, 1)[:,:,:0]
lab_image = np.dstack((L_channel, pred_img * 128))
rgb_image = lab2rgb(lab_image)
plt.imshow(np.clip(rgb_image, 0, 1))
ax.set_title("Predicted (RGB)")
ax.axis("off")

# Ground Truth
ax = plt.subplot(num_samples, 3, i * 3 + 3)
ground_truth_rgb = lab2rgb(np.dstack((X[i].reshape(120, 176, 1)[:,:,:0],
↪Y[i] * 128)))
plt.imshow(np.clip(ground_truth_rgb, 0, 1))
ax.set_title("Ground Truth (RGB)")
ax.axis("off")

plt.tight_layout()
plt.show()

```

4 Methodology

We experimented with a range of different model architectures to tackle the image colorization task, aiming to find the most accurate approach. Detailed later in the notebook are each of the 6 modeling attempts, highlighting how each approach contributes to our understanding of the colorization problem.

4.0.1 Model Evaluation

We used Mean Absolute Error (MAE) and Mean Squared Error (MSE) as both loss functions and evaluation metrics during model training. These metrics are commonly used in image regression tasks because they give us a way to quantify the difference between the predicted and true pixel values. However, we found these values not applicable to our project, as they didn't always reflect how good the results actually looked. We also experimented with custom loss functions such as perceptual loss and hybrid loss, but they performed worse than the standard losses in our case. Ultimately, we relied on the human eye evaluation to judge which model produced the most realistic and natural-looking colorization. While quantitative metrics provided a baseline, it was how natural the colors looked to a human observer that guided our judgement on the prediction quality.

4.0.2 Neural Network Overview

Model 1: Sequential RGB Model

This model has a simple structure that predicts the RGB channels. The process begins with two convolutional layers with 64 filters each, followed by another two convolutional layers with 128 filters. These layers use ReLU activation functions and apply downsampling, which reduces the spatial resolution while increasing the depth to capture more abstract features.

In the decoding stage, the model reverses this process through upsampling, followed by additional convolutional layers. A final Lambda layer is used to resize the output image to match the original input dimensions. Since the model is simple, it is efficient and easy to train and it serves as a strong baseline for comparison with more complex architectures.

Model 2: Simple CNN

This model follows a simple encoder-decoder architecture for image colorization. It's a also simplified version, designed to capture the core concept using only a few convolutional layers. The image first passes through two convolutional layers, which are Conv2D(64) and Conv2D(128), both using ReLU activation and 'same' padding. A MaxPooling2D layer then downsamples the feature map. A single Conv2D(128) layer serves as a shallow bottleneck. The decoder then upsamples the feature map using Conv2DTranspose and reconstructs the image. The final layer, Conv2D(3) with sigmoid activation, generates the 3-channel RGB output scaled between 0 and 1.

Model 3: Improved CNN

The model first passes the input through an encoder for downsampling, which reduces the spatial dimensions while capturing important features. In the first convolution block, a Conv2D (64) layer applies 64 filters to extract local patterns. This is followed by BatchNormalization to stabilize and speed up training, and a LeakyReLU activation to introduce non-linearity while allowing a small gradient when the output is negative. Finally, MaxPooling2D is applied to reduce the image size by half.

In the second convolution block, the same structure is used, but with 128 filters. After another pooling layer, the image is downsampled to a size of 30x44. At the bottleneck, a deeper Conv2D (256) layer is used to capture more abstract and high-level features. This layer serves as the compressed representation of the image. After the encoder and bottleneck, the model enters the decoder, where it up-samples the image twice using UpSampling2D and convolution layers. This restores the image to its original resolution and predicts pixel-level color in RGB format.

Model 4: U-Net Implementation

We then introduced a U-Net architecture with skip connections between the encoder and decoder to improve both training stability and output quality. First, we added deeper encoder layers to enable the model to extract more complex features. Batch normalization was applied after each convolutional layer to stabilize training and serve as a form of regularization, helping to reduce overfitting. We also incorporated skip connections to pass high-resolution features directly from the encoder to the decoder, allowing the model to better recover spatial details lost during downsampling. Additionally, we deepened the bottleneck and enhanced the decoder blocks to support more accurate and detailed color reconstruction.

Model 5: Sequential LAB Model

This model was designed to predict colors in the LAB color space using a deep sequential autoencoder architecture. The network encodes the input through multiple convolutional layers, compressing it into a lower dimensionality representation, and then decoding it back to image resolution. The final output has a tanh activation to ensure that the predicted values are normalized into the range of [-1,1] to align with the LAB color space's structure.

The training was conducted using the Adam optimizer and the Mean Absolute Error (MAE) loss function, early stopping, and model checkpointing based on validation loss. We had experimented with different learning rates including extremely small values such as 1e-14 and 1e-16. We found that

extremely low rates significantly slowed convergence and potentially prevented effective learning. We also tested both MAE and MSE loss function, MAE produced more realistic results.

Model 6: Transformers with VGG

The idea of this model is to use VGG to extract strong spatial features, Transformers to model global context, and the decoder to reconstruct a full-resolution color image. It begins by converting the grayscale input to 3 channels, then passes it through a pretrained VGG16 encoder, which extracts spatial features using filters trained on millions of natural images. Unlike basic U-Nets or autoencoders that rely purely on local convolutional context, this model integrates a Transformer block after VGG to capture global attention and long-range dependencies across the entire image. This allows it to better understand the overall scene and apply colorization based on contexts. Finally, a CNN decoder upsamples the features back to full resolution, producing a RGB output. Compared to U-Net and autoencoder models, this architecture aims to preserve fine image structure via VGG features and enhances global color consistency through attention.

While we are exploring the inclusion of VGG in this project, this was done more in a direction of exploration and learning. We understand that VGG is better suited to object identification, and that this use case requires significant retention of detail which VGG may be unable to help with. However, we would like to explore if the model can roughly identify objects and apply the right/similar colours to the same, in order to open future implementations where such a model is used in parallel with other networks that allow better/more complete retention of detail.

We will now explore the first of our implementations, utilizing the full-scene portrait images to train our neural networks.

5 Full-Scene Portrait Implementations

5.1 Image Preprocessing

```
[ ]: # Defining the required Google Drive paths for the Full Scene Portrait_
    ↪Implementation
color_image_path = '/content/drive/MyDrive/BA865/BA865 Group Project/Dataset/
    ↪Portrait/Color'
color_resized_image_path = '/content/drive/MyDrive/BA865/BA865 Group Project/
    ↪Dataset/Portrait/Color Resized'
test_color_image_path = '/content/drive/MyDrive/BA865/BA865 Group Project/
    ↪Dataset/Portrait/Test'
test_color_resized_image_path = '/content/drive/MyDrive/BA865/BA865 Group_
    ↪Project/Dataset/Portrait/Test Resized'
best_models_path = '/content/drive/MyDrive/BA865/BA865 Group Project/Best_
    ↪Models Portrait'
```

Train Color Image Preprocessing

```
[ ]: # Creating color resized image folder if it doesn't exist
os.makedirs(color_resized_image_path, exist_ok=True)

# Ensuring the color resized image folder is empty
```



```

for file_name in os.listdir(color_resized_image_path):
    file_path = os.path.join(color_resized_image_path, file_name)
    if os.path.isfile(file_path):
        os.remove(file_path)

resize_success_count = 0
resize_fail_count = 0

# Looping through all color images and applying the resizing, cropping,
↳rotation, and compression
for image in os.listdir(color_image_path):
    if image.lower().endswith(image_extensions):
        input_img_path = os.path.join(color_image_path, image)
        output_img_path = os.path.join(color_resized_image_path, image)

        # The below if statement resizes the image while also maintaining a
↳count of successful and unsuccessful conversions
        if image_resizer(input_img_path, output_img_path, (176, 120)):
            resize_success_count += 1
        else:
            resize_fail_count += 1

print(f"Successfully resized {resize_success_count} images.")
print(f"Failed to resize {resize_fail_count} images.")

```

Successfully resized 265 images.
Failed to resize 0 images.

```

[ ]: color_images = []

for image in os.listdir(color_resized_image_path):
    color_image_path = os.path.join(color_resized_image_path, image)

    # Converting the color image to numpy array
    if os.path.isfile(color_image_path):
        col_img = Image.open(color_image_path).convert('RGB')
        color_image_array = np.array(col_img).astype('float32') / 255.0
        color_images.append(color_image_array)

```

```

[ ]: # This code will use ImageDataGenerator to create additional images as a form
↳of data augmentation, helping against overfitting.
datagen = ImageDataGenerator(rotation_range=40, width_shift_range=0.2,
↳height_shift_range=0.2, rescale=1./255,
                                shear_range=0.2, zoom_range=0.2,
↳horizontal_flip=True, fill_mode='nearest')

datagen.fit(color_images)

```

```

image_batches = datagen.flow(np.stack(color_images, axis=0), batch_size=40)

n = 10
plt.figure(figsize= (20,10))

q = 0
for i, batch in enumerate(image_batches):
    for j, image in enumerate(batch):
        if j == 1:
            if i <= 5:
                ax = plt.subplot(1, 6, i+1)
                plt.imshow(tf.keras.utils.array_to_img(image))
                ax.get_xaxis().set_visible(False)
                ax.get_yaxis().set_visible(False)

                # The iterator outputs image data with pixel values between 0-1; we are
                ↪ using values between 0-255 elsewhere. We need to make the pixel values align.
                color_images.append(np.multiply(image,255))

            if q > 5:
                break

        q += 1

plt.show()

print(f'We now have a total of {len(color_images)} images.')

```



We now have a total of 530 images.

```

[ ]: # Splitting the predictor and labels in the RGB color space, for the training
    ↪ set
X = []
Y = []

for img in color_images:
    try:
        gray = tf.image.rgb_to_grayscale(img)
        X.append(gray.numpy())
        Y.append(img)

```

```

        except Exception as e:
            print("error:", e)

X = np.array(X)
Y = np.array(Y)

print("X_rgb shape:", X.shape)
print("Y_rgb shape:", Y.shape)

```

```

X_rgb shape: (530, 120, 176, 1)
Y_rgb shape: (530, 120, 176, 3)

```

```

[ ]: # Splitting the predictor and labels in the LAB color space, for the training
    ↪set
X_lab = []
Y_lab = []

for img in color_images:
    try:
        lab = rgb2lab(img)
        X_lab.append(lab[:, :, 0])
        #restrict values to be between -1 and 1
        Y_lab.append(lab[:, :, 1:] / 128)
    except Exception as e:
        print("error:", e)

X_lab = np.array(X_lab)
Y_lab = np.array(Y_lab)

#dimensions to be the same for X and Y
X_lab = X_lab.reshape(X_lab.shape+(1,))

print("X_lab shape:", X_lab.shape)
print("Y_lab shape:", Y_lab.shape)

```

```

X_lab shape: (530, 120, 176, 1)
Y_lab shape: (530, 120, 176, 2)

```

Test Color Images Preprocessing

```

[ ]: # Creating color resized image folder for test images if it doesn't exist
os.makedirs(test_color_resized_image_path, exist_ok=True)

# Ensuring the color resized image folder is empty
for file_name in os.listdir(test_color_resized_image_path):
    file_path = os.path.join(test_color_resized_image_path, file_name)
    if os.path.isfile(file_path):
        os.remove(file_path)

```

```

test_resize_success_count = 0
test_resize_fail_count = 0

# Looping through all color images and applying the resizing, cropping,
↳rotation, and compression
for test_image in os.listdir(test_color_image_path):
    if test_image.lower().endswith(image_extensions):
        input_img_path = os.path.join(test_color_image_path, test_image)
        output_img_path = os.path.join(test_color_resized_image_path,
↳test_image)

        # The below if statement resizes the image while also maintaining a
↳count of successful and unsuccessful conversions
        if image_resizer(input_img_path, output_img_path, (176, 120)):
            test_resize_success_count += 1
        else:
            test_resize_fail_count += 1

print(f"Successfully resized {test_resize_success_count} images.")
print(f"Failed to resize {test_resize_fail_count} images.")

```

Successfully resized 6 images.

Failed to resize 0 images.

```

[ ]: test_color_images = []

for image in os.listdir(test_color_resized_image_path):
    test_color_image_path = os.path.join(test_color_resized_image_path, image)

    # Converting the color image to numpy array
    if os.path.isfile(test_color_image_path):
        col_img = Image.open(test_color_image_path).convert('RGB')
        test_color_image_array = np.array(col_img).astype('float32') / 255.0
        test_color_images.append(test_color_image_array)

```

```

[ ]: # Splitting the predictor and labels in the LAB color space, for the test set
X_lab_test = []
Y_lab_test = []

for img in test_color_images:
    try:
        lab_test = rgb2lab(img)
        X_lab_test.append(lab_test[:, :, 0])
        #restrict values to be between -1 and 1
        Y_lab_test.append(lab_test[:, :, 1:] / 128)
    except Exception as e:

```

```

    print("error:", e)

X_lab_test = np.array(X_lab_test)
Y_lab_test = np.array(Y_lab_test)

#dimensions to be the same for X and Y
X_lab_test = X_lab_test.reshape(X_lab_test.shape+(1,))

print("X_lab_test shape:", X_lab_test.shape)
print("Y_lab_test shape:", Y_lab_test.shape)

```

```

X_lab_test shape: (6, 120, 176, 1)
Y_lab_test shape: (6, 120, 176, 2)

```

```

[ ]: # Splitting the predictor and labels in the RGB color space, for the test set
X_test = []
Y_test = []

for img in test_color_images:
    try:
        gray = tf.image.rgb_to_grayscale(img)
        X_test.append(gray.numpy())
        Y_test.append(img)
    except Exception as e:
        print("error:", e)

X_test = np.array(X_test)
Y_test = np.array(Y_test)

print("X_rgb_test shape:", X_test.shape) # (num_images, 120, 176, 1)
print("Y_rgb_test shape:", Y_test.shape) # (num_images, 120, 176, 3)

```

```

X_rgb_test shape: (6, 120, 176, 1)
Y_rgb_test shape: (6, 120, 176, 3)

```

```

[ ]: # Defining the early stopping callback, standard across all models
early_stop = EarlyStopping(monitor='val_loss', patience=20,
    ↪restore_best_weights=True)

```

5.2 Model 1: Sequential RGB Model

```

[ ]: # Defining the architecture for the sequential RGB model
input_1 = Input(shape=(120, 176, 1))

x = Conv2D(64, (3, 3), activation='relu', padding='same')(input_1)
x = Conv2D(64, (3, 3), activation='relu', strides=2, padding='same')(x)

```

```

x = Conv2D(128, (3, 3), activation='relu', padding='same')(x)
x = Conv2D(128, (3, 3), activation='relu', strides=2, padding='same')(x)

x = UpSampling2D((2, 2))(x)
x = Conv2D(64, (3, 3), activation='relu', padding='same')(x)

x = UpSampling2D((2, 2))(x)
x = Conv2D(32, (3, 3), activation='relu', padding='same')(x)

output_rgb = Conv2D(3, (3, 3), activation='sigmoid', padding='same')(x)
output_rgb = Lambda(lambda x: tf.image.resize_with_crop_or_pad(x, 120, 176))(output_rgb)

seq_rgb_model = Model(inputs=input_1, outputs=output_rgb)
seq_rgb_model.compile(optimizer='adam', loss='mae', metrics=['mse'])

```

```

[ ]: # Fitting the model to the training data
history_seq_rgb = seq_rgb_model.fit(X, Y, batch_size=16, epochs=200, validation_split=0.1, callbacks=[early_stop])

```

```

Epoch 1/200
30/30          25s 466ms/step -
loss: 0.2063 - mse: 0.0617 - val_loss: 0.0833 - val_mse: 0.0132
Epoch 2/200
30/30          21s 61ms/step -
loss: 0.0849 - mse: 0.0138 - val_loss: 0.0755 - val_mse: 0.0111
Epoch 3/200
30/30          2s 57ms/step -
loss: 0.0768 - mse: 0.0114 - val_loss: 0.0669 - val_mse: 0.0090
Epoch 4/200
30/30          2s 58ms/step -
loss: 0.0697 - mse: 0.0098 - val_loss: 0.0669 - val_mse: 0.0089
Epoch 5/200
30/30          2s 58ms/step -
loss: 0.0664 - mse: 0.0091 - val_loss: 0.0608 - val_mse: 0.0078
Epoch 6/200
30/30          3s 62ms/step -
loss: 0.0630 - mse: 0.0083 - val_loss: 0.0594 - val_mse: 0.0074
Epoch 7/200
30/30          2s 59ms/step -
loss: 0.0612 - mse: 0.0079 - val_loss: 0.0583 - val_mse: 0.0073
Epoch 8/200
30/30          2s 58ms/step -
loss: 0.0636 - mse: 0.0082 - val_loss: 0.0585 - val_mse: 0.0071
Epoch 9/200
30/30          3s 61ms/step -
loss: 0.0597 - mse: 0.0075 - val_loss: 0.0642 - val_mse: 0.0082

```

Epoch 10/200
30/30 2s 61ms/step -
loss: 0.0619 - mse: 0.0080 - val_loss: 0.0558 - val_mse: 0.0068
Epoch 11/200
30/30 3s 61ms/step -
loss: 0.0597 - mse: 0.0076 - val_loss: 0.0553 - val_mse: 0.0068
Epoch 12/200
30/30 2s 60ms/step -
loss: 0.0567 - mse: 0.0070 - val_loss: 0.0556 - val_mse: 0.0068
Epoch 13/200
30/30 2s 61ms/step -
loss: 0.0582 - mse: 0.0074 - val_loss: 0.0560 - val_mse: 0.0069
Epoch 14/200
30/30 3s 61ms/step -
loss: 0.0567 - mse: 0.0070 - val_loss: 0.0536 - val_mse: 0.0065
Epoch 15/200
30/30 2s 61ms/step -
loss: 0.0566 - mse: 0.0070 - val_loss: 0.0532 - val_mse: 0.0064
Epoch 16/200
30/30 2s 59ms/step -
loss: 0.0556 - mse: 0.0069 - val_loss: 0.0535 - val_mse: 0.0064
Epoch 17/200
30/30 3s 59ms/step -
loss: 0.0541 - mse: 0.0065 - val_loss: 0.0536 - val_mse: 0.0065
Epoch 18/200
30/30 3s 60ms/step -
loss: 0.0545 - mse: 0.0066 - val_loss: 0.0526 - val_mse: 0.0064
Epoch 19/200
30/30 2s 60ms/step -
loss: 0.0566 - mse: 0.0071 - val_loss: 0.0527 - val_mse: 0.0063
Epoch 20/200
30/30 3s 61ms/step -
loss: 0.0539 - mse: 0.0066 - val_loss: 0.0537 - val_mse: 0.0065
Epoch 21/200
30/30 2s 63ms/step -
loss: 0.0556 - mse: 0.0069 - val_loss: 0.0517 - val_mse: 0.0061
Epoch 22/200
30/30 2s 59ms/step -
loss: 0.0541 - mse: 0.0067 - val_loss: 0.0528 - val_mse: 0.0064
Epoch 23/200
30/30 2s 60ms/step -
loss: 0.0512 - mse: 0.0061 - val_loss: 0.0517 - val_mse: 0.0062
Epoch 24/200
30/30 3s 63ms/step -
loss: 0.0528 - mse: 0.0065 - val_loss: 0.0525 - val_mse: 0.0062
Epoch 25/200
30/30 2s 64ms/step -
loss: 0.0514 - mse: 0.0060 - val_loss: 0.0505 - val_mse: 0.0060

Epoch 26/200
30/30 2s 60ms/step -
loss: 0.0515 - mse: 0.0062 - val_loss: 0.0512 - val_mse: 0.0061
Epoch 27/200
30/30 2s 59ms/step -
loss: 0.0514 - mse: 0.0062 - val_loss: 0.0506 - val_mse: 0.0060
Epoch 28/200
30/30 3s 63ms/step -
loss: 0.0501 - mse: 0.0059 - val_loss: 0.0507 - val_mse: 0.0060
Epoch 29/200
30/30 2s 59ms/step -
loss: 0.0526 - mse: 0.0064 - val_loss: 0.0527 - val_mse: 0.0062
Epoch 30/200
30/30 2s 61ms/step -
loss: 0.0509 - mse: 0.0060 - val_loss: 0.0501 - val_mse: 0.0059
Epoch 31/200
30/30 2s 64ms/step -
loss: 0.0517 - mse: 0.0063 - val_loss: 0.0507 - val_mse: 0.0060
Epoch 32/200
30/30 2s 60ms/step -
loss: 0.0521 - mse: 0.0061 - val_loss: 0.0508 - val_mse: 0.0061
Epoch 33/200
30/30 3s 60ms/step -
loss: 0.0523 - mse: 0.0063 - val_loss: 0.0527 - val_mse: 0.0061
Epoch 34/200
30/30 2s 60ms/step -
loss: 0.0528 - mse: 0.0064 - val_loss: 0.0501 - val_mse: 0.0060
Epoch 35/200
30/30 2s 60ms/step -
loss: 0.0486 - mse: 0.0056 - val_loss: 0.0507 - val_mse: 0.0060
Epoch 36/200
30/30 2s 60ms/step -
loss: 0.0502 - mse: 0.0060 - val_loss: 0.0503 - val_mse: 0.0059
Epoch 37/200
30/30 2s 62ms/step -
loss: 0.0501 - mse: 0.0059 - val_loss: 0.0498 - val_mse: 0.0059
Epoch 38/200
30/30 3s 64ms/step -
loss: 0.0502 - mse: 0.0059 - val_loss: 0.0502 - val_mse: 0.0059
Epoch 39/200
30/30 2s 61ms/step -
loss: 0.0496 - mse: 0.0058 - val_loss: 0.0521 - val_mse: 0.0062
Epoch 40/200
30/30 2s 61ms/step -
loss: 0.0503 - mse: 0.0059 - val_loss: 0.0528 - val_mse: 0.0062
Epoch 41/200
30/30 2s 61ms/step -
loss: 0.0518 - mse: 0.0062 - val_loss: 0.0510 - val_mse: 0.0061

Epoch 42/200
30/30 2s 61ms/step -
loss: 0.0511 - mse: 0.0061 - val_loss: 0.0499 - val_mse: 0.0059
Epoch 43/200
30/30 2s 64ms/step -
loss: 0.0502 - mse: 0.0061 - val_loss: 0.0491 - val_mse: 0.0058
Epoch 44/200
30/30 3s 63ms/step -
loss: 0.0485 - mse: 0.0056 - val_loss: 0.0493 - val_mse: 0.0058
Epoch 45/200
30/30 3s 64ms/step -
loss: 0.0518 - mse: 0.0065 - val_loss: 0.0496 - val_mse: 0.0059
Epoch 46/200
30/30 2s 61ms/step -
loss: 0.0487 - mse: 0.0057 - val_loss: 0.0505 - val_mse: 0.0060
Epoch 47/200
30/30 2s 64ms/step -
loss: 0.0486 - mse: 0.0056 - val_loss: 0.0496 - val_mse: 0.0058
Epoch 48/200
30/30 2s 62ms/step -
loss: 0.0492 - mse: 0.0056 - val_loss: 0.0490 - val_mse: 0.0058
Epoch 49/200
30/30 2s 66ms/step -
loss: 0.0483 - mse: 0.0056 - val_loss: 0.0506 - val_mse: 0.0060
Epoch 50/200
30/30 2s 63ms/step -
loss: 0.0485 - mse: 0.0056 - val_loss: 0.0518 - val_mse: 0.0061
Epoch 51/200
30/30 2s 61ms/step -
loss: 0.0513 - mse: 0.0059 - val_loss: 0.0492 - val_mse: 0.0058
Epoch 52/200
30/30 2s 62ms/step -
loss: 0.0483 - mse: 0.0055 - val_loss: 0.0492 - val_mse: 0.0059
Epoch 53/200
30/30 3s 62ms/step -
loss: 0.0471 - mse: 0.0053 - val_loss: 0.0499 - val_mse: 0.0059
Epoch 54/200
30/30 2s 62ms/step -
loss: 0.0487 - mse: 0.0058 - val_loss: 0.0516 - val_mse: 0.0060
Epoch 55/200
30/30 2s 63ms/step -
loss: 0.0493 - mse: 0.0056 - val_loss: 0.0511 - val_mse: 0.0061
Epoch 56/200
30/30 2s 66ms/step -
loss: 0.0481 - mse: 0.0055 - val_loss: 0.0493 - val_mse: 0.0059
Epoch 57/200
30/30 2s 67ms/step -
loss: 0.0499 - mse: 0.0060 - val_loss: 0.0485 - val_mse: 0.0057

Epoch 58/200
30/30 2s 62ms/step -
loss: 0.0476 - mse: 0.0055 - val_loss: 0.0493 - val_mse: 0.0058
Epoch 59/200
30/30 2s 62ms/step -
loss: 0.0493 - mse: 0.0058 - val_loss: 0.0494 - val_mse: 0.0059
Epoch 60/200
30/30 3s 66ms/step -
loss: 0.0477 - mse: 0.0055 - val_loss: 0.0541 - val_mse: 0.0064
Epoch 61/200
30/30 2s 63ms/step -
loss: 0.0518 - mse: 0.0063 - val_loss: 0.0489 - val_mse: 0.0058
Epoch 62/200
30/30 3s 64ms/step -
loss: 0.0476 - mse: 0.0054 - val_loss: 0.0511 - val_mse: 0.0061
Epoch 63/200
30/30 3s 65ms/step -
loss: 0.0491 - mse: 0.0058 - val_loss: 0.0492 - val_mse: 0.0058
Epoch 64/200
30/30 2s 63ms/step -
loss: 0.0467 - mse: 0.0053 - val_loss: 0.0488 - val_mse: 0.0058
Epoch 65/200
30/30 3s 63ms/step -
loss: 0.0475 - mse: 0.0056 - val_loss: 0.0482 - val_mse: 0.0057
Epoch 66/200
30/30 2s 71ms/step -
loss: 0.0472 - mse: 0.0055 - val_loss: 0.0487 - val_mse: 0.0057
Epoch 67/200
30/30 2s 71ms/step -
loss: 0.0485 - mse: 0.0058 - val_loss: 0.0484 - val_mse: 0.0058
Epoch 68/200
30/30 3s 74ms/step -
loss: 0.0486 - mse: 0.0055 - val_loss: 0.0526 - val_mse: 0.0062
Epoch 69/200
30/30 2s 68ms/step -
loss: 0.0509 - mse: 0.0060 - val_loss: 0.0483 - val_mse: 0.0057
Epoch 70/200
30/30 2s 65ms/step -
loss: 0.0483 - mse: 0.0056 - val_loss: 0.0484 - val_mse: 0.0058
Epoch 71/200
30/30 2s 63ms/step -
loss: 0.0485 - mse: 0.0057 - val_loss: 0.0491 - val_mse: 0.0058
Epoch 72/200
30/30 2s 63ms/step -
loss: 0.0464 - mse: 0.0053 - val_loss: 0.0497 - val_mse: 0.0059
Epoch 73/200
30/30 3s 67ms/step -
loss: 0.0491 - mse: 0.0059 - val_loss: 0.0481 - val_mse: 0.0057

Epoch 74/200
30/30 2s 67ms/step -
loss: 0.0494 - mse: 0.0060 - val_loss: 0.0483 - val_mse: 0.0057
Epoch 75/200
30/30 3s 66ms/step -
loss: 0.0494 - mse: 0.0059 - val_loss: 0.0481 - val_mse: 0.0056
Epoch 76/200
30/30 2s 64ms/step -
loss: 0.0475 - mse: 0.0056 - val_loss: 0.0495 - val_mse: 0.0058
Epoch 77/200
30/30 2s 64ms/step -
loss: 0.0505 - mse: 0.0060 - val_loss: 0.0497 - val_mse: 0.0059
Epoch 78/200
30/30 2s 66ms/step -
loss: 0.0487 - mse: 0.0059 - val_loss: 0.0485 - val_mse: 0.0057
Epoch 79/200
30/30 2s 67ms/step -
loss: 0.0485 - mse: 0.0059 - val_loss: 0.0483 - val_mse: 0.0058
Epoch 80/200
30/30 2s 69ms/step -
loss: 0.0459 - mse: 0.0052 - val_loss: 0.0479 - val_mse: 0.0057
Epoch 81/200
30/30 2s 64ms/step -
loss: 0.0470 - mse: 0.0054 - val_loss: 0.0488 - val_mse: 0.0058
Epoch 82/200
30/30 3s 64ms/step -
loss: 0.0484 - mse: 0.0057 - val_loss: 0.0480 - val_mse: 0.0057
Epoch 83/200
30/30 2s 64ms/step -
loss: 0.0463 - mse: 0.0052 - val_loss: 0.0482 - val_mse: 0.0057
Epoch 84/200
30/30 2s 66ms/step -
loss: 0.0490 - mse: 0.0058 - val_loss: 0.0482 - val_mse: 0.0057
Epoch 85/200
30/30 2s 66ms/step -
loss: 0.0471 - mse: 0.0056 - val_loss: 0.0492 - val_mse: 0.0057
Epoch 86/200
30/30 2s 65ms/step -
loss: 0.0462 - mse: 0.0052 - val_loss: 0.0482 - val_mse: 0.0057
Epoch 87/200
30/30 2s 64ms/step -
loss: 0.0472 - mse: 0.0056 - val_loss: 0.0482 - val_mse: 0.0056
Epoch 88/200
30/30 2s 66ms/step -
loss: 0.0466 - mse: 0.0055 - val_loss: 0.0487 - val_mse: 0.0057
Epoch 89/200
30/30 2s 64ms/step -
loss: 0.0480 - mse: 0.0057 - val_loss: 0.0479 - val_mse: 0.0057

Epoch 90/200
30/30 3s 64ms/step -
loss: 0.0474 - mse: 0.0055 - val_loss: 0.0484 - val_mse: 0.0057
Epoch 91/200
30/30 3s 64ms/step -
loss: 0.0485 - mse: 0.0057 - val_loss: 0.0485 - val_mse: 0.0058
Epoch 92/200
30/30 3s 65ms/step -
loss: 0.0459 - mse: 0.0053 - val_loss: 0.0488 - val_mse: 0.0057
Epoch 93/200
30/30 3s 64ms/step -
loss: 0.0486 - mse: 0.0059 - val_loss: 0.0489 - val_mse: 0.0057
Epoch 94/200
30/30 3s 64ms/step -
loss: 0.0479 - mse: 0.0056 - val_loss: 0.0477 - val_mse: 0.0056
Epoch 95/200
30/30 3s 64ms/step -
loss: 0.0459 - mse: 0.0053 - val_loss: 0.0478 - val_mse: 0.0056
Epoch 96/200
30/30 3s 64ms/step -
loss: 0.0478 - mse: 0.0057 - val_loss: 0.0475 - val_mse: 0.0056
Epoch 97/200
30/30 2s 67ms/step -
loss: 0.0468 - mse: 0.0055 - val_loss: 0.0490 - val_mse: 0.0057
Epoch 98/200
30/30 2s 66ms/step -
loss: 0.0492 - mse: 0.0058 - val_loss: 0.0482 - val_mse: 0.0056
Epoch 99/200
30/30 2s 64ms/step -
loss: 0.0474 - mse: 0.0055 - val_loss: 0.0481 - val_mse: 0.0057
Epoch 100/200
30/30 2s 64ms/step -
loss: 0.0484 - mse: 0.0057 - val_loss: 0.0479 - val_mse: 0.0057
Epoch 101/200
30/30 2s 64ms/step -
loss: 0.0468 - mse: 0.0055 - val_loss: 0.0480 - val_mse: 0.0057
Epoch 102/200
30/30 3s 64ms/step -
loss: 0.0471 - mse: 0.0056 - val_loss: 0.0477 - val_mse: 0.0056
Epoch 103/200
30/30 2s 65ms/step -
loss: 0.0465 - mse: 0.0054 - val_loss: 0.0520 - val_mse: 0.0061
Epoch 104/200
30/30 3s 64ms/step -
loss: 0.0475 - mse: 0.0055 - val_loss: 0.0480 - val_mse: 0.0057
Epoch 105/200
30/30 3s 64ms/step -
loss: 0.0460 - mse: 0.0053 - val_loss: 0.0478 - val_mse: 0.0056

Epoch 106/200
30/30 3s 67ms/step -
loss: 0.0462 - mse: 0.0054 - val_loss: 0.0487 - val_mse: 0.0058
Epoch 107/200
30/30 2s 64ms/step -
loss: 0.0463 - mse: 0.0053 - val_loss: 0.0488 - val_mse: 0.0058
Epoch 108/200
30/30 2s 66ms/step -
loss: 0.0464 - mse: 0.0054 - val_loss: 0.0493 - val_mse: 0.0057
Epoch 109/200
30/30 2s 65ms/step -
loss: 0.0469 - mse: 0.0055 - val_loss: 0.0477 - val_mse: 0.0057
Epoch 110/200
30/30 2s 64ms/step -
loss: 0.0462 - mse: 0.0053 - val_loss: 0.0477 - val_mse: 0.0056
Epoch 111/200
30/30 2s 63ms/step -
loss: 0.0455 - mse: 0.0052 - val_loss: 0.0479 - val_mse: 0.0056
Epoch 112/200
30/30 3s 64ms/step -
loss: 0.0474 - mse: 0.0056 - val_loss: 0.0480 - val_mse: 0.0057
Epoch 113/200
30/30 2s 63ms/step -
loss: 0.0464 - mse: 0.0052 - val_loss: 0.0478 - val_mse: 0.0057
Epoch 114/200
30/30 2s 67ms/step -
loss: 0.0487 - mse: 0.0057 - val_loss: 0.0474 - val_mse: 0.0056
Epoch 115/200
30/30 2s 65ms/step -
loss: 0.0470 - mse: 0.0054 - val_loss: 0.0491 - val_mse: 0.0057
Epoch 116/200
30/30 3s 64ms/step -
loss: 0.0481 - mse: 0.0056 - val_loss: 0.0477 - val_mse: 0.0057
Epoch 117/200
30/30 3s 67ms/step -
loss: 0.0461 - mse: 0.0053 - val_loss: 0.0479 - val_mse: 0.0057
Epoch 118/200
30/30 2s 64ms/step -
loss: 0.0480 - mse: 0.0058 - val_loss: 0.0478 - val_mse: 0.0057
Epoch 119/200
30/30 3s 66ms/step -
loss: 0.0479 - mse: 0.0058 - val_loss: 0.0482 - val_mse: 0.0057
Epoch 120/200
30/30 2s 64ms/step -
loss: 0.0471 - mse: 0.0056 - val_loss: 0.0474 - val_mse: 0.0057
Epoch 121/200
30/30 3s 65ms/step -
loss: 0.0474 - mse: 0.0056 - val_loss: 0.0480 - val_mse: 0.0057

Epoch 122/200
30/30 3s 64ms/step -
loss: 0.0469 - mse: 0.0054 - val_loss: 0.0478 - val_mse: 0.0056
Epoch 123/200
30/30 3s 64ms/step -
loss: 0.0457 - mse: 0.0052 - val_loss: 0.0475 - val_mse: 0.0056
Epoch 124/200
30/30 3s 64ms/step -
loss: 0.0473 - mse: 0.0054 - val_loss: 0.0475 - val_mse: 0.0057
Epoch 125/200
30/30 2s 64ms/step -
loss: 0.0474 - mse: 0.0055 - val_loss: 0.0479 - val_mse: 0.0056
Epoch 126/200
30/30 2s 65ms/step -
loss: 0.0466 - mse: 0.0055 - val_loss: 0.0473 - val_mse: 0.0055
Epoch 127/200
30/30 2s 64ms/step -
loss: 0.0454 - mse: 0.0053 - val_loss: 0.0481 - val_mse: 0.0058
Epoch 128/200
30/30 3s 64ms/step -
loss: 0.0471 - mse: 0.0056 - val_loss: 0.0477 - val_mse: 0.0056
Epoch 129/200
30/30 3s 64ms/step -
loss: 0.0471 - mse: 0.0056 - val_loss: 0.0490 - val_mse: 0.0057
Epoch 130/200
30/30 3s 64ms/step -
loss: 0.0471 - mse: 0.0054 - val_loss: 0.0484 - val_mse: 0.0057
Epoch 131/200
30/30 3s 68ms/step -
loss: 0.0472 - mse: 0.0055 - val_loss: 0.0471 - val_mse: 0.0055
Epoch 132/200
30/30 2s 66ms/step -
loss: 0.0456 - mse: 0.0053 - val_loss: 0.0476 - val_mse: 0.0055
Epoch 133/200
30/30 2s 64ms/step -
loss: 0.0464 - mse: 0.0055 - val_loss: 0.0471 - val_mse: 0.0056
Epoch 134/200
30/30 3s 64ms/step -
loss: 0.0447 - mse: 0.0051 - val_loss: 0.0481 - val_mse: 0.0057
Epoch 135/200
30/30 3s 64ms/step -
loss: 0.0439 - mse: 0.0050 - val_loss: 0.0472 - val_mse: 0.0056
Epoch 136/200
30/30 2s 64ms/step -
loss: 0.0476 - mse: 0.0054 - val_loss: 0.0476 - val_mse: 0.0056
Epoch 137/200
30/30 2s 63ms/step -
loss: 0.0468 - mse: 0.0055 - val_loss: 0.0475 - val_mse: 0.0056

Epoch 138/200
30/30 3s 67ms/step -
loss: 0.0459 - mse: 0.0053 - val_loss: 0.0474 - val_mse: 0.0056
Epoch 139/200
30/30 2s 64ms/step -
loss: 0.0459 - mse: 0.0052 - val_loss: 0.0480 - val_mse: 0.0057
Epoch 140/200
30/30 3s 64ms/step -
loss: 0.0466 - mse: 0.0054 - val_loss: 0.0474 - val_mse: 0.0056
Epoch 141/200
30/30 3s 64ms/step -
loss: 0.0430 - mse: 0.0047 - val_loss: 0.0471 - val_mse: 0.0055
Epoch 142/200
30/30 3s 64ms/step -
loss: 0.0461 - mse: 0.0053 - val_loss: 0.0473 - val_mse: 0.0055
Epoch 143/200
30/30 2s 67ms/step -
loss: 0.0456 - mse: 0.0052 - val_loss: 0.0481 - val_mse: 0.0057
Epoch 144/200
30/30 2s 64ms/step -
loss: 0.0457 - mse: 0.0053 - val_loss: 0.0485 - val_mse: 0.0058
Epoch 145/200
30/30 3s 65ms/step -
loss: 0.0473 - mse: 0.0057 - val_loss: 0.0475 - val_mse: 0.0055
Epoch 146/200
30/30 3s 64ms/step -
loss: 0.0458 - mse: 0.0053 - val_loss: 0.0470 - val_mse: 0.0055
Epoch 147/200
30/30 2s 64ms/step -
loss: 0.0443 - mse: 0.0051 - val_loss: 0.0468 - val_mse: 0.0055
Epoch 148/200
30/30 3s 67ms/step -
loss: 0.0452 - mse: 0.0051 - val_loss: 0.0475 - val_mse: 0.0056
Epoch 149/200
30/30 3s 66ms/step -
loss: 0.0451 - mse: 0.0052 - val_loss: 0.0470 - val_mse: 0.0055
Epoch 150/200
30/30 2s 67ms/step -
loss: 0.0459 - mse: 0.0053 - val_loss: 0.0482 - val_mse: 0.0057
Epoch 151/200
30/30 2s 66ms/step -
loss: 0.0459 - mse: 0.0053 - val_loss: 0.0470 - val_mse: 0.0055
Epoch 152/200
30/30 2s 64ms/step -
loss: 0.0456 - mse: 0.0052 - val_loss: 0.0486 - val_mse: 0.0057
Epoch 153/200
30/30 2s 66ms/step -
loss: 0.0466 - mse: 0.0053 - val_loss: 0.0485 - val_mse: 0.0057

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Epoch 154/200
30/30          2s 64ms/step -
loss: 0.0448 - mse: 0.0051 - val_loss: 0.0481 - val_mse: 0.0057
Epoch 155/200
30/30          3s 67ms/step -
loss: 0.0458 - mse: 0.0053 - val_loss: 0.0472 - val_mse: 0.0056
Epoch 156/200
30/30          2s 64ms/step -
loss: 0.0466 - mse: 0.0054 - val_loss: 0.0472 - val_mse: 0.0056
Epoch 157/200
30/30          2s 64ms/step -
loss: 0.0449 - mse: 0.0053 - val_loss: 0.0468 - val_mse: 0.0056
Epoch 158/200
30/30          2s 64ms/step -
loss: 0.0454 - mse: 0.0053 - val_loss: 0.0479 - val_mse: 0.0057
Epoch 159/200
30/30          3s 66ms/step -
loss: 0.0480 - mse: 0.0057 - val_loss: 0.0477 - val_mse: 0.0057
Epoch 160/200
30/30          2s 64ms/step -
loss: 0.0441 - mse: 0.0051 - val_loss: 0.0475 - val_mse: 0.0056
Epoch 161/200
30/30          3s 65ms/step -
loss: 0.0456 - mse: 0.0052 - val_loss: 0.0471 - val_mse: 0.0055
Epoch 162/200
30/30          2s 64ms/step -
loss: 0.0454 - mse: 0.0053 - val_loss: 0.0468 - val_mse: 0.0055
Epoch 163/200
30/30          3s 66ms/step -
loss: 0.0452 - mse: 0.0054 - val_loss: 0.0470 - val_mse: 0.0055
Epoch 164/200
30/30          2s 64ms/step -
loss: 0.0441 - mse: 0.0050 - val_loss: 0.0482 - val_mse: 0.0057
Epoch 165/200
30/30          2s 66ms/step -
loss: 0.0467 - mse: 0.0054 - val_loss: 0.0474 - val_mse: 0.0056
Epoch 166/200
30/30          2s 64ms/step -
loss: 0.0460 - mse: 0.0054 - val_loss: 0.0485 - val_mse: 0.0056
Epoch 167/200
30/30          2s 68ms/step -
loss: 0.0453 - mse: 0.0051 - val_loss: 0.0477 - val_mse: 0.0057

```

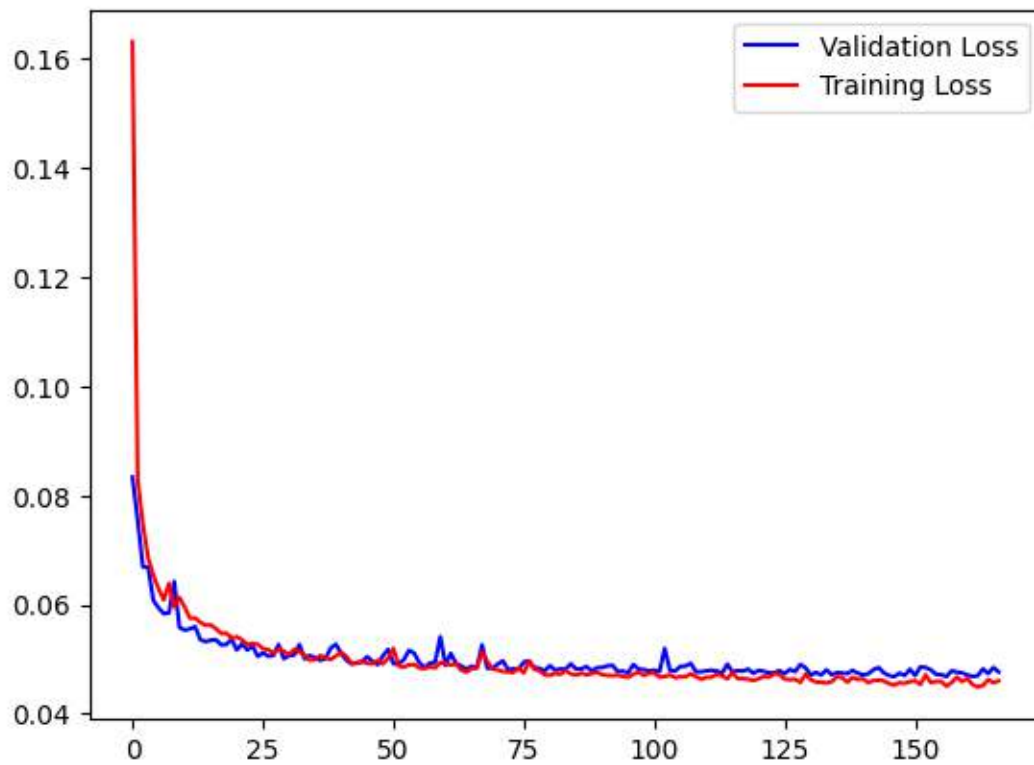
```

[ ]: # Plotting the training and validation loss
plt.plot(history_seq_rgb.history['val_loss'],c="b")
plt.plot(history_seq_rgb.history['loss'],c="r")
plt.legend(['Validation Loss','Training Loss'])

```



```
plt.show()
```



Depiction of Model Performance on Images from the Training Set

```
[ ]: show_predictions(seq_rgb_model, X, Y, num_samples=5)
```

1/1

1s 664ms/step



Depiction of Model Performance on Images from the Test Set

```
[ ]: show_predictions(seq_rgb_model, X_test, Y_test, num_samples=5)
```

1/1 0s 49ms/step



The output of the model happens to capture the details of the structure; however the overall image remains mostly gray, with beige patches appearing sporadically. In the example above, the model successfully preserved the facial features and outlines of the subject, but failed to capture the vivid colors of the clothing, flowers and background. The overall tone of the image is muted, lacking the vibrancy of the ground truth.

5.3 Model 2: Simple CNN

```
[ ]: # Defining the architecture for the simple CNN model
def build_simple_cnn(input_shape=(120, 176, 1)):
    inputs = layers.Input(shape=input_shape)

    x = layers.Conv2D(64, (3, 3), activation='relu', padding='same')(inputs)
    x = layers.Conv2D(128, (3, 3), activation='relu', padding='same')(x)
    x = layers.MaxPooling2D((2, 2))(x)

    x = layers.Conv2D(128, (3, 3), activation='relu', padding='same')(x)

    x = layers.Conv2DTranspose(64, (2, 2), strides=2, padding='same')(x)
    x = layers.ReLU()(x)
    outputs = layers.Conv2D(3, (3, 3), activation='sigmoid', padding='same')(x)

    model = models.Model(inputs=inputs, outputs=outputs)
    model.compile(optimizer='adam', loss='mse', metrics=['mae'])
    return model

model_simple_cnn = build_simple_cnn()
```

```
[ ]: # Defining the path to save the best model
model_checkpoint_path = os.path.join(best_models_path, 'simple_cnn.keras')
model_checkpoint = ModelCheckpoint(model_checkpoint_path, monitor='val_loss',
    verbose=1, save_best_only=True, mode='min')

# Fitting the model to the training data
history_simple_cnn = model_simple_cnn.fit(X, Y, batch_size=16, epochs=200,
    validation_split=0.1, callbacks=[early_stop, model_checkpoint])

best_simple_cnn = keras.models.load_model(os.path.join(best_models_path,
    'simple_cnn.keras'))
```

Epoch 1/200

30/30 0s 310ms/step -

loss: 0.0632 - mae: 0.2099

Epoch 1: val_loss improved from inf to 0.01775, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models

```

Portrait/simple_cnn.keras
30/30          25s 498ms/step -
loss: 0.0627 - mae: 0.2086 - val_loss: 0.0177 - val_mae: 0.1050
Epoch 2/200
30/30          0s 71ms/step -
loss: 0.0138 - mae: 0.0879
Epoch 2: val_loss improved from 0.01775 to 0.00890, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/simple_cnn.keras
30/30          21s 79ms/step -
loss: 0.0138 - mae: 0.0877 - val_loss: 0.0089 - val_mae: 0.0677
Epoch 3/200
30/30          0s 69ms/step -
loss: 0.0094 - mae: 0.0684
Epoch 3: val_loss improved from 0.00890 to 0.00803, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/simple_cnn.keras
30/30          2s 78ms/step -
loss: 0.0094 - mae: 0.0684 - val_loss: 0.0080 - val_mae: 0.0640
Epoch 4/200
30/30          0s 69ms/step -
loss: 0.0084 - mae: 0.0650
Epoch 4: val_loss improved from 0.00803 to 0.00758, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/simple_cnn.keras
30/30          2s 78ms/step -
loss: 0.0084 - mae: 0.0649 - val_loss: 0.0076 - val_mae: 0.0607
Epoch 5/200
30/30          0s 70ms/step -
loss: 0.0078 - mae: 0.0622
Epoch 5: val_loss improved from 0.00758 to 0.00720, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/simple_cnn.keras
30/30          3s 80ms/step -
loss: 0.0078 - mae: 0.0622 - val_loss: 0.0072 - val_mae: 0.0594
Epoch 6/200
30/30          0s 71ms/step -
loss: 0.0077 - mae: 0.0610
Epoch 6: val_loss improved from 0.00720 to 0.00667, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/simple_cnn.keras
30/30          2s 77ms/step -
loss: 0.0077 - mae: 0.0610 - val_loss: 0.0067 - val_mae: 0.0558
Epoch 7/200
30/30          0s 70ms/step -
loss: 0.0076 - mae: 0.0602
Epoch 7: val_loss improved from 0.00667 to 0.00653, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models

```

```

Portrait/simple_cnn.keras
30/30          2s 78ms/step -
loss: 0.0076 - mae: 0.0601 - val_loss: 0.0065 - val_mae: 0.0548
Epoch 8/200
30/30          0s 70ms/step -
loss: 0.0071 - mae: 0.0579
Epoch 8: val_loss improved from 0.00653 to 0.00631, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/simple_cnn.keras
30/30          2s 79ms/step -
loss: 0.0071 - mae: 0.0578 - val_loss: 0.0063 - val_mae: 0.0538
Epoch 9/200
30/30          0s 70ms/step -
loss: 0.0067 - mae: 0.0562
Epoch 9: val_loss improved from 0.00631 to 0.00630, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/simple_cnn.keras
30/30          2s 76ms/step -
loss: 0.0067 - mae: 0.0562 - val_loss: 0.0063 - val_mae: 0.0530
Epoch 10/200
30/30          0s 70ms/step -
loss: 0.0067 - mae: 0.0562
Epoch 10: val_loss did not improve from 0.00630
30/30          2s 77ms/step -
loss: 0.0067 - mae: 0.0562 - val_loss: 0.0067 - val_mae: 0.0568
Epoch 11/200
30/30          0s 72ms/step -
loss: 0.0069 - mae: 0.0565
Epoch 11: val_loss improved from 0.00630 to 0.00622, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/simple_cnn.keras
30/30          3s 81ms/step -
loss: 0.0069 - mae: 0.0565 - val_loss: 0.0062 - val_mae: 0.0522
Epoch 12/200
30/30          0s 71ms/step -
loss: 0.0063 - mae: 0.0533
Epoch 12: val_loss improved from 0.00622 to 0.00600, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/simple_cnn.keras
30/30          2s 77ms/step -
loss: 0.0063 - mae: 0.0533 - val_loss: 0.0060 - val_mae: 0.0512
Epoch 13/200
30/30          0s 71ms/step -
loss: 0.0065 - mae: 0.0549
Epoch 13: val_loss did not improve from 0.00600
30/30          2s 75ms/step -
loss: 0.0065 - mae: 0.0549 - val_loss: 0.0062 - val_mae: 0.0523
Epoch 14/200

```

```

30/30          0s 71ms/step -
loss: 0.0061 - mae: 0.0529
Epoch 14: val_loss did not improve from 0.00600
30/30          3s 75ms/step -
loss: 0.0061 - mae: 0.0530 - val_loss: 0.0064 - val_mae: 0.0554
Epoch 15/200
30/30          0s 71ms/step -
loss: 0.0063 - mae: 0.0537
Epoch 15: val_loss improved from 0.00600 to 0.00597, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/simple_cnn.keras
30/30          2s 77ms/step -
loss: 0.0063 - mae: 0.0537 - val_loss: 0.0060 - val_mae: 0.0512
Epoch 16/200
30/30          0s 72ms/step -
loss: 0.0059 - mae: 0.0512
Epoch 16: val_loss did not improve from 0.00597
30/30          2s 79ms/step -
loss: 0.0059 - mae: 0.0512 - val_loss: 0.0060 - val_mae: 0.0506
Epoch 17/200
30/30          0s 72ms/step -
loss: 0.0061 - mae: 0.0512
Epoch 17: val_loss did not improve from 0.00597
30/30          2s 78ms/step -
loss: 0.0061 - mae: 0.0512 - val_loss: 0.0060 - val_mae: 0.0524
Epoch 18/200
30/30          0s 71ms/step -
loss: 0.0060 - mae: 0.0517
Epoch 18: val_loss did not improve from 0.00597
30/30          2s 76ms/step -
loss: 0.0060 - mae: 0.0517 - val_loss: 0.0064 - val_mae: 0.0542
Epoch 19/200
30/30          0s 71ms/step -
loss: 0.0060 - mae: 0.0517
Epoch 19: val_loss improved from 0.00597 to 0.00581, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/simple_cnn.keras
30/30          2s 78ms/step -
loss: 0.0060 - mae: 0.0517 - val_loss: 0.0058 - val_mae: 0.0501
Epoch 20/200
30/30          0s 72ms/step -
loss: 0.0059 - mae: 0.0509
Epoch 20: val_loss improved from 0.00581 to 0.00573, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/simple_cnn.keras
30/30          2s 78ms/step -
loss: 0.0059 - mae: 0.0509 - val_loss: 0.0057 - val_mae: 0.0493
Epoch 21/200

```



```

30/30          0s 72ms/step -
loss: 0.0059 - mae: 0.0511
Epoch 21: val_loss did not improve from 0.00573
30/30          3s 79ms/step -
loss: 0.0059 - mae: 0.0511 - val_loss: 0.0058 - val_mae: 0.0494
Epoch 22/200
30/30          0s 73ms/step -
loss: 0.0060 - mae: 0.0511
Epoch 22: val_loss improved from 0.00573 to 0.00573, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/simple_cnn.keras
30/30          3s 83ms/step -
loss: 0.0060 - mae: 0.0510 - val_loss: 0.0057 - val_mae: 0.0492
Epoch 23/200
30/30          0s 72ms/step -
loss: 0.0059 - mae: 0.0504
Epoch 23: val_loss did not improve from 0.00573
30/30          2s 78ms/step -
loss: 0.0059 - mae: 0.0504 - val_loss: 0.0058 - val_mae: 0.0495
Epoch 24/200
30/30          0s 72ms/step -
loss: 0.0058 - mae: 0.0510
Epoch 24: val_loss did not improve from 0.00573
30/30          2s 76ms/step -
loss: 0.0058 - mae: 0.0510 - val_loss: 0.0058 - val_mae: 0.0490
Epoch 25/200
30/30          0s 72ms/step -
loss: 0.0056 - mae: 0.0490
Epoch 25: val_loss improved from 0.00573 to 0.00569, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/simple_cnn.keras
30/30          3s 80ms/step -
loss: 0.0056 - mae: 0.0490 - val_loss: 0.0057 - val_mae: 0.0494
Epoch 26/200
30/30          0s 72ms/step -
loss: 0.0065 - mae: 0.0530
Epoch 26: val_loss did not improve from 0.00569
30/30          2s 79ms/step -
loss: 0.0064 - mae: 0.0529 - val_loss: 0.0057 - val_mae: 0.0492
Epoch 27/200
30/30          0s 73ms/step -
loss: 0.0060 - mae: 0.0510
Epoch 27: val_loss did not improve from 0.00569
30/30          3s 79ms/step -
loss: 0.0060 - mae: 0.0510 - val_loss: 0.0058 - val_mae: 0.0508
Epoch 28/200
30/30          0s 73ms/step -
loss: 0.0056 - mae: 0.0493

```

Epoch 28: val_loss did not improve from 0.00569
30/30 2s 78ms/step -
loss: 0.0056 - mae: 0.0493 - val_loss: 0.0058 - val_mae: 0.0502
Epoch 29/200
30/30 0s 73ms/step -
loss: 0.0057 - mae: 0.0499
Epoch 29: val_loss did not improve from 0.00569
30/30 2s 77ms/step -
loss: 0.0057 - mae: 0.0499 - val_loss: 0.0058 - val_mae: 0.0498
Epoch 30/200
30/30 0s 73ms/step -
loss: 0.0059 - mae: 0.0501
Epoch 30: val_loss improved from 0.00569 to 0.00563, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/simple_cnn.keras
30/30 2s 79ms/step -
loss: 0.0059 - mae: 0.0501 - val_loss: 0.0056 - val_mae: 0.0482
Epoch 31/200
30/30 0s 74ms/step -
loss: 0.0055 - mae: 0.0485
Epoch 31: val_loss did not improve from 0.00563
30/30 3s 83ms/step -
loss: 0.0055 - mae: 0.0485 - val_loss: 0.0058 - val_mae: 0.0492
Epoch 32/200
30/30 0s 74ms/step -
loss: 0.0057 - mae: 0.0495
Epoch 32: val_loss did not improve from 0.00563
30/30 2s 79ms/step -
loss: 0.0057 - mae: 0.0495 - val_loss: 0.0065 - val_mae: 0.0569
Epoch 33/200
30/30 0s 74ms/step -
loss: 0.0055 - mae: 0.0490
Epoch 33: val_loss did not improve from 0.00563
30/30 2s 80ms/step -
loss: 0.0055 - mae: 0.0490 - val_loss: 0.0056 - val_mae: 0.0483
Epoch 34/200
30/30 0s 74ms/step -
loss: 0.0055 - mae: 0.0483
Epoch 34: val_loss did not improve from 0.00563
30/30 3s 80ms/step -
loss: 0.0055 - mae: 0.0483 - val_loss: 0.0057 - val_mae: 0.0499
Epoch 35/200
30/30 0s 74ms/step -
loss: 0.0061 - mae: 0.0516
Epoch 35: val_loss did not improve from 0.00563
30/30 2s 78ms/step -
loss: 0.0061 - mae: 0.0516 - val_loss: 0.0056 - val_mae: 0.0490
Epoch 36/200

```

30/30          0s 74ms/step -
loss: 0.0054 - mae: 0.0479
Epoch 36: val_loss did not improve from 0.00563
30/30          3s 78ms/step -
loss: 0.0055 - mae: 0.0479 - val_loss: 0.0059 - val_mae: 0.0496
Epoch 37/200
30/30          0s 74ms/step -
loss: 0.0057 - mae: 0.0492
Epoch 37: val_loss improved from 0.00563 to 0.00557, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/simple_cnn.keras
30/30          3s 84ms/step -
loss: 0.0057 - mae: 0.0492 - val_loss: 0.0056 - val_mae: 0.0481
Epoch 38/200
30/30          0s 75ms/step -
loss: 0.0059 - mae: 0.0491
Epoch 38: val_loss did not improve from 0.00557
30/30          2s 79ms/step -
loss: 0.0059 - mae: 0.0491 - val_loss: 0.0057 - val_mae: 0.0483
Epoch 39/200
30/30          0s 74ms/step -
loss: 0.0058 - mae: 0.0497
Epoch 39: val_loss did not improve from 0.00557
30/30          2s 80ms/step -
loss: 0.0058 - mae: 0.0497 - val_loss: 0.0059 - val_mae: 0.0509
Epoch 40/200
30/30          0s 75ms/step -
loss: 0.0056 - mae: 0.0499
Epoch 40: val_loss did not improve from 0.00557
30/30          2s 81ms/step -
loss: 0.0056 - mae: 0.0499 - val_loss: 0.0060 - val_mae: 0.0504
Epoch 41/200
30/30          0s 74ms/step -
loss: 0.0054 - mae: 0.0478
Epoch 41: val_loss improved from 0.00557 to 0.00555, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/simple_cnn.keras
30/30          3s 82ms/step -
loss: 0.0054 - mae: 0.0478 - val_loss: 0.0055 - val_mae: 0.0481
Epoch 42/200
30/30          0s 75ms/step -
loss: 0.0056 - mae: 0.0481
Epoch 42: val_loss did not improve from 0.00555
30/30          3s 81ms/step -
loss: 0.0056 - mae: 0.0481 - val_loss: 0.0056 - val_mae: 0.0478
Epoch 43/200
30/30          0s 75ms/step -
loss: 0.0052 - mae: 0.0472

```

Epoch 43: val_loss did not improve from 0.00555
30/30 3s 82ms/step -
loss: 0.0052 - mae: 0.0472 - val_loss: 0.0056 - val_mae: 0.0489
Epoch 44/200
30/30 0s 75ms/step -
loss: 0.0060 - mae: 0.0507
Epoch 44: val_loss did not improve from 0.00555
30/30 3s 81ms/step -
loss: 0.0060 - mae: 0.0507 - val_loss: 0.0056 - val_mae: 0.0478
Epoch 45/200
30/30 0s 75ms/step -
loss: 0.0058 - mae: 0.0504
Epoch 45: val_loss did not improve from 0.00555
30/30 2s 79ms/step -
loss: 0.0058 - mae: 0.0504 - val_loss: 0.0057 - val_mae: 0.0493
Epoch 46/200
30/30 0s 74ms/step -
loss: 0.0055 - mae: 0.0481
Epoch 46: val_loss did not improve from 0.00555
30/30 3s 78ms/step -
loss: 0.0055 - mae: 0.0481 - val_loss: 0.0057 - val_mae: 0.0496
Epoch 47/200
30/30 0s 75ms/step -
loss: 0.0054 - mae: 0.0475
Epoch 47: val_loss did not improve from 0.00555
30/30 3s 79ms/step -
loss: 0.0054 - mae: 0.0476 - val_loss: 0.0056 - val_mae: 0.0477
Epoch 48/200
30/30 0s 76ms/step -
loss: 0.0055 - mae: 0.0483
Epoch 48: val_loss did not improve from 0.00555
30/30 3s 83ms/step -
loss: 0.0055 - mae: 0.0482 - val_loss: 0.0056 - val_mae: 0.0477
Epoch 49/200
30/30 0s 75ms/step -
loss: 0.0053 - mae: 0.0471
Epoch 49: val_loss improved from 0.00555 to 0.00553, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/simple_cnn.keras
30/30 3s 83ms/step -
loss: 0.0053 - mae: 0.0471 - val_loss: 0.0055 - val_mae: 0.0477
Epoch 50/200
30/30 0s 75ms/step -
loss: 0.0057 - mae: 0.0484
Epoch 50: val_loss did not improve from 0.00553
30/30 2s 81ms/step -
loss: 0.0057 - mae: 0.0483 - val_loss: 0.0057 - val_mae: 0.0494
Epoch 51/200

```

30/30          0s 75ms/step -
loss: 0.0055 - mae: 0.0471
Epoch 51: val_loss improved from 0.00553 to 0.00549, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/simple_cnn.keras
30/30          2s 82ms/step -
loss: 0.0055 - mae: 0.0471 - val_loss: 0.0055 - val_mae: 0.0476
Epoch 52/200
30/30          0s 75ms/step -
loss: 0.0053 - mae: 0.0470
Epoch 52: val_loss did not improve from 0.00549
30/30          2s 79ms/step -
loss: 0.0053 - mae: 0.0470 - val_loss: 0.0057 - val_mae: 0.0487
Epoch 53/200
30/30          0s 76ms/step -
loss: 0.0054 - mae: 0.0473
Epoch 53: val_loss did not improve from 0.00549
30/30          3s 82ms/step -
loss: 0.0054 - mae: 0.0473 - val_loss: 0.0056 - val_mae: 0.0487
Epoch 54/200
30/30          0s 75ms/step -
loss: 0.0056 - mae: 0.0487
Epoch 54: val_loss did not improve from 0.00549
30/30          2s 79ms/step -
loss: 0.0056 - mae: 0.0486 - val_loss: 0.0055 - val_mae: 0.0475
Epoch 55/200
30/30          0s 75ms/step -
loss: 0.0054 - mae: 0.0471
Epoch 55: val_loss did not improve from 0.00549
30/30          2s 79ms/step -
loss: 0.0054 - mae: 0.0471 - val_loss: 0.0056 - val_mae: 0.0487
Epoch 56/200
30/30          0s 75ms/step -
loss: 0.0054 - mae: 0.0479
Epoch 56: val_loss did not improve from 0.00549
30/30          3s 79ms/step -
loss: 0.0054 - mae: 0.0479 - val_loss: 0.0055 - val_mae: 0.0473
Epoch 57/200
30/30          0s 75ms/step -
loss: 0.0052 - mae: 0.0464
Epoch 57: val_loss did not improve from 0.00549
30/30          3s 79ms/step -
loss: 0.0052 - mae: 0.0464 - val_loss: 0.0062 - val_mae: 0.0531
Epoch 58/200
30/30          0s 75ms/step -
loss: 0.0057 - mae: 0.0493
Epoch 58: val_loss did not improve from 0.00549
30/30          2s 82ms/step -

```

```

loss: 0.0057 - mae: 0.0493 - val_loss: 0.0056 - val_mae: 0.0485
Epoch 59/200
30/30          0s 75ms/step -
loss: 0.0055 - mae: 0.0479
Epoch 59: val_loss did not improve from 0.00549
30/30          2s 79ms/step -
loss: 0.0055 - mae: 0.0479 - val_loss: 0.0058 - val_mae: 0.0504
Epoch 60/200
30/30          0s 75ms/step -
loss: 0.0049 - mae: 0.0454
Epoch 60: val_loss did not improve from 0.00549
30/30          3s 79ms/step -
loss: 0.0049 - mae: 0.0455 - val_loss: 0.0055 - val_mae: 0.0477
Epoch 61/200
30/30          0s 75ms/step -
loss: 0.0053 - mae: 0.0462
Epoch 61: val_loss did not improve from 0.00549
30/30          2s 81ms/step -
loss: 0.0053 - mae: 0.0462 - val_loss: 0.0056 - val_mae: 0.0475
Epoch 62/200
30/30          0s 75ms/step -
loss: 0.0055 - mae: 0.0471
Epoch 62: val_loss did not improve from 0.00549
30/30          2s 81ms/step -
loss: 0.0055 - mae: 0.0471 - val_loss: 0.0055 - val_mae: 0.0478
Epoch 63/200
30/30          0s 74ms/step -
loss: 0.0057 - mae: 0.0491
Epoch 63: val_loss did not improve from 0.00549
30/30          3s 81ms/step -
loss: 0.0057 - mae: 0.0490 - val_loss: 0.0057 - val_mae: 0.0482
Epoch 64/200
30/30          0s 75ms/step -
loss: 0.0055 - mae: 0.0476
Epoch 64: val_loss did not improve from 0.00549
30/30          2s 80ms/step -
loss: 0.0055 - mae: 0.0476 - val_loss: 0.0060 - val_mae: 0.0512
Epoch 65/200
30/30          0s 74ms/step -
loss: 0.0053 - mae: 0.0468
Epoch 65: val_loss improved from 0.00549 to 0.00548, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/simple_cnn.keras
30/30          3s 83ms/step -
loss: 0.0053 - mae: 0.0468 - val_loss: 0.0055 - val_mae: 0.0470
Epoch 66/200
30/30          0s 75ms/step -
loss: 0.0053 - mae: 0.0460

```

Epoch 66: val_loss did not improve from 0.00548
30/30 2s 81ms/step -
loss: 0.0053 - mae: 0.0460 - val_loss: 0.0056 - val_mae: 0.0482
Epoch 67/200
30/30 0s 75ms/step -
loss: 0.0051 - mae: 0.0460
Epoch 67: val_loss did not improve from 0.00548
30/30 3s 81ms/step -
loss: 0.0051 - mae: 0.0460 - val_loss: 0.0055 - val_mae: 0.0472
Epoch 68/200
30/30 0s 75ms/step -
loss: 0.0055 - mae: 0.0467
Epoch 68: val_loss did not improve from 0.00548
30/30 3s 80ms/step -
loss: 0.0055 - mae: 0.0467 - val_loss: 0.0056 - val_mae: 0.0483
Epoch 69/200
30/30 0s 75ms/step -
loss: 0.0054 - mae: 0.0474
Epoch 69: val_loss did not improve from 0.00548
30/30 2s 83ms/step -
loss: 0.0054 - mae: 0.0473 - val_loss: 0.0055 - val_mae: 0.0476
Epoch 70/200
30/30 0s 75ms/step -
loss: 0.0053 - mae: 0.0470
Epoch 70: val_loss improved from 0.00548 to 0.00545, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/simple_cnn.keras
30/30 2s 83ms/step -
loss: 0.0053 - mae: 0.0470 - val_loss: 0.0054 - val_mae: 0.0470
Epoch 71/200
30/30 0s 75ms/step -
loss: 0.0052 - mae: 0.0460
Epoch 71: val_loss did not improve from 0.00545
30/30 2s 81ms/step -
loss: 0.0052 - mae: 0.0460 - val_loss: 0.0058 - val_mae: 0.0500
Epoch 72/200
30/30 0s 75ms/step -
loss: 0.0057 - mae: 0.0488
Epoch 72: val_loss did not improve from 0.00545
30/30 2s 79ms/step -
loss: 0.0057 - mae: 0.0487 - val_loss: 0.0055 - val_mae: 0.0473
Epoch 73/200
30/30 0s 75ms/step -
loss: 0.0052 - mae: 0.0456
Epoch 73: val_loss did not improve from 0.00545
30/30 3s 79ms/step -
loss: 0.0052 - mae: 0.0456 - val_loss: 0.0058 - val_mae: 0.0486
Epoch 74/200

```

30/30          0s 76ms/step -
loss: 0.0057 - mae: 0.0476
Epoch 74: val_loss did not improve from 0.00545
30/30          2s 82ms/step -
loss: 0.0057 - mae: 0.0476 - val_loss: 0.0055 - val_mae: 0.0471
Epoch 75/200
30/30          0s 75ms/step -
loss: 0.0053 - mae: 0.0469
Epoch 75: val_loss did not improve from 0.00545
30/30          2s 79ms/step -
loss: 0.0053 - mae: 0.0469 - val_loss: 0.0056 - val_mae: 0.0475
Epoch 76/200
30/30          0s 75ms/step -
loss: 0.0056 - mae: 0.0482
Epoch 76: val_loss did not improve from 0.00545
30/30          3s 79ms/step -
loss: 0.0055 - mae: 0.0482 - val_loss: 0.0056 - val_mae: 0.0477
Epoch 77/200
30/30          0s 75ms/step -
loss: 0.0053 - mae: 0.0462
Epoch 77: val_loss did not improve from 0.00545
30/30          3s 79ms/step -
loss: 0.0053 - mae: 0.0462 - val_loss: 0.0055 - val_mae: 0.0473
Epoch 78/200
30/30          0s 75ms/step -
loss: 0.0052 - mae: 0.0460
Epoch 78: val_loss did not improve from 0.00545
30/30          2s 81ms/step -
loss: 0.0052 - mae: 0.0460 - val_loss: 0.0057 - val_mae: 0.0482
Epoch 79/200
30/30          0s 75ms/step -
loss: 0.0055 - mae: 0.0469
Epoch 79: val_loss did not improve from 0.00545
30/30          2s 81ms/step -
loss: 0.0055 - mae: 0.0469 - val_loss: 0.0056 - val_mae: 0.0471
Epoch 80/200
30/30          0s 76ms/step -
loss: 0.0056 - mae: 0.0470
Epoch 80: val_loss did not improve from 0.00545
30/30          2s 80ms/step -
loss: 0.0056 - mae: 0.0470 - val_loss: 0.0057 - val_mae: 0.0491
Epoch 81/200
30/30          0s 75ms/step -
loss: 0.0053 - mae: 0.0466
Epoch 81: val_loss did not improve from 0.00545
30/30          2s 81ms/step -
loss: 0.0053 - mae: 0.0466 - val_loss: 0.0056 - val_mae: 0.0479
Epoch 82/200

```



```

30/30          0s 75ms/step -
loss: 0.0052 - mae: 0.0454
Epoch 82: val_loss did not improve from 0.00545
30/30          2s 79ms/step -
loss: 0.0052 - mae: 0.0454 - val_loss: 0.0058 - val_mae: 0.0488
Epoch 83/200
30/30          0s 75ms/step -
loss: 0.0051 - mae: 0.0451
Epoch 83: val_loss did not improve from 0.00545
30/30          2s 81ms/step -
loss: 0.0051 - mae: 0.0451 - val_loss: 0.0056 - val_mae: 0.0477
Epoch 84/200
30/30          0s 75ms/step -
loss: 0.0056 - mae: 0.0474
Epoch 84: val_loss did not improve from 0.00545
30/30          2s 81ms/step -
loss: 0.0056 - mae: 0.0474 - val_loss: 0.0055 - val_mae: 0.0475
Epoch 85/200
30/30          0s 75ms/step -
loss: 0.0051 - mae: 0.0452
Epoch 85: val_loss improved from 0.00545 to 0.00543, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/simple_cnn.keras
30/30          3s 85ms/step -
loss: 0.0051 - mae: 0.0452 - val_loss: 0.0054 - val_mae: 0.0471
Epoch 86/200
30/30          0s 75ms/step -
loss: 0.0053 - mae: 0.0460
Epoch 86: val_loss did not improve from 0.00543
30/30          5s 80ms/step -
loss: 0.0053 - mae: 0.0460 - val_loss: 0.0055 - val_mae: 0.0468
Epoch 87/200
30/30          0s 74ms/step -
loss: 0.0056 - mae: 0.0475
Epoch 87: val_loss did not improve from 0.00543
30/30          2s 79ms/step -
loss: 0.0056 - mae: 0.0475 - val_loss: 0.0058 - val_mae: 0.0503
Epoch 88/200
30/30          0s 74ms/step -
loss: 0.0058 - mae: 0.0483
Epoch 88: val_loss did not improve from 0.00543
30/30          3s 80ms/step -
loss: 0.0057 - mae: 0.0482 - val_loss: 0.0055 - val_mae: 0.0477
Epoch 89/200
30/30          0s 76ms/step -
loss: 0.0054 - mae: 0.0467
Epoch 89: val_loss did not improve from 0.00543
30/30          3s 82ms/step -

```

```

loss: 0.0054 - mae: 0.0466 - val_loss: 0.0055 - val_mae: 0.0469
Epoch 90/200
30/30          0s 75ms/step -
loss: 0.0056 - mae: 0.0470
Epoch 90: val_loss did not improve from 0.00543
30/30          2s 81ms/step -
loss: 0.0055 - mae: 0.0470 - val_loss: 0.0055 - val_mae: 0.0470
Epoch 91/200
30/30          0s 75ms/step -
loss: 0.0050 - mae: 0.0449
Epoch 91: val_loss did not improve from 0.00543
30/30          2s 79ms/step -
loss: 0.0050 - mae: 0.0450 - val_loss: 0.0058 - val_mae: 0.0487
Epoch 92/200
30/30          0s 75ms/step -
loss: 0.0050 - mae: 0.0457
Epoch 92: val_loss did not improve from 0.00543
30/30          3s 79ms/step -
loss: 0.0050 - mae: 0.0457 - val_loss: 0.0054 - val_mae: 0.0466
Epoch 93/200
30/30          0s 74ms/step -
loss: 0.0048 - mae: 0.0440
Epoch 93: val_loss did not improve from 0.00543
30/30          2s 80ms/step -
loss: 0.0048 - mae: 0.0440 - val_loss: 0.0058 - val_mae: 0.0493
Epoch 94/200
30/30          0s 76ms/step -
loss: 0.0049 - mae: 0.0453
Epoch 94: val_loss did not improve from 0.00543
30/30          3s 82ms/step -
loss: 0.0050 - mae: 0.0454 - val_loss: 0.0057 - val_mae: 0.0487
Epoch 95/200
30/30          0s 76ms/step -
loss: 0.0050 - mae: 0.0456
Epoch 95: val_loss did not improve from 0.00543
30/30          2s 82ms/step -
loss: 0.0050 - mae: 0.0456 - val_loss: 0.0055 - val_mae: 0.0475
Epoch 96/200
30/30          0s 74ms/step -
loss: 0.0048 - mae: 0.0444
Epoch 96: val_loss did not improve from 0.00543
30/30          2s 79ms/step -
loss: 0.0049 - mae: 0.0445 - val_loss: 0.0055 - val_mae: 0.0468
Epoch 97/200
30/30          0s 75ms/step -
loss: 0.0050 - mae: 0.0443
Epoch 97: val_loss did not improve from 0.00543
30/30          3s 81ms/step -

```

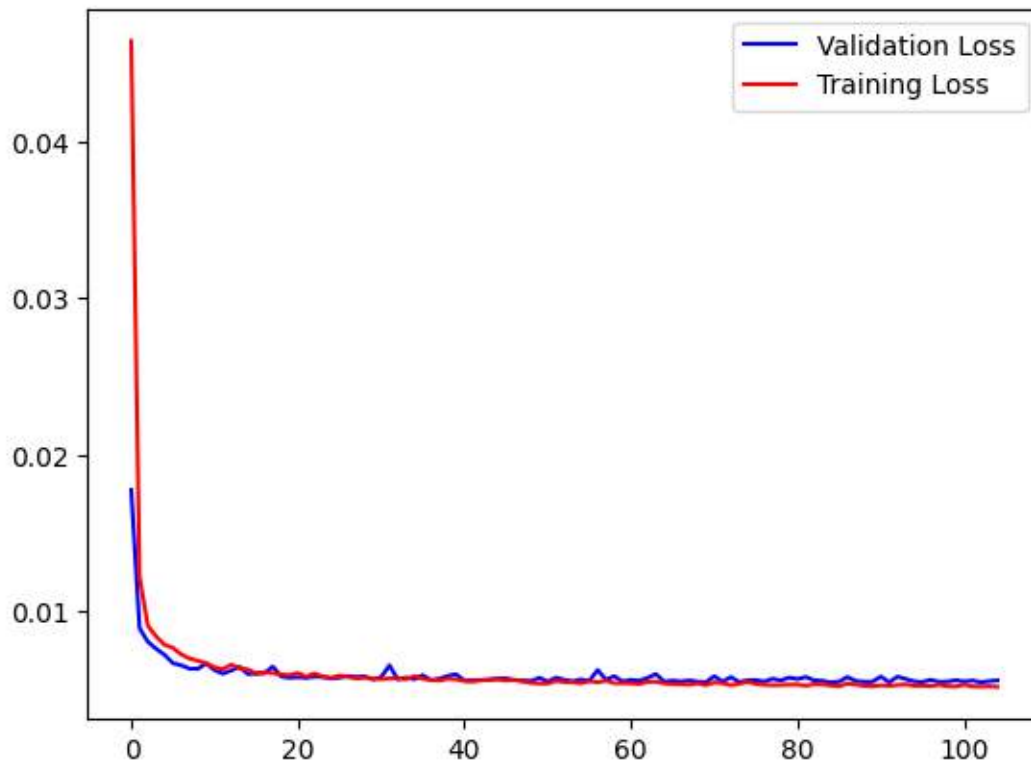
```

loss: 0.0050 - mae: 0.0444 - val_loss: 0.0056 - val_mae: 0.0486
Epoch 98/200
30/30          0s 75ms/step -
loss: 0.0052 - mae: 0.0457
Epoch 98: val_loss did not improve from 0.00543
30/30          2s 79ms/step -
loss: 0.0052 - mae: 0.0457 - val_loss: 0.0055 - val_mae: 0.0468
Epoch 99/200
30/30          0s 75ms/step -
loss: 0.0052 - mae: 0.0455
Epoch 99: val_loss did not improve from 0.00543
30/30          3s 82ms/step -
loss: 0.0052 - mae: 0.0455 - val_loss: 0.0055 - val_mae: 0.0476
Epoch 100/200
30/30          0s 75ms/step -
loss: 0.0054 - mae: 0.0464
Epoch 100: val_loss did not improve from 0.00543
30/30          2s 80ms/step -
loss: 0.0054 - mae: 0.0464 - val_loss: 0.0056 - val_mae: 0.0488
Epoch 101/200
30/30          0s 75ms/step -
loss: 0.0054 - mae: 0.0469
Epoch 101: val_loss did not improve from 0.00543
30/30          3s 81ms/step -
loss: 0.0054 - mae: 0.0469 - val_loss: 0.0055 - val_mae: 0.0474
Epoch 102/200
30/30          0s 75ms/step -
loss: 0.0053 - mae: 0.0463
Epoch 102: val_loss did not improve from 0.00543
30/30          2s 79ms/step -
loss: 0.0053 - mae: 0.0463 - val_loss: 0.0056 - val_mae: 0.0474
Epoch 103/200
30/30          0s 75ms/step -
loss: 0.0052 - mae: 0.0459
Epoch 103: val_loss did not improve from 0.00543
30/30          2s 79ms/step -
loss: 0.0052 - mae: 0.0459 - val_loss: 0.0054 - val_mae: 0.0471
Epoch 104/200
30/30          0s 75ms/step -
loss: 0.0050 - mae: 0.0451
Epoch 104: val_loss did not improve from 0.00543
30/30          3s 80ms/step -
loss: 0.0050 - mae: 0.0451 - val_loss: 0.0055 - val_mae: 0.0472
Epoch 105/200
30/30          0s 75ms/step -
loss: 0.0051 - mae: 0.0453
Epoch 105: val_loss did not improve from 0.00543
30/30          2s 81ms/step -

```

loss: 0.0051 - mae: 0.0453 - val_loss: 0.0056 - val_mae: 0.0484

```
[ ]: # Plotting the training and validation loss
plt.plot(history_simple_cnn.history['val_loss'],c="b")
plt.plot(history_simple_cnn.history['loss'],c="r")
plt.legend(['Validation Loss','Training Loss'])
plt.show()
```



Depiction of Model Performance on Images from the Training Set

```
[ ]: show_predictions(best_simple_cnn, X, Y, num_samples=5)
```

1/1 0s 269ms/step



Depiction of Model Performance on Images from the Test Set

```
[ ]: show_predictions(best_simple_cnn, X_test, Y_test, num_samples=5)
```

1/1 0s 74ms/step



The model produces clean and spatially coherent outputs. However, the predicted colors are often washed out and overly sepia-toned. It struggles to differentiate between regions such as skin, background, and clothing, revealing a limited ability to infer diverse or contextually appropriate colors.

5.4 Model 3: Improved CNN

```
[ ]: # Defining the architecture for the improved CNN model
def build_improved_cnn():
    inputs = layers.Input(shape=(120, 176, 1), name="grayscale_input")
    x1 = layers.Conv2D(64, (3, 3), padding='same')(inputs)
    x1 = layers.BatchNormalization()(x1)
    x1 = layers.LeakyReLU(negative_slope=0.1)(x1)
    p1 = layers.MaxPooling2D((2, 2))(x1)
    x2 = layers.Conv2D(128, (3, 3), padding='same')(p1)
    x2 = layers.BatchNormalization()(x2)
    x2 = layers.LeakyReLU(negative_slope=0.1)(x2)
    p2 = layers.MaxPooling2D((2, 2))(x2)
    x = layers.Conv2D(256, (3, 3), padding='same')(p2)
    x = layers.BatchNormalization()(x)
    x = layers.LeakyReLU(negative_slope=0.1)(x)
    x = layers.UpSampling2D((2, 2))(x)
    x = layers.Conv2D(128, (3, 3), padding='same')(x)
    x = layers.BatchNormalization()(x)
    x = layers.LeakyReLU(negative_slope=0.1)(x)
    x = layers.UpSampling2D((2, 2))(x)
    x = layers.Conv2D(64, (3, 3), padding='same')(x)
    x = layers.BatchNormalization()(x)
    x = layers.LeakyReLU(negative_slope=0.1)(x)
    outputs = layers.Conv2D(3, (3, 3), activation='tanh', padding='same',
    ↪name="ab_output")(x)
    return models.Model(inputs=inputs, outputs=outputs,
    ↪name="ColorizationModel")

[ ]: improved_cnn = build_improved_cnn()
improved_cnn.compile(optimizer='adam', loss='mse', metrics=['mae'])

# Defining the path to save the best model
model_checkpoint_path = os.path.join(best_models_path, 'improved_cnn.keras')
model_checkpoint = ModelCheckpoint(model_checkpoint_path, monitor='val_loss',
    ↪verbose=1, save_best_only=True, mode='min')

# Fitting the model to the training data
history_improved_cnn = improved_cnn.fit(X, Y, batch_size=16, epochs=1000,
    ↪validation_split=0.1, callbacks=[early_stop, model_checkpoint])
```



```
best_improved_cnn = keras.models.load_model(os.path.join(best_models_path,
↳ 'improved_cnn.keras'))
```

```
Epoch 1/1000
30/30          0s 275ms/step -
loss: 0.2068 - mae: 0.3204
Epoch 1: val_loss improved from inf to 0.06849, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/improved_cnn.keras
30/30          19s 388ms/step -
loss: 0.2032 - mae: 0.3168 - val_loss: 0.0685 - val_mae: 0.2137
Epoch 2/1000
30/30          0s 92ms/step -
loss: 0.0220 - mae: 0.1142
Epoch 2: val_loss improved from 0.06849 to 0.04086, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/improved_cnn.keras
30/30          9s 105ms/step -
loss: 0.0220 - mae: 0.1140 - val_loss: 0.0409 - val_mae: 0.1714
Epoch 3/1000
30/30          0s 94ms/step -
loss: 0.0154 - mae: 0.0944
Epoch 3: val_loss did not improve from 0.04086
30/30          5s 99ms/step -
loss: 0.0154 - mae: 0.0943 - val_loss: 0.0567 - val_mae: 0.1982
Epoch 4/1000
30/30          0s 91ms/step -
loss: 0.0145 - mae: 0.0909
Epoch 4: val_loss did not improve from 0.04086
30/30          5s 97ms/step -
loss: 0.0145 - mae: 0.0907 - val_loss: 0.0494 - val_mae: 0.1882
Epoch 5/1000
30/30          0s 94ms/step -
loss: 0.0112 - mae: 0.0780
Epoch 5: val_loss did not improve from 0.04086
30/30          5s 99ms/step -
loss: 0.0112 - mae: 0.0781 - val_loss: 0.0502 - val_mae: 0.1890
Epoch 6/1000
30/30          0s 91ms/step -
loss: 0.0113 - mae: 0.0781
Epoch 6: val_loss did not improve from 0.04086
30/30          5s 98ms/step -
loss: 0.0113 - mae: 0.0780 - val_loss: 0.0436 - val_mae: 0.1776
Epoch 7/1000
30/30          0s 91ms/step -
loss: 0.0117 - mae: 0.0803
```

Epoch 7: val_loss did not improve from 0.04086
30/30 3s 96ms/step -
loss: 0.0117 - mae: 0.0803 - val_loss: 0.0442 - val_mae: 0.1783
Epoch 8/1000
30/30 0s 93ms/step -
loss: 0.0109 - mae: 0.0777
Epoch 8: val_loss improved from 0.04086 to 0.03930, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/improved_cnn.keras
30/30 5s 103ms/step -
loss: 0.0109 - mae: 0.0777 - val_loss: 0.0393 - val_mae: 0.1681
Epoch 9/1000
30/30 0s 92ms/step -
loss: 0.0110 - mae: 0.0785
Epoch 9: val_loss improved from 0.03930 to 0.03606, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/improved_cnn.keras
30/30 5s 102ms/step -
loss: 0.0110 - mae: 0.0786 - val_loss: 0.0361 - val_mae: 0.1609
Epoch 10/1000
30/30 0s 93ms/step -
loss: 0.0102 - mae: 0.0754
Epoch 10: val_loss improved from 0.03606 to 0.03551, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/improved_cnn.keras
30/30 5s 106ms/step -
loss: 0.0103 - mae: 0.0755 - val_loss: 0.0355 - val_mae: 0.1600
Epoch 11/1000
30/30 0s 92ms/step -
loss: 0.0133 - mae: 0.0881
Epoch 11: val_loss did not improve from 0.03551
30/30 3s 97ms/step -
loss: 0.0132 - mae: 0.0881 - val_loss: 0.0363 - val_mae: 0.1613
Epoch 12/1000
30/30 0s 92ms/step -
loss: 0.0113 - mae: 0.0796
Epoch 12: val_loss improved from 0.03551 to 0.02960, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/improved_cnn.keras
30/30 3s 102ms/step -
loss: 0.0112 - mae: 0.0795 - val_loss: 0.0296 - val_mae: 0.1453
Epoch 13/1000
30/30 0s 92ms/step -
loss: 0.0097 - mae: 0.0728
Epoch 13: val_loss improved from 0.02960 to 0.02856, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/improved_cnn.keras
30/30 5s 102ms/step -

```

loss: 0.0097 - mae: 0.0729 - val_loss: 0.0286 - val_mae: 0.1424
Epoch 14/1000
30/30          0s 93ms/step -
loss: 0.0097 - mae: 0.0727
Epoch 14: val_loss improved from 0.02856 to 0.02211, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/improved_cnn.keras
30/30          3s 105ms/step -
loss: 0.0097 - mae: 0.0728 - val_loss: 0.0221 - val_mae: 0.1239
Epoch 15/1000
30/30          0s 93ms/step -
loss: 0.0101 - mae: 0.0760
Epoch 15: val_loss improved from 0.02211 to 0.01982, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/improved_cnn.keras
30/30          3s 103ms/step -
loss: 0.0101 - mae: 0.0760 - val_loss: 0.0198 - val_mae: 0.1173
Epoch 16/1000
30/30          0s 93ms/step -
loss: 0.0088 - mae: 0.0676
Epoch 16: val_loss improved from 0.01982 to 0.01793, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/improved_cnn.keras
30/30          3s 104ms/step -
loss: 0.0088 - mae: 0.0676 - val_loss: 0.0179 - val_mae: 0.1113
Epoch 17/1000
30/30          0s 94ms/step -
loss: 0.0095 - mae: 0.0711
Epoch 17: val_loss improved from 0.01793 to 0.01421, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/improved_cnn.keras
30/30          5s 107ms/step -
loss: 0.0095 - mae: 0.0711 - val_loss: 0.0142 - val_mae: 0.0975
Epoch 18/1000
30/30          0s 94ms/step -
loss: 0.0092 - mae: 0.0703
Epoch 18: val_loss improved from 0.01421 to 0.01356, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/improved_cnn.keras
30/30          3s 104ms/step -
loss: 0.0092 - mae: 0.0703 - val_loss: 0.0136 - val_mae: 0.0952
Epoch 19/1000
30/30          0s 93ms/step -
loss: 0.0090 - mae: 0.0691
Epoch 19: val_loss improved from 0.01356 to 0.01068, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/improved_cnn.keras
30/30          5s 103ms/step -

```

```

loss: 0.0090 - mae: 0.0691 - val_loss: 0.0107 - val_mae: 0.0814
Epoch 20/1000
30/30          0s 95ms/step -
loss: 0.0094 - mae: 0.0710
Epoch 20: val_loss improved from 0.01068 to 0.01066, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/improved_cnn.keras
30/30          5s 108ms/step -
loss: 0.0094 - mae: 0.0709 - val_loss: 0.0107 - val_mae: 0.0800
Epoch 21/1000
30/30          0s 95ms/step -
loss: 0.0086 - mae: 0.0676
Epoch 21: val_loss improved from 0.01066 to 0.01031, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/improved_cnn.keras
30/30          3s 104ms/step -
loss: 0.0086 - mae: 0.0677 - val_loss: 0.0103 - val_mae: 0.0802
Epoch 22/1000
30/30          0s 95ms/step -
loss: 0.0088 - mae: 0.0684
Epoch 22: val_loss improved from 0.01031 to 0.00900, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/improved_cnn.keras
30/30          3s 105ms/step -
loss: 0.0088 - mae: 0.0684 - val_loss: 0.0090 - val_mae: 0.0717
Epoch 23/1000
30/30          0s 95ms/step -
loss: 0.0095 - mae: 0.0708
Epoch 23: val_loss improved from 0.00900 to 0.00853, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/improved_cnn.keras
30/30          3s 105ms/step -
loss: 0.0095 - mae: 0.0707 - val_loss: 0.0085 - val_mae: 0.0687
Epoch 24/1000
30/30          0s 95ms/step -
loss: 0.0097 - mae: 0.0736
Epoch 24: val_loss improved from 0.00853 to 0.00819, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/improved_cnn.keras
30/30          5s 105ms/step -
loss: 0.0097 - mae: 0.0736 - val_loss: 0.0082 - val_mae: 0.0671
Epoch 25/1000
30/30          0s 95ms/step -
loss: 0.0100 - mae: 0.0750
Epoch 25: val_loss did not improve from 0.00819
30/30          3s 101ms/step -
loss: 0.0100 - mae: 0.0750 - val_loss: 0.0087 - val_mae: 0.0699
Epoch 26/1000

```

```

30/30          0s 95ms/step -
loss: 0.0097 - mae: 0.0727
Epoch 26: val_loss improved from 0.00819 to 0.00817, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/improved_cnn.keras
30/30          5s 105ms/step -
loss: 0.0096 - mae: 0.0727 - val_loss: 0.0082 - val_mae: 0.0650
Epoch 27/1000
30/30          0s 95ms/step -
loss: 0.0086 - mae: 0.0676
Epoch 27: val_loss did not improve from 0.00817
30/30          5s 102ms/step -
loss: 0.0086 - mae: 0.0676 - val_loss: 0.0088 - val_mae: 0.0713
Epoch 28/1000
30/30          0s 95ms/step -
loss: 0.0097 - mae: 0.0735
Epoch 28: val_loss improved from 0.00817 to 0.00785, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/improved_cnn.keras
30/30          5s 108ms/step -
loss: 0.0097 - mae: 0.0736 - val_loss: 0.0079 - val_mae: 0.0628
Epoch 29/1000
30/30          0s 94ms/step -
loss: 0.0085 - mae: 0.0676
Epoch 29: val_loss improved from 0.00785 to 0.00770, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/improved_cnn.keras
30/30          3s 104ms/step -
loss: 0.0085 - mae: 0.0677 - val_loss: 0.0077 - val_mae: 0.0635
Epoch 30/1000
30/30          0s 96ms/step -
loss: 0.0095 - mae: 0.0722
Epoch 30: val_loss did not improve from 0.00770
30/30          5s 102ms/step -
loss: 0.0095 - mae: 0.0722 - val_loss: 0.0085 - val_mae: 0.0677
Epoch 31/1000
30/30          0s 95ms/step -
loss: 0.0088 - mae: 0.0683
Epoch 31: val_loss did not improve from 0.00770
30/30          3s 102ms/step -
loss: 0.0087 - mae: 0.0682 - val_loss: 0.0080 - val_mae: 0.0649
Epoch 32/1000
30/30          0s 95ms/step -
loss: 0.0088 - mae: 0.0696
Epoch 32: val_loss improved from 0.00770 to 0.00766, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/improved_cnn.keras
30/30          3s 105ms/step -

```

```

loss: 0.0088 - mae: 0.0696 - val_loss: 0.0077 - val_mae: 0.0616
Epoch 33/1000
30/30          0s 96ms/step -
loss: 0.0087 - mae: 0.0684
Epoch 33: val_loss improved from 0.00766 to 0.00742, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/improved_cnn.keras
30/30          3s 109ms/step -
loss: 0.0087 - mae: 0.0683 - val_loss: 0.0074 - val_mae: 0.0613
Epoch 34/1000
30/30          0s 96ms/step -
loss: 0.0080 - mae: 0.0645
Epoch 34: val_loss did not improve from 0.00742
30/30          5s 101ms/step -
loss: 0.0080 - mae: 0.0645 - val_loss: 0.0077 - val_mae: 0.0621
Epoch 35/1000
30/30          0s 95ms/step -
loss: 0.0085 - mae: 0.0662
Epoch 35: val_loss improved from 0.00742 to 0.00726, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/improved_cnn.keras
30/30          5s 105ms/step -
loss: 0.0085 - mae: 0.0662 - val_loss: 0.0073 - val_mae: 0.0611
Epoch 36/1000
30/30          0s 96ms/step -
loss: 0.0082 - mae: 0.0657
Epoch 36: val_loss did not improve from 0.00726
30/30          5s 102ms/step -
loss: 0.0082 - mae: 0.0658 - val_loss: 0.0074 - val_mae: 0.0604
Epoch 37/1000
30/30          0s 96ms/step -
loss: 0.0087 - mae: 0.0690
Epoch 37: val_loss improved from 0.00726 to 0.00723, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/improved_cnn.keras
30/30          5s 107ms/step -
loss: 0.0087 - mae: 0.0689 - val_loss: 0.0072 - val_mae: 0.0605
Epoch 38/1000
30/30          0s 96ms/step -
loss: 0.0081 - mae: 0.0643
Epoch 38: val_loss improved from 0.00723 to 0.00714, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/improved_cnn.keras
30/30          5s 108ms/step -
loss: 0.0081 - mae: 0.0643 - val_loss: 0.0071 - val_mae: 0.0593
Epoch 39/1000
30/30          0s 97ms/step -
loss: 0.0077 - mae: 0.0634

```

Epoch 39: val_loss improved from 0.00714 to 0.00704, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/improved_cnn.keras

30/30 3s 108ms/step -
loss: 0.0077 - mae: 0.0634 - val_loss: 0.0070 - val_mae: 0.0589

Epoch 40/1000

30/30 0s 97ms/step -
loss: 0.0084 - mae: 0.0666

Epoch 40: val_loss did not improve from 0.00704

30/30 3s 102ms/step -
loss: 0.0084 - mae: 0.0666 - val_loss: 0.0083 - val_mae: 0.0639

Epoch 41/1000

30/30 0s 96ms/step -
loss: 0.0078 - mae: 0.0634

Epoch 41: val_loss did not improve from 0.00704

30/30 3s 102ms/step -
loss: 0.0078 - mae: 0.0634 - val_loss: 0.0071 - val_mae: 0.0587

Epoch 42/1000

30/30 0s 97ms/step -
loss: 0.0086 - mae: 0.0680

Epoch 42: val_loss did not improve from 0.00704

30/30 5s 103ms/step -
loss: 0.0086 - mae: 0.0681 - val_loss: 0.0071 - val_mae: 0.0600

Epoch 43/1000

30/30 0s 96ms/step -
loss: 0.0078 - mae: 0.0626

Epoch 43: val_loss did not improve from 0.00704

30/30 5s 102ms/step -
loss: 0.0078 - mae: 0.0627 - val_loss: 0.0079 - val_mae: 0.0650

Epoch 44/1000

30/30 0s 96ms/step -
loss: 0.0087 - mae: 0.0679

Epoch 44: val_loss did not improve from 0.00704

30/30 3s 102ms/step -
loss: 0.0087 - mae: 0.0678 - val_loss: 0.0074 - val_mae: 0.0610

Epoch 45/1000

30/30 0s 97ms/step -
loss: 0.0081 - mae: 0.0645

Epoch 45: val_loss did not improve from 0.00704

30/30 5s 102ms/step -
loss: 0.0081 - mae: 0.0645 - val_loss: 0.0083 - val_mae: 0.0664

Epoch 46/1000

30/30 0s 96ms/step -
loss: 0.0078 - mae: 0.0635

Epoch 46: val_loss did not improve from 0.00704

30/30 5s 102ms/step -
loss: 0.0078 - mae: 0.0635 - val_loss: 0.0084 - val_mae: 0.0687

Epoch 47/1000

```

30/30          0s 95ms/step -
loss: 0.0076 - mae: 0.0631
Epoch 47: val_loss did not improve from 0.00704
30/30          3s 102ms/step -
loss: 0.0076 - mae: 0.0631 - val_loss: 0.0083 - val_mae: 0.0660
Epoch 48/1000
30/30          0s 96ms/step -
loss: 0.0074 - mae: 0.0623
Epoch 48: val_loss did not improve from 0.00704
30/30          5s 102ms/step -
loss: 0.0074 - mae: 0.0623 - val_loss: 0.0071 - val_mae: 0.0583
Epoch 49/1000
30/30          0s 96ms/step -
loss: 0.0076 - mae: 0.0625
Epoch 49: val_loss did not improve from 0.00704
30/30          5s 101ms/step -
loss: 0.0076 - mae: 0.0625 - val_loss: 0.0071 - val_mae: 0.0597
Epoch 50/1000
30/30          0s 96ms/step -
loss: 0.0076 - mae: 0.0627
Epoch 50: val_loss did not improve from 0.00704
30/30          5s 103ms/step -
loss: 0.0076 - mae: 0.0628 - val_loss: 0.0095 - val_mae: 0.0744
Epoch 51/1000
30/30          0s 96ms/step -
loss: 0.0084 - mae: 0.0679
Epoch 51: val_loss did not improve from 0.00704
30/30          5s 101ms/step -
loss: 0.0084 - mae: 0.0679 - val_loss: 0.0094 - val_mae: 0.0741
Epoch 52/1000
30/30          0s 95ms/step -
loss: 0.0076 - mae: 0.0633
Epoch 52: val_loss did not improve from 0.00704
30/30          5s 101ms/step -
loss: 0.0076 - mae: 0.0634 - val_loss: 0.0074 - val_mae: 0.0612
Epoch 53/1000
30/30          0s 96ms/step -
loss: 0.0079 - mae: 0.0643
Epoch 53: val_loss did not improve from 0.00704
30/30          3s 102ms/step -
loss: 0.0079 - mae: 0.0643 - val_loss: 0.0076 - val_mae: 0.0622
Epoch 54/1000
30/30          0s 96ms/step -
loss: 0.0079 - mae: 0.0639
Epoch 54: val_loss did not improve from 0.00704
30/30          3s 101ms/step -
loss: 0.0079 - mae: 0.0639 - val_loss: 0.0080 - val_mae: 0.0660
Epoch 55/1000

```



```

30/30          0s 96ms/step -
loss: 0.0073 - mae: 0.0615
Epoch 55: val_loss did not improve from 0.00704
30/30          5s 101ms/step -
loss: 0.0073 - mae: 0.0615 - val_loss: 0.0078 - val_mae: 0.0651
Epoch 56/1000
30/30          0s 96ms/step -
loss: 0.0072 - mae: 0.0599
Epoch 56: val_loss improved from 0.00704 to 0.00702, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/improved_cnn.keras
30/30          3s 108ms/step -
loss: 0.0072 - mae: 0.0600 - val_loss: 0.0070 - val_mae: 0.0584
Epoch 57/1000
30/30          0s 96ms/step -
loss: 0.0075 - mae: 0.0626
Epoch 57: val_loss improved from 0.00702 to 0.00673, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/improved_cnn.keras
30/30          5s 106ms/step -
loss: 0.0075 - mae: 0.0627 - val_loss: 0.0067 - val_mae: 0.0569
Epoch 58/1000
30/30          0s 96ms/step -
loss: 0.0073 - mae: 0.0619
Epoch 58: val_loss did not improve from 0.00673
30/30          3s 102ms/step -
loss: 0.0073 - mae: 0.0619 - val_loss: 0.0077 - val_mae: 0.0635
Epoch 59/1000
30/30          0s 96ms/step -
loss: 0.0075 - mae: 0.0618
Epoch 59: val_loss did not improve from 0.00673
30/30          3s 102ms/step -
loss: 0.0075 - mae: 0.0619 - val_loss: 0.0103 - val_mae: 0.0741
Epoch 60/1000
30/30          0s 97ms/step -
loss: 0.0079 - mae: 0.0644
Epoch 60: val_loss did not improve from 0.00673
30/30          5s 102ms/step -
loss: 0.0079 - mae: 0.0645 - val_loss: 0.0079 - val_mae: 0.0636
Epoch 61/1000
30/30          0s 96ms/step -
loss: 0.0080 - mae: 0.0645
Epoch 61: val_loss improved from 0.00673 to 0.00670, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/improved_cnn.keras
30/30          5s 106ms/step -
loss: 0.0080 - mae: 0.0645 - val_loss: 0.0067 - val_mae: 0.0569
Epoch 62/1000

```

```

30/30          0s 97ms/step -
loss: 0.0082 - mae: 0.0655
Epoch 62: val_loss did not improve from 0.00670
30/30          5s 103ms/step -
loss: 0.0082 - mae: 0.0655 - val_loss: 0.0082 - val_mae: 0.0637
Epoch 63/1000
30/30          0s 96ms/step -
loss: 0.0069 - mae: 0.0587
Epoch 63: val_loss did not improve from 0.00670
30/30          5s 101ms/step -
loss: 0.0069 - mae: 0.0588 - val_loss: 0.0088 - val_mae: 0.0694
Epoch 64/1000
30/30          0s 95ms/step -
loss: 0.0079 - mae: 0.0637
Epoch 64: val_loss improved from 0.00670 to 0.00645, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/improved_cnn.keras
30/30          3s 105ms/step -
loss: 0.0078 - mae: 0.0636 - val_loss: 0.0065 - val_mae: 0.0552
Epoch 65/1000
30/30          0s 96ms/step -
loss: 0.0073 - mae: 0.0619
Epoch 65: val_loss did not improve from 0.00645
30/30          5s 102ms/step -
loss: 0.0073 - mae: 0.0619 - val_loss: 0.0071 - val_mae: 0.0597
Epoch 66/1000
30/30          0s 95ms/step -
loss: 0.0074 - mae: 0.0624
Epoch 66: val_loss did not improve from 0.00645
30/30          3s 102ms/step -
loss: 0.0074 - mae: 0.0623 - val_loss: 0.0084 - val_mae: 0.0678
Epoch 67/1000
30/30          0s 96ms/step -
loss: 0.0076 - mae: 0.0634
Epoch 67: val_loss did not improve from 0.00645
30/30          3s 102ms/step -
loss: 0.0076 - mae: 0.0634 - val_loss: 0.0081 - val_mae: 0.0671
Epoch 68/1000
30/30          0s 97ms/step -
loss: 0.0077 - mae: 0.0621
Epoch 68: val_loss did not improve from 0.00645
30/30          5s 103ms/step -
loss: 0.0077 - mae: 0.0620 - val_loss: 0.0078 - val_mae: 0.0645
Epoch 69/1000
30/30          0s 95ms/step -
loss: 0.0070 - mae: 0.0592
Epoch 69: val_loss did not improve from 0.00645
30/30          5s 101ms/step -

```

```

loss: 0.0070 - mae: 0.0592 - val_loss: 0.0067 - val_mae: 0.0555
Epoch 70/1000
30/30          0s 96ms/step -
loss: 0.0075 - mae: 0.0610
Epoch 70: val_loss did not improve from 0.00645
30/30          5s 102ms/step -
loss: 0.0075 - mae: 0.0610 - val_loss: 0.0072 - val_mae: 0.0608
Epoch 71/1000
30/30          0s 96ms/step -
loss: 0.0070 - mae: 0.0591
Epoch 71: val_loss did not improve from 0.00645
30/30          3s 103ms/step -
loss: 0.0070 - mae: 0.0591 - val_loss: 0.0073 - val_mae: 0.0616
Epoch 72/1000
30/30          0s 96ms/step -
loss: 0.0078 - mae: 0.0637
Epoch 72: val_loss did not improve from 0.00645
30/30          3s 102ms/step -
loss: 0.0078 - mae: 0.0637 - val_loss: 0.0073 - val_mae: 0.0604
Epoch 73/1000
30/30          0s 97ms/step -
loss: 0.0084 - mae: 0.0667
Epoch 73: val_loss did not improve from 0.00645
30/30          5s 103ms/step -
loss: 0.0084 - mae: 0.0666 - val_loss: 0.0153 - val_mae: 0.0884
Epoch 74/1000
30/30          0s 96ms/step -
loss: 0.0074 - mae: 0.0607
Epoch 74: val_loss did not improve from 0.00645
30/30          3s 103ms/step -
loss: 0.0074 - mae: 0.0607 - val_loss: 0.0074 - val_mae: 0.0608
Epoch 75/1000
30/30          0s 96ms/step -
loss: 0.0070 - mae: 0.0594
Epoch 75: val_loss did not improve from 0.00645
30/30          5s 102ms/step -
loss: 0.0071 - mae: 0.0594 - val_loss: 0.0069 - val_mae: 0.0583
Epoch 76/1000
30/30          0s 96ms/step -
loss: 0.0075 - mae: 0.0628
Epoch 76: val_loss did not improve from 0.00645
30/30          3s 102ms/step -
loss: 0.0075 - mae: 0.0627 - val_loss: 0.0069 - val_mae: 0.0585
Epoch 77/1000
30/30          0s 96ms/step -
loss: 0.0067 - mae: 0.0584
Epoch 77: val_loss did not improve from 0.00645
30/30          5s 103ms/step -

```

```

loss: 0.0068 - mae: 0.0584 - val_loss: 0.0069 - val_mae: 0.0582
Epoch 78/1000
30/30          0s 96ms/step -
loss: 0.0067 - mae: 0.0572
Epoch 78: val_loss did not improve from 0.00645
30/30          3s 102ms/step -
loss: 0.0067 - mae: 0.0572 - val_loss: 0.0072 - val_mae: 0.0613
Epoch 79/1000
30/30          0s 95ms/step -
loss: 0.0075 - mae: 0.0618
Epoch 79: val_loss did not improve from 0.00645
30/30          3s 102ms/step -
loss: 0.0075 - mae: 0.0618 - val_loss: 0.0079 - val_mae: 0.0653
Epoch 80/1000
30/30          0s 96ms/step -
loss: 0.0072 - mae: 0.0600
Epoch 80: val_loss did not improve from 0.00645
30/30          3s 101ms/step -
loss: 0.0072 - mae: 0.0600 - val_loss: 0.0073 - val_mae: 0.0622
Epoch 81/1000
30/30          0s 96ms/step -
loss: 0.0072 - mae: 0.0608
Epoch 81: val_loss did not improve from 0.00645
30/30          3s 103ms/step -
loss: 0.0072 - mae: 0.0608 - val_loss: 0.0071 - val_mae: 0.0603
Epoch 82/1000
30/30          0s 96ms/step -
loss: 0.0071 - mae: 0.0607
Epoch 82: val_loss did not improve from 0.00645
30/30          3s 103ms/step -
loss: 0.0071 - mae: 0.0607 - val_loss: 0.0069 - val_mae: 0.0587
Epoch 83/1000
30/30          0s 96ms/step -
loss: 0.0072 - mae: 0.0611
Epoch 83: val_loss did not improve from 0.00645
30/30          5s 101ms/step -
loss: 0.0072 - mae: 0.0611 - val_loss: 0.0068 - val_mae: 0.0588
Epoch 84/1000
30/30          0s 96ms/step -
loss: 0.0072 - mae: 0.0611
Epoch 84: val_loss did not improve from 0.00645
30/30          3s 102ms/step -
loss: 0.0072 - mae: 0.0612 - val_loss: 0.0072 - val_mae: 0.0605

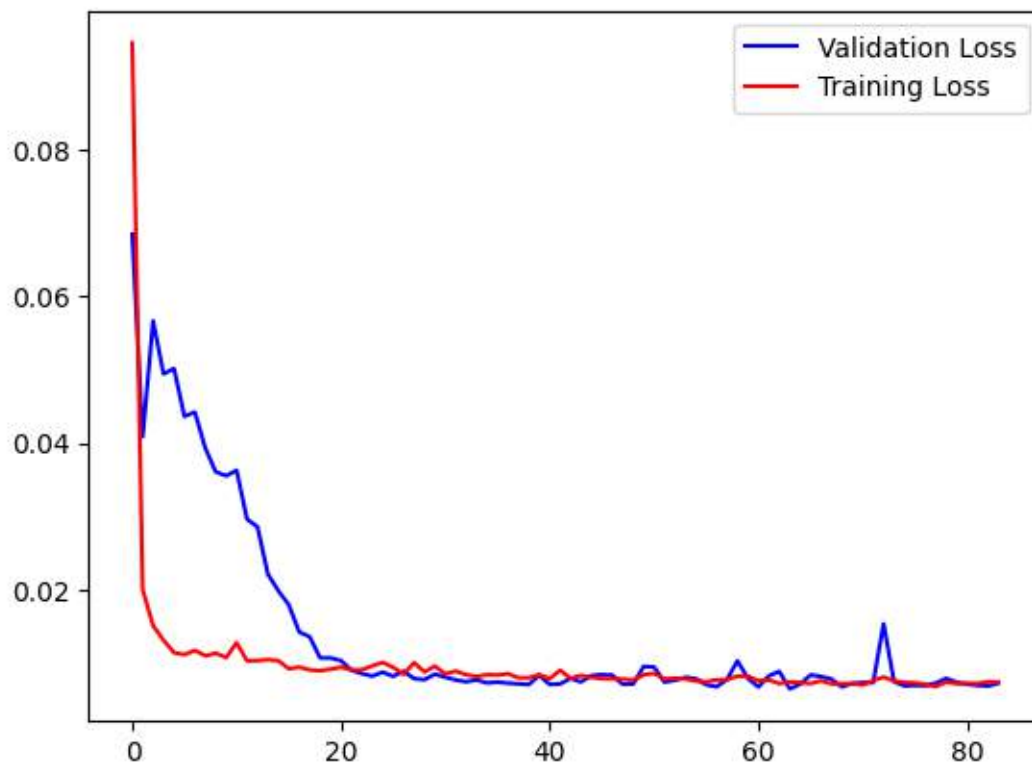
```

```

[ ]: # Plotting the training and validation loss
plt.plot(history_improved_cnn.history['val_loss'],c="b")
plt.plot(history_improved_cnn.history['loss'],c="r")

```

```
plt.legend(['Validation Loss', 'Training Loss'])  
plt.show()
```



Depiction of Model Performance on Images from the Training Set

```
[ ]: show_predictions(best_improved_cnn, X, Y, num_samples=5)
```

1/1 1s 964ms/step



Depiction of Model Performance on Images from the Test Set

```
[ ]: show_predictions(best_improved_cnn, X_test, Y_test, num_samples=5)
```

1/1 0s 59ms/step



While the validation loss dropped sharply after 20-30 epochs, the model failed to fully capture both color and resolution. The output resulted in a mix of beige and grey tones, and the entire image appears blurry. It is clear that the model struggled with both semantic understanding and spatial detail, missing vivid elements in the ground truth. Instead of assigning meaningful colors, it defaulted to desaturated tones across the entire image.

5.5 Model 4: U-Net Implementation

```
[ ]: # Defining the architecture for the U-Net model
def build_unet_model(input_shape=(120, 176, 1)):
    inputs = layers.Input(shape=input_shape)

    # Encoder
    conv1 = layers.Conv2D(64, (3, 3), padding='same')(inputs)
    conv1 = layers.BatchNormalization()(conv1)
    conv1 = layers.ReLU()(conv1)
    conv1 = layers.Conv2D(64, (3, 3), padding='same')(conv1)
    conv1 = layers.BatchNormalization()(conv1)
    conv1 = layers.ReLU()(conv1)
    pool1 = layers.MaxPooling2D((2, 2))(conv1)

    conv2 = layers.Conv2D(128, (3, 3), padding='same')(pool1)
    conv2 = layers.BatchNormalization()(conv2)
    conv2 = layers.ReLU()(conv2)
    conv2 = layers.Conv2D(128, (3, 3), padding='same')(conv2)
    conv2 = layers.BatchNormalization()(conv2)
    conv2 = layers.ReLU()(conv2)
    pool2 = layers.MaxPooling2D((2, 2))(conv2)

    # Bottleneck
    bottleneck = layers.Conv2D(256, (3, 3), padding='same')(pool2)
    bottleneck = layers.BatchNormalization()(bottleneck)
    bottleneck = layers.ReLU()(bottleneck)

    # Decoder
    up2 = layers.Conv2DTranspose(128, (2, 2), strides=2, padding='same')(bottleneck)
    up2 = layers.Concatenate()([up2, conv2])
    up2 = layers.Conv2D(128, (3, 3), padding='same')(up2)
    up2 = layers.BatchNormalization()(up2)
    up2 = layers.ReLU()(up2)

    up1 = layers.Conv2DTranspose(64, (2, 2), strides=2, padding='same')(up2)
    up1 = layers.Concatenate()([up1, conv1])
    up1 = layers.Conv2D(64, (3, 3), padding='same')(up1)
```

```

up1 = layers.BatchNormalization()(up1)
up1 = layers.ReLU()(up1)

outputs = layers.Conv2D(3, (3, 3), activation='sigmoid',
↳padding='same')(up1)

model = models.Model(inputs=inputs, outputs=outputs)
model.compile(optimizer='adam', loss='mse', metrics=['mae'])
return model

UNET_MODEL = build_UNET_MODEL()

```

```

[ ]: # Defining the path to save the best model
model_checkpoint_path = os.path.join(best_models_path, 'UNET_MODEL.keras')
model_checkpoint = ModelCheckpoint(model_checkpoint_path, monitor='val_loss',
↳verbose=1, save_best_only=True, mode='min')

# fitting the model to the training data
history_UNET_MODEL = UNET_MODEL.fit(X, Y, batch_size=16, epochs=200,
↳validation_split=0.1, callbacks=[early_stop, model_checkpoint])

best_UNET_MODEL = keras.models.load_model(os.path.join(best_models_path,
↳'UNET_MODEL.keras'))

```

```

Epoch 1/200
30/30          0s 569ms/step -
loss: 0.0297 - mae: 0.1139
Epoch 1: val_loss improved from inf to 0.06086, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/UNET_MODEL.keras
30/30          42s 715ms/step -
loss: 0.0292 - mae: 0.1128 - val_loss: 0.0609 - val_mae: 0.2085
Epoch 2/200
30/30          0s 136ms/step -
loss: 0.0069 - mae: 0.0585
Epoch 2: val_loss improved from 0.06086 to 0.05717, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/UNET_MODEL.keras
30/30          5s 150ms/step -
loss: 0.0069 - mae: 0.0585 - val_loss: 0.0572 - val_mae: 0.2022
Epoch 3/200
30/30          0s 136ms/step -
loss: 0.0068 - mae: 0.0573
Epoch 3: val_loss improved from 0.05717 to 0.05512, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/UNET_MODEL.keras
30/30          4s 150ms/step -

```

```

loss: 0.0068 - mae: 0.0573 - val_loss: 0.0551 - val_mae: 0.1991
Epoch 4/200
30/30          0s 138ms/step -
loss: 0.0057 - mae: 0.0524
Epoch 4: val_loss improved from 0.05512 to 0.05224, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/unet_model.keras
30/30          5s 156ms/step -
loss: 0.0057 - mae: 0.0524 - val_loss: 0.0522 - val_mae: 0.1942
Epoch 5/200
30/30          0s 138ms/step -
loss: 0.0057 - mae: 0.0508
Epoch 5: val_loss improved from 0.05224 to 0.04996, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/unet_model.keras
30/30          5s 152ms/step -
loss: 0.0057 - mae: 0.0509 - val_loss: 0.0500 - val_mae: 0.1901
Epoch 6/200
30/30          0s 138ms/step -
loss: 0.0058 - mae: 0.0526
Epoch 6: val_loss improved from 0.04996 to 0.04879, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/unet_model.keras
30/30          5s 158ms/step -
loss: 0.0058 - mae: 0.0527 - val_loss: 0.0488 - val_mae: 0.1879
Epoch 7/200
30/30          0s 140ms/step -
loss: 0.0059 - mae: 0.0519
Epoch 7: val_loss improved from 0.04879 to 0.04815, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/unet_model.keras
30/30          5s 154ms/step -
loss: 0.0059 - mae: 0.0518 - val_loss: 0.0481 - val_mae: 0.1871
Epoch 8/200
30/30          0s 140ms/step -
loss: 0.0059 - mae: 0.0540
Epoch 8: val_loss improved from 0.04815 to 0.04351, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/unet_model.keras
30/30          5s 154ms/step -
loss: 0.0060 - mae: 0.0541 - val_loss: 0.0435 - val_mae: 0.1778
Epoch 9/200
30/30          0s 141ms/step -
loss: 0.0071 - mae: 0.0586
Epoch 9: val_loss improved from 0.04351 to 0.04177, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/unet_model.keras
30/30          5s 157ms/step -

```

```

loss: 0.0070 - mae: 0.0586 - val_loss: 0.0418 - val_mae: 0.1747
Epoch 10/200
30/30          0s 141ms/step -
loss: 0.0057 - mae: 0.0523
Epoch 10: val_loss improved from 0.04177 to 0.03875, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/unet_model.keras
30/30          5s 155ms/step -
loss: 0.0058 - mae: 0.0524 - val_loss: 0.0388 - val_mae: 0.1681
Epoch 11/200
30/30          0s 141ms/step -
loss: 0.0063 - mae: 0.0549
Epoch 11: val_loss improved from 0.03875 to 0.03687, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/unet_model.keras
30/30          5s 159ms/step -
loss: 0.0063 - mae: 0.0549 - val_loss: 0.0369 - val_mae: 0.1644
Epoch 12/200
30/30          0s 142ms/step -
loss: 0.0063 - mae: 0.0551
Epoch 12: val_loss improved from 0.03687 to 0.03466, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/unet_model.keras
30/30          5s 157ms/step -
loss: 0.0063 - mae: 0.0550 - val_loss: 0.0347 - val_mae: 0.1593
Epoch 13/200
30/30          0s 142ms/step -
loss: 0.0061 - mae: 0.0541
Epoch 13: val_loss improved from 0.03466 to 0.03167, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/unet_model.keras
30/30          5s 156ms/step -
loss: 0.0061 - mae: 0.0541 - val_loss: 0.0317 - val_mae: 0.1522
Epoch 14/200
30/30          0s 143ms/step -
loss: 0.0059 - mae: 0.0524
Epoch 14: val_loss improved from 0.03167 to 0.02722, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/unet_model.keras
30/30          5s 162ms/step -
loss: 0.0059 - mae: 0.0524 - val_loss: 0.0272 - val_mae: 0.1398
Epoch 15/200
30/30          0s 145ms/step -
loss: 0.0059 - mae: 0.0536
Epoch 15: val_loss did not improve from 0.02722
30/30          5s 156ms/step -
loss: 0.0059 - mae: 0.0536 - val_loss: 0.0272 - val_mae: 0.1400
Epoch 16/200

```

```

30/30          0s 144ms/step -
loss: 0.0051 - mae: 0.0481
Epoch 16: val_loss improved from 0.02722 to 0.02281, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/unet_model.keras
30/30          5s 158ms/step -
loss: 0.0051 - mae: 0.0482 - val_loss: 0.0228 - val_mae: 0.1275
Epoch 17/200
30/30          0s 144ms/step -
loss: 0.0060 - mae: 0.0534
Epoch 17: val_loss improved from 0.02281 to 0.02031, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/unet_model.keras
30/30          5s 161ms/step -
loss: 0.0060 - mae: 0.0534 - val_loss: 0.0203 - val_mae: 0.1195
Epoch 18/200
30/30          0s 144ms/step -
loss: 0.0053 - mae: 0.0496
Epoch 18: val_loss improved from 0.02031 to 0.01561, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/unet_model.keras
30/30          5s 163ms/step -
loss: 0.0053 - mae: 0.0496 - val_loss: 0.0156 - val_mae: 0.1031
Epoch 19/200
30/30          0s 144ms/step -
loss: 0.0057 - mae: 0.0509
Epoch 19: val_loss improved from 0.01561 to 0.01236, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/unet_model.keras
30/30          5s 158ms/step -
loss: 0.0057 - mae: 0.0509 - val_loss: 0.0124 - val_mae: 0.0896
Epoch 20/200
30/30          0s 146ms/step -
loss: 0.0056 - mae: 0.0504
Epoch 20: val_loss improved from 0.01236 to 0.00923, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/unet_model.keras
30/30          5s 163ms/step -
loss: 0.0056 - mae: 0.0504 - val_loss: 0.0092 - val_mae: 0.0769
Epoch 21/200
30/30          0s 146ms/step -
loss: 0.0054 - mae: 0.0500
Epoch 21: val_loss improved from 0.00923 to 0.00908, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/unet_model.keras
30/30          5s 164ms/step -
loss: 0.0054 - mae: 0.0500 - val_loss: 0.0091 - val_mae: 0.0753
Epoch 22/200

```

```

30/30          0s 145ms/step -
loss: 0.0058 - mae: 0.0521
Epoch 22: val_loss improved from 0.00908 to 0.00776, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/unet_model.keras
30/30          5s 164ms/step -
loss: 0.0058 - mae: 0.0521 - val_loss: 0.0078 - val_mae: 0.0654
Epoch 23/200
30/30          0s 148ms/step -
loss: 0.0057 - mae: 0.0525
Epoch 23: val_loss improved from 0.00776 to 0.00578, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/unet_model.keras
30/30          5s 162ms/step -
loss: 0.0057 - mae: 0.0525 - val_loss: 0.0058 - val_mae: 0.0504
Epoch 24/200
30/30          0s 147ms/step -
loss: 0.0057 - mae: 0.0510
Epoch 24: val_loss improved from 0.00578 to 0.00554, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/unet_model.keras
30/30          5s 161ms/step -
loss: 0.0057 - mae: 0.0511 - val_loss: 0.0055 - val_mae: 0.0488
Epoch 25/200
30/30          0s 149ms/step -
loss: 0.0058 - mae: 0.0512
Epoch 25: val_loss did not improve from 0.00554
30/30          5s 161ms/step -
loss: 0.0058 - mae: 0.0512 - val_loss: 0.0059 - val_mae: 0.0511
Epoch 26/200
30/30          0s 148ms/step -
loss: 0.0055 - mae: 0.0494
Epoch 26: val_loss did not improve from 0.00554
30/30          5s 155ms/step -
loss: 0.0055 - mae: 0.0494 - val_loss: 0.0059 - val_mae: 0.0532
Epoch 27/200
30/30          0s 147ms/step -
loss: 0.0060 - mae: 0.0527
Epoch 27: val_loss did not improve from 0.00554
30/30          5s 159ms/step -
loss: 0.0060 - mae: 0.0526 - val_loss: 0.0062 - val_mae: 0.0541
Epoch 28/200
30/30          0s 148ms/step -
loss: 0.0056 - mae: 0.0487
Epoch 28: val_loss did not improve from 0.00554
30/30          5s 157ms/step -
loss: 0.0056 - mae: 0.0486 - val_loss: 0.0056 - val_mae: 0.0483
Epoch 29/200

```

```

30/30          0s 146ms/step -
loss: 0.0059 - mae: 0.0520
Epoch 29: val_loss did not improve from 0.00554
30/30          5s 158ms/step -
loss: 0.0059 - mae: 0.0520 - val_loss: 0.0056 - val_mae: 0.0495
Epoch 30/200
30/30          0s 147ms/step -
loss: 0.0057 - mae: 0.0509
Epoch 30: val_loss did not improve from 0.00554
30/30          5s 155ms/step -
loss: 0.0057 - mae: 0.0509 - val_loss: 0.0066 - val_mae: 0.0578
Epoch 31/200
30/30          0s 147ms/step -
loss: 0.0056 - mae: 0.0499
Epoch 31: val_loss did not improve from 0.00554
30/30          5s 155ms/step -
loss: 0.0056 - mae: 0.0500 - val_loss: 0.0080 - val_mae: 0.0656
Epoch 32/200
30/30          0s 146ms/step -
loss: 0.0057 - mae: 0.0517
Epoch 32: val_loss did not improve from 0.00554
30/30          5s 158ms/step -
loss: 0.0057 - mae: 0.0517 - val_loss: 0.0062 - val_mae: 0.0529
Epoch 33/200
30/30          0s 146ms/step -
loss: 0.0061 - mae: 0.0528
Epoch 33: val_loss did not improve from 0.00554
30/30          5s 158ms/step -
loss: 0.0060 - mae: 0.0527 - val_loss: 0.0077 - val_mae: 0.0618
Epoch 34/200
30/30          0s 146ms/step -
loss: 0.0056 - mae: 0.0516
Epoch 34: val_loss improved from 0.00554 to 0.00549, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/unet_model.keras
30/30          5s 160ms/step -
loss: 0.0056 - mae: 0.0516 - val_loss: 0.0055 - val_mae: 0.0487
Epoch 35/200
30/30          0s 146ms/step -
loss: 0.0049 - mae: 0.0467
Epoch 35: val_loss did not improve from 0.00549
30/30          5s 154ms/step -
loss: 0.0049 - mae: 0.0468 - val_loss: 0.0056 - val_mae: 0.0485
Epoch 36/200
30/30          0s 148ms/step -
loss: 0.0061 - mae: 0.0526
Epoch 36: val_loss did not improve from 0.00549
30/30          5s 159ms/step -

```

```

loss: 0.0061 - mae: 0.0525 - val_loss: 0.0058 - val_mae: 0.0521
Epoch 37/200
30/30          0s 147ms/step -
loss: 0.0051 - mae: 0.0473
Epoch 37: val_loss did not improve from 0.00549
30/30          5s 154ms/step -
loss: 0.0051 - mae: 0.0473 - val_loss: 0.0056 - val_mae: 0.0490
Epoch 38/200
30/30          0s 147ms/step -
loss: 0.0057 - mae: 0.0523
Epoch 38: val_loss did not improve from 0.00549
30/30          5s 160ms/step -
loss: 0.0057 - mae: 0.0523 - val_loss: 0.0058 - val_mae: 0.0511
Epoch 39/200
30/30          0s 147ms/step -
loss: 0.0056 - mae: 0.0512
Epoch 39: val_loss did not improve from 0.00549
30/30          5s 155ms/step -
loss: 0.0056 - mae: 0.0512 - val_loss: 0.0061 - val_mae: 0.0539
Epoch 40/200
30/30          0s 147ms/step -
loss: 0.0057 - mae: 0.0521
Epoch 40: val_loss did not improve from 0.00549
30/30          5s 155ms/step -
loss: 0.0058 - mae: 0.0521 - val_loss: 0.0081 - val_mae: 0.0660
Epoch 41/200
30/30          0s 148ms/step -
loss: 0.0055 - mae: 0.0499
Epoch 41: val_loss did not improve from 0.00549
30/30          5s 157ms/step -
loss: 0.0055 - mae: 0.0500 - val_loss: 0.0079 - val_mae: 0.0621
Epoch 42/200
30/30          0s 147ms/step -
loss: 0.0051 - mae: 0.0481
Epoch 42: val_loss did not improve from 0.00549
30/30          5s 154ms/step -
loss: 0.0051 - mae: 0.0481 - val_loss: 0.0059 - val_mae: 0.0516
Epoch 43/200
30/30          0s 147ms/step -
loss: 0.0056 - mae: 0.0508
Epoch 43: val_loss did not improve from 0.00549
30/30          5s 155ms/step -
loss: 0.0056 - mae: 0.0508 - val_loss: 0.0056 - val_mae: 0.0495
Epoch 44/200
30/30          0s 147ms/step -
loss: 0.0059 - mae: 0.0520
Epoch 44: val_loss did not improve from 0.00549
30/30          5s 155ms/step -

```



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loss: 0.0059 - mae: 0.0519 - val_loss: 0.0057 - val_mae: 0.0497
Epoch 45/200
30/30          0s 147ms/step -
loss: 0.0057 - mae: 0.0514
Epoch 45: val_loss did not improve from 0.00549
30/30          5s 155ms/step -
loss: 0.0057 - mae: 0.0513 - val_loss: 0.0055 - val_mae: 0.0489
Epoch 46/200
30/30          0s 148ms/step -
loss: 0.0054 - mae: 0.0490
Epoch 46: val_loss did not improve from 0.00549
30/30          5s 161ms/step -
loss: 0.0054 - mae: 0.0490 - val_loss: 0.0064 - val_mae: 0.0545
Epoch 47/200
30/30          0s 148ms/step -
loss: 0.0055 - mae: 0.0506
Epoch 47: val_loss did not improve from 0.00549
30/30          5s 159ms/step -
loss: 0.0055 - mae: 0.0506 - val_loss: 0.0057 - val_mae: 0.0498
Epoch 48/200
30/30          0s 147ms/step -
loss: 0.0051 - mae: 0.0465
Epoch 48: val_loss improved from 0.00549 to 0.00547, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/unet_model.keras
30/30          5s 165ms/step -
loss: 0.0051 - mae: 0.0465 - val_loss: 0.0055 - val_mae: 0.0485
Epoch 49/200
30/30          0s 149ms/step -
loss: 0.0054 - mae: 0.0497
Epoch 49: val_loss did not improve from 0.00547
30/30          5s 161ms/step -
loss: 0.0054 - mae: 0.0497 - val_loss: 0.0058 - val_mae: 0.0521
Epoch 50/200
30/30          0s 148ms/step -
loss: 0.0054 - mae: 0.0497
Epoch 50: val_loss did not improve from 0.00547
30/30          5s 156ms/step -
loss: 0.0055 - mae: 0.0497 - val_loss: 0.0057 - val_mae: 0.0497
Epoch 51/200
30/30          0s 148ms/step -
loss: 0.0051 - mae: 0.0468
Epoch 51: val_loss did not improve from 0.00547
30/30          5s 160ms/step -
loss: 0.0051 - mae: 0.0468 - val_loss: 0.0056 - val_mae: 0.0491
Epoch 52/200
30/30          0s 149ms/step -
loss: 0.0057 - mae: 0.0509

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Epoch 52: val_loss did not improve from 0.00547
30/30 5s 160ms/step -
loss: 0.0057 - mae: 0.0508 - val_loss: 0.0057 - val_mae: 0.0489
Epoch 53/200
30/30 0s 148ms/step -
loss: 0.0051 - mae: 0.0468
Epoch 53: val_loss did not improve from 0.00547
30/30 5s 156ms/step -
loss: 0.0051 - mae: 0.0469 - val_loss: 0.0065 - val_mae: 0.0542
Epoch 54/200
30/30 0s 146ms/step -
loss: 0.0051 - mae: 0.0478
Epoch 54: val_loss did not improve from 0.00547
30/30 5s 155ms/step -
loss: 0.0051 - mae: 0.0478 - val_loss: 0.0056 - val_mae: 0.0494
Epoch 55/200
30/30 0s 148ms/step -
loss: 0.0050 - mae: 0.0467
Epoch 55: val_loss did not improve from 0.00547
30/30 5s 159ms/step -
loss: 0.0050 - mae: 0.0467 - val_loss: 0.0057 - val_mae: 0.0499
Epoch 56/200
30/30 0s 146ms/step -
loss: 0.0056 - mae: 0.0486
Epoch 56: val_loss did not improve from 0.00547
30/30 5s 154ms/step -
loss: 0.0056 - mae: 0.0485 - val_loss: 0.0055 - val_mae: 0.0490
Epoch 57/200
30/30 0s 147ms/step -
loss: 0.0053 - mae: 0.0478
Epoch 57: val_loss did not improve from 0.00547
30/30 5s 156ms/step -
loss: 0.0053 - mae: 0.0478 - val_loss: 0.0061 - val_mae: 0.0518
Epoch 58/200
30/30 0s 146ms/step -
loss: 0.0048 - mae: 0.0460
Epoch 58: val_loss did not improve from 0.00547
30/30 5s 153ms/step -
loss: 0.0048 - mae: 0.0461 - val_loss: 0.0058 - val_mae: 0.0500
Epoch 59/200
30/30 0s 147ms/step -
loss: 0.0056 - mae: 0.0504
Epoch 59: val_loss did not improve from 0.00547
30/30 5s 154ms/step -
loss: 0.0056 - mae: 0.0503 - val_loss: 0.0062 - val_mae: 0.0516
Epoch 60/200
30/30 0s 148ms/step -
loss: 0.0050 - mae: 0.0482

Epoch 60: val_loss improved from 0.00547 to 0.00541, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/unet_model.keras
30/30 5s 162ms/step -
loss: 0.0050 - mae: 0.0482 - val_loss: 0.0054 - val_mae: 0.0473
Epoch 61/200
30/30 0s 146ms/step -
loss: 0.0051 - mae: 0.0472
Epoch 61: val_loss did not improve from 0.00541
30/30 5s 153ms/step -
loss: 0.0051 - mae: 0.0472 - val_loss: 0.0059 - val_mae: 0.0513
Epoch 62/200
30/30 0s 147ms/step -
loss: 0.0054 - mae: 0.0490
Epoch 62: val_loss did not improve from 0.00541
30/30 5s 155ms/step -
loss: 0.0054 - mae: 0.0491 - val_loss: 0.0063 - val_mae: 0.0550
Epoch 63/200
30/30 0s 147ms/step -
loss: 0.0051 - mae: 0.0462
Epoch 63: val_loss did not improve from 0.00541
30/30 5s 159ms/step -
loss: 0.0051 - mae: 0.0463 - val_loss: 0.0065 - val_mae: 0.0564
Epoch 64/200
30/30 0s 147ms/step -
loss: 0.0054 - mae: 0.0490
Epoch 64: val_loss did not improve from 0.00541
30/30 5s 159ms/step -
loss: 0.0054 - mae: 0.0490 - val_loss: 0.0083 - val_mae: 0.0681
Epoch 65/200
30/30 0s 147ms/step -
loss: 0.0054 - mae: 0.0501
Epoch 65: val_loss did not improve from 0.00541
30/30 5s 159ms/step -
loss: 0.0054 - mae: 0.0501 - val_loss: 0.0071 - val_mae: 0.0600
Epoch 66/200
30/30 0s 147ms/step -
loss: 0.0052 - mae: 0.0473
Epoch 66: val_loss did not improve from 0.00541
30/30 5s 155ms/step -
loss: 0.0052 - mae: 0.0473 - val_loss: 0.0061 - val_mae: 0.0532
Epoch 67/200
30/30 0s 148ms/step -
loss: 0.0054 - mae: 0.0487
Epoch 67: val_loss did not improve from 0.00541
30/30 5s 160ms/step -
loss: 0.0054 - mae: 0.0487 - val_loss: 0.0056 - val_mae: 0.0489
Epoch 68/200

```

30/30          0s 148ms/step -
loss: 0.0049 - mae: 0.0458
Epoch 68: val_loss improved from 0.00541 to 0.00532, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/unet_model.keras
30/30          5s 166ms/step -
loss: 0.0049 - mae: 0.0459 - val_loss: 0.0053 - val_mae: 0.0467
Epoch 69/200
30/30          0s 148ms/step -
loss: 0.0053 - mae: 0.0491
Epoch 69: val_loss did not improve from 0.00532
30/30          5s 156ms/step -
loss: 0.0053 - mae: 0.0491 - val_loss: 0.0056 - val_mae: 0.0502
Epoch 70/200
30/30          0s 148ms/step -
loss: 0.0053 - mae: 0.0491
Epoch 70: val_loss did not improve from 0.00532
30/30          5s 161ms/step -
loss: 0.0053 - mae: 0.0491 - val_loss: 0.0054 - val_mae: 0.0468
Epoch 71/200
30/30          0s 148ms/step -
loss: 0.0049 - mae: 0.0463
Epoch 71: val_loss did not improve from 0.00532
30/30          5s 156ms/step -
loss: 0.0049 - mae: 0.0463 - val_loss: 0.0064 - val_mae: 0.0565
Epoch 72/200
30/30          0s 148ms/step -
loss: 0.0049 - mae: 0.0457
Epoch 72: val_loss did not improve from 0.00532
30/30          5s 160ms/step -
loss: 0.0049 - mae: 0.0457 - val_loss: 0.0054 - val_mae: 0.0464
Epoch 73/200
30/30          0s 148ms/step -
loss: 0.0049 - mae: 0.0468
Epoch 73: val_loss did not improve from 0.00532
30/30          5s 160ms/step -
loss: 0.0049 - mae: 0.0468 - val_loss: 0.0054 - val_mae: 0.0478
Epoch 74/200
30/30          0s 147ms/step -
loss: 0.0050 - mae: 0.0468
Epoch 74: val_loss did not improve from 0.00532
30/30          5s 155ms/step -
loss: 0.0050 - mae: 0.0468 - val_loss: 0.0057 - val_mae: 0.0506
Epoch 75/200
30/30          0s 148ms/step -
loss: 0.0056 - mae: 0.0485
Epoch 75: val_loss did not improve from 0.00532
30/30          5s 160ms/step -

```

```

loss: 0.0055 - mae: 0.0484 - val_loss: 0.0056 - val_mae: 0.0472
Epoch 76/200
30/30          0s 147ms/step -
loss: 0.0051 - mae: 0.0473
Epoch 76: val_loss improved from 0.00532 to 0.00531, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/unet_model.keras
30/30          5s 161ms/step -
loss: 0.0051 - mae: 0.0473 - val_loss: 0.0053 - val_mae: 0.0471
Epoch 77/200
30/30          0s 148ms/step -
loss: 0.0052 - mae: 0.0491
Epoch 77: val_loss did not improve from 0.00531
30/30          5s 156ms/step -
loss: 0.0052 - mae: 0.0491 - val_loss: 0.0060 - val_mae: 0.0520
Epoch 78/200
30/30          0s 147ms/step -
loss: 0.0049 - mae: 0.0458
Epoch 78: val_loss did not improve from 0.00531
30/30          5s 155ms/step -
loss: 0.0049 - mae: 0.0458 - val_loss: 0.0054 - val_mae: 0.0478
Epoch 79/200
30/30          0s 146ms/step -
loss: 0.0046 - mae: 0.0434
Epoch 79: val_loss did not improve from 0.00531
30/30          5s 154ms/step -
loss: 0.0046 - mae: 0.0434 - val_loss: 0.0054 - val_mae: 0.0477
Epoch 80/200
30/30          0s 146ms/step -
loss: 0.0047 - mae: 0.0452
Epoch 80: val_loss did not improve from 0.00531
30/30          5s 154ms/step -
loss: 0.0047 - mae: 0.0453 - val_loss: 0.0072 - val_mae: 0.0617
Epoch 81/200
30/30          0s 148ms/step -
loss: 0.0051 - mae: 0.0490
Epoch 81: val_loss did not improve from 0.00531
30/30          5s 155ms/step -
loss: 0.0051 - mae: 0.0489 - val_loss: 0.0056 - val_mae: 0.0483
Epoch 82/200
30/30          0s 147ms/step -
loss: 0.0052 - mae: 0.0471
Epoch 82: val_loss did not improve from 0.00531
30/30          5s 158ms/step -
loss: 0.0052 - mae: 0.0471 - val_loss: 0.0057 - val_mae: 0.0500
Epoch 83/200
30/30          0s 146ms/step -
loss: 0.0054 - mae: 0.0480

```

Epoch 83: val_loss did not improve from 0.00531
30/30 5s 155ms/step -
loss: 0.0053 - mae: 0.0479 - val_loss: 0.0061 - val_mae: 0.0529
Epoch 84/200
30/30 0s 148ms/step -
loss: 0.0056 - mae: 0.0498
Epoch 84: val_loss did not improve from 0.00531
30/30 5s 156ms/step -
loss: 0.0056 - mae: 0.0497 - val_loss: 0.0057 - val_mae: 0.0501
Epoch 85/200
30/30 0s 147ms/step -
loss: 0.0055 - mae: 0.0501
Epoch 85: val_loss did not improve from 0.00531
30/30 5s 154ms/step -
loss: 0.0055 - mae: 0.0501 - val_loss: 0.0061 - val_mae: 0.0542
Epoch 86/200
30/30 0s 147ms/step -
loss: 0.0052 - mae: 0.0485
Epoch 86: val_loss did not improve from 0.00531
30/30 5s 156ms/step -
loss: 0.0052 - mae: 0.0485 - val_loss: 0.0055 - val_mae: 0.0476
Epoch 87/200
30/30 0s 146ms/step -
loss: 0.0052 - mae: 0.0486
Epoch 87: val_loss did not improve from 0.00531
30/30 5s 154ms/step -
loss: 0.0052 - mae: 0.0485 - val_loss: 0.0058 - val_mae: 0.0499
Epoch 88/200
30/30 0s 146ms/step -
loss: 0.0051 - mae: 0.0478
Epoch 88: val_loss did not improve from 0.00531
30/30 5s 158ms/step -
loss: 0.0051 - mae: 0.0478 - val_loss: 0.0058 - val_mae: 0.0491
Epoch 89/200
30/30 0s 147ms/step -
loss: 0.0047 - mae: 0.0443
Epoch 89: val_loss did not improve from 0.00531
30/30 5s 159ms/step -
loss: 0.0047 - mae: 0.0443 - val_loss: 0.0066 - val_mae: 0.0553
Epoch 90/200
30/30 0s 147ms/step -
loss: 0.0052 - mae: 0.0471
Epoch 90: val_loss did not improve from 0.00531
30/30 5s 155ms/step -
loss: 0.0052 - mae: 0.0471 - val_loss: 0.0070 - val_mae: 0.0571
Epoch 91/200
30/30 0s 148ms/step -
loss: 0.0052 - mae: 0.0491

```

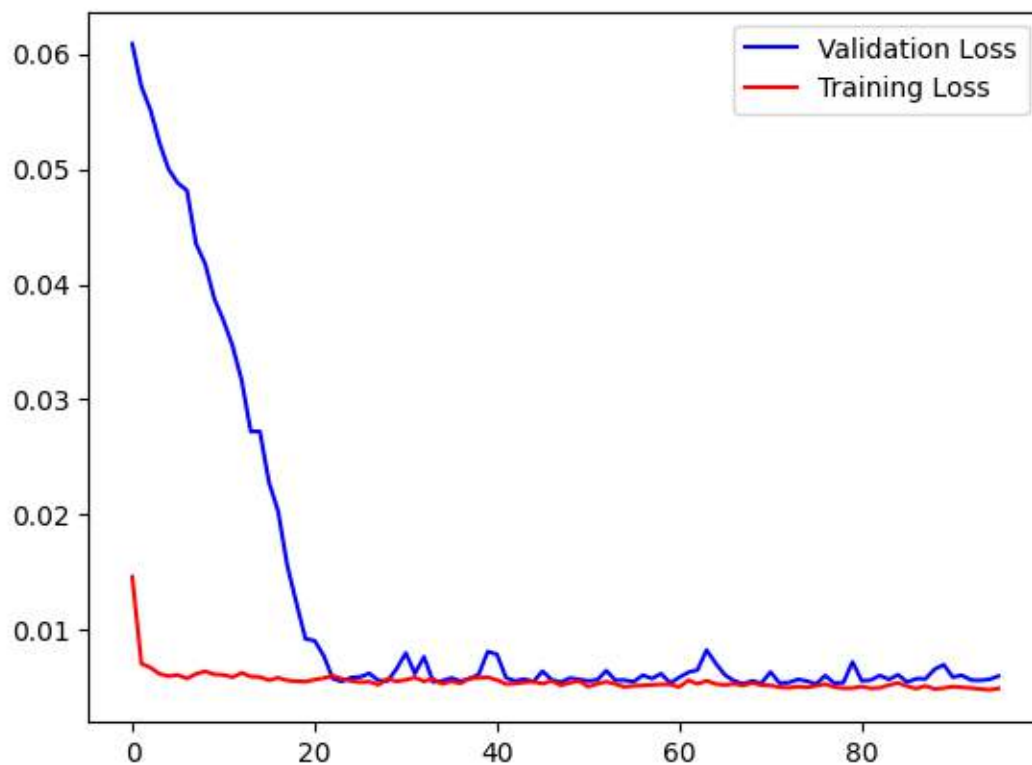
Epoch 91: val_loss did not improve from 0.00531
30/30          5s 160ms/step -
loss: 0.0052 - mae: 0.0491 - val_loss: 0.0059 - val_mae: 0.0523
Epoch 92/200
30/30          0s 148ms/step -
loss: 0.0051 - mae: 0.0474
Epoch 92: val_loss did not improve from 0.00531
30/30          5s 160ms/step -
loss: 0.0051 - mae: 0.0474 - val_loss: 0.0061 - val_mae: 0.0550
Epoch 93/200
30/30          0s 147ms/step -
loss: 0.0048 - mae: 0.0463
Epoch 93: val_loss did not improve from 0.00531
30/30          5s 158ms/step -
loss: 0.0048 - mae: 0.0463 - val_loss: 0.0057 - val_mae: 0.0504
Epoch 94/200
30/30          0s 148ms/step -
loss: 0.0046 - mae: 0.0448
Epoch 94: val_loss did not improve from 0.00531
30/30          5s 157ms/step -
loss: 0.0046 - mae: 0.0448 - val_loss: 0.0056 - val_mae: 0.0498
Epoch 95/200
30/30          0s 148ms/step -
loss: 0.0047 - mae: 0.0443
Epoch 95: val_loss did not improve from 0.00531
30/30          5s 160ms/step -
loss: 0.0047 - mae: 0.0443 - val_loss: 0.0057 - val_mae: 0.0499
Epoch 96/200
30/30          0s 147ms/step -
loss: 0.0048 - mae: 0.0453
Epoch 96: val_loss did not improve from 0.00531
30/30          5s 155ms/step -
loss: 0.0048 - mae: 0.0454 - val_loss: 0.0060 - val_mae: 0.0515

```

```

[ ]: # Plotting the training and validation loss
plt.plot(history_unet_model.history['val_loss'],c="b")
plt.plot(history_unet_model.history['loss'],c="r")
plt.legend(['Validation Loss','Training Loss'])
plt.show()

```



Depiction of Model Performance on Images from the Training Set

```
[ ]: show_predictions(best_unet_model, X, Y, num_samples=5)
```

1/1

1s 997ms/step



Depiction of Model Performance on Images from the Test Set

```
[ ]: show_predictions(best_unet_model, X_test, Y_test, num_samples=5)
```

1/1 0s 63ms/step



The outputs didn't change significantly from the previous model. While the spatial structure and main forms are still well preserved, the model continues to default to warm, beige-toned hues across the entire image, applying similar shades to skin, clothing, and background pavement. Despite the addition of more layers and a reinforced architecture, the overall colorization quality remains largely the same, highlighting the model's ongoing limitations in capturing semantic context and color diversity.

5.6 Model 5: Sequential LAB Model

```
[ ]: # Defining the architecture for the sequential LAB model
# Encoder
seq_lab_model = Sequential()
seq_lab_model.add(Conv2D(64, (3, 3), activation='relu', padding='same',
    ↳ strides=2, input_shape=(120, 176, 1)))
seq_lab_model.add(Conv2D(128, (3, 3), activation='relu', padding='same'))
seq_lab_model.add(Conv2D(128, (3,3), activation='relu', padding='same',
    ↳ strides=2))
seq_lab_model.add(Conv2D(256, (3,3), activation='relu', padding='same'))
seq_lab_model.add(Conv2D(512, (3,3), activation='relu', padding='same'))
seq_lab_model.add(Conv2D(512, (3,3), activation='relu', padding='same'))
seq_lab_model.add(Conv2D(256, (3,3), activation='relu', padding='same',
    ↳ strides=2))
seq_lab_model.add(Conv2D(512, (3,3), activation='relu', padding='same'))
seq_lab_model.add(Conv2D(512, (3,3), activation='relu', padding='same'))
seq_lab_model.add(Conv2D(256, (3,3), activation='relu', padding='same'))
# Decoder
seq_lab_model.add(Conv2D(128, (3,3), activation='relu', padding='same'))
seq_lab_model.add(UpSampling2D((2, 2)))
seq_lab_model.add(Conv2D(64, (3,3), activation='relu', padding='same'))
seq_lab_model.add(UpSampling2D((2, 2)))
seq_lab_model.add(Conv2D(32, (3,3), activation='relu', padding='same'))
seq_lab_model.add(Conv2D(16, (3,3), activation='relu', padding='same'))
seq_lab_model.add(Conv2D(2, (3, 3), activation='tanh', padding='same'))
seq_lab_model.add(UpSampling2D((2, 2)))
seq_lab_model.compile(optimizer=Adam(learning_rate=1e-4), loss='mae',
    ↳ metrics=['mse'])
seq_lab_model.summary()
```

```
/usr/local/lib/python3.11/dist-
packages/keras/src/layers/convolutional/base_conv.py:107: UserWarning: Do not
pass an `input_shape`/`input_dim` argument to a layer. When using Sequential
models, prefer using an `Input(shape)` object as the first layer in the model
instead.
```

```
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d_24 (Conv2D)	(None, 60, 88, 64)	640
conv2d_25 (Conv2D)	(None, 60, 88, 128)	73,856
conv2d_26 (Conv2D)	(None, 30, 44, 128)	147,584
conv2d_27 (Conv2D)	(None, 30, 44, 256)	295,168
conv2d_28 (Conv2D)	(None, 30, 44, 512)	1,180,160
conv2d_29 (Conv2D)	(None, 30, 44, 512)	2,359,808
conv2d_30 (Conv2D)	(None, 15, 22, 256)	1,179,904
conv2d_31 (Conv2D)	(None, 15, 22, 512)	1,180,160
conv2d_32 (Conv2D)	(None, 15, 22, 512)	2,359,808
conv2d_33 (Conv2D)	(None, 15, 22, 256)	1,179,904
conv2d_34 (Conv2D)	(None, 15, 22, 128)	295,040
up_sampling2d_4 (UpSampling2D)	(None, 30, 44, 128)	0
conv2d_35 (Conv2D)	(None, 30, 44, 64)	73,792
up_sampling2d_5 (UpSampling2D)	(None, 60, 88, 64)	0
conv2d_36 (Conv2D)	(None, 60, 88, 32)	18,464
conv2d_37 (Conv2D)	(None, 60, 88, 16)	4,624
conv2d_38 (Conv2D)	(None, 60, 88, 2)	290
up_sampling2d_6 (UpSampling2D)	(None, 120, 176, 2)	0

Total params: 10,349,202 (39.48 MB)

Trainable params: 10,349,202 (39.48 MB)

Non-trainable params: 0 (0.00 B)

```
[ ]: # Defining the path to save the best model
model_checkpoint_path = os.path.join(best_models_path, 'seq_lab_model.keras')
model_checkpoint = ModelCheckpoint(model_checkpoint_path, monitor='val_loss',
    ↳ verbose=1, save_best_only=True, mode='min')

# Fitting the model to the training data
history_seq_lab_model = seq_lab_model.fit(X_lab,Y_lab,batch_size=16,
    ↳ epochs=200, validation_split=0.1, callbacks=[early_stop, model_checkpoint])

best_seq_lab_model = keras.models.load_model(os.path.join(best_models_path,
    ↳ 'seq_lab_model.keras'))
```

```
Epoch 1/200
30/30          0s 676ms/step -
loss: 0.0637 - mse: 0.0091
Epoch 1: val_loss improved from inf to 0.06261, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/seq_lab_model.keras
30/30          53s 994ms/step -
loss: 0.0636 - mse: 0.0091 - val_loss: 0.0626 - val_mse: 0.0095
Epoch 2/200
30/30          0s 141ms/step -
loss: 0.0583 - mse: 0.0085
Epoch 2: val_loss improved from 0.06261 to 0.06218, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/seq_lab_model.keras
30/30          40s 180ms/step -
loss: 0.0583 - mse: 0.0085 - val_loss: 0.0622 - val_mse: 0.0095
Epoch 3/200
30/30          0s 141ms/step -
loss: 0.0580 - mse: 0.0085
Epoch 3: val_loss improved from 0.06218 to 0.06203, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/seq_lab_model.keras
30/30          10s 174ms/step -
loss: 0.0580 - mse: 0.0085 - val_loss: 0.0620 - val_mse: 0.0095
Epoch 4/200
30/30          0s 143ms/step -
loss: 0.0577 - mse: 0.0086
Epoch 4: val_loss improved from 0.06203 to 0.06159, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/seq_lab_model.keras
30/30          5s 174ms/step -
loss: 0.0577 - mse: 0.0086 - val_loss: 0.0616 - val_mse: 0.0093
Epoch 5/200
```

```

30/30          0s 144ms/step -
loss: 0.0584 - mse: 0.0086
Epoch 5: val_loss improved from 0.06159 to 0.06131, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/seq_lab_model.keras
30/30          5s 174ms/step -
loss: 0.0584 - mse: 0.0086 - val_loss: 0.0613 - val_mse: 0.0092
Epoch 6/200
30/30          0s 144ms/step -
loss: 0.0547 - mse: 0.0078
Epoch 6: val_loss improved from 0.06131 to 0.06114, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/seq_lab_model.keras
30/30         10s 176ms/step -
loss: 0.0547 - mse: 0.0078 - val_loss: 0.0611 - val_mse: 0.0091
Epoch 7/200
30/30          0s 144ms/step -
loss: 0.0566 - mse: 0.0083
Epoch 7: val_loss did not improve from 0.06114
30/30          5s 156ms/step -
loss: 0.0566 - mse: 0.0082 - val_loss: 0.0612 - val_mse: 0.0092
Epoch 8/200
30/30          0s 147ms/step -
loss: 0.0576 - mse: 0.0085
Epoch 8: val_loss improved from 0.06114 to 0.06068, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/seq_lab_model.keras
30/30          6s 194ms/step -
loss: 0.0576 - mse: 0.0085 - val_loss: 0.0607 - val_mse: 0.0090
Epoch 9/200
30/30          0s 147ms/step -
loss: 0.0568 - mse: 0.0083
Epoch 9: val_loss improved from 0.06068 to 0.06063, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/seq_lab_model.keras
30/30          5s 181ms/step -
loss: 0.0568 - mse: 0.0083 - val_loss: 0.0606 - val_mse: 0.0090
Epoch 10/200
30/30          0s 147ms/step -
loss: 0.0542 - mse: 0.0078
Epoch 10: val_loss did not improve from 0.06063
30/30         10s 155ms/step -
loss: 0.0543 - mse: 0.0078 - val_loss: 0.0608 - val_mse: 0.0091
Epoch 11/200
30/30          0s 150ms/step -
loss: 0.0556 - mse: 0.0084
Epoch 11: val_loss did not improve from 0.06063
30/30          5s 164ms/step -

```

```

loss: 0.0556 - mse: 0.0083 - val_loss: 0.0607 - val_mse: 0.0091
Epoch 12/200
30/30          0s 150ms/step -
loss: 0.0570 - mse: 0.0084
Epoch 12: val_loss improved from 0.06063 to 0.06037, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/seq_lab_model.keras
30/30          5s 181ms/step -
loss: 0.0569 - mse: 0.0084 - val_loss: 0.0604 - val_mse: 0.0089
Epoch 13/200
30/30          0s 149ms/step -
loss: 0.0545 - mse: 0.0076
Epoch 13: val_loss did not improve from 0.06037
30/30          10s 158ms/step -
loss: 0.0545 - mse: 0.0076 - val_loss: 0.0605 - val_mse: 0.0088
Epoch 14/200
30/30          0s 152ms/step -
loss: 0.0563 - mse: 0.0079
Epoch 14: val_loss did not improve from 0.06037
30/30          5s 164ms/step -
loss: 0.0563 - mse: 0.0079 - val_loss: 0.0614 - val_mse: 0.0093
Epoch 15/200
30/30          0s 152ms/step -
loss: 0.0561 - mse: 0.0083
Epoch 15: val_loss improved from 0.06037 to 0.06029, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/seq_lab_model.keras
30/30          6s 190ms/step -
loss: 0.0561 - mse: 0.0083 - val_loss: 0.0603 - val_mse: 0.0090
Epoch 16/200
30/30          0s 151ms/step -
loss: 0.0536 - mse: 0.0075
Epoch 16: val_loss improved from 0.06029 to 0.06020, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/seq_lab_model.keras
30/30          10s 186ms/step -
loss: 0.0536 - mse: 0.0076 - val_loss: 0.0602 - val_mse: 0.0090
Epoch 17/200
30/30          0s 153ms/step -
loss: 0.0571 - mse: 0.0085
Epoch 17: val_loss did not improve from 0.06020
30/30          5s 162ms/step -
loss: 0.0570 - mse: 0.0085 - val_loss: 0.0607 - val_mse: 0.0092
Epoch 18/200
30/30          0s 153ms/step -
loss: 0.0541 - mse: 0.0078
Epoch 18: val_loss did not improve from 0.06020
30/30          5s 162ms/step -

```



```

loss: 0.0541 - mse: 0.0078 - val_loss: 0.0604 - val_mse: 0.0089
Epoch 19/200
30/30          0s 155ms/step -
loss: 0.0536 - mse: 0.0076
Epoch 19: val_loss did not improve from 0.06020
30/30          5s 166ms/step -
loss: 0.0536 - mse: 0.0076 - val_loss: 0.0604 - val_mse: 0.0088
Epoch 20/200
30/30          0s 155ms/step -
loss: 0.0515 - mse: 0.0070
Epoch 20: val_loss did not improve from 0.06020
30/30          5s 167ms/step -
loss: 0.0515 - mse: 0.0071 - val_loss: 0.0614 - val_mse: 0.0088
Epoch 21/200
30/30          0s 156ms/step -
loss: 0.0520 - mse: 0.0070
Epoch 21: val_loss did not improve from 0.06020
30/30          5s 164ms/step -
loss: 0.0520 - mse: 0.0070 - val_loss: 0.0606 - val_mse: 0.0091
Epoch 22/200
30/30          0s 157ms/step -
loss: 0.0515 - mse: 0.0071
Epoch 22: val_loss improved from 0.06020 to 0.05989, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
Portrait/seq_lab_model.keras
30/30          6s 197ms/step -
loss: 0.0515 - mse: 0.0071 - val_loss: 0.0599 - val_mse: 0.0089
Epoch 23/200
30/30          0s 158ms/step -
loss: 0.0481 - mse: 0.0064
Epoch 23: val_loss did not improve from 0.05989
30/30          5s 166ms/step -
loss: 0.0482 - mse: 0.0064 - val_loss: 0.0613 - val_mse: 0.0092
Epoch 24/200
30/30          0s 159ms/step -
loss: 0.0491 - mse: 0.0064
Epoch 24: val_loss did not improve from 0.05989
30/30          5s 167ms/step -
loss: 0.0491 - mse: 0.0064 - val_loss: 0.0605 - val_mse: 0.0089
Epoch 25/200
30/30          0s 160ms/step -
loss: 0.0469 - mse: 0.0062
Epoch 25: val_loss did not improve from 0.05989
30/30          5s 169ms/step -
loss: 0.0469 - mse: 0.0062 - val_loss: 0.0609 - val_mse: 0.0089
Epoch 26/200
30/30          0s 161ms/step -
loss: 0.0474 - mse: 0.0065

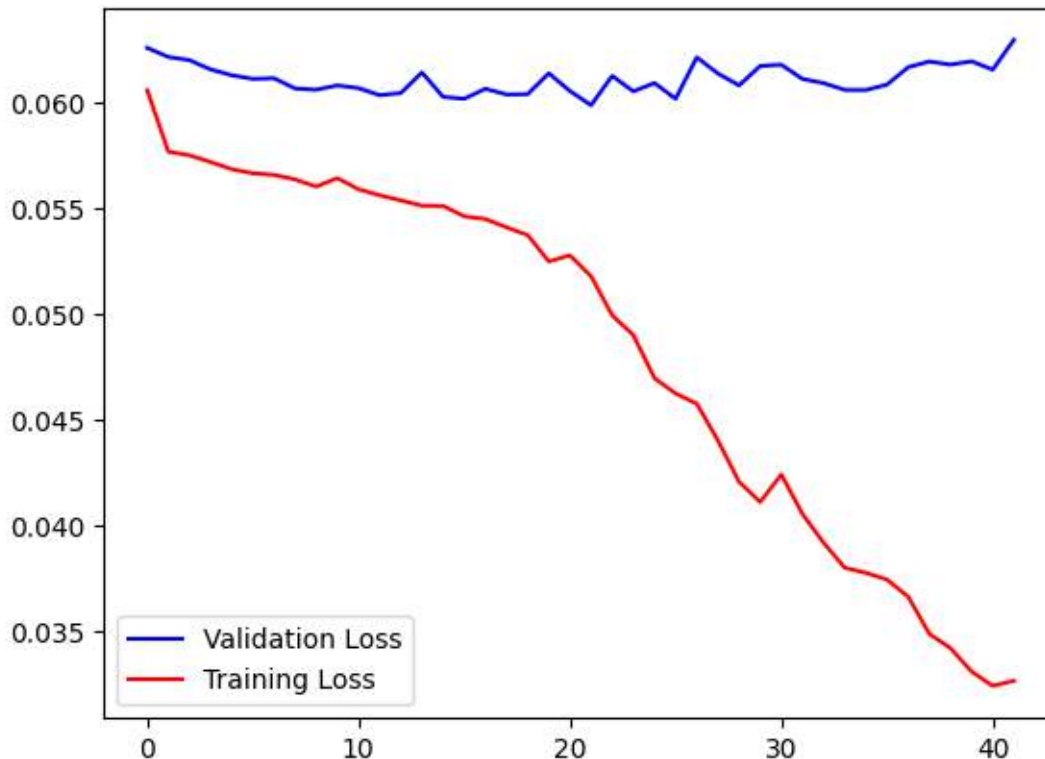
```

Epoch 26: val_loss did not improve from 0.05989
30/30 5s 173ms/step -
loss: 0.0473 - mse: 0.0064 - val_loss: 0.0602 - val_mse: 0.0089
Epoch 27/200
30/30 0s 160ms/step -
loss: 0.0457 - mse: 0.0056
Epoch 27: val_loss did not improve from 0.05989
30/30 5s 173ms/step -
loss: 0.0457 - mse: 0.0056 - val_loss: 0.0622 - val_mse: 0.0088
Epoch 28/200
30/30 0s 159ms/step -
loss: 0.0446 - mse: 0.0058
Epoch 28: val_loss did not improve from 0.05989
30/30 5s 168ms/step -
loss: 0.0445 - mse: 0.0058 - val_loss: 0.0614 - val_mse: 0.0089
Epoch 29/200
30/30 0s 159ms/step -
loss: 0.0412 - mse: 0.0047
Epoch 29: val_loss did not improve from 0.05989
30/30 5s 168ms/step -
loss: 0.0412 - mse: 0.0047 - val_loss: 0.0608 - val_mse: 0.0089
Epoch 30/200
30/30 0s 160ms/step -
loss: 0.0416 - mse: 0.0051
Epoch 30: val_loss did not improve from 0.05989
30/30 5s 172ms/step -
loss: 0.0415 - mse: 0.0051 - val_loss: 0.0618 - val_mse: 0.0090
Epoch 31/200
30/30 0s 159ms/step -
loss: 0.0418 - mse: 0.0047
Epoch 31: val_loss did not improve from 0.05989
30/30 5s 171ms/step -
loss: 0.0418 - mse: 0.0047 - val_loss: 0.0618 - val_mse: 0.0091
Epoch 32/200
30/30 0s 157ms/step -
loss: 0.0426 - mse: 0.0051
Epoch 32: val_loss did not improve from 0.05989
30/30 5s 170ms/step -
loss: 0.0425 - mse: 0.0051 - val_loss: 0.0611 - val_mse: 0.0089
Epoch 33/200
30/30 0s 158ms/step -
loss: 0.0395 - mse: 0.0044
Epoch 33: val_loss did not improve from 0.05989
30/30 5s 169ms/step -
loss: 0.0395 - mse: 0.0044 - val_loss: 0.0609 - val_mse: 0.0089
Epoch 34/200
30/30 0s 157ms/step -
loss: 0.0391 - mse: 0.0043

Epoch 34: val_loss did not improve from 0.05989
30/30 5s 169ms/step -
loss: 0.0390 - mse: 0.0043 - val_loss: 0.0606 - val_mse: 0.0088
Epoch 35/200
30/30 0s 158ms/step -
loss: 0.0369 - mse: 0.0038
Epoch 35: val_loss did not improve from 0.05989
30/30 5s 170ms/step -
loss: 0.0369 - mse: 0.0038 - val_loss: 0.0606 - val_mse: 0.0089
Epoch 36/200
30/30 0s 157ms/step -
loss: 0.0376 - mse: 0.0039
Epoch 36: val_loss did not improve from 0.05989
30/30 5s 166ms/step -
loss: 0.0376 - mse: 0.0039 - val_loss: 0.0609 - val_mse: 0.0088
Epoch 37/200
30/30 0s 158ms/step -
loss: 0.0372 - mse: 0.0040
Epoch 37: val_loss did not improve from 0.05989
30/30 5s 170ms/step -
loss: 0.0371 - mse: 0.0039 - val_loss: 0.0617 - val_mse: 0.0092
Epoch 38/200
30/30 0s 155ms/step -
loss: 0.0347 - mse: 0.0034
Epoch 38: val_loss did not improve from 0.05989
30/30 10s 164ms/step -
loss: 0.0347 - mse: 0.0034 - val_loss: 0.0620 - val_mse: 0.0093
Epoch 39/200
30/30 0s 157ms/step -
loss: 0.0341 - mse: 0.0032
Epoch 39: val_loss did not improve from 0.05989
30/30 5s 169ms/step -
loss: 0.0341 - mse: 0.0032 - val_loss: 0.0618 - val_mse: 0.0091
Epoch 40/200
30/30 0s 159ms/step -
loss: 0.0329 - mse: 0.0030
Epoch 40: val_loss did not improve from 0.05989
30/30 5s 170ms/step -
loss: 0.0329 - mse: 0.0030 - val_loss: 0.0620 - val_mse: 0.0092
Epoch 41/200
30/30 0s 159ms/step -
loss: 0.0322 - mse: 0.0029
Epoch 41: val_loss did not improve from 0.05989
30/30 10s 168ms/step -
loss: 0.0322 - mse: 0.0029 - val_loss: 0.0616 - val_mse: 0.0090
Epoch 42/200
30/30 0s 161ms/step -
loss: 0.0336 - mse: 0.0031

```
Epoch 42: val_loss did not improve from 0.05989
30/30          5s 169ms/step -
loss: 0.0335 - mse: 0.0031 - val_loss: 0.0630 - val_mse: 0.0096
```

```
[ ]: # Plotting the training and validation loss
plt.plot(history_seq_lab_model.history['val_loss'],c="b")
plt.plot(history_seq_lab_model.history['loss'],c="r")
plt.legend(['Validation Loss','Training Loss'])
plt.show()
```



Depiction of Model Performance on images from the Training Set

```
[ ]: show_predictions_lab(best_seq_lab_model, X_lab, Y_lab, num_samples=5)
```

WARNING:tensorflow:5 out of the last 9 calls to <function TensorFlowTrainer.make_predict_function.<locals>.one_step_on_data_distributed at 0x7e2f56b11080> triggered tf.function retracing. Tracing is expensive and the excessive number of tracings could be due to (1) creating @tf.function repeatedly in a loop, (2) passing tensors with different shapes, (3) passing Python objects instead of tensors. For (1), please define your @tf.function outside of the loop. For (2), @tf.function has reduce_retracing=True option that can avoid unnecessary retracing. For (3), please refer to https://www.tensorflow.org/guide/function#controlling_retracing and

https://www.tensorflow.org/api_docs/python/tf/function for more details.

1/1 1s 752ms/step



Depiction of Model Performance on images from the Test Set

```
[ ]: show_predictions_lab(best_seq_lab_model, X_lab_test, Y_lab_test,num_samples=5)
```

1/1 0s 69ms/step



The training and validation curves showed clear overfitting since the training loss decreased significantly and the validation loss plateaued early on. This resulted in the model performing well on training images, almost matching the ground truth. But when the model is applied to unseen test images, the color predictions become inaccurate as the model applies a beige tone across the scene, failing to distinguish between skin, slothing, and background elements. The lack of color highlights the model's difficulty generalizing beyond the training set. This resulted in the model performing well on training images, almost matching the ground truth when ran with a lower patience and higher epoch.

The overfitting of the training data is a promising sign, as it shows the model has enough learning capacity to map grayscale input. This is the first model where we can clearly see that learning is happening, and the model is starting to predict color in patches. We assumed that this architecture has strong potential to improve.

5.7 Model 6: Transformer with VGG

```
[ ]: # Defining the custom loss functions that we experimented with
def perceptual_loss(y_true, y_pred):
    # Scale from [0, 1] to [0, 255] before using VGG
    y_true_proc = preprocess_input(y_true * 255.0)
    y_pred_proc = preprocess_input(y_pred * 255.0)

    y_true_features = feature_extractor(y_true_proc)
    y_pred_features = feature_extractor(y_pred_proc)

    return tf.reduce_mean(tf.square(y_true_features - y_pred_features))

def hybrid_loss(y_true, y_pred):
    mae_loss = tf.reduce_mean(tf.abs(y_true - y_pred))
    p_loss = perceptual_loss(y_true, y_pred)
    return 0.3 * p_loss + 0.7 * mae_loss
```

```
[ ]: # Defining the transformer encoder and decoder blocks
def transformer_encoder(inputs, num_heads=4, ff_dim=256):
    # Flatten spatial dims into sequence
    _, H, W, C = inputs.shape
    x = Reshape((H * W, C))(inputs)

    # LayerNorm + MHA
    x_norm = LayerNormalization(epsilon=1e-6)(x)
    attn_output = MultiHeadAttention(num_heads=num_heads, key_dim=C)(x_norm, x_norm)
    ↪ x_norm)
    x = Add()([x, attn_output])

    # Feed-forward network
```

```

x_norm = LayerNormalization(epsilon=1e-6)(x)
ff_output = Dense(ff_dim, activation='relu')(x_norm)
ff_output = Dense(C)(ff_output)
x = Add()([x, ff_output])

x = Reshape((H, W, C))(x)
return x

def build_decoder(vgg_output):
    # 15x22x256 → 30x44x256
    x = UpSampling2D((2, 2))(vgg_output)
    x = Conv2D(256, (3, 3), padding='same')(x)
    x = BatchNormalization()(x)
    x = Activation('relu')(x)

    # 30x44x256 → 60x88x128
    x = UpSampling2D((2, 2))(x)
    x = Conv2D(128, (3, 3), padding='same')(x)
    x = BatchNormalization()(x)
    x = Activation('relu')(x)

    # 60x88x128 → 120x176x64
    x = UpSampling2D((2, 2))(x)
    x = Conv2D(64, (3, 3), padding='same')(x)
    x = BatchNormalization()(x)
    x = Activation('relu')(x)

    x = Conv2D(32, (3, 3), padding='same')(x)
    x = BatchNormalization()(x)
    x = Activation('relu')(x)

    output_rgb = Conv2D(3, (3, 3), activation='sigmoid', padding='same')(x)
    return output_rgb

```

```

[ ]: # Defining the transformer + vgg model architecture
# Input grayscale image
input_1 = Input(shape=(120, 176, 1), name='grayscale_input')
x_rgb = Lambda(lambda x: tf.image.grayscale_to_rgb(x))(input_1)

# VGG encoder
vgg_base = VGG16(include_top=False, weights='imagenet', input_tensor=x_rgb)
for layer in vgg_base.layers:
    layer.trainable = False

vgg_features = vgg_base.get_layer('block3_conv3').output # (15, 22, 256)

# Transformer block

```

```

x = transformer_encoder(vgg_features, num_heads=4, ff_dim=512)

# Decoder block
x = build_decoder(x)

output_rgb = Lambda(lambda x: tf.image.resize_with_crop_or_pad(x, 120, 176))(x)

trans_vgg_model = Model(inputs=input_1, outputs=output_rgb)

# Feature extractor for perceptual loss
vgg_feat_model = VGG16(include_top=False, weights='imagenet', input_shape=(120, 176, 3))
vgg_feat_model.trainable = False
feature_extractor = Model(inputs=vgg_feat_model.input, outputs=vgg_feat_model.get_layer('block3_conv3').output)

trans_vgg_model.compile(optimizer='adam', loss='mse', metrics=['mae'])
trans_vgg_model.summary()

```

Model: "functional_82"

Layer (type)	Output Shape	Param #	Connected to
grayscale_input (InputLayer)	(None, 120, 176, 1)	0	-
lambda_6 (Lambda)	(None, 120, 176, 3)	0	grayscale_input[...]
block1_conv1 (Conv2D)	(None, 120, 176, 64)	1,792	lambda_6[0][0]
block1_conv2 (Conv2D)	(None, 120, 176, 64)	36,928	block1_conv1[0][...]
block1_pool (MaxPooling2D)	(None, 60, 88, 64)	0	block1_conv2[0][...]
block2_conv1 (Conv2D)	(None, 60, 88, 128)	73,856	block1_pool[0][0]
block2_conv2 (Conv2D)	(None, 60, 88, 128)	147,584	block2_conv1[0][...]
block2_pool (MaxPooling2D)	(None, 30, 44, 128)	0	block2_conv2[0][...]

block3_conv1 (Conv2D)	(None, 30, 44, 256)	295,168	block2_pool[0][0]
block3_conv2 (Conv2D)	(None, 30, 44, 256)	590,080	block3_conv1[0][...]
block3_conv3 (Conv2D)	(None, 30, 44, 256)	590,080	block3_conv2[0][...]
reshape_4 (Reshape)	(None, 1320, 256)	0	block3_conv3[0][...]
layer_normalizatio... (LayerNormalizatio...	(None, 1320, 256)	512	reshape_4[0][0]
multi_head_attenti... (MultiHeadAttentio...	(None, 1320, 256)	1,051,904	layer_normalizat... layer_normalizat...
add_4 (Add)	(None, 1320, 256)	0	reshape_4[0][0], multi_head_atten...
layer_normalizatio... (LayerNormalizatio...	(None, 1320, 256)	512	add_4[0][0]
dense_4 (Dense)	(None, 1320, 512)	131,584	layer_normalizat...
dense_5 (Dense)	(None, 1320, 256)	131,328	dense_4[0][0]
add_5 (Add)	(None, 1320, 256)	0	add_4[0][0], dense_5[0][0]
reshape_5 (Reshape)	(None, 30, 44, 256)	0	add_5[0][0]
up_sampling2d_20 (UpSampling2D)	(None, 60, 88, 256)	0	reshape_5[0][0]
conv2d_88 (Conv2D)	(None, 60, 88, 256)	590,080	up_sampling2d_20...
batch_normalizatio... (BatchNormalizatio...	(None, 60, 88, 256)	1,024	conv2d_88[0][0]
activation_8 (Activation)	(None, 60, 88, 256)	0	batch_normalizat...
up_sampling2d_21 (UpSampling2D)	(None, 120, 176, 256)	0	activation_8[0][...]

conv2d_89 (Conv2D)	(None, 120, 176, 128)	295,040	up_sampling2d_21...
batch_normalizatio... (BatchNormalizatio...	(None, 120, 176, 128)	512	conv2d_89[0][0]
activation_9 (Activation)	(None, 120, 176, 128)	0	batch_normalizat...
up_sampling2d_22 (UpSampling2D)	(None, 240, 352, 128)	0	activation_9[0][...
conv2d_90 (Conv2D)	(None, 240, 352, 64)	73,792	up_sampling2d_22...
batch_normalizatio... (BatchNormalizatio...	(None, 240, 352, 64)	256	conv2d_90[0][0]
activation_10 (Activation)	(None, 240, 352, 64)	0	batch_normalizat...
conv2d_91 (Conv2D)	(None, 240, 352, 32)	18,464	activation_10[0]...
batch_normalizatio... (BatchNormalizatio...	(None, 240, 352, 32)	128	conv2d_91[0][0]
activation_11 (Activation)	(None, 240, 352, 32)	0	batch_normalizat...
conv2d_92 (Conv2D)	(None, 240, 352, 3)	867	activation_11[0]...
lambda_7 (Lambda)	(None, 120, 176, 3)	0	conv2d_92[0][0]

Total params: 4,031,491 (15.38 MB)

Trainable params: 2,295,043 (8.75 MB)

Non-trainable params: 1,736,448 (6.62 MB)

```
[ ]: # Fitting the model to the training data
history_trans_vgg = trans_vgg_model.fit(X, Y, batch_size=16, epochs=200,
    ↪ validation_split=0.20, callbacks=[early_stop])
```

```
Epoch 1/200
27/27          39s 1s/step - loss:
0.0810 - mae: 0.2375 - val_loss: 0.2669 - val_mae: 0.4396
Epoch 2/200
27/27          24s 567ms/step -
loss: 0.0639 - mae: 0.2121 - val_loss: 0.1952 - val_mae: 0.3660
Epoch 3/200
27/27          22s 619ms/step -
loss: 0.0616 - mae: 0.2074 - val_loss: 0.2376 - val_mae: 0.4241
Epoch 4/200
27/27          21s 631ms/step -
loss: 0.0567 - mae: 0.1960 - val_loss: 0.2637 - val_mae: 0.4497
Epoch 5/200
27/27          20s 594ms/step -
loss: 0.0563 - mae: 0.1953 - val_loss: 0.1584 - val_mae: 0.3338
Epoch 6/200
27/27          16s 607ms/step -
loss: 0.0553 - mae: 0.1941 - val_loss: 0.1340 - val_mae: 0.3006
Epoch 7/200
27/27          18s 661ms/step -
loss: 0.0528 - mae: 0.1885 - val_loss: 0.1085 - val_mae: 0.2691
Epoch 8/200
27/27          21s 670ms/step -
loss: 0.0521 - mae: 0.1865 - val_loss: 0.0611 - val_mae: 0.1964
Epoch 9/200
27/27          21s 675ms/step -
loss: 0.0499 - mae: 0.1817 - val_loss: 0.0645 - val_mae: 0.2026
Epoch 10/200
27/27          21s 681ms/step -
loss: 0.0496 - mae: 0.1806 - val_loss: 0.0665 - val_mae: 0.2097
Epoch 11/200
27/27          20s 682ms/step -
loss: 0.0482 - mae: 0.1785 - val_loss: 0.0614 - val_mae: 0.2018
Epoch 12/200
27/27          17s 636ms/step -
loss: 0.0462 - mae: 0.1747 - val_loss: 0.0655 - val_mae: 0.1989
Epoch 13/200
27/27          21s 674ms/step -
loss: 0.0450 - mae: 0.1703 - val_loss: 0.0649 - val_mae: 0.1996
Epoch 14/200
27/27          21s 681ms/step -
loss: 0.0439 - mae: 0.1674 - val_loss: 0.0616 - val_mae: 0.2000
Epoch 15/200
```

27/27 20s 645ms/step -
 loss: 0.0414 - mae: 0.1625 - val_loss: 0.0634 - val_mae: 0.1966
 Epoch 16/200
 27/27 17s 643ms/step -
 loss: 0.0411 - mae: 0.1618 - val_loss: 0.0619 - val_mae: 0.2010
 Epoch 17/200
 27/27 18s 677ms/step -
 loss: 0.0405 - mae: 0.1608 - val_loss: 0.0626 - val_mae: 0.1977
 Epoch 18/200
 27/27 20s 675ms/step -
 loss: 0.0391 - mae: 0.1573 - val_loss: 0.0611 - val_mae: 0.1965
 Epoch 19/200
 27/27 19s 637ms/step -
 loss: 0.0375 - mae: 0.1538 - val_loss: 0.0663 - val_mae: 0.2027
 Epoch 20/200
 27/27 21s 649ms/step -
 loss: 0.0373 - mae: 0.1529 - val_loss: 0.0640 - val_mae: 0.1994
 Epoch 21/200
 27/27 21s 683ms/step -
 loss: 0.0361 - mae: 0.1498 - val_loss: 0.0674 - val_mae: 0.2067
 Epoch 22/200
 27/27 20s 676ms/step -
 loss: 0.0351 - mae: 0.1470 - val_loss: 0.0635 - val_mae: 0.1995
 Epoch 23/200
 27/27 19s 639ms/step -
 loss: 0.0329 - mae: 0.1412 - val_loss: 0.0688 - val_mae: 0.2056
 Epoch 24/200
 27/27 17s 649ms/step -
 loss: 0.0313 - mae: 0.1381 - val_loss: 0.0653 - val_mae: 0.2010
 Epoch 25/200
 27/27 21s 681ms/step -
 loss: 0.0296 - mae: 0.1329 - val_loss: 0.0660 - val_mae: 0.2020
 Epoch 26/200
 27/27 20s 677ms/step -
 loss: 0.0307 - mae: 0.1359 - val_loss: 0.0671 - val_mae: 0.2028
 Epoch 27/200
 27/27 19s 639ms/step -
 loss: 0.0306 - mae: 0.1358 - val_loss: 0.0647 - val_mae: 0.2017
 Epoch 28/200
 27/27 17s 646ms/step -
 loss: 0.0293 - mae: 0.1323 - val_loss: 0.0651 - val_mae: 0.2010
 Epoch 29/200
 27/27 18s 678ms/step -
 loss: 0.0276 - mae: 0.1281 - val_loss: 0.0648 - val_mae: 0.2024
 Epoch 30/200
 27/27 19s 634ms/step -
 loss: 0.0275 - mae: 0.1274 - val_loss: 0.0672 - val_mae: 0.2039
 Epoch 31/200

```

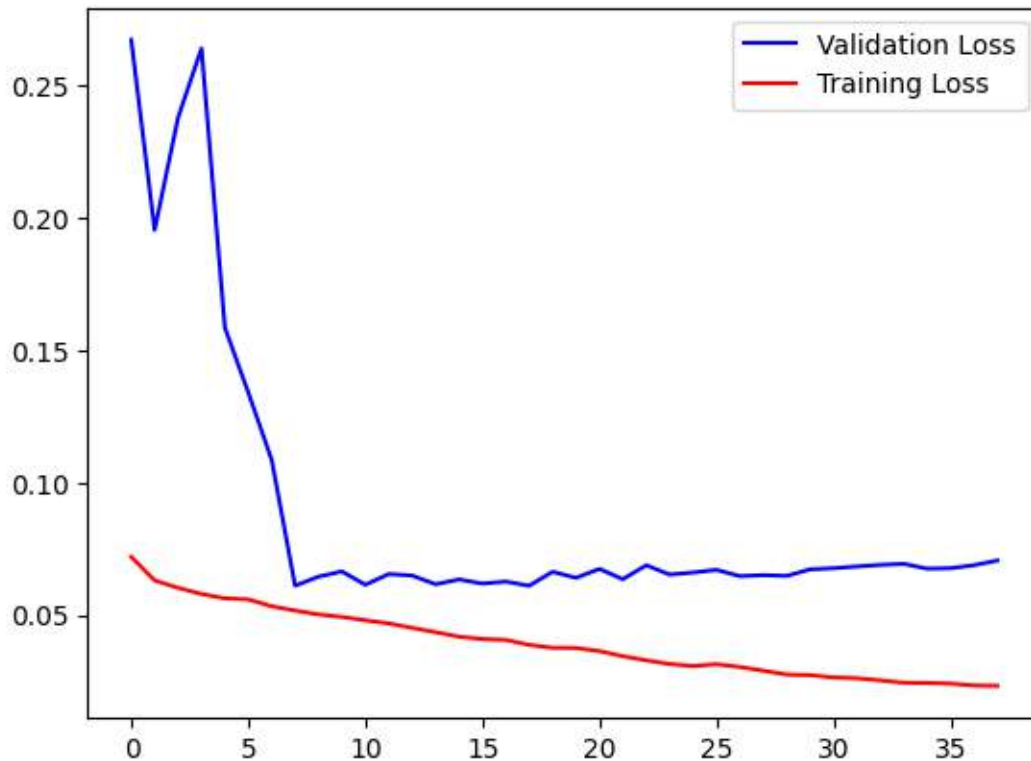
27/27          21s 640ms/step -
loss: 0.0261 - mae: 0.1238 - val_loss: 0.0677 - val_mae: 0.2039
Epoch 32/200
27/27          17s 646ms/step -
loss: 0.0258 - mae: 0.1232 - val_loss: 0.0683 - val_mae: 0.2064
Epoch 33/200
27/27          20s 637ms/step -
loss: 0.0248 - mae: 0.1206 - val_loss: 0.0689 - val_mae: 0.2070
Epoch 34/200
27/27          22s 677ms/step -
loss: 0.0242 - mae: 0.1189 - val_loss: 0.0693 - val_mae: 0.2077
Epoch 35/200
27/27          20s 641ms/step -
loss: 0.0243 - mae: 0.1190 - val_loss: 0.0675 - val_mae: 0.2048
Epoch 36/200
27/27          22s 686ms/step -
loss: 0.0240 - mae: 0.1187 - val_loss: 0.0677 - val_mae: 0.2045
Epoch 37/200
27/27          20s 677ms/step -
loss: 0.0233 - mae: 0.1161 - val_loss: 0.0688 - val_mae: 0.2067
Epoch 38/200
27/27          20s 676ms/step -
loss: 0.0233 - mae: 0.1165 - val_loss: 0.0706 - val_mae: 0.2087

```

```

[ ]: # Plotting the training and validation loss
plt.plot(history_trans_vgg.history['val_loss'],c="b")
plt.plot(history_trans_vgg.history['loss'],c="r")
plt.legend(['Validation Loss','Training Loss'])
plt.show()

```

Depiction of Model Performance on images from the Training Set

```
[ ]: show_predictions(trans_vgg_model, X, Y, num_samples=5)
```

WARNING:tensorflow:6 out of the last 11 calls to <function TensorFlowTrainer.make_predict_function.<locals>.one_step_on_data_distributed at 0x7e2f3770ca40> triggered tf.function retracing. Tracing is expensive and the excessive number of tracings could be due to (1) creating @tf.function repeatedly in a loop, (2) passing tensors with different shapes, (3) passing Python objects instead of tensors. For (1), please define your @tf.function outside of the loop. For (2), @tf.function has reduce_retracing=True option that can avoid unnecessary retracing. For (3), please refer to https://www.tensorflow.org/guide/function#controlling_retracing and https://www.tensorflow.org/api_docs/python/tf/function for more details.

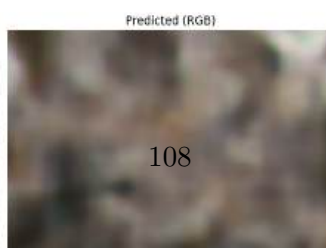
1/1 9s 9s/step



Depiction of Model Performance on images from the Test Set

```
[ ]: show_predictions(trans_vgg_model, X_test, Y_test, num_samples=6)
```

1/1 10s 10s/step



This loss plot shows that the training loss decreases consistently, indicating the model is learning from the training data. The validation loss drops sharply at first, then flattens out, remaining higher than the training loss. This gap suggests the model has limited generalization and may be overfitting slightly.

In the test images, the predicted color output is heavily blurred and lacks details of objects and people. The model fails to reconstruct realistic colors or distinguish between regions, producing a noisy result. This indicates that, despite a deeper architecture, the model is not effectively capturing fine features or context needed for accurate colorization, learning patterns that don't fully transfer to unseen data.

6 MSBA Headshot Implementations

Having explored colorization using full scene portraits, as shown above, we decided to pivot to a new dataset with a consistent background and a more standardized color range. We used headshots taken during the MSBA (Master of Science in Business Analytics) orientation to rerun our model and observe the results. The original dataset included 186 headshots, and we applied data augmentation to increase the total to 452 images. This was done to reduce the risk of overfitting and to provide the model with more data to learn from.

6.1 Image Preprocessing

```
[ ]: # Defining the required Google Drive paths for the MSBA Headshot Implementation
color_image_path = '/content/drive/MyDrive/BA865/BA865 Group Project/Dataset/
↳MSBA/Color'
color_resized_image_path = '/content/drive/MyDrive/BA865/BA865 Group Project/
↳Dataset/MSBA/Color Resized'
test_color_image_path = '/content/drive/MyDrive/BA865/BA865 Group Project/
↳Dataset/MSBA/Test'
test_color_resized_image_path = '/content/drive/MyDrive/BA865/BA865 Group
↳Project/Dataset/MSBA/Test Resized'
best_models_path = '/content/drive/MyDrive/BA865/BA865 Group Project/Best
↳Models MSBA'
```

Train Color Images Preprocessing

```
[ ]: # Creating color resized image folder if it doesn't exist
os.makedirs(color_resized_image_path, exist_ok=True)

# Ensuring the color resized image folder is empty
for file_name in os.listdir(color_resized_image_path):
    file_path = os.path.join(color_resized_image_path, file_name)
    if os.path.isfile(file_path):
        os.remove(file_path)
```



```

resize_success_count = 0
resize_fail_count = 0

# Looping through all color images and applying the resizing, cropping,
↪rotation, and compression
for image in os.listdir(color_image_path):
    if image.lower().endswith(image_extensions):
        input_img_path = os.path.join(color_image_path, image)
        output_img_path = os.path.join(color_resized_image_path, image)

        # The below if statement resizes the image while also maintaining a
↪count of successful and unsuccessful conversions
        if image_resizer(input_img_path, output_img_path, (176, 120)):
            resize_success_count += 1
        else:
            resize_fail_count += 1

print(f"Successfully resized {resize_success_count} images.")
print(f"Failed to resize {resize_fail_count} images.")

```

Successfully resized 186 images.
Failed to resize 0 images.

```

[ ]: color_images = []

for image in os.listdir(color_resized_image_path):
    color_image_path = os.path.join(color_resized_image_path, image)

    # Converting the color image to numpy array
    if os.path.isfile(color_image_path):
        col_img = Image.open(color_image_path).convert('RGB')
        color_image_array = np.array(col_img).astype('float32') / 255.0
        color_images.append(color_image_array)

```

```

[ ]: # This code will use ImageDataGenerator to create additional images as a form
↪of data augmentation, helping against overfitting.
datagen = ImageDataGenerator(rotation_range=40, width_shift_range=0.2,
↪height_shift_range=0.2, rescale=1./255,
                                shear_range=0.2, zoom_range=0.2,
↪horizontal_flip=True, fill_mode='nearest')

datagen.fit(color_images)

image_batches = datagen.flow(np.stack(color_images, axis=0), batch_size=40)

n = 10
plt.figure(figsize= (20,10))

```

```

q = 0
for i, batch in enumerate(image_batches):
    for j, image in enumerate(batch):
        if j == 1:
            if i <= 5:
                ax = plt.subplot(1, 6, i+1)
                plt.imshow(tf.keras.utils.array_to_img(image))
                ax.get_xaxis().set_visible(False)
                ax.get_yaxis().set_visible(False)

                # The iterator outputs image data with pixel values between 0-1; we are
                ↪ using values between 0-255 elsewhere. We need to make the pixel values align.
                color_images.append(np.multiply(image, 255))

            if q > 5:
                break

        q += 1

plt.show()

print(f'We now have a total of {len(color_images)} images.')

```



We now have a total of 452 images.

```

[ ]: # Splitting the predictor and labels in the RGB color space, for the training
    ↪ set
X = []
Y = []

for img in color_images:
    try:
        gray = tf.image.rgb_to_grayscale(img)
        X.append(gray.numpy())
        Y.append(img)
    except Exception as e:
        print("error:", e)

X = np.array(X)
Y = np.array(Y)

```

```
print("X_rgb shape:", X.shape)
print("Y_rgb shape:", Y.shape)
```

```
X_rgb shape: (452, 120, 176, 1)
Y_rgb shape: (452, 120, 176, 3)
```

```
[ ]: # Splitting the predictor and labels in the LAB color space, for the training
      ↪ set
X_lab = []
Y_lab = []

for img in color_images:
    try:
        lab = rgb2lab(img)
        X_lab.append(lab[:, :, 0])
        #restrict values to be between -1 and 1
        Y_lab.append(lab[:, :, 1:] / 128)
    except Exception as e:
        print("error:", e)

X_lab = np.array(X_lab)
Y_lab = np.array(Y_lab)

#dimensions to be the same for X and Y
X_lab = X_lab.reshape(X_lab.shape+(1,))

print("X_lab shape:", X_lab.shape)
print("Y_lab shape:", Y_lab.shape)
```

```
X_lab shape: (452, 120, 176, 1)
Y_lab shape: (452, 120, 176, 2)
```

Test Color Images Preprocessing

```
[ ]: # Creating color resized image folder for the test set if it doesn't exist
os.makedirs(test_color_resized_image_path, exist_ok=True)

# Ensuring the color resized image folder is empty
for file_name in os.listdir(test_color_resized_image_path):
    file_path = os.path.join(test_color_resized_image_path, file_name)
    if os.path.isfile(file_path):
        os.remove(file_path)

test_resize_success_count = 0
test_resize_fail_count = 0
```



```

# Looping through all color images and applying the resizing, cropping,
↪rotation, and compression
for test_image in os.listdir(test_color_image_path):
    if test_image.lower().endswith(image_extensions):
        input_img_path = os.path.join(test_color_image_path, test_image)
        output_img_path = os.path.join(test_color_resized_image_path,
↪test_image)

        # The below if statement resizes the image while also maintaining a
↪count of successful and unsuccessful conversions
        if image_resizer(input_img_path, output_img_path, (176, 120)):
            test_resize_success_count += 1
        else:
            test_resize_fail_count += 1

print(f"Successfully resized {test_resize_success_count} images.")
print(f"Failed to resize {test_resize_fail_count} images.")

```

Successfully resized 8 images.

Failed to resize 0 images.

```

[ ]: test_color_images = []

for image in os.listdir(test_color_resized_image_path):
    test_color_image_path = os.path.join(test_color_resized_image_path, image)

    # Converting the color image to numpy array
    if os.path.isfile(test_color_image_path):
        col_img = Image.open(test_color_image_path).convert('RGB')
        test_color_image_array = np.array(col_img).astype('float32') / 255.0
        test_color_images.append(test_color_image_array)

```

```

[ ]: # Splitting the predictor and labels in the LAB color space, for the test set
X_lab_test = []
Y_lab_test = []

for img in test_color_images:
    try:
        lab_test = rgb2lab(img)
        X_lab_test.append(lab_test[:, :, 0])
        #restrict values to be between -1 and 1
        Y_lab_test.append(lab_test[:, :, 1:] / 128)
    except Exception as e:
        print("error:", e)

X_lab_test = np.array(X_lab_test)
Y_lab_test = np.array(Y_lab_test)

```

```
#dimensions to be the same for X and Y
X_lab_test = X_lab_test.reshape(X_lab_test.shape+(1,))

print("X_lab_test shape:", X_lab_test.shape)
print("Y_lab_test shape:", Y_lab_test.shape)
```

```
X_lab_test shape: (8, 120, 176, 1)
Y_lab_test shape: (8, 120, 176, 2)
```

```
[ ]: # Splitting the predictor and labels in the LAB color space, for the test set
X_test = []
Y_test = []

for img in test_color_images:
    try:
        gray = tf.image.rgb_to_grayscale(img)
        X_test.append(gray.numpy())
        Y_test.append(img)
    except Exception as e:
        print("error:", e)

X_test = np.array(X_test)
Y_test = np.array(Y_test)

print("X_rgb_test shape:", X_test.shape)
print("Y_rgb_test shape:", Y_test.shape)
```

```
X_rgb_test shape: (8, 120, 176, 1)
Y_rgb_test shape: (8, 120, 176, 3)
```

6.2 Model 1: Sequential RGB Model

```
[ ]: # Defining the architecture for the sequential RGB model
input_1 = Input(shape=(120, 176, 1))

x = Conv2D(64, (3, 3), activation='relu', padding='same')(input_1)
x = Conv2D(64, (3, 3), activation='relu', strides=2, padding='same')(x)

x = Conv2D(128, (3, 3), activation='relu', padding='same')(x)
x = Conv2D(128, (3, 3), activation='relu', strides=2, padding='same')(x)

x = UpSampling2D((2, 2))(x)
x = Conv2D(64, (3, 3), activation='relu', padding='same')(x)

x = UpSampling2D((2, 2))(x)
x = Conv2D(32, (3, 3), activation='relu', padding='same')(x)
```

```

output_rgb = Conv2D(3, (3, 3), activation='sigmoid', padding='same')(x)
output_rgb = Lambda(lambda x: tf.image.resize_with_crop_or_pad(x, 120, 176))(output_rgb)

seq_rgb_model = Model(inputs=input_1, outputs=output_rgb)
seq_rgb_model.compile(optimizer='adam', loss='mae', metrics=['mse'])

```

```

[ ]: # Fitting the model to the training data
history_seq_rgb = seq_rgb_model.fit(X, Y, batch_size=16, epochs=200, validation_split=0.1, callbacks=[early_stop])

```

```

Epoch 1/200
26/26          15s 475ms/step -
loss: 0.2283 - mse: 0.0750 - val_loss: 0.0735 - val_mse: 0.0106
Epoch 2/200
26/26          9s 64ms/step -
loss: 0.0718 - mse: 0.0098 - val_loss: 0.0466 - val_mse: 0.0048
Epoch 3/200
26/26          2s 80ms/step -
loss: 0.0506 - mse: 0.0054 - val_loss: 0.0388 - val_mse: 0.0035
Epoch 4/200
26/26          2s 65ms/step -
loss: 0.0408 - mse: 0.0038 - val_loss: 0.0361 - val_mse: 0.0029
Epoch 5/200
26/26          2s 63ms/step -
loss: 0.0372 - mse: 0.0031 - val_loss: 0.0404 - val_mse: 0.0032
Epoch 6/200
26/26          2s 58ms/step -
loss: 0.0407 - mse: 0.0034 - val_loss: 0.0346 - val_mse: 0.0025
Epoch 7/200
26/26          2s 58ms/step -
loss: 0.0353 - mse: 0.0027 - val_loss: 0.0290 - val_mse: 0.0019
Epoch 8/200
26/26          2s 58ms/step -
loss: 0.0327 - mse: 0.0024 - val_loss: 0.0279 - val_mse: 0.0018
Epoch 9/200
26/26          3s 63ms/step -
loss: 0.0327 - mse: 0.0023 - val_loss: 0.0296 - val_mse: 0.0019
Epoch 10/200
26/26          2s 65ms/step -
loss: 0.0309 - mse: 0.0021 - val_loss: 0.0266 - val_mse: 0.0016
Epoch 11/200
26/26          2s 58ms/step -
loss: 0.0306 - mse: 0.0020 - val_loss: 0.0306 - val_mse: 0.0018
Epoch 12/200
26/26          2s 58ms/step -

```

```

loss: 0.0294 - mse: 0.0019 - val_loss: 0.0256 - val_mse: 0.0015
Epoch 13/200
26/26          3s 59ms/step -
loss: 0.0283 - mse: 0.0018 - val_loss: 0.0247 - val_mse: 0.0013
Epoch 14/200
26/26          3s 59ms/step -
loss: 0.0301 - mse: 0.0020 - val_loss: 0.0317 - val_mse: 0.0020
Epoch 15/200
26/26          3s 61ms/step -
loss: 0.0312 - mse: 0.0020 - val_loss: 0.0247 - val_mse: 0.0013
Epoch 16/200
26/26          2s 59ms/step -
loss: 0.0279 - mse: 0.0017 - val_loss: 0.0230 - val_mse: 0.0012
Epoch 17/200
26/26          2s 58ms/step -
loss: 0.0273 - mse: 0.0017 - val_loss: 0.0247 - val_mse: 0.0014
Epoch 18/200
26/26          3s 59ms/step -
loss: 0.0260 - mse: 0.0016 - val_loss: 0.0231 - val_mse: 0.0012
Epoch 19/200
26/26          2s 59ms/step -
loss: 0.0255 - mse: 0.0015 - val_loss: 0.0229 - val_mse: 0.0012
Epoch 20/200
26/26          3s 59ms/step -
loss: 0.0250 - mse: 0.0015 - val_loss: 0.0237 - val_mse: 0.0013
Epoch 21/200
26/26          2s 63ms/step -
loss: 0.0254 - mse: 0.0015 - val_loss: 0.0235 - val_mse: 0.0012
Epoch 22/200
26/26          2s 64ms/step -
loss: 0.0271 - mse: 0.0016 - val_loss: 0.0246 - val_mse: 0.0014
Epoch 23/200
26/26          2s 59ms/step -
loss: 0.0251 - mse: 0.0015 - val_loss: 0.0214 - val_mse: 0.0011
Epoch 24/200
26/26          3s 60ms/step -
loss: 0.0245 - mse: 0.0014 - val_loss: 0.0210 - val_mse: 0.0011
Epoch 25/200
26/26          2s 59ms/step -
loss: 0.0243 - mse: 0.0014 - val_loss: 0.0206 - val_mse: 0.0010
Epoch 26/200
26/26          2s 60ms/step -
loss: 0.0232 - mse: 0.0013 - val_loss: 0.0206 - val_mse: 0.0010
Epoch 27/200
26/26          3s 68ms/step -
loss: 0.0226 - mse: 0.0012 - val_loss: 0.0277 - val_mse: 0.0015
Epoch 28/200
26/26          2s 67ms/step -

```

```

loss: 0.0272 - mse: 0.0016 - val_loss: 0.0227 - val_mse: 0.0012
Epoch 29/200
26/26          2s 65ms/step -
loss: 0.0241 - mse: 0.0013 - val_loss: 0.0207 - val_mse: 0.0010
Epoch 30/200
26/26          2s 60ms/step -
loss: 0.0230 - mse: 0.0013 - val_loss: 0.0215 - val_mse: 0.0011
Epoch 31/200
26/26          3s 59ms/step -
loss: 0.0238 - mse: 0.0013 - val_loss: 0.0218 - val_mse: 0.0011
Epoch 32/200
26/26          2s 60ms/step -
loss: 0.0232 - mse: 0.0013 - val_loss: 0.0206 - val_mse: 9.7312e-04
Epoch 33/200
26/26          3s 61ms/step -
loss: 0.0229 - mse: 0.0012 - val_loss: 0.0210 - val_mse: 0.0010
Epoch 34/200
26/26          2s 64ms/step -
loss: 0.0251 - mse: 0.0014 - val_loss: 0.0212 - val_mse: 0.0010
Epoch 35/200
26/26          2s 64ms/step -
loss: 0.0236 - mse: 0.0013 - val_loss: 0.0195 - val_mse: 9.0936e-04
Epoch 36/200
26/26          2s 60ms/step -
loss: 0.0225 - mse: 0.0012 - val_loss: 0.0206 - val_mse: 9.8167e-04
Epoch 37/200
26/26          2s 60ms/step -
loss: 0.0225 - mse: 0.0012 - val_loss: 0.0196 - val_mse: 9.0993e-04
Epoch 38/200
26/26          2s 60ms/step -
loss: 0.0216 - mse: 0.0012 - val_loss: 0.0214 - val_mse: 0.0010
Epoch 39/200
26/26          3s 60ms/step -
loss: 0.0244 - mse: 0.0013 - val_loss: 0.0214 - val_mse: 0.0010
Epoch 40/200
26/26          2s 60ms/step -
loss: 0.0260 - mse: 0.0014 - val_loss: 0.0221 - val_mse: 0.0011
Epoch 41/200
26/26          3s 66ms/step -
loss: 0.0233 - mse: 0.0012 - val_loss: 0.0201 - val_mse: 9.6190e-04
Epoch 42/200
26/26          2s 62ms/step -
loss: 0.0209 - mse: 0.0011 - val_loss: 0.0193 - val_mse: 8.8430e-04
Epoch 43/200
26/26          3s 70ms/step -
loss: 0.0211 - mse: 0.0011 - val_loss: 0.0193 - val_mse: 9.2006e-04
Epoch 44/200
26/26          2s 63ms/step -

```

```

loss: 0.0220 - mse: 0.0012 - val_loss: 0.0197 - val_mse: 9.0402e-04
Epoch 45/200
26/26          3s 68ms/step -
loss: 0.0213 - mse: 0.0011 - val_loss: 0.0206 - val_mse: 9.2901e-04
Epoch 46/200
26/26          2s 65ms/step -
loss: 0.0247 - mse: 0.0013 - val_loss: 0.0224 - val_mse: 0.0010
Epoch 47/200
26/26          2s 66ms/step -
loss: 0.0228 - mse: 0.0011 - val_loss: 0.0187 - val_mse: 8.5165e-04
Epoch 48/200
26/26          2s 61ms/step -
loss: 0.0219 - mse: 0.0011 - val_loss: 0.0194 - val_mse: 9.3610e-04
Epoch 49/200
26/26          2s 61ms/step -
loss: 0.0208 - mse: 0.0011 - val_loss: 0.0196 - val_mse: 9.2274e-04
Epoch 50/200
26/26          2s 61ms/step -
loss: 0.0219 - mse: 0.0011 - val_loss: 0.0186 - val_mse: 8.3098e-04
Epoch 51/200
26/26          2s 61ms/step -
loss: 0.0201 - mse: 0.0010 - val_loss: 0.0193 - val_mse: 9.0099e-04
Epoch 52/200
26/26          3s 63ms/step -
loss: 0.0205 - mse: 0.0011 - val_loss: 0.0203 - val_mse: 9.1935e-04
Epoch 53/200
26/26          2s 64ms/step -
loss: 0.0213 - mse: 0.0011 - val_loss: 0.0186 - val_mse: 8.1961e-04
Epoch 54/200
26/26          2s 68ms/step -
loss: 0.0206 - mse: 0.0010 - val_loss: 0.0194 - val_mse: 8.6630e-04
Epoch 55/200
26/26          3s 67ms/step -
loss: 0.0206 - mse: 0.0010 - val_loss: 0.0194 - val_mse: 8.7814e-04
Epoch 56/200
26/26          2s 65ms/step -
loss: 0.0212 - mse: 0.0011 - val_loss: 0.0187 - val_mse: 8.7480e-04
Epoch 57/200
26/26          2s 62ms/step -
loss: 0.0198 - mse: 9.9622e-04 - val_loss: 0.0189 - val_mse: 8.2899e-04
Epoch 58/200
26/26          2s 62ms/step -
loss: 0.0207 - mse: 0.0010 - val_loss: 0.0186 - val_mse: 8.2358e-04
Epoch 59/200
26/26          2s 70ms/step -
loss: 0.0194 - mse: 9.3308e-04 - val_loss: 0.0184 - val_mse: 8.2089e-04
Epoch 60/200
26/26          2s 67ms/step -

```

```

loss: 0.0199 - mse: 9.6970e-04 - val_loss: 0.0204 - val_mse: 9.1547e-04
Epoch 61/200
26/26          2s 62ms/step -
loss: 0.0212 - mse: 0.0010 - val_loss: 0.0179 - val_mse: 7.7747e-04
Epoch 62/200
26/26          2s 62ms/step -
loss: 0.0194 - mse: 9.4444e-04 - val_loss: 0.0201 - val_mse: 8.7980e-04
Epoch 63/200
26/26          3s 68ms/step -
loss: 0.0213 - mse: 0.0011 - val_loss: 0.0299 - val_mse: 0.0016
Epoch 64/200
26/26          2s 62ms/step -
loss: 0.0257 - mse: 0.0013 - val_loss: 0.0180 - val_mse: 7.9202e-04
Epoch 65/200
26/26          2s 62ms/step -
loss: 0.0197 - mse: 9.4920e-04 - val_loss: 0.0180 - val_mse: 7.8277e-04
Epoch 66/200
26/26          2s 67ms/step -
loss: 0.0192 - mse: 9.5308e-04 - val_loss: 0.0173 - val_mse: 7.4815e-04
Epoch 67/200
26/26          3s 66ms/step -
loss: 0.0192 - mse: 9.3777e-04 - val_loss: 0.0187 - val_mse: 8.3677e-04
Epoch 68/200
26/26          2s 68ms/step -
loss: 0.0199 - mse: 9.6581e-04 - val_loss: 0.0182 - val_mse: 8.0159e-04
Epoch 69/200
26/26          2s 63ms/step -
loss: 0.0199 - mse: 9.6453e-04 - val_loss: 0.0238 - val_mse: 0.0012
Epoch 70/200
26/26          2s 63ms/step -
loss: 0.0223 - mse: 0.0011 - val_loss: 0.0201 - val_mse: 9.0385e-04
Epoch 71/200
26/26          3s 62ms/step -
loss: 0.0210 - mse: 0.0010 - val_loss: 0.0193 - val_mse: 8.5337e-04
Epoch 72/200
26/26          3s 64ms/step -
loss: 0.0198 - mse: 9.3883e-04 - val_loss: 0.0194 - val_mse: 8.4134e-04
Epoch 73/200
26/26          2s 63ms/step -
loss: 0.0206 - mse: 9.7549e-04 - val_loss: 0.0197 - val_mse: 8.8741e-04
Epoch 74/200
26/26          2s 62ms/step -
loss: 0.0200 - mse: 9.7876e-04 - val_loss: 0.0199 - val_mse: 8.6019e-04
Epoch 75/200
26/26          3s 68ms/step -
loss: 0.0202 - mse: 9.7855e-04 - val_loss: 0.0183 - val_mse: 7.8101e-04
Epoch 76/200
26/26          3s 67ms/step -

```

```

loss: 0.0191 - mse: 9.2197e-04 - val_loss: 0.0185 - val_mse: 8.0319e-04
Epoch 77/200
26/26          2s 63ms/step -
loss: 0.0194 - mse: 9.0640e-04 - val_loss: 0.0181 - val_mse: 7.8047e-04
Epoch 78/200
26/26          2s 85ms/step -
loss: 0.0199 - mse: 9.9311e-04 - val_loss: 0.0188 - val_mse: 8.3178e-04
Epoch 79/200
26/26          2s 76ms/step -
loss: 0.0198 - mse: 9.6133e-04 - val_loss: 0.0176 - val_mse: 7.6251e-04
Epoch 80/200
26/26          2s 64ms/step -
loss: 0.0192 - mse: 9.1587e-04 - val_loss: 0.0177 - val_mse: 7.8911e-04
Epoch 81/200
26/26          2s 63ms/step -
loss: 0.0189 - mse: 8.9384e-04 - val_loss: 0.0173 - val_mse: 7.3696e-04
Epoch 82/200
26/26          2s 67ms/step -
loss: 0.0192 - mse: 9.1603e-04 - val_loss: 0.0177 - val_mse: 7.7382e-04
Epoch 83/200
26/26          2s 66ms/step -
loss: 0.0191 - mse: 9.1285e-04 - val_loss: 0.0173 - val_mse: 7.3426e-04
Epoch 84/200
26/26          2s 63ms/step -
loss: 0.0194 - mse: 9.0898e-04 - val_loss: 0.0181 - val_mse: 7.5962e-04
Epoch 85/200
26/26          2s 67ms/step -
loss: 0.0196 - mse: 9.2266e-04 - val_loss: 0.0181 - val_mse: 7.9126e-04
Epoch 86/200
26/26          2s 64ms/step -
loss: 0.0189 - mse: 8.9908e-04 - val_loss: 0.0172 - val_mse: 7.2495e-04
Epoch 87/200
26/26          3s 63ms/step -
loss: 0.0185 - mse: 8.5408e-04 - val_loss: 0.0178 - val_mse: 7.6457e-04
Epoch 88/200
26/26          3s 63ms/step -
loss: 0.0189 - mse: 9.0952e-04 - val_loss: 0.0171 - val_mse: 7.3558e-04
Epoch 89/200
26/26          3s 63ms/step -
loss: 0.0189 - mse: 8.6756e-04 - val_loss: 0.0173 - val_mse: 7.4477e-04
Epoch 90/200
26/26          2s 63ms/step -
loss: 0.0186 - mse: 8.5919e-04 - val_loss: 0.0187 - val_mse: 7.8840e-04
Epoch 91/200
26/26          2s 69ms/step -
loss: 0.0196 - mse: 9.0728e-04 - val_loss: 0.0194 - val_mse: 8.3634e-04
Epoch 92/200
26/26          2s 65ms/step -

```



```

loss: 0.0198 - mse: 9.4084e-04 - val_loss: 0.0173 - val_mse: 7.2759e-04
Epoch 93/200
26/26          2s 63ms/step -
loss: 0.0188 - mse: 8.5160e-04 - val_loss: 0.0196 - val_mse: 8.5053e-04
Epoch 94/200
26/26          2s 63ms/step -
loss: 0.0194 - mse: 9.2011e-04 - val_loss: 0.0185 - val_mse: 7.6009e-04
Epoch 95/200
26/26          2s 63ms/step -
loss: 0.0195 - mse: 8.8601e-04 - val_loss: 0.0171 - val_mse: 7.2173e-04
Epoch 96/200
26/26          3s 63ms/step -
loss: 0.0185 - mse: 8.6255e-04 - val_loss: 0.0201 - val_mse: 9.0655e-04
Epoch 97/200
26/26          3s 68ms/step -
loss: 0.0194 - mse: 8.9653e-04 - val_loss: 0.0201 - val_mse: 8.6170e-04
Epoch 98/200
26/26          2s 70ms/step -
loss: 0.0194 - mse: 9.2321e-04 - val_loss: 0.0169 - val_mse: 7.1702e-04
Epoch 99/200
26/26          2s 63ms/step -
loss: 0.0181 - mse: 8.3525e-04 - val_loss: 0.0175 - val_mse: 7.4653e-04
Epoch 100/200
26/26          2s 67ms/step -
loss: 0.0187 - mse: 8.4574e-04 - val_loss: 0.0172 - val_mse: 7.1489e-04
Epoch 101/200
26/26          2s 70ms/step -
loss: 0.0193 - mse: 8.7985e-04 - val_loss: 0.0219 - val_mse: 9.5536e-04
Epoch 102/200
26/26          2s 63ms/step -
loss: 0.0211 - mse: 9.4473e-04 - val_loss: 0.0215 - val_mse: 9.0637e-04
Epoch 103/200
26/26          2s 65ms/step -
loss: 0.0218 - mse: 0.0010 - val_loss: 0.0194 - val_mse: 7.9326e-04
Epoch 104/200
26/26          3s 70ms/step -
loss: 0.0187 - mse: 8.1323e-04 - val_loss: 0.0166 - val_mse: 6.8812e-04
Epoch 105/200
26/26          2s 70ms/step -
loss: 0.0184 - mse: 8.3625e-04 - val_loss: 0.0173 - val_mse: 7.1268e-04
Epoch 106/200
26/26          2s 64ms/step -
loss: 0.0185 - mse: 8.4214e-04 - val_loss: 0.0171 - val_mse: 7.3102e-04
Epoch 107/200
26/26          3s 67ms/step -
loss: 0.0182 - mse: 8.4368e-04 - val_loss: 0.0164 - val_mse: 6.7703e-04
Epoch 108/200
26/26          3s 67ms/step -

```

```

loss: 0.0177 - mse: 8.0444e-04 - val_loss: 0.0168 - val_mse: 6.9000e-04
Epoch 109/200
26/26          3s 67ms/step -
loss: 0.0178 - mse: 8.0762e-04 - val_loss: 0.0175 - val_mse: 7.3193e-04
Epoch 110/200
26/26          2s 65ms/step -
loss: 0.0179 - mse: 8.0584e-04 - val_loss: 0.0169 - val_mse: 7.0977e-04
Epoch 111/200
26/26          2s 70ms/step -
loss: 0.0182 - mse: 8.3388e-04 - val_loss: 0.0169 - val_mse: 7.1062e-04
Epoch 112/200
26/26          2s 63ms/step -
loss: 0.0179 - mse: 8.1417e-04 - val_loss: 0.0179 - val_mse: 7.6653e-04
Epoch 113/200
26/26          3s 63ms/step -
loss: 0.0183 - mse: 8.1616e-04 - val_loss: 0.0182 - val_mse: 7.6782e-04
Epoch 114/200
26/26          3s 63ms/step -
loss: 0.0188 - mse: 8.5954e-04 - val_loss: 0.0163 - val_mse: 6.8414e-04
Epoch 115/200
26/26          2s 63ms/step -
loss: 0.0177 - mse: 7.8714e-04 - val_loss: 0.0172 - val_mse: 7.1809e-04
Epoch 116/200
26/26          2s 63ms/step -
loss: 0.0180 - mse: 7.8437e-04 - val_loss: 0.0185 - val_mse: 7.5914e-04
Epoch 117/200
26/26          3s 68ms/step -
loss: 0.0187 - mse: 8.2245e-04 - val_loss: 0.0168 - val_mse: 6.8706e-04
Epoch 118/200
26/26          2s 63ms/step -
loss: 0.0181 - mse: 8.0307e-04 - val_loss: 0.0178 - val_mse: 7.8736e-04
Epoch 119/200
26/26          2s 63ms/step -
loss: 0.0176 - mse: 7.9037e-04 - val_loss: 0.0165 - val_mse: 6.7853e-04
Epoch 120/200
26/26          2s 63ms/step -
loss: 0.0176 - mse: 7.7576e-04 - val_loss: 0.0176 - val_mse: 7.3307e-04
Epoch 121/200
26/26          2s 63ms/step -
loss: 0.0180 - mse: 7.8846e-04 - val_loss: 0.0172 - val_mse: 7.1479e-04
Epoch 122/200
26/26          3s 63ms/step -
loss: 0.0176 - mse: 7.9415e-04 - val_loss: 0.0170 - val_mse: 7.0512e-04
Epoch 123/200
26/26          3s 68ms/step -
loss: 0.0174 - mse: 7.5458e-04 - val_loss: 0.0181 - val_mse: 7.3065e-04
Epoch 124/200
26/26          2s 73ms/step -

```

```

loss: 0.0176 - mse: 7.5666e-04 - val_loss: 0.0175 - val_mse: 7.2223e-04
Epoch 125/200
26/26          2s 64ms/step -
loss: 0.0186 - mse: 7.9666e-04 - val_loss: 0.0165 - val_mse: 6.7773e-04
Epoch 126/200
26/26          2s 63ms/step -
loss: 0.0179 - mse: 7.7269e-04 - val_loss: 0.0167 - val_mse: 6.7683e-04
Epoch 127/200
26/26          3s 64ms/step -
loss: 0.0176 - mse: 7.9357e-04 - val_loss: 0.0164 - val_mse: 6.7692e-04
Epoch 128/200
26/26          3s 66ms/step -
loss: 0.0176 - mse: 7.6553e-04 - val_loss: 0.0164 - val_mse: 6.7211e-04
Epoch 129/200
26/26          2s 64ms/step -
loss: 0.0173 - mse: 7.4645e-04 - val_loss: 0.0176 - val_mse: 7.1950e-04
Epoch 130/200
26/26          2s 64ms/step -
loss: 0.0174 - mse: 7.6241e-04 - val_loss: 0.0172 - val_mse: 7.0496e-04
Epoch 131/200
26/26          2s 63ms/step -
loss: 0.0177 - mse: 7.6888e-04 - val_loss: 0.0164 - val_mse: 6.9053e-04
Epoch 132/200
26/26          3s 63ms/step -
loss: 0.0176 - mse: 7.6761e-04 - val_loss: 0.0165 - val_mse: 6.6671e-04
Epoch 133/200
26/26          2s 63ms/step -
loss: 0.0174 - mse: 7.4268e-04 - val_loss: 0.0161 - val_mse: 6.5220e-04
Epoch 134/200
26/26          3s 63ms/step -
loss: 0.0182 - mse: 8.1416e-04 - val_loss: 0.0169 - val_mse: 6.8882e-04
Epoch 135/200
26/26          3s 63ms/step -
loss: 0.0179 - mse: 7.9544e-04 - val_loss: 0.0164 - val_mse: 6.6435e-04
Epoch 136/200
26/26          3s 72ms/step -
loss: 0.0173 - mse: 7.4007e-04 - val_loss: 0.0166 - val_mse: 6.8237e-04
Epoch 137/200
26/26          2s 65ms/step -
loss: 0.0177 - mse: 7.7847e-04 - val_loss: 0.0176 - val_mse: 7.2322e-04
Epoch 138/200
26/26          2s 68ms/step -
loss: 0.0177 - mse: 7.7137e-04 - val_loss: 0.0175 - val_mse: 7.1358e-04
Epoch 139/200
26/26          2s 63ms/step -
loss: 0.0174 - mse: 7.2553e-04 - val_loss: 0.0204 - val_mse: 8.3761e-04
Epoch 140/200
26/26          2s 63ms/step -

```

```

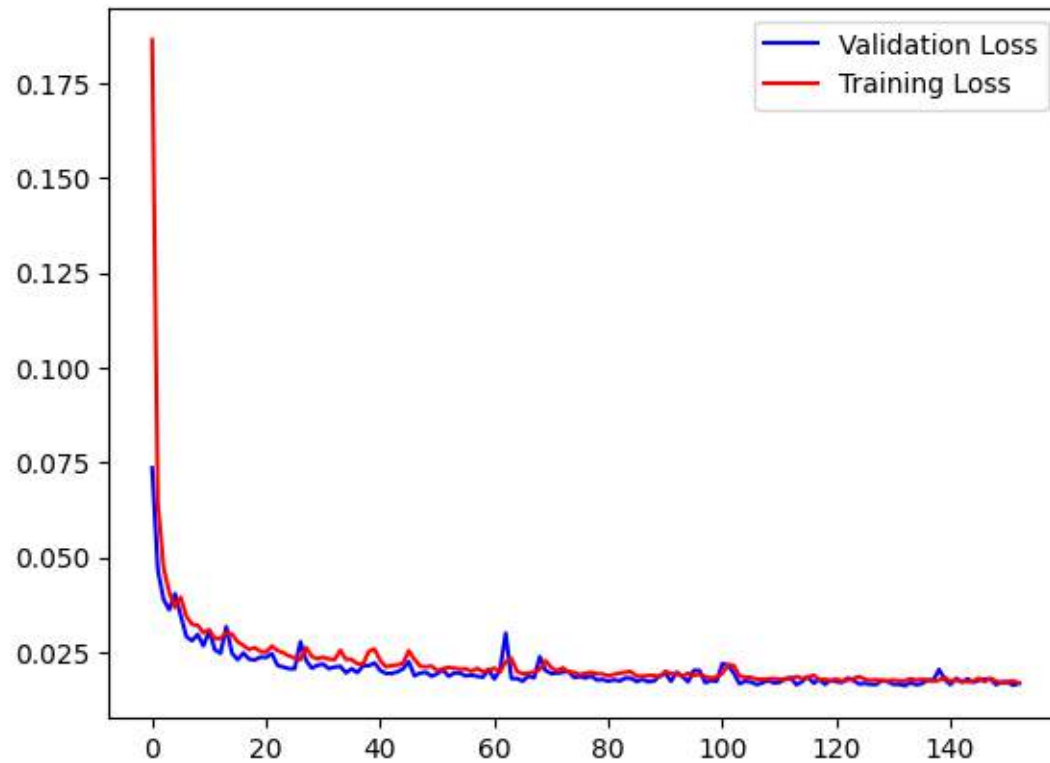
loss: 0.0187 - mse: 7.8832e-04 - val_loss: 0.0177 - val_mse: 7.3689e-04
Epoch 141/200
26/26          3s 63ms/step -
loss: 0.0176 - mse: 7.6490e-04 - val_loss: 0.0164 - val_mse: 6.6659e-04
Epoch 142/200
26/26          3s 69ms/step -
loss: 0.0174 - mse: 7.5604e-04 - val_loss: 0.0180 - val_mse: 7.4984e-04
Epoch 143/200
26/26          2s 71ms/step -
loss: 0.0175 - mse: 7.4724e-04 - val_loss: 0.0170 - val_mse: 6.8622e-04
Epoch 144/200
26/26          2s 68ms/step -
loss: 0.0171 - mse: 7.3936e-04 - val_loss: 0.0179 - val_mse: 7.1533e-04
Epoch 145/200
26/26          2s 63ms/step -
loss: 0.0175 - mse: 7.4581e-04 - val_loss: 0.0170 - val_mse: 7.1094e-04
Epoch 146/200
26/26          3s 63ms/step -
loss: 0.0183 - mse: 7.9604e-04 - val_loss: 0.0175 - val_mse: 7.0360e-04
Epoch 147/200
26/26          3s 70ms/step -
loss: 0.0173 - mse: 7.2815e-04 - val_loss: 0.0179 - val_mse: 7.1730e-04
Epoch 148/200
26/26          2s 70ms/step -
loss: 0.0174 - mse: 7.1588e-04 - val_loss: 0.0178 - val_mse: 7.0466e-04
Epoch 149/200
26/26          2s 68ms/step -
loss: 0.0177 - mse: 7.3020e-04 - val_loss: 0.0164 - val_mse: 6.5609e-04
Epoch 150/200
26/26          3s 68ms/step -
loss: 0.0169 - mse: 7.0380e-04 - val_loss: 0.0171 - val_mse: 7.0876e-04
Epoch 151/200
26/26          2s 62ms/step -
loss: 0.0170 - mse: 7.2769e-04 - val_loss: 0.0170 - val_mse: 6.9190e-04
Epoch 152/200
26/26          3s 63ms/step -
loss: 0.0178 - mse: 7.7081e-04 - val_loss: 0.0163 - val_mse: 6.5526e-04
Epoch 153/200
26/26          2s 63ms/step -
loss: 0.0166 - mse: 6.9298e-04 - val_loss: 0.0166 - val_mse: 6.7523e-04

```

```

[ ]: # Plotting the training and validation loss
plt.plot(history_seq_rgb.history['val_loss'],c="b")
plt.plot(history_seq_rgb.history['loss'],c="r")
plt.legend(['Validation Loss','Training Loss'])
plt.show()

```



Depiction of Model Performance on images from the Training Set

```
[ ]: show_predictions(seq_rgb_model, X, Y, num_samples=5)
```

1/1

1s 716ms/step



Depiction of Model Performance on images from the Test Set

```
[ ]: show_predictions(seq_rgb_model, X_test, Y_test, num_samples=5)
```

1/1 0s 46ms/step



For the MSBA headshots, the model performed noticeably better due to the structured background and uniform lighting. It was able to accurately colorize the images compared to the more complex full-scene portraits. The model successfully captured overall skin tones, background gradients, and clothing contrast, producing outputs that are quite close to the ground truth. However, the predictions still show slight blurriness in finer facial details and appear less saturated than the original images.

6.3 Model 2: Simple CNN

```
[ ]: # Defining the architecture for the simple CNN model
def build_simple_cnn(input_shape=(120, 176, 1)):
    inputs = layers.Input(shape=input_shape)

    x = layers.Conv2D(64, (3, 3), activation='relu', padding='same')(inputs)
    x = layers.Conv2D(128, (3, 3), activation='relu', padding='same')(x)
    x = layers.MaxPooling2D((2, 2))(x)

    x = layers.Conv2D(128, (3, 3), activation='relu', padding='same')(x)

    x = layers.Conv2DTranspose(64, (2, 2), strides=2, padding='same')(x)
    x = layers.ReLU()(x)
    outputs = layers.Conv2D(3, (3, 3), activation='sigmoid', padding='same')(x)

    model = models.Model(inputs=inputs, outputs=outputs)
    model.compile(optimizer='adam', loss='mse', metrics=['mae'])
    return model

model_simple_cnn = build_simple_cnn()
```

```
[ ]: # Defining the path to save the best model
model_checkpoint_path = os.path.join(best_models_path, 'simple_cnn.keras')
model_checkpoint = ModelCheckpoint(model_checkpoint_path, monitor='val_loss',
    ↳ verbose=1, save_best_only=True, mode='min')

# Fitting the model to the training data
history_simple_cnn = model_simple_cnn.fit(X, Y, batch_size=16, epochs=200,
    ↳ validation_split=0.1, callbacks=[early_stop, model_checkpoint])

best_simple_cnn = keras.models.load_model(os.path.join(best_models_path,
    ↳ 'simple_cnn.keras'))
```

Epoch 1/200

26/26 0s 253ms/step -

loss: 0.0780 - mae: 0.2413

Epoch 1: val_loss improved from inf to 0.01737, saving model to

```

/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          15s 475ms/step -
loss: 0.0774 - mae: 0.2401 - val_loss: 0.0174 - val_mae: 0.1139
Epoch 2/200
25/26          0s 70ms/step -
loss: 0.0119 - mae: 0.0842
Epoch 2: val_loss improved from 0.01737 to 0.00441, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          9s 78ms/step -
loss: 0.0116 - mae: 0.0830 - val_loss: 0.0044 - val_mae: 0.0490
Epoch 3/200
25/26          0s 69ms/step -
loss: 0.0045 - mae: 0.0473
Epoch 3: val_loss improved from 0.00441 to 0.00328, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          2s 75ms/step -
loss: 0.0044 - mae: 0.0471 - val_loss: 0.0033 - val_mae: 0.0395
Epoch 4/200
25/26          0s 69ms/step -
loss: 0.0034 - mae: 0.0402
Epoch 4: val_loss improved from 0.00328 to 0.00270, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          3s 75ms/step -
loss: 0.0034 - mae: 0.0402 - val_loss: 0.0027 - val_mae: 0.0358
Epoch 5/200
25/26          0s 75ms/step -
loss: 0.0030 - mae: 0.0379
Epoch 5: val_loss improved from 0.00270 to 0.00235, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          3s 86ms/step -
loss: 0.0030 - mae: 0.0379 - val_loss: 0.0023 - val_mae: 0.0331
Epoch 6/200
25/26          0s 72ms/step -
loss: 0.0028 - mae: 0.0361
Epoch 6: val_loss improved from 0.00235 to 0.00226, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          3s 86ms/step -
loss: 0.0028 - mae: 0.0360 - val_loss: 0.0023 - val_mae: 0.0328
Epoch 7/200
25/26          0s 71ms/step -
loss: 0.0024 - mae: 0.0339
Epoch 7: val_loss did not improve from 0.00226

```

```

26/26          2s 77ms/step -
loss: 0.0024 - mae: 0.0340 - val_loss: 0.0024 - val_mae: 0.0342
Epoch 8/200
25/26          0s 71ms/step -
loss: 0.0024 - mae: 0.0337
Epoch 8: val_loss improved from 0.00226 to 0.00197, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          2s 86ms/step -
loss: 0.0024 - mae: 0.0337 - val_loss: 0.0020 - val_mae: 0.0303
Epoch 9/200
25/26          0s 74ms/step -
loss: 0.0024 - mae: 0.0338
Epoch 9: val_loss did not improve from 0.00197
26/26          2s 79ms/step -
loss: 0.0024 - mae: 0.0339 - val_loss: 0.0022 - val_mae: 0.0331
Epoch 10/200
25/26          0s 70ms/step -
loss: 0.0022 - mae: 0.0321
Epoch 10: val_loss improved from 0.00197 to 0.00180, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          2s 75ms/step -
loss: 0.0022 - mae: 0.0321 - val_loss: 0.0018 - val_mae: 0.0290
Epoch 11/200
25/26          0s 71ms/step -
loss: 0.0021 - mae: 0.0311
Epoch 11: val_loss improved from 0.00180 to 0.00173, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          3s 80ms/step -
loss: 0.0021 - mae: 0.0311 - val_loss: 0.0017 - val_mae: 0.0283
Epoch 12/200
25/26          0s 70ms/step -
loss: 0.0019 - mae: 0.0297
Epoch 12: val_loss did not improve from 0.00173
26/26          2s 73ms/step -
loss: 0.0020 - mae: 0.0297 - val_loss: 0.0019 - val_mae: 0.0307
Epoch 13/200
25/26          0s 70ms/step -
loss: 0.0021 - mae: 0.0312
Epoch 13: val_loss did not improve from 0.00173
26/26          3s 73ms/step -
loss: 0.0021 - mae: 0.0312 - val_loss: 0.0020 - val_mae: 0.0311
Epoch 14/200
25/26          0s 70ms/step -
loss: 0.0020 - mae: 0.0306
Epoch 14: val_loss did not improve from 0.00173

```

```

26/26          3s 73ms/step -
loss: 0.0020 - mae: 0.0307 - val_loss: 0.0020 - val_mae: 0.0318
Epoch 15/200
25/26          0s 71ms/step -
loss: 0.0020 - mae: 0.0308
Epoch 15: val_loss improved from 0.00173 to 0.00160, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          3s 80ms/step -
loss: 0.0020 - mae: 0.0307 - val_loss: 0.0016 - val_mae: 0.0272
Epoch 16/200
25/26          0s 72ms/step -
loss: 0.0019 - mae: 0.0293
Epoch 16: val_loss did not improve from 0.00160
26/26          2s 76ms/step -
loss: 0.0019 - mae: 0.0293 - val_loss: 0.0016 - val_mae: 0.0282
Epoch 17/200
25/26          0s 72ms/step -
loss: 0.0019 - mae: 0.0301
Epoch 17: val_loss improved from 0.00160 to 0.00150, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          2s 77ms/step -
loss: 0.0019 - mae: 0.0300 - val_loss: 0.0015 - val_mae: 0.0266
Epoch 18/200
25/26          0s 71ms/step -
loss: 0.0017 - mae: 0.0284
Epoch 18: val_loss did not improve from 0.00150
26/26          2s 74ms/step -
loss: 0.0017 - mae: 0.0284 - val_loss: 0.0015 - val_mae: 0.0263
Epoch 19/200
25/26          0s 71ms/step -
loss: 0.0018 - mae: 0.0287
Epoch 19: val_loss did not improve from 0.00150
26/26          2s 74ms/step -
loss: 0.0018 - mae: 0.0287 - val_loss: 0.0016 - val_mae: 0.0292
Epoch 20/200
25/26          0s 71ms/step -
loss: 0.0019 - mae: 0.0303
Epoch 20: val_loss improved from 0.00150 to 0.00148, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          3s 77ms/step -
loss: 0.0019 - mae: 0.0303 - val_loss: 0.0015 - val_mae: 0.0261
Epoch 21/200
25/26          0s 71ms/step -
loss: 0.0017 - mae: 0.0277
Epoch 21: val_loss did not improve from 0.00148

```

```

26/26          2s 74ms/step -
loss: 0.0017 - mae: 0.0277 - val_loss: 0.0015 - val_mae: 0.0270
Epoch 22/200
25/26          0s 72ms/step -
loss: 0.0017 - mae: 0.0280
Epoch 22: val_loss improved from 0.00148 to 0.00136, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          3s 82ms/step -
loss: 0.0017 - mae: 0.0280 - val_loss: 0.0014 - val_mae: 0.0250
Epoch 23/200
25/26          0s 73ms/step -
loss: 0.0016 - mae: 0.0269
Epoch 23: val_loss improved from 0.00136 to 0.00130, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          2s 79ms/step -
loss: 0.0016 - mae: 0.0269 - val_loss: 0.0013 - val_mae: 0.0245
Epoch 24/200
25/26          0s 72ms/step -
loss: 0.0015 - mae: 0.0261
Epoch 24: val_loss did not improve from 0.00130
26/26          2s 77ms/step -
loss: 0.0015 - mae: 0.0261 - val_loss: 0.0015 - val_mae: 0.0271
Epoch 25/200
25/26          0s 72ms/step -
loss: 0.0017 - mae: 0.0280
Epoch 25: val_loss improved from 0.00130 to 0.00127, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          2s 77ms/step -
loss: 0.0017 - mae: 0.0279 - val_loss: 0.0013 - val_mae: 0.0240
Epoch 26/200
25/26          0s 72ms/step -
loss: 0.0016 - mae: 0.0270
Epoch 26: val_loss did not improve from 0.00127
26/26          2s 75ms/step -
loss: 0.0016 - mae: 0.0271 - val_loss: 0.0013 - val_mae: 0.0241
Epoch 27/200
25/26          0s 72ms/step -
loss: 0.0015 - mae: 0.0266
Epoch 27: val_loss did not improve from 0.00127
26/26          2s 77ms/step -
loss: 0.0015 - mae: 0.0266 - val_loss: 0.0015 - val_mae: 0.0274
Epoch 28/200
25/26          0s 73ms/step -
loss: 0.0016 - mae: 0.0272
Epoch 28: val_loss did not improve from 0.00127

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26/26          2s 77ms/step -
loss: 0.0016 - mae: 0.0271 - val_loss: 0.0013 - val_mae: 0.0242
Epoch 29/200
25/26          0s 73ms/step -
loss: 0.0015 - mae: 0.0256
Epoch 29: val_loss did not improve from 0.00127
26/26          3s 78ms/step -
loss: 0.0015 - mae: 0.0255 - val_loss: 0.0013 - val_mae: 0.0253
Epoch 30/200
25/26          0s 72ms/step -
loss: 0.0016 - mae: 0.0269
Epoch 30: val_loss did not improve from 0.00127
26/26          2s 75ms/step -
loss: 0.0016 - mae: 0.0269 - val_loss: 0.0014 - val_mae: 0.0261
Epoch 31/200
25/26          0s 72ms/step -
loss: 0.0015 - mae: 0.0263
Epoch 31: val_loss improved from 0.00127 to 0.00126, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          3s 78ms/step -
loss: 0.0015 - mae: 0.0262 - val_loss: 0.0013 - val_mae: 0.0236
Epoch 32/200
25/26          0s 73ms/step -
loss: 0.0013 - mae: 0.0246
Epoch 32: val_loss did not improve from 0.00126
26/26          2s 75ms/step -
loss: 0.0013 - mae: 0.0247 - val_loss: 0.0013 - val_mae: 0.0251
Epoch 33/200
25/26          0s 72ms/step -
loss: 0.0015 - mae: 0.0264
Epoch 33: val_loss did not improve from 0.00126
26/26          2s 79ms/step -
loss: 0.0015 - mae: 0.0264 - val_loss: 0.0013 - val_mae: 0.0248
Epoch 34/200
25/26          0s 73ms/step -
loss: 0.0015 - mae: 0.0264
Epoch 34: val_loss did not improve from 0.00126
26/26          2s 80ms/step -
loss: 0.0015 - mae: 0.0264 - val_loss: 0.0013 - val_mae: 0.0252
Epoch 35/200
25/26          0s 72ms/step -
loss: 0.0014 - mae: 0.0252
Epoch 35: val_loss improved from 0.00126 to 0.00121, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          2s 78ms/step -
loss: 0.0014 - mae: 0.0252 - val_loss: 0.0012 - val_mae: 0.0231

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Epoch 36/200
25/26          0s 72ms/step -
loss: 0.0014 - mae: 0.0255
Epoch 36: val_loss did not improve from 0.00121
26/26          2s 75ms/step -
loss: 0.0014 - mae: 0.0255 - val_loss: 0.0013 - val_mae: 0.0252
Epoch 37/200
25/26          0s 72ms/step -
loss: 0.0015 - mae: 0.0265
Epoch 37: val_loss improved from 0.00121 to 0.00115, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          3s 78ms/step -
loss: 0.0015 - mae: 0.0264 - val_loss: 0.0012 - val_mae: 0.0228
Epoch 38/200
25/26          0s 73ms/step -
loss: 0.0014 - mae: 0.0250
Epoch 38: val_loss did not improve from 0.00115
26/26          2s 76ms/step -
loss: 0.0014 - mae: 0.0250 - val_loss: 0.0013 - val_mae: 0.0246
Epoch 39/200
25/26          0s 74ms/step -
loss: 0.0014 - mae: 0.0255
Epoch 39: val_loss did not improve from 0.00115
26/26          3s 77ms/step -
loss: 0.0014 - mae: 0.0255 - val_loss: 0.0012 - val_mae: 0.0232
Epoch 40/200
25/26          0s 73ms/step -
loss: 0.0014 - mae: 0.0251
Epoch 40: val_loss did not improve from 0.00115
26/26          3s 76ms/step -
loss: 0.0014 - mae: 0.0251 - val_loss: 0.0012 - val_mae: 0.0231
Epoch 41/200
25/26          0s 73ms/step -
loss: 0.0015 - mae: 0.0263
Epoch 41: val_loss improved from 0.00115 to 0.00114, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          2s 78ms/step -
loss: 0.0015 - mae: 0.0263 - val_loss: 0.0011 - val_mae: 0.0226
Epoch 42/200
25/26          0s 73ms/step -
loss: 0.0014 - mae: 0.0251
Epoch 42: val_loss improved from 0.00114 to 0.00112, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          2s 81ms/step -
loss: 0.0014 - mae: 0.0251 - val_loss: 0.0011 - val_mae: 0.0223

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Epoch 43/200
25/26 0s 73ms/step -
loss: 0.0013 - mae: 0.0240
Epoch 43: val_loss improved from 0.00112 to 0.00108, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26 2s 79ms/step -
loss: 0.0013 - mae: 0.0240 - val_loss: 0.0011 - val_mae: 0.0222
Epoch 44/200
25/26 0s 74ms/step -
loss: 0.0013 - mae: 0.0238
Epoch 44: val_loss did not improve from 0.00108
26/26 2s 80ms/step -
loss: 0.0013 - mae: 0.0238 - val_loss: 0.0012 - val_mae: 0.0230
Epoch 45/200
25/26 0s 74ms/step -
loss: 0.0013 - mae: 0.0240
Epoch 45: val_loss did not improve from 0.00108
26/26 3s 79ms/step -
loss: 0.0013 - mae: 0.0240 - val_loss: 0.0012 - val_mae: 0.0231
Epoch 46/200
25/26 0s 74ms/step -
loss: 0.0013 - mae: 0.0247
Epoch 46: val_loss did not improve from 0.00108
26/26 2s 76ms/step -
loss: 0.0013 - mae: 0.0247 - val_loss: 0.0012 - val_mae: 0.0232
Epoch 47/200
25/26 0s 73ms/step -
loss: 0.0013 - mae: 0.0245
Epoch 47: val_loss did not improve from 0.00108
26/26 2s 79ms/step -
loss: 0.0013 - mae: 0.0245 - val_loss: 0.0014 - val_mae: 0.0253
Epoch 48/200
25/26 0s 74ms/step -
loss: 0.0014 - mae: 0.0251
Epoch 48: val_loss did not improve from 0.00108
26/26 2s 76ms/step -
loss: 0.0014 - mae: 0.0251 - val_loss: 0.0011 - val_mae: 0.0219
Epoch 49/200
25/26 0s 74ms/step -
loss: 0.0014 - mae: 0.0247
Epoch 49: val_loss did not improve from 0.00108
26/26 2s 77ms/step -
loss: 0.0014 - mae: 0.0246 - val_loss: 0.0011 - val_mae: 0.0218
Epoch 50/200
25/26 0s 74ms/step -
loss: 0.0013 - mae: 0.0242
Epoch 50: val_loss did not improve from 0.00108


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26/26          2s 80ms/step -
loss: 0.0013 - mae: 0.0242 - val_loss: 0.0011 - val_mae: 0.0220
Epoch 51/200
25/26          0s 75ms/step -
loss: 0.0014 - mae: 0.0247
Epoch 51: val_loss improved from 0.00108 to 0.00100, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          2s 83ms/step -
loss: 0.0014 - mae: 0.0246 - val_loss: 9.9779e-04 - val_mae: 0.0211
Epoch 52/200
25/26          0s 74ms/step -
loss: 0.0012 - mae: 0.0232
Epoch 52: val_loss did not improve from 0.00100
26/26          2s 80ms/step -
loss: 0.0012 - mae: 0.0232 - val_loss: 0.0011 - val_mae: 0.0223
Epoch 53/200
25/26          0s 74ms/step -
loss: 0.0013 - mae: 0.0241
Epoch 53: val_loss did not improve from 0.00100
26/26          3s 80ms/step -
loss: 0.0013 - mae: 0.0241 - val_loss: 0.0012 - val_mae: 0.0234
Epoch 54/200
25/26          0s 75ms/step -
loss: 0.0013 - mae: 0.0247
Epoch 54: val_loss did not improve from 0.00100
26/26          2s 78ms/step -
loss: 0.0013 - mae: 0.0246 - val_loss: 0.0012 - val_mae: 0.0250
Epoch 55/200
25/26          0s 74ms/step -
loss: 0.0013 - mae: 0.0246
Epoch 55: val_loss did not improve from 0.00100
26/26          2s 77ms/step -
loss: 0.0013 - mae: 0.0246 - val_loss: 0.0010 - val_mae: 0.0213
Epoch 56/200
25/26          0s 74ms/step -
loss: 0.0012 - mae: 0.0234
Epoch 56: val_loss did not improve from 0.00100
26/26          2s 78ms/step -
loss: 0.0012 - mae: 0.0235 - val_loss: 0.0010 - val_mae: 0.0215
Epoch 57/200
25/26          0s 75ms/step -
loss: 0.0013 - mae: 0.0243
Epoch 57: val_loss did not improve from 0.00100
26/26          3s 80ms/step -
loss: 0.0013 - mae: 0.0244 - val_loss: 0.0012 - val_mae: 0.0241
Epoch 58/200
25/26          0s 75ms/step -

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loss: 0.0013 - mae: 0.0239
Epoch 58: val_loss did not improve from 0.00100
26/26          2s 77ms/step -
loss: 0.0013 - mae: 0.0239 - val_loss: 0.0010 - val_mae: 0.0211
Epoch 59/200
25/26          0s 74ms/step -
loss: 0.0013 - mae: 0.0239
Epoch 59: val_loss did not improve from 0.00100
26/26          3s 80ms/step -
loss: 0.0013 - mae: 0.0239 - val_loss: 0.0010 - val_mae: 0.0217
Epoch 60/200
25/26          0s 75ms/step -
loss: 0.0012 - mae: 0.0235
Epoch 60: val_loss did not improve from 0.00100
26/26          2s 77ms/step -
loss: 0.0012 - mae: 0.0236 - val_loss: 0.0012 - val_mae: 0.0235
Epoch 61/200
25/26          0s 75ms/step -
loss: 0.0013 - mae: 0.0238
Epoch 61: val_loss improved from 0.00100 to 0.00095, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          3s 80ms/step -
loss: 0.0013 - mae: 0.0238 - val_loss: 9.5257e-04 - val_mae: 0.0206
Epoch 62/200
25/26          0s 75ms/step -
loss: 0.0012 - mae: 0.0230
Epoch 62: val_loss did not improve from 0.00095
26/26          2s 81ms/step -
loss: 0.0012 - mae: 0.0231 - val_loss: 9.9751e-04 - val_mae: 0.0211
Epoch 63/200
25/26          0s 76ms/step -
loss: 0.0012 - mae: 0.0233
Epoch 63: val_loss improved from 0.00095 to 0.00093, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          2s 81ms/step -
loss: 0.0012 - mae: 0.0233 - val_loss: 9.3415e-04 - val_mae: 0.0204
Epoch 64/200
25/26          0s 76ms/step -
loss: 0.0011 - mae: 0.0225
Epoch 64: val_loss did not improve from 0.00093
26/26          2s 79ms/step -
loss: 0.0011 - mae: 0.0225 - val_loss: 9.6443e-04 - val_mae: 0.0208
Epoch 65/200
25/26          0s 75ms/step -
loss: 0.0012 - mae: 0.0233
Epoch 65: val_loss did not improve from 0.00093

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26/26          3s 78ms/step -
loss: 0.0012 - mae: 0.0233 - val_loss: 0.0010 - val_mae: 0.0218
Epoch 66/200
25/26          0s 75ms/step -
loss: 0.0012 - mae: 0.0228
Epoch 66: val_loss did not improve from 0.00093
26/26          3s 78ms/step -
loss: 0.0012 - mae: 0.0228 - val_loss: 9.9507e-04 - val_mae: 0.0214
Epoch 67/200
25/26          0s 75ms/step -
loss: 0.0011 - mae: 0.0224
Epoch 67: val_loss did not improve from 0.00093
26/26          2s 81ms/step -
loss: 0.0011 - mae: 0.0224 - val_loss: 9.7788e-04 - val_mae: 0.0214
Epoch 68/200
25/26          0s 76ms/step -
loss: 0.0012 - mae: 0.0234
Epoch 68: val_loss did not improve from 0.00093
26/26          3s 82ms/step -
loss: 0.0012 - mae: 0.0235 - val_loss: 0.0010 - val_mae: 0.0224
Epoch 69/200
25/26          0s 76ms/step -
loss: 0.0012 - mae: 0.0231
Epoch 69: val_loss did not improve from 0.00093
26/26          2s 79ms/step -
loss: 0.0012 - mae: 0.0231 - val_loss: 0.0010 - val_mae: 0.0218
Epoch 70/200
25/26          0s 75ms/step -
loss: 0.0013 - mae: 0.0240
Epoch 70: val_loss did not improve from 0.00093
26/26          3s 78ms/step -
loss: 0.0013 - mae: 0.0240 - val_loss: 0.0011 - val_mae: 0.0224
Epoch 71/200
25/26          0s 75ms/step -
loss: 0.0012 - mae: 0.0231
Epoch 71: val_loss did not improve from 0.00093
26/26          3s 80ms/step -
loss: 0.0012 - mae: 0.0230 - val_loss: 9.7306e-04 - val_mae: 0.0207
Epoch 72/200
25/26          0s 75ms/step -
loss: 0.0012 - mae: 0.0230
Epoch 72: val_loss did not improve from 0.00093
26/26          2s 78ms/step -
loss: 0.0012 - mae: 0.0230 - val_loss: 9.8344e-04 - val_mae: 0.0213
Epoch 73/200
25/26          0s 76ms/step -
loss: 0.0011 - mae: 0.0225
Epoch 73: val_loss did not improve from 0.00093

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26/26          2s 82ms/step -
loss: 0.0011 - mae: 0.0225 - val_loss: 9.8568e-04 - val_mae: 0.0213
Epoch 74/200
25/26          0s 76ms/step -
loss: 0.0011 - mae: 0.0221
Epoch 74: val_loss did not improve from 0.00093
26/26          2s 79ms/step -
loss: 0.0011 - mae: 0.0221 - val_loss: 9.8019e-04 - val_mae: 0.0212
Epoch 75/200
25/26          0s 75ms/step -
loss: 0.0011 - mae: 0.0225
Epoch 75: val_loss improved from 0.00093 to 0.00092, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          2s 80ms/step -
loss: 0.0011 - mae: 0.0225 - val_loss: 9.2482e-04 - val_mae: 0.0206
Epoch 76/200
25/26          0s 75ms/step -
loss: 0.0011 - mae: 0.0223
Epoch 76: val_loss improved from 0.00092 to 0.00090, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          3s 80ms/step -
loss: 0.0011 - mae: 0.0223 - val_loss: 8.9530e-04 - val_mae: 0.0199
Epoch 77/200
25/26          0s 75ms/step -
loss: 0.0011 - mae: 0.0221
Epoch 77: val_loss did not improve from 0.00090
26/26          2s 78ms/step -
loss: 0.0011 - mae: 0.0221 - val_loss: 9.2108e-04 - val_mae: 0.0201
Epoch 78/200
25/26          0s 75ms/step -
loss: 0.0011 - mae: 0.0217
Epoch 78: val_loss did not improve from 0.00090
26/26          2s 78ms/step -
loss: 0.0011 - mae: 0.0217 - val_loss: 9.2623e-04 - val_mae: 0.0204
Epoch 79/200
25/26          0s 76ms/step -
loss: 0.0011 - mae: 0.0220
Epoch 79: val_loss did not improve from 0.00090
26/26          3s 82ms/step -
loss: 0.0011 - mae: 0.0219 - val_loss: 9.1745e-04 - val_mae: 0.0201
Epoch 80/200
25/26          0s 75ms/step -
loss: 0.0011 - mae: 0.0220
Epoch 80: val_loss improved from 0.00090 to 0.00087, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras

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26/26                2s 80ms/step -
loss: 0.0011 - mae: 0.0220 - val_loss: 8.6835e-04 - val_mae: 0.0196
Epoch 81/200
25/26                0s 74ms/step -
loss: 0.0010 - mae: 0.0212
Epoch 81: val_loss did not improve from 0.00087
26/26                2s 77ms/step -
loss: 0.0010 - mae: 0.0212 - val_loss: 8.9045e-04 - val_mae: 0.0201
Epoch 82/200
25/26                0s 75ms/step -
loss: 0.0010 - mae: 0.0209
Epoch 82: val_loss improved from 0.00087 to 0.00085, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26                2s 81ms/step -
loss: 0.0010 - mae: 0.0209 - val_loss: 8.4990e-04 - val_mae: 0.0192
Epoch 83/200
25/26                0s 75ms/step -
loss: 0.0011 - mae: 0.0215
Epoch 83: val_loss did not improve from 0.00085
26/26                2s 78ms/step -
loss: 0.0011 - mae: 0.0216 - val_loss: 0.0011 - val_mae: 0.0224
Epoch 84/200
25/26                0s 75ms/step -
loss: 0.0011 - mae: 0.0226
Epoch 84: val_loss did not improve from 0.00085
26/26                2s 81ms/step -
loss: 0.0011 - mae: 0.0226 - val_loss: 0.0010 - val_mae: 0.0220
Epoch 85/200
25/26                0s 75ms/step -
loss: 0.0011 - mae: 0.0226
Epoch 85: val_loss did not improve from 0.00085
26/26                2s 78ms/step -
loss: 0.0011 - mae: 0.0226 - val_loss: 0.0010 - val_mae: 0.0216
Epoch 86/200
25/26                0s 74ms/step -
loss: 0.0011 - mae: 0.0226
Epoch 86: val_loss did not improve from 0.00085
26/26                2s 77ms/step -
loss: 0.0011 - mae: 0.0226 - val_loss: 9.0451e-04 - val_mae: 0.0199
Epoch 87/200
25/26                0s 74ms/step -
loss: 0.0010 - mae: 0.0212
Epoch 87: val_loss did not improve from 0.00085
26/26                2s 77ms/step -
loss: 0.0010 - mae: 0.0213 - val_loss: 8.5789e-04 - val_mae: 0.0195
Epoch 88/200
25/26                0s 75ms/step -

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loss: 0.0010 - mae: 0.0214
Epoch 88: val_loss did not improve from 0.00085
26/26          3s 78ms/step -
loss: 0.0010 - mae: 0.0215 - val_loss: 9.9100e-04 - val_mae: 0.0211
Epoch 89/200
25/26          0s 75ms/step -
loss: 0.0010 - mae: 0.0216
Epoch 89: val_loss improved from 0.00085 to 0.00085, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          2s 83ms/step -
loss: 0.0010 - mae: 0.0216 - val_loss: 8.4682e-04 - val_mae: 0.0196
Epoch 90/200
25/26          0s 75ms/step -
loss: 0.0011 - mae: 0.0220
Epoch 90: val_loss did not improve from 0.00085
26/26          3s 82ms/step -
loss: 0.0011 - mae: 0.0220 - val_loss: 9.4330e-04 - val_mae: 0.0204
Epoch 91/200
25/26          0s 75ms/step -
loss: 0.0011 - mae: 0.0219
Epoch 91: val_loss did not improve from 0.00085
26/26          3s 80ms/step -
loss: 0.0011 - mae: 0.0219 - val_loss: 8.4885e-04 - val_mae: 0.0194
Epoch 92/200
25/26          0s 75ms/step -
loss: 9.4545e-04 - mae: 0.0204
Epoch 92: val_loss did not improve from 0.00085
26/26          2s 77ms/step -
loss: 9.5054e-04 - mae: 0.0204 - val_loss: 9.2014e-04 - val_mae: 0.0202
Epoch 93/200
25/26          0s 75ms/step -
loss: 0.0010 - mae: 0.0214
Epoch 93: val_loss improved from 0.00085 to 0.00084, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          2s 80ms/step -
loss: 0.0010 - mae: 0.0214 - val_loss: 8.4429e-04 - val_mae: 0.0194
Epoch 94/200
25/26          0s 75ms/step -
loss: 0.0011 - mae: 0.0218
Epoch 94: val_loss did not improve from 0.00084
26/26          2s 78ms/step -
loss: 0.0011 - mae: 0.0218 - val_loss: 8.4735e-04 - val_mae: 0.0192
Epoch 95/200
25/26          0s 75ms/step -
loss: 0.0010 - mae: 0.0209
Epoch 95: val_loss did not improve from 0.00084

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26/26          3s 80ms/step -
loss: 0.0010 - mae: 0.0209 - val_loss: 8.5866e-04 - val_mae: 0.0194
Epoch 96/200
25/26          0s 75ms/step -
loss: 0.0011 - mae: 0.0224
Epoch 96: val_loss did not improve from 0.00084
26/26          2s 79ms/step -
loss: 0.0011 - mae: 0.0224 - val_loss: 8.9528e-04 - val_mae: 0.0201
Epoch 97/200
25/26          0s 75ms/step -
loss: 0.0010 - mae: 0.0215
Epoch 97: val_loss improved from 0.00084 to 0.00082, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          2s 81ms/step -
loss: 0.0010 - mae: 0.0215 - val_loss: 8.2449e-04 - val_mae: 0.0190
Epoch 98/200
25/26          0s 75ms/step -
loss: 9.6984e-04 - mae: 0.0206
Epoch 98: val_loss did not improve from 0.00082
26/26          2s 78ms/step -
loss: 9.7273e-04 - mae: 0.0207 - val_loss: 8.9019e-04 - val_mae: 0.0198
Epoch 99/200
25/26          0s 75ms/step -
loss: 0.0010 - mae: 0.0215
Epoch 99: val_loss did not improve from 0.00082
26/26          3s 78ms/step -
loss: 0.0010 - mae: 0.0215 - val_loss: 8.3561e-04 - val_mae: 0.0193
Epoch 100/200
25/26          0s 75ms/step -
loss: 9.5477e-04 - mae: 0.0205
Epoch 100: val_loss did not improve from 0.00082
26/26          2s 78ms/step -
loss: 9.5810e-04 - mae: 0.0206 - val_loss: 8.7235e-04 - val_mae: 0.0196
Epoch 101/200
25/26          0s 75ms/step -
loss: 0.0011 - mae: 0.0221
Epoch 101: val_loss did not improve from 0.00082
26/26          2s 81ms/step -
loss: 0.0011 - mae: 0.0220 - val_loss: 8.3566e-04 - val_mae: 0.0195
Epoch 102/200
25/26          0s 76ms/step -
loss: 9.9450e-04 - mae: 0.0211
Epoch 102: val_loss did not improve from 0.00082
26/26          3s 83ms/step -
loss: 9.9568e-04 - mae: 0.0211 - val_loss: 8.3216e-04 - val_mae: 0.0191
Epoch 103/200
25/26          0s 76ms/step -

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loss: 9.9601e-04 - mae: 0.0212
Epoch 103: val_loss improved from 0.00082 to 0.00080, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          2s 81ms/step -
loss: 9.9736e-04 - mae: 0.0212 - val_loss: 7.9801e-04 - val_mae: 0.0184
Epoch 104/200
25/26          0s 75ms/step -
loss: 9.3231e-04 - mae: 0.0201
Epoch 104: val_loss did not improve from 0.00080
26/26          3s 81ms/step -
loss: 9.3503e-04 - mae: 0.0201 - val_loss: 7.9982e-04 - val_mae: 0.0187
Epoch 105/200
25/26          0s 75ms/step -
loss: 9.6084e-04 - mae: 0.0203
Epoch 105: val_loss did not improve from 0.00080
26/26          2s 78ms/step -
loss: 9.6061e-04 - mae: 0.0203 - val_loss: 8.5101e-04 - val_mae: 0.0194
Epoch 106/200
25/26          0s 75ms/step -
loss: 0.0010 - mae: 0.0213
Epoch 106: val_loss did not improve from 0.00080
26/26          2s 78ms/step -
loss: 0.0010 - mae: 0.0213 - val_loss: 9.2080e-04 - val_mae: 0.0204
Epoch 107/200
25/26          0s 76ms/step -
loss: 9.5712e-04 - mae: 0.0205
Epoch 107: val_loss did not improve from 0.00080
26/26          2s 79ms/step -
loss: 9.5706e-04 - mae: 0.0205 - val_loss: 8.2760e-04 - val_mae: 0.0189
Epoch 108/200
25/26          0s 76ms/step -
loss: 9.4301e-04 - mae: 0.0202
Epoch 108: val_loss did not improve from 0.00080
26/26          2s 81ms/step -
loss: 9.4452e-04 - mae: 0.0202 - val_loss: 7.9984e-04 - val_mae: 0.0188
Epoch 109/200
25/26          0s 75ms/step -
loss: 9.7015e-04 - mae: 0.0209
Epoch 109: val_loss did not improve from 0.00080
26/26          2s 78ms/step -
loss: 9.7213e-04 - mae: 0.0209 - val_loss: 8.0801e-04 - val_mae: 0.0188
Epoch 110/200
25/26          0s 75ms/step -
loss: 9.9943e-04 - mae: 0.0210
Epoch 110: val_loss did not improve from 0.00080
26/26          3s 78ms/step -
loss: 9.9975e-04 - mae: 0.0210 - val_loss: 8.3108e-04 - val_mae: 0.0190

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Epoch 111/200
25/26          0s 75ms/step -
loss: 9.8611e-04 - mae: 0.0207
Epoch 111: val_loss did not improve from 0.00080
26/26          2s 80ms/step -
loss: 9.8393e-04 - mae: 0.0207 - val_loss: 8.7618e-04 - val_mae: 0.0200
Epoch 112/200
25/26          0s 75ms/step -
loss: 0.0010 - mae: 0.0215
Epoch 112: val_loss did not improve from 0.00080
26/26          2s 78ms/step -
loss: 0.0010 - mae: 0.0215 - val_loss: 8.1940e-04 - val_mae: 0.0193
Epoch 113/200
25/26          0s 76ms/step -
loss: 9.5722e-04 - mae: 0.0206
Epoch 113: val_loss improved from 0.00080 to 0.00079, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          2s 85ms/step -
loss: 9.6014e-04 - mae: 0.0206 - val_loss: 7.9048e-04 - val_mae: 0.0183
Epoch 114/200
25/26          0s 76ms/step -
loss: 9.3251e-04 - mae: 0.0202
Epoch 114: val_loss did not improve from 0.00079
26/26          2s 78ms/step -
loss: 9.3621e-04 - mae: 0.0203 - val_loss: 8.8542e-04 - val_mae: 0.0204
Epoch 115/200
25/26          0s 75ms/step -
loss: 0.0010 - mae: 0.0217
Epoch 115: val_loss improved from 0.00079 to 0.00079, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          3s 83ms/step -
loss: 0.0010 - mae: 0.0217 - val_loss: 7.8752e-04 - val_mae: 0.0185
Epoch 116/200
25/26          0s 75ms/step -
loss: 9.6048e-04 - mae: 0.0205
Epoch 116: val_loss did not improve from 0.00079
26/26          2s 78ms/step -
loss: 9.6181e-04 - mae: 0.0205 - val_loss: 8.9038e-04 - val_mae: 0.0204
Epoch 117/200
25/26          0s 75ms/step -
loss: 0.0010 - mae: 0.0216
Epoch 117: val_loss did not improve from 0.00079
26/26          2s 78ms/step -
loss: 0.0010 - mae: 0.0215 - val_loss: 8.8919e-04 - val_mae: 0.0202
Epoch 118/200
25/26          0s 75ms/step -

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loss: 0.0010 - mae: 0.0214
Epoch 118: val_loss improved from 0.00079 to 0.00078, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          3s 83ms/step -
loss: 0.0010 - mae: 0.0213 - val_loss: 7.8351e-04 - val_mae: 0.0187
Epoch 119/200
25/26          0s 76ms/step -
loss: 9.5592e-04 - mae: 0.0203
Epoch 119: val_loss improved from 0.00078 to 0.00077, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          2s 83ms/step -
loss: 9.5537e-04 - mae: 0.0203 - val_loss: 7.6768e-04 - val_mae: 0.0182
Epoch 120/200
25/26          0s 75ms/step -
loss: 8.7255e-04 - mae: 0.0194
Epoch 120: val_loss did not improve from 0.00077
26/26          2s 79ms/step -
loss: 8.7578e-04 - mae: 0.0194 - val_loss: 7.9148e-04 - val_mae: 0.0187
Epoch 121/200
25/26          0s 75ms/step -
loss: 9.0349e-04 - mae: 0.0198
Epoch 121: val_loss did not improve from 0.00077
26/26          2s 78ms/step -
loss: 9.0643e-04 - mae: 0.0199 - val_loss: 7.9437e-04 - val_mae: 0.0187
Epoch 122/200
25/26          0s 75ms/step -
loss: 9.7781e-04 - mae: 0.0207
Epoch 122: val_loss did not improve from 0.00077
26/26          3s 80ms/step -
loss: 9.7615e-04 - mae: 0.0207 - val_loss: 7.7668e-04 - val_mae: 0.0181
Epoch 123/200
25/26          0s 75ms/step -
loss: 9.5064e-04 - mae: 0.0200
Epoch 123: val_loss did not improve from 0.00077
26/26          2s 78ms/step -
loss: 9.5024e-04 - mae: 0.0200 - val_loss: 8.0593e-04 - val_mae: 0.0191
Epoch 124/200
25/26          0s 75ms/step -
loss: 9.4698e-04 - mae: 0.0201
Epoch 124: val_loss did not improve from 0.00077
26/26          2s 79ms/step -
loss: 9.4748e-04 - mae: 0.0201 - val_loss: 8.5493e-04 - val_mae: 0.0195
Epoch 125/200
25/26          0s 76ms/step -
loss: 9.3163e-04 - mae: 0.0201
Epoch 125: val_loss improved from 0.00077 to 0.00075, saving model to

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/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          2s 85ms/step -
loss: 9.3116e-04 - mae: 0.0201 - val_loss: 7.4575e-04 - val_mae: 0.0178
Epoch 126/200
25/26          0s 75ms/step -
loss: 9.0757e-04 - mae: 0.0198
Epoch 126: val_loss did not improve from 0.00075
26/26          2s 78ms/step -
loss: 9.0778e-04 - mae: 0.0198 - val_loss: 7.6255e-04 - val_mae: 0.0183
Epoch 127/200
25/26          0s 76ms/step -
loss: 8.5821e-04 - mae: 0.0191
Epoch 127: val_loss did not improve from 0.00075
26/26          2s 79ms/step -
loss: 8.6239e-04 - mae: 0.0192 - val_loss: 7.7181e-04 - val_mae: 0.0182
Epoch 128/200
25/26          0s 75ms/step -
loss: 9.5421e-04 - mae: 0.0206
Epoch 128: val_loss did not improve from 0.00075
26/26          2s 81ms/step -
loss: 9.5302e-04 - mae: 0.0206 - val_loss: 7.4726e-04 - val_mae: 0.0181
Epoch 129/200
25/26          0s 75ms/step -
loss: 9.3186e-04 - mae: 0.0199
Epoch 129: val_loss did not improve from 0.00075
26/26          2s 78ms/step -
loss: 9.3104e-04 - mae: 0.0199 - val_loss: 8.2308e-04 - val_mae: 0.0189
Epoch 130/200
25/26          0s 75ms/step -
loss: 9.5006e-04 - mae: 0.0199
Epoch 130: val_loss did not improve from 0.00075
26/26          3s 81ms/step -
loss: 9.4636e-04 - mae: 0.0199 - val_loss: 7.7070e-04 - val_mae: 0.0181
Epoch 131/200
25/26          0s 76ms/step -
loss: 9.0174e-04 - mae: 0.0195
Epoch 131: val_loss did not improve from 0.00075
26/26          2s 79ms/step -
loss: 9.0229e-04 - mae: 0.0195 - val_loss: 9.2659e-04 - val_mae: 0.0218
Epoch 132/200
25/26          0s 75ms/step -
loss: 0.0010 - mae: 0.0216
Epoch 132: val_loss did not improve from 0.00075
26/26          2s 78ms/step -
loss: 0.0010 - mae: 0.0215 - val_loss: 0.0011 - val_mae: 0.0232
Epoch 133/200
25/26          0s 75ms/step -

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loss: 0.0010 - mae: 0.0217
Epoch 133: val_loss improved from 0.00075 to 0.00074, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          3s 81ms/step -
loss: 0.0010 - mae: 0.0216 - val_loss: 7.4081e-04 - val_mae: 0.0180
Epoch 134/200
25/26          0s 75ms/step -
loss: 9.2885e-04 - mae: 0.0200
Epoch 134: val_loss did not improve from 0.00074
26/26          2s 78ms/step -
loss: 9.2803e-04 - mae: 0.0200 - val_loss: 7.5429e-04 - val_mae: 0.0180
Epoch 135/200
25/26          0s 75ms/step -
loss: 9.5114e-04 - mae: 0.0202
Epoch 135: val_loss did not improve from 0.00074
26/26          3s 80ms/step -
loss: 9.5294e-04 - mae: 0.0202 - val_loss: 9.2281e-04 - val_mae: 0.0211
Epoch 136/200
25/26          0s 75ms/step -
loss: 9.5372e-04 - mae: 0.0207
Epoch 136: val_loss did not improve from 0.00074
26/26          2s 82ms/step -
loss: 9.5159e-04 - mae: 0.0207 - val_loss: 7.7840e-04 - val_mae: 0.0183
Epoch 137/200
25/26          0s 76ms/step -
loss: 9.8728e-04 - mae: 0.0207
Epoch 137: val_loss did not improve from 0.00074
26/26          2s 78ms/step -
loss: 9.8410e-04 - mae: 0.0207 - val_loss: 9.5853e-04 - val_mae: 0.0218
Epoch 138/200
25/26          0s 74ms/step -
loss: 0.0010 - mae: 0.0217
Epoch 138: val_loss did not improve from 0.00074
26/26          2s 77ms/step -
loss: 0.0010 - mae: 0.0217 - val_loss: 7.7150e-04 - val_mae: 0.0182
Epoch 139/200
25/26          0s 75ms/step -
loss: 8.7066e-04 - mae: 0.0194
Epoch 139: val_loss did not improve from 0.00074
26/26          3s 78ms/step -
loss: 8.7292e-04 - mae: 0.0194 - val_loss: 8.0541e-04 - val_mae: 0.0194
Epoch 140/200
25/26          0s 74ms/step -
loss: 9.4615e-04 - mae: 0.0204
Epoch 140: val_loss did not improve from 0.00074
26/26          2s 77ms/step -
loss: 9.4494e-04 - mae: 0.0204 - val_loss: 8.4070e-04 - val_mae: 0.0194

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Epoch 141/200
25/26          0s 75ms/step -
loss: 9.4659e-04 - mae: 0.0202
Epoch 141: val_loss did not improve from 0.00074
26/26          3s 80ms/step -
loss: 9.4496e-04 - mae: 0.0202 - val_loss: 7.6938e-04 - val_mae: 0.0186
Epoch 142/200
25/26          0s 75ms/step -
loss: 9.2002e-04 - mae: 0.0199
Epoch 142: val_loss improved from 0.00074 to 0.00074, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          2s 85ms/step -
loss: 9.1825e-04 - mae: 0.0199 - val_loss: 7.3696e-04 - val_mae: 0.0179
Epoch 143/200
25/26          0s 75ms/step -
loss: 8.6105e-04 - mae: 0.0191
Epoch 143: val_loss did not improve from 0.00074
26/26          2s 78ms/step -
loss: 8.6143e-04 - mae: 0.0191 - val_loss: 8.5341e-04 - val_mae: 0.0193
Epoch 144/200
25/26          0s 75ms/step -
loss: 9.5409e-04 - mae: 0.0205
Epoch 144: val_loss did not improve from 0.00074
26/26          2s 78ms/step -
loss: 9.5049e-04 - mae: 0.0205 - val_loss: 7.8631e-04 - val_mae: 0.0182
Epoch 145/200
25/26          0s 75ms/step -
loss: 9.3302e-04 - mae: 0.0198
Epoch 145: val_loss did not improve from 0.00074
26/26          3s 80ms/step -
loss: 9.3258e-04 - mae: 0.0198 - val_loss: 8.1420e-04 - val_mae: 0.0188
Epoch 146/200
25/26          0s 75ms/step -
loss: 9.6261e-04 - mae: 0.0204
Epoch 146: val_loss did not improve from 0.00074
26/26          2s 78ms/step -
loss: 9.6147e-04 - mae: 0.0204 - val_loss: 7.6124e-04 - val_mae: 0.0182
Epoch 147/200
25/26          0s 75ms/step -
loss: 9.2440e-04 - mae: 0.0202
Epoch 147: val_loss improved from 0.00074 to 0.00072, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          2s 84ms/step -
loss: 9.2288e-04 - mae: 0.0202 - val_loss: 7.2108e-04 - val_mae: 0.0174
Epoch 148/200
25/26          0s 76ms/step -

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loss: 8.5379e-04 - mae: 0.0190
Epoch 148: val_loss did not improve from 0.00072
26/26          2s 82ms/step -
loss: 8.5477e-04 - mae: 0.0190 - val_loss: 8.4392e-04 - val_mae: 0.0191
Epoch 149/200
25/26          0s 75ms/step -
loss: 9.4488e-04 - mae: 0.0203
Epoch 149: val_loss did not improve from 0.00072
26/26          2s 78ms/step -
loss: 9.4350e-04 - mae: 0.0203 - val_loss: 8.2796e-04 - val_mae: 0.0192
Epoch 150/200
25/26          0s 75ms/step -
loss: 8.7083e-04 - mae: 0.0195
Epoch 150: val_loss did not improve from 0.00072
26/26          3s 78ms/step -
loss: 8.7257e-04 - mae: 0.0195 - val_loss: 7.4947e-04 - val_mae: 0.0183
Epoch 151/200
25/26          0s 75ms/step -
loss: 9.4886e-04 - mae: 0.0202
Epoch 151: val_loss did not improve from 0.00072
26/26          2s 78ms/step -
loss: 9.4718e-04 - mae: 0.0202 - val_loss: 7.6835e-04 - val_mae: 0.0185
Epoch 152/200
25/26          0s 75ms/step -
loss: 8.4671e-04 - mae: 0.0192
Epoch 152: val_loss improved from 0.00072 to 0.00072, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          3s 83ms/step -
loss: 8.4915e-04 - mae: 0.0192 - val_loss: 7.1582e-04 - val_mae: 0.0174
Epoch 153/200
25/26          0s 75ms/step -
loss: 8.8563e-04 - mae: 0.0194
Epoch 153: val_loss did not improve from 0.00072
26/26          2s 81ms/step -
loss: 8.8475e-04 - mae: 0.0193 - val_loss: 7.1939e-04 - val_mae: 0.0174
Epoch 154/200
25/26          0s 76ms/step -
loss: 8.8159e-04 - mae: 0.0193
Epoch 154: val_loss did not improve from 0.00072
26/26          2s 79ms/step -
loss: 8.7960e-04 - mae: 0.0192 - val_loss: 7.5604e-04 - val_mae: 0.0181
Epoch 155/200
25/26          0s 75ms/step -
loss: 9.0599e-04 - mae: 0.0197
Epoch 155: val_loss did not improve from 0.00072
26/26          2s 78ms/step -
loss: 9.0678e-04 - mae: 0.0197 - val_loss: 7.7747e-04 - val_mae: 0.0183

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Epoch 156/200
25/26 0s 75ms/step -
loss: 8.9825e-04 - mae: 0.0196
Epoch 156: val_loss did not improve from 0.00072
26/26 3s 78ms/step -
loss: 8.9776e-04 - mae: 0.0196 - val_loss: 7.6107e-04 - val_mae: 0.0180
Epoch 157/200
25/26 0s 75ms/step -
loss: 9.9509e-04 - mae: 0.0209
Epoch 157: val_loss did not improve from 0.00072
26/26 3s 78ms/step -
loss: 9.9325e-04 - mae: 0.0209 - val_loss: 8.4220e-04 - val_mae: 0.0195
Epoch 158/200
25/26 0s 75ms/step -
loss: 9.9483e-04 - mae: 0.0211
Epoch 158: val_loss did not improve from 0.00072
26/26 3s 81ms/step -
loss: 9.9696e-04 - mae: 0.0211 - val_loss: 9.2351e-04 - val_mae: 0.0211
Epoch 159/200
25/26 0s 75ms/step -
loss: 9.0378e-04 - mae: 0.0202
Epoch 159: val_loss did not improve from 0.00072
26/26 2s 81ms/step -
loss: 9.0515e-04 - mae: 0.0202 - val_loss: 8.1684e-04 - val_mae: 0.0191
Epoch 160/200
25/26 0s 75ms/step -
loss: 8.9393e-04 - mae: 0.0197
Epoch 160: val_loss improved from 0.00072 to 0.00071, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26 3s 82ms/step -
loss: 8.9176e-04 - mae: 0.0197 - val_loss: 7.1442e-04 - val_mae: 0.0173
Epoch 161/200
25/26 0s 75ms/step -
loss: 8.2979e-04 - mae: 0.0185
Epoch 161: val_loss did not improve from 0.00071
26/26 2s 77ms/step -
loss: 8.3075e-04 - mae: 0.0185 - val_loss: 7.4996e-04 - val_mae: 0.0179
Epoch 162/200
25/26 0s 75ms/step -
loss: 8.7293e-04 - mae: 0.0192
Epoch 162: val_loss did not improve from 0.00071
26/26 3s 78ms/step -
loss: 8.7281e-04 - mae: 0.0192 - val_loss: 9.1854e-04 - val_mae: 0.0200
Epoch 163/200
25/26 0s 75ms/step -
loss: 9.6697e-04 - mae: 0.0204
Epoch 163: val_loss did not improve from 0.00071

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26/26          2s 78ms/step -
loss: 9.6312e-04 - mae: 0.0203 - val_loss: 7.2629e-04 - val_mae: 0.0175
Epoch 164/200
25/26          0s 75ms/step -
loss: 8.7968e-04 - mae: 0.0191
Epoch 164: val_loss did not improve from 0.00071
26/26          2s 79ms/step -
loss: 8.7925e-04 - mae: 0.0191 - val_loss: 7.4722e-04 - val_mae: 0.0180
Epoch 165/200
25/26          0s 75ms/step -
loss: 8.7133e-04 - mae: 0.0192
Epoch 165: val_loss did not improve from 0.00071
26/26          3s 79ms/step -
loss: 8.7181e-04 - mae: 0.0192 - val_loss: 7.6411e-04 - val_mae: 0.0183
Epoch 166/200
25/26          0s 75ms/step -
loss: 8.6466e-04 - mae: 0.0192
Epoch 166: val_loss did not improve from 0.00071
26/26          2s 78ms/step -
loss: 8.6440e-04 - mae: 0.0192 - val_loss: 7.1969e-04 - val_mae: 0.0175
Epoch 167/200
25/26          0s 75ms/step -
loss: 8.8118e-04 - mae: 0.0193
Epoch 167: val_loss did not improve from 0.00071
26/26          3s 78ms/step -
loss: 8.8119e-04 - mae: 0.0193 - val_loss: 7.3415e-04 - val_mae: 0.0178
Epoch 168/200
25/26          0s 75ms/step -
loss: 8.4353e-04 - mae: 0.0191
Epoch 168: val_loss did not improve from 0.00071
26/26          2s 81ms/step -
loss: 8.4444e-04 - mae: 0.0191 - val_loss: 7.2570e-04 - val_mae: 0.0176
Epoch 169/200
25/26          0s 75ms/step -
loss: 8.4500e-04 - mae: 0.0189
Epoch 169: val_loss did not improve from 0.00071
26/26          3s 81ms/step -
loss: 8.4698e-04 - mae: 0.0189 - val_loss: 7.4287e-04 - val_mae: 0.0179
Epoch 170/200
25/26          0s 76ms/step -
loss: 8.6061e-04 - mae: 0.0194
Epoch 170: val_loss did not improve from 0.00071
26/26          2s 82ms/step -
loss: 8.6072e-04 - mae: 0.0194 - val_loss: 7.3402e-04 - val_mae: 0.0179
Epoch 171/200
25/26          0s 75ms/step -
loss: 8.8230e-04 - mae: 0.0197
Epoch 171: val_loss did not improve from 0.00071

```



```

26/26          2s 78ms/step -
loss: 8.8273e-04 - mae: 0.0197 - val_loss: 7.6872e-04 - val_mae: 0.0183
Epoch 172/200
25/26          0s 75ms/step -
loss: 9.0624e-04 - mae: 0.0198
Epoch 172: val_loss did not improve from 0.00071
26/26          2s 78ms/step -
loss: 9.0498e-04 - mae: 0.0198 - val_loss: 7.2768e-04 - val_mae: 0.0175
Epoch 173/200
25/26          0s 75ms/step -
loss: 8.3431e-04 - mae: 0.0190
Epoch 173: val_loss did not improve from 0.00071
26/26          3s 78ms/step -
loss: 8.3530e-04 - mae: 0.0190 - val_loss: 7.4193e-04 - val_mae: 0.0177
Epoch 174/200
25/26          0s 75ms/step -
loss: 8.7497e-04 - mae: 0.0195
Epoch 174: val_loss did not improve from 0.00071
26/26          2s 78ms/step -
loss: 8.7495e-04 - mae: 0.0195 - val_loss: 7.2267e-04 - val_mae: 0.0174
Epoch 175/200
25/26          0s 75ms/step -
loss: 8.4003e-04 - mae: 0.0188
Epoch 175: val_loss improved from 0.00071 to 0.00071, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          3s 83ms/step -
loss: 8.3971e-04 - mae: 0.0188 - val_loss: 7.1268e-04 - val_mae: 0.0172
Epoch 176/200
25/26          0s 76ms/step -
loss: 8.0929e-04 - mae: 0.0186
Epoch 176: val_loss did not improve from 0.00071
26/26          3s 82ms/step -
loss: 8.1218e-04 - mae: 0.0186 - val_loss: 7.5069e-04 - val_mae: 0.0184
Epoch 177/200
25/26          0s 76ms/step -
loss: 8.3246e-04 - mae: 0.0188
Epoch 177: val_loss improved from 0.00071 to 0.00070, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          3s 87ms/step -
loss: 8.3193e-04 - mae: 0.0188 - val_loss: 6.9773e-04 - val_mae: 0.0172
Epoch 178/200
25/26          0s 76ms/step -
loss: 8.1309e-04 - mae: 0.0186
Epoch 178: val_loss did not improve from 0.00070
26/26          2s 79ms/step -
loss: 8.1616e-04 - mae: 0.0186 - val_loss: 7.2780e-04 - val_mae: 0.0177

```

Epoch 179/200
25/26 0s 75ms/step -
loss: 9.0955e-04 - mae: 0.0201
Epoch 179: val_loss did not improve from 0.00070
26/26 2s 78ms/step -
loss: 9.1260e-04 - mae: 0.0201 - val_loss: 7.9513e-04 - val_mae: 0.0187
Epoch 180/200
25/26 0s 74ms/step -
loss: 8.2258e-04 - mae: 0.0188
Epoch 180: val_loss did not improve from 0.00070
26/26 2s 80ms/step -
loss: 8.2483e-04 - mae: 0.0188 - val_loss: 7.2655e-04 - val_mae: 0.0177
Epoch 181/200
25/26 0s 75ms/step -
loss: 8.9403e-04 - mae: 0.0194
Epoch 181: val_loss did not improve from 0.00070
26/26 3s 81ms/step -
loss: 8.9217e-04 - mae: 0.0194 - val_loss: 7.4172e-04 - val_mae: 0.0185
Epoch 182/200
25/26 0s 75ms/step -
loss: 8.6611e-04 - mae: 0.0193
Epoch 182: val_loss did not improve from 0.00070
26/26 2s 78ms/step -
loss: 8.6637e-04 - mae: 0.0193 - val_loss: 7.9873e-04 - val_mae: 0.0192
Epoch 183/200
25/26 0s 75ms/step -
loss: 8.3876e-04 - mae: 0.0190
Epoch 183: val_loss did not improve from 0.00070
26/26 2s 78ms/step -
loss: 8.3911e-04 - mae: 0.0190 - val_loss: 7.3017e-04 - val_mae: 0.0177
Epoch 184/200
25/26 0s 75ms/step -
loss: 8.3402e-04 - mae: 0.0188
Epoch 184: val_loss did not improve from 0.00070
26/26 2s 81ms/step -
loss: 8.3448e-04 - mae: 0.0188 - val_loss: 8.5325e-04 - val_mae: 0.0202
Epoch 185/200
25/26 0s 75ms/step -
loss: 8.8019e-04 - mae: 0.0194
Epoch 185: val_loss did not improve from 0.00070
26/26 2s 80ms/step -
loss: 8.7714e-04 - mae: 0.0194 - val_loss: 7.1754e-04 - val_mae: 0.0173
Epoch 186/200
25/26 0s 74ms/step -
loss: 7.9382e-04 - mae: 0.0185
Epoch 186: val_loss did not improve from 0.00070
26/26 2s 77ms/step -
loss: 7.9674e-04 - mae: 0.0185 - val_loss: 7.4163e-04 - val_mae: 0.0180

```

Epoch 187/200
25/26          0s 75ms/step -
loss: 8.2171e-04 - mae: 0.0188
Epoch 187: val_loss did not improve from 0.00070
26/26          3s 81ms/step -
loss: 8.2381e-04 - mae: 0.0188 - val_loss: 7.0290e-04 - val_mae: 0.0171
Epoch 188/200
25/26          0s 75ms/step -
loss: 7.8859e-04 - mae: 0.0182
Epoch 188: val_loss did not improve from 0.00070
26/26          2s 78ms/step -
loss: 7.9115e-04 - mae: 0.0183 - val_loss: 7.8326e-04 - val_mae: 0.0189
Epoch 189/200
25/26          0s 75ms/step -
loss: 8.2494e-04 - mae: 0.0188
Epoch 189: val_loss did not improve from 0.00070
26/26          2s 78ms/step -
loss: 8.2517e-04 - mae: 0.0188 - val_loss: 6.9839e-04 - val_mae: 0.0171
Epoch 190/200
25/26          0s 75ms/step -
loss: 8.9576e-04 - mae: 0.0196
Epoch 190: val_loss did not improve from 0.00070
26/26          3s 80ms/step -
loss: 8.9628e-04 - mae: 0.0197 - val_loss: 7.6844e-04 - val_mae: 0.0184
Epoch 191/200
25/26          0s 74ms/step -
loss: 8.9312e-04 - mae: 0.0197
Epoch 191: val_loss did not improve from 0.00070
26/26          3s 80ms/step -
loss: 8.9164e-04 - mae: 0.0197 - val_loss: 7.5058e-04 - val_mae: 0.0185
Epoch 192/200
25/26          0s 74ms/step -
loss: 8.9349e-04 - mae: 0.0197
Epoch 192: val_loss did not improve from 0.00070
26/26          2s 77ms/step -
loss: 8.9381e-04 - mae: 0.0197 - val_loss: 7.5012e-04 - val_mae: 0.0182
Epoch 193/200
25/26          0s 76ms/step -
loss: 8.4741e-04 - mae: 0.0191
Epoch 193: val_loss did not improve from 0.00070
26/26          3s 81ms/step -
loss: 8.4769e-04 - mae: 0.0191 - val_loss: 8.4044e-04 - val_mae: 0.0201
Epoch 194/200
25/26          0s 75ms/step -
loss: 8.9825e-04 - mae: 0.0201
Epoch 194: val_loss did not improve from 0.00070
26/26          2s 78ms/step -
loss: 8.9596e-04 - mae: 0.0200 - val_loss: 7.6454e-04 - val_mae: 0.0187

```

```

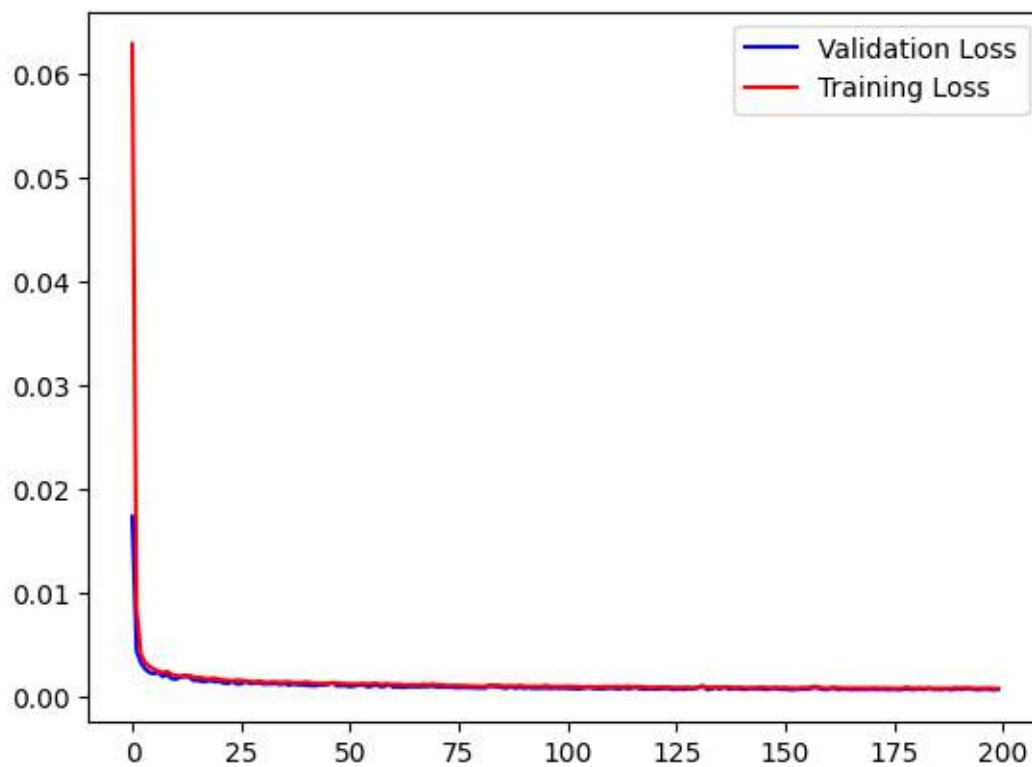
Epoch 195/200
25/26          0s 74ms/step -
loss: 8.2998e-04 - mae: 0.0191
Epoch 195: val_loss improved from 0.00070 to 0.00069, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          3s 79ms/step -
loss: 8.2941e-04 - mae: 0.0191 - val_loss: 6.9058e-04 - val_mae: 0.0170
Epoch 196/200
25/26          0s 75ms/step -
loss: 8.1846e-04 - mae: 0.0185
Epoch 196: val_loss did not improve from 0.00069
26/26          2s 78ms/step -
loss: 8.2089e-04 - mae: 0.0185 - val_loss: 7.1565e-04 - val_mae: 0.0178
Epoch 197/200
25/26          0s 74ms/step -
loss: 8.3693e-04 - mae: 0.0189
Epoch 197: val_loss did not improve from 0.00069
26/26          2s 80ms/step -
loss: 8.3752e-04 - mae: 0.0189 - val_loss: 7.4928e-04 - val_mae: 0.0177
Epoch 198/200
25/26          0s 75ms/step -
loss: 8.6849e-04 - mae: 0.0192
Epoch 198: val_loss did not improve from 0.00069
26/26          2s 81ms/step -
loss: 8.6843e-04 - mae: 0.0192 - val_loss: 7.2286e-04 - val_mae: 0.0176
Epoch 199/200
25/26          0s 77ms/step -
loss: 8.1072e-04 - mae: 0.0188
Epoch 199: val_loss improved from 0.00069 to 0.00069, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/simple_cnn.keras
26/26          2s 84ms/step -
loss: 8.1164e-04 - mae: 0.0188 - val_loss: 6.9015e-04 - val_mae: 0.0169
Epoch 200/200
25/26          0s 75ms/step -
loss: 8.0994e-04 - mae: 0.0183
Epoch 200: val_loss did not improve from 0.00069
26/26          2s 78ms/step -
loss: 8.1109e-04 - mae: 0.0183 - val_loss: 7.2976e-04 - val_mae: 0.0181

```

```

[ ]: # Plotting the training and validation loss
plt.plot(history_simple_cnn.history['val_loss'],c="b")
plt.plot(history_simple_cnn.history['loss'],c="r")
plt.legend(['Validation Loss','Training Loss'])
plt.show()

```



Depiction of Model Performance on images from the Training Set

```
[ ]: show_predictions(best_simple_cnn, X, Y, num_samples=5)
```

1/1

0s 243ms/step



Depiction of Model Performance on images from the Test Set

```
[ ]: show_predictions(best_simple_cnn, X_test, Y_test, num_samples=5)
```

1/1 0s 61ms/step



The predicted image captures the overall structure and lighting quite well for headshots compared to full scene portraits. However, the colors are slightly muted and less vibrant than the ground truth. This suggests that the model learned the general color distribution but struggles with fine color accuracy, especially for clothing or skin tones.

6.4 Model 3: Improved CNN

```
[ ]: # Defining the architecture for the improved CNN model
def build_improved_cnn():
    inputs = layers.Input(shape=(120, 176, 1), name="grayscale_input")
    x1 = layers.Conv2D(64, (3, 3), padding='same')(inputs)
    x1 = layers.BatchNormalization()(x1)
    x1 = layers.LeakyReLU(negative_slope=0.1)(x1)
    p1 = layers.MaxPooling2D((2, 2))(x1)
    x2 = layers.Conv2D(128, (3, 3), padding='same')(p1)
    x2 = layers.BatchNormalization()(x2)
    x2 = layers.LeakyReLU(negative_slope=0.1)(x2)
    p2 = layers.MaxPooling2D((2, 2))(x2)
    x = layers.Conv2D(256, (3, 3), padding='same')(p2)
    x = layers.BatchNormalization()(x)
    x = layers.LeakyReLU(negative_slope=0.1)(x)
    x = layers.UpSampling2D((2, 2))(x)
    x = layers.Conv2D(128, (3, 3), padding='same')(x)
    x = layers.BatchNormalization()(x)
    x = layers.LeakyReLU(negative_slope=0.1)(x)
    x = layers.UpSampling2D((2, 2))(x)
    x = layers.Conv2D(64, (3, 3), padding='same')(x)
    x = layers.BatchNormalization()(x)
    x = layers.LeakyReLU(negative_slope=0.1)(x)
    outputs = layers.Conv2D(3, (3, 3), activation='tanh', padding='same',
↪name="ab_output")(x)
    return models.Model(inputs=inputs, outputs=outputs,
↪name="ColorizationModel")

[ ]: improved_cnn = build_improved_cnn()
improved_cnn.compile(optimizer='adam', loss='mse', metrics=['mae'])

# Defining the path to save the best model
model_checkpoint_path = os.path.join(best_models_path, 'improved_cnn.keras')
model_checkpoint = ModelCheckpoint(model_checkpoint_path, monitor='val_loss',
↪verbose=1, save_best_only=True, mode='min')

# Fitting the model to the training data
history_improved_cnn = improved_cnn.fit(X, Y, batch_size=16, epochs=1000,
↪validation_split=0.1, callbacks=[early_stop, model_checkpoint])
```

```
best_improved_cnn = keras.models.load_model(os.path.join(best_models_path,
↳ 'improved_cnn.keras'))
```

```
Epoch 1/1000
26/26          0s 372ms/step -
loss: 0.3606 - mae: 0.4161
Epoch 1: val_loss improved from inf to 0.07506, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/improved_cnn.keras
26/26          20s 497ms/step -
loss: 0.3531 - mae: 0.4102 - val_loss: 0.0751 - val_mae: 0.2411
Epoch 2/1000
25/26          0s 95ms/step -
loss: 0.0192 - mae: 0.1100
Epoch 2: val_loss improved from 0.07506 to 0.06820, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/improved_cnn.keras
26/26          3s 107ms/step -
loss: 0.0190 - mae: 0.1092 - val_loss: 0.0682 - val_mae: 0.2290
Epoch 3/1000
25/26          0s 97ms/step -
loss: 0.0100 - mae: 0.0764
Epoch 3: val_loss did not improve from 0.06820
26/26          5s 103ms/step -
loss: 0.0099 - mae: 0.0760 - val_loss: 0.0717 - val_mae: 0.2362
Epoch 4/1000
25/26          0s 96ms/step -
loss: 0.0060 - mae: 0.0560
Epoch 4: val_loss did not improve from 0.06820
26/26          5s 101ms/step -
loss: 0.0060 - mae: 0.0559 - val_loss: 0.0724 - val_mae: 0.2418
Epoch 5/1000
25/26          0s 95ms/step -
loss: 0.0050 - mae: 0.0504
Epoch 5: val_loss improved from 0.06820 to 0.06559, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/improved_cnn.keras
26/26          5s 106ms/step -
loss: 0.0050 - mae: 0.0505 - val_loss: 0.0656 - val_mae: 0.2316
Epoch 6/1000
25/26          0s 100ms/step -
loss: 0.0047 - mae: 0.0495
Epoch 6: val_loss did not improve from 0.06559
26/26          5s 105ms/step -
loss: 0.0047 - mae: 0.0496 - val_loss: 0.0679 - val_mae: 0.2353
Epoch 7/1000
```

```

25/26          0s 95ms/step -
loss: 0.0048 - mae: 0.0504
Epoch 7: val_loss did not improve from 0.06559
26/26          5s 101ms/step -
loss: 0.0047 - mae: 0.0502 - val_loss: 0.0657 - val_mae: 0.2313
Epoch 8/1000
25/26          0s 95ms/step -
loss: 0.0038 - mae: 0.0443
Epoch 8: val_loss improved from 0.06559 to 0.05880, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/Improved_cnn.keras
26/26          5s 107ms/step -
loss: 0.0038 - mae: 0.0443 - val_loss: 0.0588 - val_mae: 0.2202
Epoch 9/1000
25/26          0s 94ms/step -
loss: 0.0040 - mae: 0.0459
Epoch 9: val_loss improved from 0.05880 to 0.05317, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/Improved_cnn.keras
26/26          3s 103ms/step -
loss: 0.0040 - mae: 0.0459 - val_loss: 0.0532 - val_mae: 0.2096
Epoch 10/1000
25/26          0s 95ms/step -
loss: 0.0035 - mae: 0.0425
Epoch 10: val_loss improved from 0.05317 to 0.05167, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/Improved_cnn.keras
26/26          5s 105ms/step -
loss: 0.0035 - mae: 0.0425 - val_loss: 0.0517 - val_mae: 0.2070
Epoch 11/1000
25/26          0s 94ms/step -
loss: 0.0034 - mae: 0.0422
Epoch 11: val_loss improved from 0.05167 to 0.04880, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/Improved_cnn.keras
26/26          3s 105ms/step -
loss: 0.0034 - mae: 0.0422 - val_loss: 0.0488 - val_mae: 0.2010
Epoch 12/1000
25/26          0s 95ms/step -
loss: 0.0036 - mae: 0.0439
Epoch 12: val_loss did not improve from 0.04880
26/26          3s 99ms/step -
loss: 0.0036 - mae: 0.0440 - val_loss: 0.0492 - val_mae: 0.2020
Epoch 13/1000
25/26          0s 97ms/step -
loss: 0.0037 - mae: 0.0455
Epoch 13: val_loss improved from 0.04880 to 0.04581, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models

```

```

MSBA/improved_cnn.keras
26/26          5s 110ms/step -
loss: 0.0037 - mae: 0.0455 - val_loss: 0.0458 - val_mae: 0.1949
Epoch 14/1000
25/26          0s 99ms/step -
loss: 0.0032 - mae: 0.0410
Epoch 14: val_loss improved from 0.04581 to 0.03554, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/improved_cnn.keras
26/26          5s 110ms/step -
loss: 0.0032 - mae: 0.0411 - val_loss: 0.0355 - val_mae: 0.1684
Epoch 15/1000
25/26          0s 95ms/step -
loss: 0.0037 - mae: 0.0448
Epoch 15: val_loss improved from 0.03554 to 0.03159, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/improved_cnn.keras
26/26          5s 104ms/step -
loss: 0.0036 - mae: 0.0446 - val_loss: 0.0316 - val_mae: 0.1593
Epoch 16/1000
25/26          0s 97ms/step -
loss: 0.0032 - mae: 0.0413
Epoch 16: val_loss improved from 0.03159 to 0.03150, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/improved_cnn.keras
26/26          5s 109ms/step -
loss: 0.0032 - mae: 0.0412 - val_loss: 0.0315 - val_mae: 0.1602
Epoch 17/1000
25/26          0s 98ms/step -
loss: 0.0031 - mae: 0.0404
Epoch 17: val_loss improved from 0.03150 to 0.02707, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/improved_cnn.keras
26/26          3s 109ms/step -
loss: 0.0031 - mae: 0.0406 - val_loss: 0.0271 - val_mae: 0.1467
Epoch 18/1000
25/26          0s 97ms/step -
loss: 0.0041 - mae: 0.0482
Epoch 18: val_loss improved from 0.02707 to 0.02208, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/improved_cnn.keras
26/26          3s 110ms/step -
loss: 0.0041 - mae: 0.0481 - val_loss: 0.0221 - val_mae: 0.1298
Epoch 19/1000
25/26          0s 97ms/step -
loss: 0.0032 - mae: 0.0416
Epoch 19: val_loss improved from 0.02208 to 0.01396, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models

```

```

MSBA/improved_cnn.keras
26/26          3s 107ms/step -
loss: 0.0032 - mae: 0.0417 - val_loss: 0.0140 - val_mae: 0.1034
Epoch 20/1000
25/26          0s 98ms/step -
loss: 0.0032 - mae: 0.0418
Epoch 20: val_loss did not improve from 0.01396
26/26          3s 101ms/step -
loss: 0.0032 - mae: 0.0417 - val_loss: 0.0141 - val_mae: 0.1008
Epoch 21/1000
25/26          0s 100ms/step -
loss: 0.0029 - mae: 0.0395
Epoch 21: val_loss did not improve from 0.01396
26/26          5s 106ms/step -
loss: 0.0029 - mae: 0.0397 - val_loss: 0.0169 - val_mae: 0.1166
Epoch 22/1000
26/26          0s 99ms/step -
loss: 0.0031 - mae: 0.0415
Epoch 22: val_loss improved from 0.01396 to 0.01140, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/improved_cnn.keras
26/26          4s 141ms/step -
loss: 0.0031 - mae: 0.0415 - val_loss: 0.0114 - val_mae: 0.0914
Epoch 23/1000
25/26          0s 101ms/step -
loss: 0.0029 - mae: 0.0389
Epoch 23: val_loss improved from 0.01140 to 0.00660, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/improved_cnn.keras
26/26          3s 115ms/step -
loss: 0.0029 - mae: 0.0390 - val_loss: 0.0066 - val_mae: 0.0676
Epoch 24/1000
25/26          0s 99ms/step -
loss: 0.0025 - mae: 0.0356
Epoch 24: val_loss improved from 0.00660 to 0.00602, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/improved_cnn.keras
26/26          3s 110ms/step -
loss: 0.0025 - mae: 0.0358 - val_loss: 0.0060 - val_mae: 0.0624
Epoch 25/1000
25/26          0s 98ms/step -
loss: 0.0032 - mae: 0.0429
Epoch 25: val_loss did not improve from 0.00602
26/26          3s 101ms/step -
loss: 0.0032 - mae: 0.0429 - val_loss: 0.0084 - val_mae: 0.0762
Epoch 26/1000
25/26          0s 96ms/step -
loss: 0.0035 - mae: 0.0452

```

Epoch 26: val_loss improved from 0.00602 to 0.00366, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/improved_cnn.keras
26/26 5s 105ms/step -
loss: 0.0035 - mae: 0.0452 - val_loss: 0.0037 - val_mae: 0.0473
Epoch 27/1000
25/26 0s 96ms/step -
loss: 0.0033 - mae: 0.0433
Epoch 27: val_loss improved from 0.00366 to 0.00262, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/improved_cnn.keras
26/26 5s 105ms/step -
loss: 0.0033 - mae: 0.0433 - val_loss: 0.0026 - val_mae: 0.0395
Epoch 28/1000
25/26 0s 102ms/step -
loss: 0.0031 - mae: 0.0422
Epoch 28: val_loss did not improve from 0.00262
26/26 5s 111ms/step -
loss: 0.0031 - mae: 0.0420 - val_loss: 0.0031 - val_mae: 0.0437
Epoch 29/1000
25/26 0s 99ms/step -
loss: 0.0024 - mae: 0.0345
Epoch 29: val_loss did not improve from 0.00262
26/26 3s 103ms/step -
loss: 0.0024 - mae: 0.0345 - val_loss: 0.0031 - val_mae: 0.0454
Epoch 30/1000
25/26 0s 98ms/step -
loss: 0.0024 - mae: 0.0351
Epoch 30: val_loss improved from 0.00262 to 0.00225, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/improved_cnn.keras
26/26 3s 110ms/step -
loss: 0.0024 - mae: 0.0351 - val_loss: 0.0023 - val_mae: 0.0356
Epoch 31/1000
25/26 0s 97ms/step -
loss: 0.0026 - mae: 0.0377
Epoch 31: val_loss did not improve from 0.00225
26/26 5s 103ms/step -
loss: 0.0026 - mae: 0.0380 - val_loss: 0.0059 - val_mae: 0.0608
Epoch 32/1000
26/26 0s 96ms/step -
loss: 0.0037 - mae: 0.0470
Epoch 32: val_loss did not improve from 0.00225
26/26 5s 104ms/step -
loss: 0.0036 - mae: 0.0469 - val_loss: 0.0034 - val_mae: 0.0473
Epoch 33/1000
25/26 0s 98ms/step -
loss: 0.0033 - mae: 0.0420

Epoch 33: val_loss improved from 0.00225 to 0.00225, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/improved_cnn.keras

26/26 3s 109ms/step -
loss: 0.0033 - mae: 0.0419 - val_loss: 0.0022 - val_mae: 0.0359

Epoch 34/1000

25/26 0s 96ms/step -
loss: 0.0039 - mae: 0.0486

Epoch 34: val_loss did not improve from 0.00225

26/26 3s 100ms/step -
loss: 0.0039 - mae: 0.0483 - val_loss: 0.0025 - val_mae: 0.0380

Epoch 35/1000

25/26 0s 98ms/step -
loss: 0.0027 - mae: 0.0385

Epoch 35: val_loss did not improve from 0.00225

26/26 5s 110ms/step -
loss: 0.0027 - mae: 0.0385 - val_loss: 0.0023 - val_mae: 0.0356

Epoch 36/1000

25/26 0s 96ms/step -
loss: 0.0026 - mae: 0.0381

Epoch 36: val_loss did not improve from 0.00225

26/26 5s 99ms/step -
loss: 0.0026 - mae: 0.0380 - val_loss: 0.0023 - val_mae: 0.0365

Epoch 37/1000

25/26 0s 96ms/step -
loss: 0.0024 - mae: 0.0361

Epoch 37: val_loss improved from 0.00225 to 0.00205, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/improved_cnn.keras

26/26 5s 109ms/step -
loss: 0.0024 - mae: 0.0362 - val_loss: 0.0021 - val_mae: 0.0343

Epoch 38/1000

25/26 0s 101ms/step -
loss: 0.0025 - mae: 0.0368

Epoch 38: val_loss did not improve from 0.00205

26/26 3s 107ms/step -
loss: 0.0024 - mae: 0.0366 - val_loss: 0.0031 - val_mae: 0.0418

Epoch 39/1000

25/26 0s 98ms/step -
loss: 0.0029 - mae: 0.0409

Epoch 39: val_loss improved from 0.00205 to 0.00201, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/improved_cnn.keras

26/26 3s 108ms/step -
loss: 0.0029 - mae: 0.0408 - val_loss: 0.0020 - val_mae: 0.0342

Epoch 40/1000

25/26 0s 98ms/step -
loss: 0.0032 - mae: 0.0441

Epoch 40: val_loss did not improve from 0.00201
26/26 3s 103ms/step -
loss: 0.0032 - mae: 0.0441 - val_loss: 0.0021 - val_mae: 0.0363
Epoch 41/1000
25/26 0s 101ms/step -
loss: 0.0028 - mae: 0.0404
Epoch 41: val_loss improved from 0.00201 to 0.00178, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/improved_cnn.keras
26/26 5s 114ms/step -
loss: 0.0028 - mae: 0.0404 - val_loss: 0.0018 - val_mae: 0.0316
Epoch 42/1000
25/26 0s 101ms/step -
loss: 0.0029 - mae: 0.0403
Epoch 42: val_loss did not improve from 0.00178
26/26 5s 108ms/step -
loss: 0.0029 - mae: 0.0403 - val_loss: 0.0021 - val_mae: 0.0354
Epoch 43/1000
25/26 0s 101ms/step -
loss: 0.0023 - mae: 0.0356
Epoch 43: val_loss did not improve from 0.00178
26/26 3s 107ms/step -
loss: 0.0023 - mae: 0.0358 - val_loss: 0.0022 - val_mae: 0.0354
Epoch 44/1000
25/26 0s 99ms/step -
loss: 0.0024 - mae: 0.0365
Epoch 44: val_loss did not improve from 0.00178
26/26 3s 106ms/step -
loss: 0.0024 - mae: 0.0365 - val_loss: 0.0018 - val_mae: 0.0326
Epoch 45/1000
25/26 0s 100ms/step -
loss: 0.0022 - mae: 0.0340
Epoch 45: val_loss improved from 0.00178 to 0.00175, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/improved_cnn.keras
26/26 5s 118ms/step -
loss: 0.0022 - mae: 0.0339 - val_loss: 0.0017 - val_mae: 0.0310
Epoch 46/1000
25/26 0s 97ms/step -
loss: 0.0024 - mae: 0.0366
Epoch 46: val_loss improved from 0.00175 to 0.00174, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/improved_cnn.keras
26/26 5s 108ms/step -
loss: 0.0024 - mae: 0.0366 - val_loss: 0.0017 - val_mae: 0.0313
Epoch 47/1000
25/26 0s 97ms/step -
loss: 0.0022 - mae: 0.0345

Epoch 47: val_loss improved from 0.00174 to 0.00164, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/improved_cnn.keras
26/26 5s 108ms/step -
loss: 0.0022 - mae: 0.0346 - val_loss: 0.0016 - val_mae: 0.0299
Epoch 48/1000
25/26 0s 96ms/step -
loss: 0.0027 - mae: 0.0392
Epoch 48: val_loss did not improve from 0.00164
26/26 5s 99ms/step -
loss: 0.0027 - mae: 0.0392 - val_loss: 0.0019 - val_mae: 0.0321
Epoch 49/1000
25/26 0s 95ms/step -
loss: 0.0024 - mae: 0.0367
Epoch 49: val_loss improved from 0.00164 to 0.00149, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/improved_cnn.keras
26/26 3s 104ms/step -
loss: 0.0024 - mae: 0.0366 - val_loss: 0.0015 - val_mae: 0.0276
Epoch 50/1000
25/26 0s 97ms/step -
loss: 0.0024 - mae: 0.0364
Epoch 50: val_loss did not improve from 0.00149
26/26 5s 103ms/step -
loss: 0.0024 - mae: 0.0363 - val_loss: 0.0022 - val_mae: 0.0354
Epoch 51/1000
25/26 0s 96ms/step -
loss: 0.0020 - mae: 0.0330
Epoch 51: val_loss improved from 0.00149 to 0.00146, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/improved_cnn.keras
26/26 3s 105ms/step -
loss: 0.0020 - mae: 0.0330 - val_loss: 0.0015 - val_mae: 0.0276
Epoch 52/1000
25/26 0s 96ms/step -
loss: 0.0022 - mae: 0.0349
Epoch 52: val_loss did not improve from 0.00146
26/26 3s 99ms/step -
loss: 0.0023 - mae: 0.0351 - val_loss: 0.0027 - val_mae: 0.0413
Epoch 53/1000
25/26 0s 96ms/step -
loss: 0.0027 - mae: 0.0395
Epoch 53: val_loss did not improve from 0.00146
26/26 3s 99ms/step -
loss: 0.0027 - mae: 0.0396 - val_loss: 0.0027 - val_mae: 0.0418
Epoch 54/1000
25/26 0s 96ms/step -
loss: 0.0032 - mae: 0.0427

Epoch 54: val_loss did not improve from 0.00146
26/26 3s 99ms/step -
loss: 0.0032 - mae: 0.0425 - val_loss: 0.0019 - val_mae: 0.0337
Epoch 55/1000
25/26 0s 100ms/step -
loss: 0.0022 - mae: 0.0350
Epoch 55: val_loss did not improve from 0.00146
26/26 5s 103ms/step -
loss: 0.0022 - mae: 0.0350 - val_loss: 0.0032 - val_mae: 0.0447
Epoch 56/1000
25/26 0s 101ms/step -
loss: 0.0024 - mae: 0.0367
Epoch 56: val_loss did not improve from 0.00146
26/26 3s 104ms/step -
loss: 0.0024 - mae: 0.0366 - val_loss: 0.0019 - val_mae: 0.0328
Epoch 57/1000
25/26 0s 100ms/step -
loss: 0.0023 - mae: 0.0357
Epoch 57: val_loss did not improve from 0.00146
26/26 3s 103ms/step -
loss: 0.0023 - mae: 0.0358 - val_loss: 0.0024 - val_mae: 0.0374
Epoch 58/1000
25/26 0s 97ms/step -
loss: 0.0022 - mae: 0.0352
Epoch 58: val_loss did not improve from 0.00146
26/26 3s 102ms/step -
loss: 0.0022 - mae: 0.0352 - val_loss: 0.0051 - val_mae: 0.0559
Epoch 59/1000
25/26 0s 97ms/step -
loss: 0.0033 - mae: 0.0426
Epoch 59: val_loss did not improve from 0.00146
26/26 3s 101ms/step -
loss: 0.0032 - mae: 0.0423 - val_loss: 0.0017 - val_mae: 0.0314
Epoch 60/1000
25/26 0s 97ms/step -
loss: 0.0024 - mae: 0.0373
Epoch 60: val_loss did not improve from 0.00146
26/26 3s 102ms/step -
loss: 0.0024 - mae: 0.0371 - val_loss: 0.0016 - val_mae: 0.0282
Epoch 61/1000
25/26 0s 97ms/step -
loss: 0.0019 - mae: 0.0310
Epoch 61: val_loss did not improve from 0.00146
26/26 3s 100ms/step -
loss: 0.0019 - mae: 0.0311 - val_loss: 0.0017 - val_mae: 0.0299
Epoch 62/1000
25/26 0s 97ms/step -
loss: 0.0018 - mae: 0.0300

Epoch 62: val_loss did not improve from 0.00146
26/26 5s 103ms/step -
loss: 0.0018 - mae: 0.0303 - val_loss: 0.0035 - val_mae: 0.0457
Epoch 63/1000
25/26 0s 102ms/step -
loss: 0.0027 - mae: 0.0393
Epoch 63: val_loss did not improve from 0.00146
26/26 3s 117ms/step -
loss: 0.0027 - mae: 0.0392 - val_loss: 0.0017 - val_mae: 0.0317
Epoch 64/1000
25/26 0s 98ms/step -
loss: 0.0022 - mae: 0.0345
Epoch 64: val_loss did not improve from 0.00146
26/26 5s 104ms/step -
loss: 0.0022 - mae: 0.0347 - val_loss: 0.0019 - val_mae: 0.0335
Epoch 65/1000
25/26 0s 96ms/step -
loss: 0.0027 - mae: 0.0398
Epoch 65: val_loss did not improve from 0.00146
26/26 5s 100ms/step -
loss: 0.0027 - mae: 0.0397 - val_loss: 0.0021 - val_mae: 0.0348
Epoch 66/1000
26/26 0s 95ms/step -
loss: 0.0020 - mae: 0.0330
Epoch 66: val_loss did not improve from 0.00146
26/26 5s 104ms/step -
loss: 0.0020 - mae: 0.0329 - val_loss: 0.0017 - val_mae: 0.0310
Epoch 67/1000
25/26 0s 98ms/step -
loss: 0.0018 - mae: 0.0300
Epoch 67: val_loss did not improve from 0.00146
26/26 3s 106ms/step -
loss: 0.0018 - mae: 0.0301 - val_loss: 0.0015 - val_mae: 0.0279
Epoch 68/1000
25/26 0s 100ms/step -
loss: 0.0021 - mae: 0.0337
Epoch 68: val_loss did not improve from 0.00146
26/26 3s 116ms/step -
loss: 0.0021 - mae: 0.0337 - val_loss: 0.0015 - val_mae: 0.0276
Epoch 69/1000
26/26 0s 100ms/step -
loss: 0.0019 - mae: 0.0327
Epoch 69: val_loss did not improve from 0.00146
26/26 3s 114ms/step -
loss: 0.0019 - mae: 0.0326 - val_loss: 0.0015 - val_mae: 0.0281
Epoch 70/1000
25/26 0s 99ms/step -
loss: 0.0022 - mae: 0.0355

Epoch 70: val_loss improved from 0.00146 to 0.00143, saving model to /content/drive/MyDrive/BA865/BA865 Group Project/Best Models MSBA/improved_cnn.keras

26/26 3s 115ms/step -
loss: 0.0022 - mae: 0.0354 - val_loss: 0.0014 - val_mae: 0.0272

Epoch 71/1000

25/26 0s 100ms/step -
loss: 0.0019 - mae: 0.0314

Epoch 71: val_loss did not improve from 0.00143

26/26 3s 105ms/step -
loss: 0.0019 - mae: 0.0315 - val_loss: 0.0025 - val_mae: 0.0376

Epoch 72/1000

25/26 0s 99ms/step -
loss: 0.0021 - mae: 0.0342

Epoch 72: val_loss did not improve from 0.00143

26/26 3s 105ms/step -
loss: 0.0021 - mae: 0.0344 - val_loss: 0.0023 - val_mae: 0.0380

Epoch 73/1000

25/26 0s 98ms/step -
loss: 0.0024 - mae: 0.0358

Epoch 73: val_loss did not improve from 0.00143

26/26 5s 104ms/step -
loss: 0.0024 - mae: 0.0358 - val_loss: 0.0017 - val_mae: 0.0318

Epoch 74/1000

25/26 0s 99ms/step -
loss: 0.0019 - mae: 0.0322

Epoch 74: val_loss did not improve from 0.00143

26/26 5s 105ms/step -
loss: 0.0019 - mae: 0.0322 - val_loss: 0.0015 - val_mae: 0.0281

Epoch 75/1000

25/26 0s 99ms/step -
loss: 0.0017 - mae: 0.0292

Epoch 75: val_loss did not improve from 0.00143

26/26 3s 102ms/step -
loss: 0.0017 - mae: 0.0292 - val_loss: 0.0015 - val_mae: 0.0295

Epoch 76/1000

25/26 0s 99ms/step -
loss: 0.0020 - mae: 0.0335

Epoch 76: val_loss did not improve from 0.00143

26/26 5s 109ms/step -
loss: 0.0020 - mae: 0.0335 - val_loss: 0.0037 - val_mae: 0.0459

Epoch 77/1000

25/26 0s 98ms/step -
loss: 0.0020 - mae: 0.0327

Epoch 77: val_loss did not improve from 0.00143

26/26 3s 101ms/step -
loss: 0.0020 - mae: 0.0327 - val_loss: 0.0017 - val_mae: 0.0322

Epoch 78/1000

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25/26          0s 97ms/step -
loss: 0.0018 - mae: 0.0309
Epoch 78: val_loss improved from 0.00143 to 0.00126, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/improved_cnn.keras
26/26          3s 106ms/step -
loss: 0.0018 - mae: 0.0310 - val_loss: 0.0013 - val_mae: 0.0251
Epoch 79/1000
25/26          0s 101ms/step -
loss: 0.0023 - mae: 0.0353
Epoch 79: val_loss did not improve from 0.00126
26/26          3s 107ms/step -
loss: 0.0023 - mae: 0.0353 - val_loss: 0.0020 - val_mae: 0.0346
Epoch 80/1000
25/26          0s 102ms/step -
loss: 0.0018 - mae: 0.0304
Epoch 80: val_loss did not improve from 0.00126
26/26          5s 110ms/step -
loss: 0.0018 - mae: 0.0305 - val_loss: 0.0014 - val_mae: 0.0281
Epoch 81/1000
25/26          0s 97ms/step -
loss: 0.0021 - mae: 0.0343
Epoch 81: val_loss did not improve from 0.00126
26/26          5s 102ms/step -
loss: 0.0021 - mae: 0.0344 - val_loss: 0.0036 - val_mae: 0.0467
Epoch 82/1000
25/26          0s 95ms/step -
loss: 0.0024 - mae: 0.0363
Epoch 82: val_loss did not improve from 0.00126
26/26          3s 99ms/step -
loss: 0.0024 - mae: 0.0361 - val_loss: 0.0017 - val_mae: 0.0301
Epoch 83/1000
25/26          0s 96ms/step -
loss: 0.0020 - mae: 0.0329
Epoch 83: val_loss did not improve from 0.00126
26/26          5s 100ms/step -
loss: 0.0020 - mae: 0.0329 - val_loss: 0.0018 - val_mae: 0.0324
Epoch 84/1000
25/26          0s 97ms/step -
loss: 0.0022 - mae: 0.0347
Epoch 84: val_loss did not improve from 0.00126
26/26          5s 102ms/step -
loss: 0.0022 - mae: 0.0346 - val_loss: 0.0016 - val_mae: 0.0294
Epoch 85/1000
25/26          0s 95ms/step -
loss: 0.0020 - mae: 0.0333
Epoch 85: val_loss did not improve from 0.00126
26/26          3s 98ms/step -

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loss: 0.0020 - mae: 0.0332 - val_loss: 0.0016 - val_mae: 0.0294
Epoch 86/1000
25/26          0s 97ms/step -
loss: 0.0021 - mae: 0.0335
Epoch 86: val_loss did not improve from 0.00126
26/26          5s 103ms/step -
loss: 0.0021 - mae: 0.0336 - val_loss: 0.0027 - val_mae: 0.0398
Epoch 87/1000
25/26          0s 97ms/step -
loss: 0.0018 - mae: 0.0308
Epoch 87: val_loss did not improve from 0.00126
26/26          5s 100ms/step -
loss: 0.0018 - mae: 0.0308 - val_loss: 0.0016 - val_mae: 0.0308
Epoch 88/1000
25/26          0s 96ms/step -
loss: 0.0020 - mae: 0.0334
Epoch 88: val_loss did not improve from 0.00126
26/26          5s 99ms/step -
loss: 0.0020 - mae: 0.0333 - val_loss: 0.0014 - val_mae: 0.0277
Epoch 89/1000
25/26          0s 96ms/step -
loss: 0.0021 - mae: 0.0341
Epoch 89: val_loss did not improve from 0.00126
26/26          3s 102ms/step -
loss: 0.0021 - mae: 0.0339 - val_loss: 0.0025 - val_mae: 0.0386
Epoch 90/1000
25/26          0s 97ms/step -
loss: 0.0018 - mae: 0.0315
Epoch 90: val_loss did not improve from 0.00126
26/26          3s 102ms/step -
loss: 0.0018 - mae: 0.0315 - val_loss: 0.0021 - val_mae: 0.0348
Epoch 91/1000
25/26          0s 96ms/step -
loss: 0.0020 - mae: 0.0332
Epoch 91: val_loss did not improve from 0.00126
26/26          3s 99ms/step -
loss: 0.0020 - mae: 0.0333 - val_loss: 0.0014 - val_mae: 0.0281
Epoch 92/1000
25/26          0s 97ms/step -
loss: 0.0018 - mae: 0.0309
Epoch 92: val_loss did not improve from 0.00126
26/26          5s 100ms/step -
loss: 0.0018 - mae: 0.0309 - val_loss: 0.0019 - val_mae: 0.0328
Epoch 93/1000
25/26          0s 98ms/step -
loss: 0.0019 - mae: 0.0325
Epoch 93: val_loss did not improve from 0.00126
26/26          5s 101ms/step -

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loss: 0.0019 - mae: 0.0325 - val_loss: 0.0013 - val_mae: 0.0257
Epoch 94/1000
25/26          0s 96ms/step -
loss: 0.0017 - mae: 0.0295
Epoch 94: val_loss did not improve from 0.00126
26/26          3s 100ms/step -
loss: 0.0017 - mae: 0.0296 - val_loss: 0.0019 - val_mae: 0.0320
Epoch 95/1000
25/26          0s 97ms/step -
loss: 0.0020 - mae: 0.0329
Epoch 95: val_loss improved from 0.00126 to 0.00124, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/improved_cnn.keras
26/26          5s 105ms/step -
loss: 0.0019 - mae: 0.0329 - val_loss: 0.0012 - val_mae: 0.0249
Epoch 96/1000
25/26          0s 98ms/step -
loss: 0.0017 - mae: 0.0301
Epoch 96: val_loss did not improve from 0.00124
26/26          5s 103ms/step -
loss: 0.0017 - mae: 0.0301 - val_loss: 0.0015 - val_mae: 0.0299
Epoch 97/1000
25/26          0s 97ms/step -
loss: 0.0020 - mae: 0.0326
Epoch 97: val_loss did not improve from 0.00124
26/26          3s 100ms/step -
loss: 0.0020 - mae: 0.0327 - val_loss: 0.0014 - val_mae: 0.0285
Epoch 98/1000
25/26          0s 98ms/step -
loss: 0.0017 - mae: 0.0306
Epoch 98: val_loss did not improve from 0.00124
26/26          3s 101ms/step -
loss: 0.0017 - mae: 0.0305 - val_loss: 0.0024 - val_mae: 0.0379
Epoch 99/1000
25/26          0s 98ms/step -
loss: 0.0018 - mae: 0.0306
Epoch 99: val_loss did not improve from 0.00124
26/26          3s 103ms/step -
loss: 0.0018 - mae: 0.0307 - val_loss: 0.0033 - val_mae: 0.0459
Epoch 100/1000
25/26          0s 98ms/step -
loss: 0.0028 - mae: 0.0407
Epoch 100: val_loss did not improve from 0.00124
26/26          3s 101ms/step -
loss: 0.0028 - mae: 0.0403 - val_loss: 0.0016 - val_mae: 0.0291
Epoch 101/1000
25/26          0s 98ms/step -
loss: 0.0019 - mae: 0.0323

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Epoch 101: val_loss did not improve from 0.00124
26/26 5s 101ms/step -
loss: 0.0019 - mae: 0.0323 - val_loss: 0.0040 - val_mae: 0.0472
Epoch 102/1000
25/26 0s 96ms/step -
loss: 0.0024 - mae: 0.0375
Epoch 102: val_loss did not improve from 0.00124
26/26 3s 100ms/step -
loss: 0.0024 - mae: 0.0373 - val_loss: 0.0018 - val_mae: 0.0343
Epoch 103/1000
25/26 0s 96ms/step -
loss: 0.0020 - mae: 0.0334
Epoch 103: val_loss did not improve from 0.00124
26/26 3s 102ms/step -
loss: 0.0020 - mae: 0.0334 - val_loss: 0.0014 - val_mae: 0.0271
Epoch 104/1000
25/26 0s 96ms/step -
loss: 0.0020 - mae: 0.0330
Epoch 104: val_loss did not improve from 0.00124
26/26 5s 99ms/step -
loss: 0.0020 - mae: 0.0332 - val_loss: 0.0038 - val_mae: 0.0462
Epoch 105/1000
25/26 0s 96ms/step -
loss: 0.0029 - mae: 0.0419
Epoch 105: val_loss did not improve from 0.00124
26/26 3s 100ms/step -
loss: 0.0029 - mae: 0.0416 - val_loss: 0.0047 - val_mae: 0.0496
Epoch 106/1000
25/26 0s 97ms/step -
loss: 0.0018 - mae: 0.0316
Epoch 106: val_loss did not improve from 0.00124
26/26 5s 103ms/step -
loss: 0.0018 - mae: 0.0316 - val_loss: 0.0018 - val_mae: 0.0306
Epoch 107/1000
25/26 0s 96ms/step -
loss: 0.0019 - mae: 0.0319
Epoch 107: val_loss did not improve from 0.00124
26/26 3s 99ms/step -
loss: 0.0019 - mae: 0.0320 - val_loss: 0.0014 - val_mae: 0.0276
Epoch 108/1000
25/26 0s 95ms/step -
loss: 0.0016 - mae: 0.0289
Epoch 108: val_loss did not improve from 0.00124
26/26 5s 98ms/step -
loss: 0.0016 - mae: 0.0292 - val_loss: 0.0024 - val_mae: 0.0366
Epoch 109/1000
25/26 0s 95ms/step -
loss: 0.0020 - mae: 0.0334

Epoch 109: val_loss did not improve from 0.00124
26/26 3s 101ms/step -
loss: 0.0020 - mae: 0.0333 - val_loss: 0.0017 - val_mae: 0.0310
Epoch 110/1000
25/26 0s 96ms/step -
loss: 0.0018 - mae: 0.0321
Epoch 110: val_loss improved from 0.00124 to 0.00121, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/improved_cnn.keras
26/26 5s 107ms/step -
loss: 0.0018 - mae: 0.0321 - val_loss: 0.0012 - val_mae: 0.0247
Epoch 111/1000
25/26 0s 96ms/step -
loss: 0.0019 - mae: 0.0316
Epoch 111: val_loss did not improve from 0.00121
26/26 5s 100ms/step -
loss: 0.0019 - mae: 0.0315 - val_loss: 0.0013 - val_mae: 0.0252
Epoch 112/1000
26/26 0s 94ms/step -
loss: 0.0020 - mae: 0.0338
Epoch 112: val_loss did not improve from 0.00121
26/26 5s 103ms/step -
loss: 0.0020 - mae: 0.0339 - val_loss: 0.0020 - val_mae: 0.0321
Epoch 113/1000
25/26 0s 97ms/step -
loss: 0.0020 - mae: 0.0332
Epoch 113: val_loss did not improve from 0.00121
26/26 5s 100ms/step -
loss: 0.0020 - mae: 0.0333 - val_loss: 0.0027 - val_mae: 0.0438
Epoch 114/1000
25/26 0s 97ms/step -
loss: 0.0017 - mae: 0.0304
Epoch 114: val_loss did not improve from 0.00121
26/26 3s 100ms/step -
loss: 0.0017 - mae: 0.0304 - val_loss: 0.0025 - val_mae: 0.0411
Epoch 115/1000
25/26 0s 98ms/step -
loss: 0.0017 - mae: 0.0303
Epoch 115: val_loss did not improve from 0.00121
26/26 5s 104ms/step -
loss: 0.0017 - mae: 0.0302 - val_loss: 0.0013 - val_mae: 0.0272
Epoch 116/1000
25/26 0s 98ms/step -
loss: 0.0015 - mae: 0.0278
Epoch 116: val_loss did not improve from 0.00121
26/26 5s 101ms/step -
loss: 0.0015 - mae: 0.0279 - val_loss: 0.0014 - val_mae: 0.0285
Epoch 117/1000

```

25/26          0s 98ms/step -
loss: 0.0018 - mae: 0.0319
Epoch 117: val_loss did not improve from 0.00121
26/26          3s 101ms/step -
loss: 0.0018 - mae: 0.0317 - val_loss: 0.0012 - val_mae: 0.0252
Epoch 118/1000
25/26          0s 99ms/step -
loss: 0.0015 - mae: 0.0277
Epoch 118: val_loss did not improve from 0.00121
26/26          5s 105ms/step -
loss: 0.0015 - mae: 0.0279 - val_loss: 0.0020 - val_mae: 0.0356
Epoch 119/1000
25/26          0s 97ms/step -
loss: 0.0020 - mae: 0.0336
Epoch 119: val_loss did not improve from 0.00121
26/26          5s 100ms/step -
loss: 0.0020 - mae: 0.0335 - val_loss: 0.0019 - val_mae: 0.0332
Epoch 120/1000
25/26          0s 97ms/step -
loss: 0.0019 - mae: 0.0317
Epoch 120: val_loss did not improve from 0.00121
26/26          3s 100ms/step -
loss: 0.0018 - mae: 0.0316 - val_loss: 0.0016 - val_mae: 0.0310
Epoch 121/1000
25/26          0s 98ms/step -
loss: 0.0018 - mae: 0.0313
Epoch 121: val_loss did not improve from 0.00121
26/26          5s 102ms/step -
loss: 0.0018 - mae: 0.0312 - val_loss: 0.0023 - val_mae: 0.0389
Epoch 122/1000
25/26          0s 97ms/step -
loss: 0.0019 - mae: 0.0331
Epoch 122: val_loss did not improve from 0.00121
26/26          5s 100ms/step -
loss: 0.0019 - mae: 0.0328 - val_loss: 0.0013 - val_mae: 0.0263
Epoch 123/1000
25/26          0s 96ms/step -
loss: 0.0018 - mae: 0.0313
Epoch 123: val_loss did not improve from 0.00121
26/26          5s 99ms/step -
loss: 0.0018 - mae: 0.0313 - val_loss: 0.0014 - val_mae: 0.0286
Epoch 124/1000
25/26          0s 96ms/step -
loss: 0.0015 - mae: 0.0282
Epoch 124: val_loss did not improve from 0.00121
26/26          5s 99ms/step -
loss: 0.0015 - mae: 0.0282 - val_loss: 0.0021 - val_mae: 0.0367
Epoch 125/1000

```

```

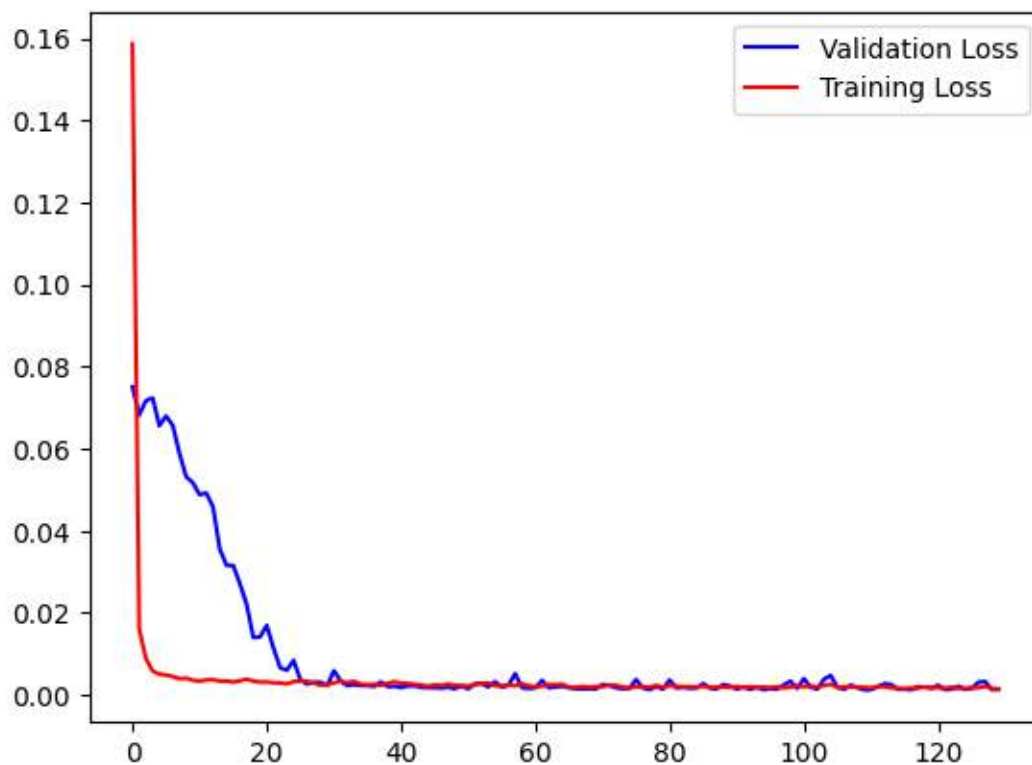
25/26          0s 96ms/step -
loss: 0.0016 - mae: 0.0298
Epoch 125: val_loss did not improve from 0.00121
26/26          3s 101ms/step -
loss: 0.0016 - mae: 0.0297 - val_loss: 0.0013 - val_mae: 0.0274
Epoch 126/1000
25/26          0s 96ms/step -
loss: 0.0017 - mae: 0.0311
Epoch 126: val_loss did not improve from 0.00121
26/26          5s 102ms/step -
loss: 0.0017 - mae: 0.0311 - val_loss: 0.0016 - val_mae: 0.0294
Epoch 127/1000
25/26          0s 95ms/step -
loss: 0.0018 - mae: 0.0309
Epoch 127: val_loss did not improve from 0.00121
26/26          5s 98ms/step -
loss: 0.0018 - mae: 0.0309 - val_loss: 0.0031 - val_mae: 0.0441
Epoch 128/1000
25/26          0s 96ms/step -
loss: 0.0021 - mae: 0.0351
Epoch 128: val_loss did not improve from 0.00121
26/26          5s 99ms/step -
loss: 0.0021 - mae: 0.0350 - val_loss: 0.0032 - val_mae: 0.0447
Epoch 129/1000
25/26          0s 96ms/step -
loss: 0.0015 - mae: 0.0280
Epoch 129: val_loss did not improve from 0.00121
26/26          5s 101ms/step -
loss: 0.0015 - mae: 0.0281 - val_loss: 0.0013 - val_mae: 0.0257
Epoch 130/1000
25/26          0s 95ms/step -
loss: 0.0014 - mae: 0.0263
Epoch 130: val_loss did not improve from 0.00121
26/26          5s 99ms/step -
loss: 0.0014 - mae: 0.0263 - val_loss: 0.0013 - val_mae: 0.0261

```

```

[ ]: # Plotting the training and validation loss
plt.plot(history_improved_cnn.history['val_loss'],c="b")
plt.plot(history_improved_cnn.history['loss'],c="r")
plt.legend(['Validation Loss','Training Loss'])
plt.show()

```



Depiction of Model Performance on images from the Training Set

```
[ ]: show_predictions(best_improved_cnn, X, Y, num_samples=5)
```

1/1

1s 608ms/step



Depiction of Model Performance on images from the Test Set

```
[ ]: show_predictions(best_improved_cnn, X_test, Y_test, num_samples=5)
```

1/1 0s 55ms/step



The model performed slightly better on the MSBA headshots, likely due to the structured and uniform composition of the photos. While it was able to estimate general skin tones and background colors more accurately, the results still lacked detail and vibrancy, appearing overly smoothed and blurred.

6.5 Model 4: U-Net Implementation

```
[ ]: # Defining the architecture for the U-Net model
def build_unet_model(input_shape=(120, 176, 1)):
    inputs = layers.Input(shape=input_shape)

    # Encoder
    conv1 = layers.Conv2D(64, (3, 3), padding='same')(inputs)
    conv1 = layers.BatchNormalization()(conv1)
    conv1 = layers.ReLU()(conv1)
    conv1 = layers.Conv2D(64, (3, 3), padding='same')(conv1)
    conv1 = layers.BatchNormalization()(conv1)
    conv1 = layers.ReLU()(conv1)
    pool1 = layers.MaxPooling2D((2, 2))(conv1)

    conv2 = layers.Conv2D(128, (3, 3), padding='same')(pool1)
    conv2 = layers.BatchNormalization()(conv2)
    conv2 = layers.ReLU()(conv2)
    conv2 = layers.Conv2D(128, (3, 3), padding='same')(conv2)
    conv2 = layers.BatchNormalization()(conv2)
    conv2 = layers.ReLU()(conv2)
    pool2 = layers.MaxPooling2D((2, 2))(conv2)

    # Bottleneck
    bottleneck = layers.Conv2D(256, (3, 3), padding='same')(pool2)
    bottleneck = layers.BatchNormalization()(bottleneck)
    bottleneck = layers.ReLU()(bottleneck)

    # Decoder
    up2 = layers.Conv2DTranspose(128, (2, 2), strides=2, padding='same')(bottleneck)
    up2 = layers.Concatenate()([up2, conv2])
    up2 = layers.Conv2D(128, (3, 3), padding='same')(up2)
    up2 = layers.BatchNormalization()(up2)
    up2 = layers.ReLU()(up2)

    up1 = layers.Conv2DTranspose(64, (2, 2), strides=2, padding='same')(up2)
    up1 = layers.Concatenate()([up1, conv1])
    up1 = layers.Conv2D(64, (3, 3), padding='same')(up1)
    up1 = layers.BatchNormalization()(up1)
```



```

up1 = layers.ReLU()(up1)

outputs = layers.Conv2D(3, (3, 3), activation='sigmoid',
↳padding='same')(up1)

model = models.Model(inputs=inputs, outputs=outputs)
model.compile(optimizer='adam', loss='mse', metrics=['mae'])
return model

UNET_MODEL = build_UNET_MODEL()

```

```

[ ]: # Defining the path to save the best model
model_checkpoint_path = os.path.join(best_models_path, 'UNET_MODEL.keras')
model_checkpoint = ModelCheckpoint(model_checkpoint_path, monitor='val_loss',
↳verbose=1, save_best_only=True, mode='min')

# fitting the model to the training data
history_UNET_MODEL = UNET_MODEL.fit(X, Y, batch_size=16, epochs=200,
↳validation_split=0.1, callbacks=[early_stop, model_checkpoint])

best_UNET_MODEL = keras.models.load_model(os.path.join(best_models_path,
↳'UNET_MODEL.keras'))

```

```

Epoch 1/200
26/26          0s 656ms/step -
loss: 0.0307 - mae: 0.1103
Epoch 1: val_loss improved from inf to 0.07734, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/UNET_MODEL.keras
26/26          41s 1s/step - loss:
0.0300 - mae: 0.1086 - val_loss: 0.0773 - val_mae: 0.2498
Epoch 2/200
26/26          0s 144ms/step -
loss: 0.0031 - mae: 0.0407
Epoch 2: val_loss improved from 0.07734 to 0.07580, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/UNET_MODEL.keras
26/26          10s 159ms/step -
loss: 0.0031 - mae: 0.0407 - val_loss: 0.0758 - val_mae: 0.2488
Epoch 3/200
26/26          0s 144ms/step -
loss: 0.0023 - mae: 0.0354
Epoch 3: val_loss improved from 0.07580 to 0.07336, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/UNET_MODEL.keras
26/26          4s 168ms/step -
loss: 0.0023 - mae: 0.0353 - val_loss: 0.0734 - val_mae: 0.2453

```

Epoch 4/200
26/26 0s 147ms/step -
loss: 0.0017 - mae: 0.0297
Epoch 4: val_loss improved from 0.07336 to 0.06890, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/unet_model.keras
26/26 4s 163ms/step -
loss: 0.0017 - mae: 0.0297 - val_loss: 0.0689 - val_mae: 0.2385
Epoch 5/200
26/26 0s 150ms/step -
loss: 0.0018 - mae: 0.0310
Epoch 5: val_loss improved from 0.06890 to 0.06733, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/unet_model.keras
26/26 4s 165ms/step -
loss: 0.0018 - mae: 0.0309 - val_loss: 0.0673 - val_mae: 0.2367
Epoch 6/200
26/26 0s 151ms/step -
loss: 0.0014 - mae: 0.0267
Epoch 6: val_loss improved from 0.06733 to 0.06430, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/unet_model.keras
26/26 5s 176ms/step -
loss: 0.0014 - mae: 0.0267 - val_loss: 0.0643 - val_mae: 0.2314
Epoch 7/200
26/26 0s 150ms/step -
loss: 0.0015 - mae: 0.0280
Epoch 7: val_loss did not improve from 0.06430
26/26 4s 159ms/step -
loss: 0.0015 - mae: 0.0280 - val_loss: 0.0644 - val_mae: 0.2318
Epoch 8/200
26/26 0s 151ms/step -
loss: 0.0016 - mae: 0.0297
Epoch 8: val_loss improved from 0.06430 to 0.05954, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/unet_model.keras
26/26 5s 178ms/step -
loss: 0.0016 - mae: 0.0296 - val_loss: 0.0595 - val_mae: 0.2224
Epoch 9/200
26/26 0s 151ms/step -
loss: 0.0014 - mae: 0.0270
Epoch 9: val_loss improved from 0.05954 to 0.05717, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/unet_model.keras
26/26 5s 176ms/step -
loss: 0.0014 - mae: 0.0270 - val_loss: 0.0572 - val_mae: 0.2179
Epoch 10/200
26/26 0s 147ms/step -

```

loss: 0.0015 - mae: 0.0283
Epoch 10: val_loss improved from 0.05717 to 0.05200, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/unet_model.keras
26/26          4s 163ms/step -
loss: 0.0015 - mae: 0.0282 - val_loss: 0.0520 - val_mae: 0.2076
Epoch 11/200
26/26          0s 146ms/step -
loss: 0.0014 - mae: 0.0277
Epoch 11: val_loss improved from 0.05200 to 0.04937, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/unet_model.keras
26/26          4s 167ms/step -
loss: 0.0014 - mae: 0.0277 - val_loss: 0.0494 - val_mae: 0.2019
Epoch 12/200
26/26          0s 143ms/step -
loss: 0.0013 - mae: 0.0266
Epoch 12: val_loss improved from 0.04937 to 0.04606, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/unet_model.keras
26/26          5s 166ms/step -
loss: 0.0013 - mae: 0.0266 - val_loss: 0.0461 - val_mae: 0.1951
Epoch 13/200
26/26          0s 142ms/step -
loss: 0.0012 - mae: 0.0254
Epoch 13: val_loss improved from 0.04606 to 0.04078, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/unet_model.keras
26/26          4s 158ms/step -
loss: 0.0012 - mae: 0.0254 - val_loss: 0.0408 - val_mae: 0.1840
Epoch 14/200
26/26          0s 142ms/step -
loss: 0.0015 - mae: 0.0283
Epoch 14: val_loss improved from 0.04078 to 0.03688, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/unet_model.keras
26/26          4s 158ms/step -
loss: 0.0015 - mae: 0.0282 - val_loss: 0.0369 - val_mae: 0.1756
Epoch 15/200
26/26          0s 146ms/step -
loss: 0.0011 - mae: 0.0236
Epoch 15: val_loss improved from 0.03688 to 0.03320, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/unet_model.keras
26/26          5s 171ms/step -
loss: 0.0011 - mae: 0.0236 - val_loss: 0.0332 - val_mae: 0.1667
Epoch 16/200
26/26          0s 142ms/step -

```

```

loss: 0.0012 - mae: 0.0252
Epoch 16: val_loss improved from 0.03320 to 0.02902, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/unet_model.keras
26/26          4s 158ms/step -
loss: 0.0012 - mae: 0.0253 - val_loss: 0.0290 - val_mae: 0.1545
Epoch 17/200
26/26          0s 143ms/step -
loss: 0.0012 - mae: 0.0257
Epoch 17: val_loss improved from 0.02902 to 0.02050, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/unet_model.keras
26/26          4s 158ms/step -
loss: 0.0012 - mae: 0.0257 - val_loss: 0.0205 - val_mae: 0.1295
Epoch 18/200
26/26          0s 143ms/step -
loss: 0.0012 - mae: 0.0255
Epoch 18: val_loss improved from 0.02050 to 0.01998, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/unet_model.keras
26/26          4s 166ms/step -
loss: 0.0012 - mae: 0.0255 - val_loss: 0.0200 - val_mae: 0.1279
Epoch 19/200
26/26          0s 142ms/step -
loss: 0.0014 - mae: 0.0286
Epoch 19: val_loss improved from 0.01998 to 0.01275, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/unet_model.keras
26/26          4s 157ms/step -
loss: 0.0014 - mae: 0.0286 - val_loss: 0.0127 - val_mae: 0.1022
Epoch 20/200
26/26          0s 143ms/step -
loss: 0.0011 - mae: 0.0243
Epoch 20: val_loss did not improve from 0.01275
26/26          5s 151ms/step -
loss: 0.0011 - mae: 0.0242 - val_loss: 0.0128 - val_mae: 0.1018
Epoch 21/200
26/26          0s 144ms/step -
loss: 0.0011 - mae: 0.0237
Epoch 21: val_loss improved from 0.01275 to 0.00687, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/unet_model.keras
26/26          5s 160ms/step -
loss: 0.0011 - mae: 0.0239 - val_loss: 0.0069 - val_mae: 0.0746
Epoch 22/200
26/26          0s 145ms/step -
loss: 0.0013 - mae: 0.0267
Epoch 22: val_loss improved from 0.00687 to 0.00568, saving model to

```

```

/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/unet_model.keras
26/26          4s 161ms/step -
loss: 0.0013 - mae: 0.0267 - val_loss: 0.0057 - val_mae: 0.0664
Epoch 23/200
26/26          0s 145ms/step -
loss: 0.0011 - mae: 0.0239
Epoch 23: val_loss improved from 0.00568 to 0.00545, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/unet_model.keras
26/26          5s 168ms/step -
loss: 0.0011 - mae: 0.0239 - val_loss: 0.0054 - val_mae: 0.0654
Epoch 24/200
26/26          0s 147ms/step -
loss: 0.0011 - mae: 0.0239
Epoch 24: val_loss improved from 0.00545 to 0.00330, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/unet_model.keras
26/26          4s 163ms/step -
loss: 0.0011 - mae: 0.0239 - val_loss: 0.0033 - val_mae: 0.0474
Epoch 25/200
26/26          0s 148ms/step -
loss: 0.0011 - mae: 0.0236
Epoch 25: val_loss improved from 0.00330 to 0.00234, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/unet_model.keras
26/26          5s 163ms/step -
loss: 0.0011 - mae: 0.0236 - val_loss: 0.0023 - val_mae: 0.0425
Epoch 26/200
26/26          0s 149ms/step -
loss: 9.9758e-04 - mae: 0.0226
Epoch 26: val_loss did not improve from 0.00234
26/26          6s 180ms/step -
loss: 9.9718e-04 - mae: 0.0226 - val_loss: 0.0025 - val_mae: 0.0422
Epoch 27/200
26/26          0s 147ms/step -
loss: 9.0181e-04 - mae: 0.0216
Epoch 27: val_loss improved from 0.00234 to 0.00208, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/unet_model.keras
26/26          4s 162ms/step -
loss: 9.0370e-04 - mae: 0.0217 - val_loss: 0.0021 - val_mae: 0.0373
Epoch 28/200
26/26          0s 145ms/step -
loss: 8.9946e-04 - mae: 0.0215
Epoch 28: val_loss improved from 0.00208 to 0.00109, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/unet_model.keras

```

26/26 4s 166ms/step -
 loss: 9.0091e-04 - mae: 0.0215 - val_loss: 0.0011 - val_mae: 0.0264
 Epoch 29/200
 26/26 0s 146ms/step -
 loss: 0.0011 - mae: 0.0243
 Epoch 29: val_loss improved from 0.00109 to 0.00094, saving model to
 /content/drive/MyDrive/BA865/BA865 Group Project/Best Models
 MSBA/unet_model.keras
 26/26 4s 165ms/step -
 loss: 0.0011 - mae: 0.0243 - val_loss: 9.4367e-04 - val_mae: 0.0225
 Epoch 30/200
 26/26 0s 144ms/step -
 loss: 9.3477e-04 - mae: 0.0220
 Epoch 30: val_loss improved from 0.00094 to 0.00078, saving model to
 /content/drive/MyDrive/BA865/BA865 Group Project/Best Models
 MSBA/unet_model.keras
 26/26 5s 160ms/step -
 loss: 9.3608e-04 - mae: 0.0220 - val_loss: 7.8146e-04 - val_mae: 0.0196
 Epoch 31/200
 26/26 0s 145ms/step -
 loss: 0.0014 - mae: 0.0288
 Epoch 31: val_loss did not improve from 0.00078
 26/26 4s 153ms/step -
 loss: 0.0014 - mae: 0.0287 - val_loss: 0.0023 - val_mae: 0.0344
 Epoch 32/200
 26/26 0s 144ms/step -
 loss: 0.0012 - mae: 0.0251
 Epoch 32: val_loss did not improve from 0.00078
 26/26 4s 158ms/step -
 loss: 0.0012 - mae: 0.0251 - val_loss: 0.0010 - val_mae: 0.0248
 Epoch 33/200
 26/26 0s 143ms/step -
 loss: 9.6828e-04 - mae: 0.0227
 Epoch 33: val_loss did not improve from 0.00078
 26/26 5s 151ms/step -
 loss: 9.6914e-04 - mae: 0.0227 - val_loss: 0.0013 - val_mae: 0.0273
 Epoch 34/200
 26/26 0s 143ms/step -
 loss: 9.3103e-04 - mae: 0.0221
 Epoch 34: val_loss did not improve from 0.00078
 26/26 4s 151ms/step -
 loss: 9.3060e-04 - mae: 0.0221 - val_loss: 8.6018e-04 - val_mae: 0.0219
 Epoch 35/200
 26/26 0s 143ms/step -
 loss: 8.8111e-04 - mae: 0.0211
 Epoch 35: val_loss did not improve from 0.00078
 26/26 5s 151ms/step -
 loss: 8.8411e-04 - mae: 0.0212 - val_loss: 8.5022e-04 - val_mae: 0.0214

Epoch 36/200
26/26 0s 143ms/step -
loss: 9.0272e-04 - mae: 0.0223
Epoch 36: val_loss did not improve from 0.00078
26/26 4s 151ms/step -
loss: 9.0381e-04 - mae: 0.0223 - val_loss: 9.5463e-04 - val_mae: 0.0231
Epoch 37/200
26/26 0s 143ms/step -
loss: 7.9763e-04 - mae: 0.0199
Epoch 37: val_loss did not improve from 0.00078
26/26 5s 151ms/step -
loss: 7.9915e-04 - mae: 0.0199 - val_loss: 9.4939e-04 - val_mae: 0.0234
Epoch 38/200
26/26 0s 143ms/step -
loss: 8.3012e-04 - mae: 0.0205
Epoch 38: val_loss did not improve from 0.00078
26/26 5s 152ms/step -
loss: 8.3213e-04 - mae: 0.0206 - val_loss: 7.8563e-04 - val_mae: 0.0206
Epoch 39/200
26/26 0s 144ms/step -
loss: 8.6940e-04 - mae: 0.0213
Epoch 39: val_loss did not improve from 0.00078
26/26 5s 158ms/step -
loss: 8.6965e-04 - mae: 0.0213 - val_loss: 0.0013 - val_mae: 0.0257
Epoch 40/200
26/26 0s 145ms/step -
loss: 7.9062e-04 - mae: 0.0201
Epoch 40: val_loss did not improve from 0.00078
26/26 4s 154ms/step -
loss: 7.9087e-04 - mae: 0.0201 - val_loss: 0.0011 - val_mae: 0.0243
Epoch 41/200
26/26 0s 146ms/step -
loss: 9.5943e-04 - mae: 0.0224
Epoch 41: val_loss did not improve from 0.00078
26/26 4s 154ms/step -
loss: 9.6502e-04 - mae: 0.0225 - val_loss: 0.0016 - val_mae: 0.0311
Epoch 42/200
26/26 0s 146ms/step -
loss: 9.5943e-04 - mae: 0.0225
Epoch 42: val_loss did not improve from 0.00078
26/26 5s 158ms/step -
loss: 9.5886e-04 - mae: 0.0225 - val_loss: 0.0010 - val_mae: 0.0232
Epoch 43/200
26/26 0s 146ms/step -
loss: 0.0012 - mae: 0.0261
Epoch 43: val_loss did not improve from 0.00078
26/26 4s 155ms/step -
loss: 0.0012 - mae: 0.0261 - val_loss: 0.0023 - val_mae: 0.0342

```

Epoch 44/200
26/26          0s 150ms/step -
loss: 9.6591e-04 - mae: 0.0229
Epoch 44: val_loss did not improve from 0.00078
26/26          4s 158ms/step -
loss: 9.6266e-04 - mae: 0.0228 - val_loss: 0.0011 - val_mae: 0.0244
Epoch 45/200
26/26          0s 145ms/step -
loss: 8.3200e-04 - mae: 0.0207
Epoch 45: val_loss did not improve from 0.00078
26/26          4s 153ms/step -
loss: 8.3175e-04 - mae: 0.0207 - val_loss: 0.0013 - val_mae: 0.0278
Epoch 46/200
26/26          0s 143ms/step -
loss: 8.0295e-04 - mae: 0.0200
Epoch 46: val_loss did not improve from 0.00078
26/26          5s 158ms/step -
loss: 8.0176e-04 - mae: 0.0200 - val_loss: 8.3613e-04 - val_mae: 0.0212
Epoch 47/200
26/26          0s 145ms/step -
loss: 0.0010 - mae: 0.0234
Epoch 47: val_loss did not improve from 0.00078
26/26          5s 160ms/step -
loss: 0.0010 - mae: 0.0235 - val_loss: 0.0013 - val_mae: 0.0269
Epoch 48/200
26/26          0s 143ms/step -
loss: 7.3110e-04 - mae: 0.0192
Epoch 48: val_loss did not improve from 0.00078
26/26          4s 157ms/step -
loss: 7.3106e-04 - mae: 0.0192 - val_loss: 9.6743e-04 - val_mae: 0.0242
Epoch 49/200
26/26          0s 145ms/step -
loss: 8.0010e-04 - mae: 0.0205
Epoch 49: val_loss did not improve from 0.00078
26/26          4s 159ms/step -
loss: 7.9894e-04 - mae: 0.0205 - val_loss: 0.0012 - val_mae: 0.0257
Epoch 50/200
26/26          0s 146ms/step -
loss: 7.1493e-04 - mae: 0.0187
Epoch 50: val_loss did not improve from 0.00078
26/26          5s 154ms/step -
loss: 7.1591e-04 - mae: 0.0188 - val_loss: 9.9279e-04 - val_mae: 0.0237

```

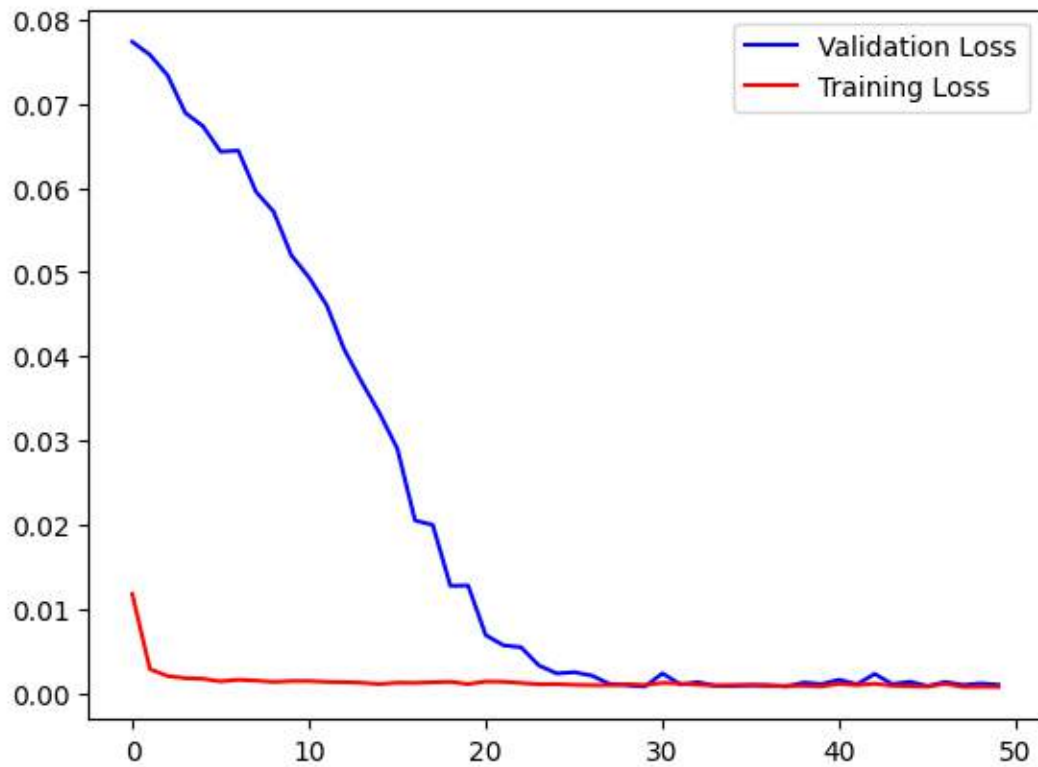
```

[ ]: # Plotting the training and validation loss
plt.plot(history_unet_model.history['val_loss'],c="b")
plt.plot(history_unet_model.history['loss'],c="r")
plt.legend(['Validation Loss','Training Loss'])

```



```
plt.show()
```



Depiction of Model Performance on images from the Training Set

```
[ ]: show_predictions(best_unet_model, X, Y, num_samples=5)
```

1/1

1s 606ms/step



Depiction of Model Performance on images from the Test Set

```
[ ]: show_predictions(best_unet_model, X_test, Y_test, num_samples=5)
```

1/1 0s 74ms/step



Although the enhanced model shows improvement in structural details, its color predictions remain limited compared to the simple CNN model. While the overall shape and shading are accurate, the colors appear overly warm and unrealistic as if a color filter has been applied. This suggests that the model has learned the general tone of color but still struggles with where and what color to apply. It likely relies on average color patterns from the training data, resulting in tinted outputs.

6.6 Model 5: Sequential LAB Model

```
[ ]: # Defining the architecture for the sequential LAB model
# Encoder
seq_lab_model = Sequential()
seq_lab_model.add(Conv2D(64, (3, 3), activation='relu', padding='same',
    ↳strides=2, input_shape=(120, 176, 1)))
seq_lab_model.add(Conv2D(128, (3, 3), activation='relu', padding='same'))
seq_lab_model.add(Conv2D(128, (3,3), activation='relu', padding='same',
    ↳strides=2))
seq_lab_model.add(Conv2D(256, (3,3), activation='relu', padding='same'))
seq_lab_model.add(Conv2D(512, (3,3), activation='relu', padding='same'))
seq_lab_model.add(Conv2D(512, (3,3), activation='relu', padding='same'))
seq_lab_model.add(Conv2D(256, (3,3), activation='relu', padding='same',
    ↳strides=2))
seq_lab_model.add(Conv2D(512, (3,3), activation='relu', padding='same'))
seq_lab_model.add(Conv2D(512, (3,3), activation='relu', padding='same'))
seq_lab_model.add(Conv2D(256, (3,3), activation='relu', padding='same'))
# Decoder
seq_lab_model.add(Conv2D(128, (3,3), activation='relu', padding='same'))
seq_lab_model.add(UpSampling2D((2, 2)))
seq_lab_model.add(Conv2D(64, (3,3), activation='relu', padding='same'))
seq_lab_model.add(UpSampling2D((2, 2)))
seq_lab_model.add(Conv2D(32, (3,3), activation='relu', padding='same'))
seq_lab_model.add(Conv2D(16, (3,3), activation='relu', padding='same'))
seq_lab_model.add(Conv2D(2, (3, 3), activation='tanh', padding='same'))
seq_lab_model.add(UpSampling2D((2, 2)))
seq_lab_model.compile(optimizer=Adam(learning_rate=1e-4), loss='mae',
    ↳metrics=['mse'])
seq_lab_model.summary()
```

```
/usr/local/lib/python3.11/dist-
packages/keras/src/layers/convolutional/base_conv.py:107: UserWarning: Do not
pass an `input_shape`/`input_dim` argument to a layer. When using Sequential
models, prefer using an `Input(shape)` object as the first layer in the model
instead.
```

```
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

```
Model: "sequential_1"
```

Layer (type)	Output Shape	Param #
conv2d_68 (Conv2D)	(None, 60, 88, 64)	640
conv2d_69 (Conv2D)	(None, 60, 88, 128)	73,856
conv2d_70 (Conv2D)	(None, 30, 44, 128)	147,584
conv2d_71 (Conv2D)	(None, 30, 44, 256)	295,168
conv2d_72 (Conv2D)	(None, 30, 44, 512)	1,180,160
conv2d_73 (Conv2D)	(None, 30, 44, 512)	2,359,808
conv2d_74 (Conv2D)	(None, 15, 22, 256)	1,179,904
conv2d_75 (Conv2D)	(None, 15, 22, 512)	1,180,160
conv2d_76 (Conv2D)	(None, 15, 22, 512)	2,359,808
conv2d_77 (Conv2D)	(None, 15, 22, 256)	1,179,904
conv2d_78 (Conv2D)	(None, 15, 22, 128)	295,040
up_sampling2d_14 (UpSampling2D)	(None, 30, 44, 128)	0
conv2d_79 (Conv2D)	(None, 30, 44, 64)	73,792
up_sampling2d_15 (UpSampling2D)	(None, 60, 88, 64)	0
conv2d_80 (Conv2D)	(None, 60, 88, 32)	18,464
conv2d_81 (Conv2D)	(None, 60, 88, 16)	4,624
conv2d_82 (Conv2D)	(None, 60, 88, 2)	290
up_sampling2d_16 (UpSampling2D)	(None, 120, 176, 2)	0

Total params: 10,349,202 (39.48 MB)

Trainable params: 10,349,202 (39.48 MB)

Non-trainable params: 0 (0.00 B)

```
[ ]: # Defining the path to save the best model
model_checkpoint_path = os.path.join(best_models_path, 'seq_lab_model.keras')
model_checkpoint = ModelCheckpoint(model_checkpoint_path, monitor='val_loss',
    verbose=1, save_best_only=True, mode='min')

# Fitting the model to the training data
history_seq_lab_model = seq_lab_model.fit(X_lab,Y_lab,batch_size=16,
    epochs=200, validation_split=0.1, callbacks=[early_stop, model_checkpoint])

best_seq_lab_model = keras.models.load_model(os.path.join(best_models_path,
    'seq_lab_model.keras'))
```

```
Epoch 1/200
26/26          0s 696ms/step -
loss: 0.0659 - mse: 0.0097
Epoch 1: val_loss improved from inf to 0.03299, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/seq_lab_model.keras
26/26          39s 1s/step - loss:
0.0652 - mse: 0.0095 - val_loss: 0.0330 - val_mse: 0.0025
Epoch 2/200
26/26          0s 150ms/step -
loss: 0.0337 - mse: 0.0025
Epoch 2: val_loss did not improve from 0.03299
26/26          4s 163ms/step -
loss: 0.0337 - mse: 0.0025 - val_loss: 0.0353 - val_mse: 0.0026
Epoch 3/200
26/26          0s 152ms/step -
loss: 0.0331 - mse: 0.0024
Epoch 3: val_loss improved from 0.03299 to 0.03038, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/seq_lab_model.keras
26/26          5s 199ms/step -
loss: 0.0331 - mse: 0.0024 - val_loss: 0.0304 - val_mse: 0.0021
Epoch 4/200
26/26          0s 148ms/step -
loss: 0.0317 - mse: 0.0022
Epoch 4: val_loss improved from 0.03038 to 0.03018, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/seq_lab_model.keras
26/26         10s 189ms/step -
loss: 0.0317 - mse: 0.0022 - val_loss: 0.0302 - val_mse: 0.0020
Epoch 5/200
26/26          0s 150ms/step -
loss: 0.0300 - mse: 0.0020
Epoch 5: val_loss improved from 0.03018 to 0.02847, saving model to
```

```

/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/seq_lab_model.keras
26/26          5s 190ms/step -
loss: 0.0300 - mse: 0.0020 - val_loss: 0.0285 - val_mse: 0.0018
Epoch 6/200
26/26          0s 149ms/step -
loss: 0.0296 - mse: 0.0019
Epoch 6: val_loss did not improve from 0.02847
26/26          4s 163ms/step -
loss: 0.0296 - mse: 0.0019 - val_loss: 0.0299 - val_mse: 0.0020
Epoch 7/200
26/26          0s 151ms/step -
loss: 0.0293 - mse: 0.0019
Epoch 7: val_loss improved from 0.02847 to 0.02672, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/seq_lab_model.keras
26/26          7s 260ms/step -
loss: 0.0293 - mse: 0.0019 - val_loss: 0.0267 - val_mse: 0.0015
Epoch 8/200
26/26          0s 149ms/step -
loss: 0.0276 - mse: 0.0017
Epoch 8: val_loss improved from 0.02672 to 0.02586, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/seq_lab_model.keras
26/26          9s 190ms/step -
loss: 0.0276 - mse: 0.0017 - val_loss: 0.0259 - val_mse: 0.0015
Epoch 9/200
26/26          0s 151ms/step -
loss: 0.0263 - mse: 0.0016
Epoch 9: val_loss improved from 0.02586 to 0.02570, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/seq_lab_model.keras
26/26          5s 195ms/step -
loss: 0.0263 - mse: 0.0016 - val_loss: 0.0257 - val_mse: 0.0015
Epoch 10/200
26/26          0s 153ms/step -
loss: 0.0250 - mse: 0.0014
Epoch 10: val_loss did not improve from 0.02570
26/26          4s 161ms/step -
loss: 0.0250 - mse: 0.0014 - val_loss: 0.0266 - val_mse: 0.0015
Epoch 11/200
26/26          0s 155ms/step -
loss: 0.0268 - mse: 0.0016
Epoch 11: val_loss improved from 0.02570 to 0.02561, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/seq_lab_model.keras
26/26          6s 241ms/step -
loss: 0.0268 - mse: 0.0016 - val_loss: 0.0256 - val_mse: 0.0014

```



```

Epoch 12/200
26/26          0s 155ms/step -
loss: 0.0259 - mse: 0.0015
Epoch 12: val_loss did not improve from 0.02561
26/26          4s 164ms/step -
loss: 0.0259 - mse: 0.0015 - val_loss: 0.0316 - val_mse: 0.0021
Epoch 13/200
26/26          0s 156ms/step -
loss: 0.0273 - mse: 0.0016
Epoch 13: val_loss did not improve from 0.02561
26/26          5s 171ms/step -
loss: 0.0273 - mse: 0.0016 - val_loss: 0.0265 - val_mse: 0.0015
Epoch 14/200
26/26          0s 157ms/step -
loss: 0.0244 - mse: 0.0013
Epoch 14: val_loss improved from 0.02561 to 0.02480, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/seq_lab_model.keras
26/26          5s 204ms/step -
loss: 0.0244 - mse: 0.0013 - val_loss: 0.0248 - val_mse: 0.0013
Epoch 15/200
26/26          0s 157ms/step -
loss: 0.0245 - mse: 0.0014
Epoch 15: val_loss improved from 0.02480 to 0.02399, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/seq_lab_model.keras
26/26          5s 194ms/step -
loss: 0.0245 - mse: 0.0014 - val_loss: 0.0240 - val_mse: 0.0013
Epoch 16/200
26/26          0s 156ms/step -
loss: 0.0237 - mse: 0.0013
Epoch 16: val_loss did not improve from 0.02399
26/26          4s 165ms/step -
loss: 0.0237 - mse: 0.0013 - val_loss: 0.0240 - val_mse: 0.0013
Epoch 17/200
26/26          0s 158ms/step -
loss: 0.0236 - mse: 0.0013
Epoch 17: val_loss did not improve from 0.02399
26/26          5s 172ms/step -
loss: 0.0236 - mse: 0.0013 - val_loss: 0.0248 - val_mse: 0.0014
Epoch 18/200
26/26          0s 159ms/step -
loss: 0.0240 - mse: 0.0013
Epoch 18: val_loss improved from 0.02399 to 0.02346, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/seq_lab_model.keras
26/26          6s 202ms/step -
loss: 0.0240 - mse: 0.0013 - val_loss: 0.0235 - val_mse: 0.0012

```

```

Epoch 19/200
26/26          0s 154ms/step -
loss: 0.0225 - mse: 0.0012
Epoch 19: val_loss did not improve from 0.02346
26/26          9s 163ms/step -
loss: 0.0225 - mse: 0.0012 - val_loss: 0.0244 - val_mse: 0.0013
Epoch 20/200
26/26          0s 157ms/step -
loss: 0.0225 - mse: 0.0012
Epoch 20: val_loss improved from 0.02346 to 0.02277, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/seq_lab_model.keras
26/26          5s 208ms/step -
loss: 0.0225 - mse: 0.0012 - val_loss: 0.0228 - val_mse: 0.0012
Epoch 21/200
26/26          0s 154ms/step -
loss: 0.0227 - mse: 0.0012
Epoch 21: val_loss did not improve from 0.02277
26/26          9s 168ms/step -
loss: 0.0227 - mse: 0.0012 - val_loss: 0.0256 - val_mse: 0.0015
Epoch 22/200
26/26          0s 158ms/step -
loss: 0.0222 - mse: 0.0012
Epoch 22: val_loss did not improve from 0.02277
26/26          5s 175ms/step -
loss: 0.0222 - mse: 0.0012 - val_loss: 0.0244 - val_mse: 0.0014
Epoch 23/200
26/26          0s 159ms/step -
loss: 0.0226 - mse: 0.0012
Epoch 23: val_loss improved from 0.02277 to 0.02159, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/seq_lab_model.keras
26/26          5s 195ms/step -
loss: 0.0226 - mse: 0.0012 - val_loss: 0.0216 - val_mse: 0.0011
Epoch 24/200
26/26          0s 158ms/step -
loss: 0.0205 - mse: 0.0010
Epoch 24: val_loss improved from 0.02159 to 0.02089, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/seq_lab_model.keras
26/26          5s 197ms/step -
loss: 0.0205 - mse: 0.0010 - val_loss: 0.0209 - val_mse: 0.0010
Epoch 25/200
26/26          0s 160ms/step -
loss: 0.0203 - mse: 0.0010
Epoch 25: val_loss improved from 0.02089 to 0.02050, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/seq_lab_model.keras

```

```

26/26          6s 222ms/step -
loss: 0.0202 - mse: 0.0010 - val_loss: 0.0205 - val_mse: 9.7263e-04
Epoch 26/200
26/26          0s 159ms/step -
loss: 0.0189 - mse: 8.7050e-04
Epoch 26: val_loss did not improve from 0.02050
26/26          4s 168ms/step -
loss: 0.0189 - mse: 8.7036e-04 - val_loss: 0.0208 - val_mse: 9.7334e-04
Epoch 27/200
26/26          0s 158ms/step -
loss: 0.0186 - mse: 8.6798e-04
Epoch 27: val_loss improved from 0.02050 to 0.01987, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/seq_lab_model.keras
26/26          5s 194ms/step -
loss: 0.0186 - mse: 8.6726e-04 - val_loss: 0.0199 - val_mse: 9.1909e-04
Epoch 28/200
26/26          0s 158ms/step -
loss: 0.0174 - mse: 7.4870e-04
Epoch 28: val_loss did not improve from 0.01987
26/26          4s 167ms/step -
loss: 0.0174 - mse: 7.5133e-04 - val_loss: 0.0204 - val_mse: 9.6417e-04
Epoch 29/200
26/26          0s 157ms/step -
loss: 0.0183 - mse: 8.3832e-04
Epoch 29: val_loss improved from 0.01987 to 0.01970, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/seq_lab_model.keras
26/26          5s 199ms/step -
loss: 0.0183 - mse: 8.3647e-04 - val_loss: 0.0197 - val_mse: 8.9791e-04
Epoch 30/200
26/26          0s 156ms/step -
loss: 0.0166 - mse: 6.9661e-04
Epoch 30: val_loss did not improve from 0.01970
26/26          4s 166ms/step -
loss: 0.0166 - mse: 6.9742e-04 - val_loss: 0.0203 - val_mse: 9.4038e-04
Epoch 31/200
26/26          0s 156ms/step -
loss: 0.0174 - mse: 7.5533e-04
Epoch 31: val_loss did not improve from 0.01970
26/26          5s 166ms/step -
loss: 0.0174 - mse: 7.5457e-04 - val_loss: 0.0198 - val_mse: 9.2286e-04
Epoch 32/200
26/26          0s 157ms/step -
loss: 0.0164 - mse: 6.7717e-04
Epoch 32: val_loss improved from 0.01970 to 0.01891, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/seq_lab_model.keras

```

```

26/26          5s 197ms/step -
loss: 0.0164 - mse: 6.7708e-04 - val_loss: 0.0189 - val_mse: 8.3730e-04
Epoch 33/200
26/26          0s 157ms/step -
loss: 0.0156 - mse: 6.0771e-04
Epoch 33: val_loss did not improve from 0.01891
26/26          4s 171ms/step -
loss: 0.0156 - mse: 6.0948e-04 - val_loss: 0.0190 - val_mse: 8.5029e-04
Epoch 34/200
26/26          0s 155ms/step -
loss: 0.0154 - mse: 6.0641e-04
Epoch 34: val_loss improved from 0.01891 to 0.01871, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/seq_lab_model.keras
26/26          6s 192ms/step -
loss: 0.0154 - mse: 6.0626e-04 - val_loss: 0.0187 - val_mse: 8.1686e-04
Epoch 35/200
26/26          0s 156ms/step -
loss: 0.0146 - mse: 5.4289e-04
Epoch 35: val_loss did not improve from 0.01871
26/26          4s 170ms/step -
loss: 0.0146 - mse: 5.4417e-04 - val_loss: 0.0193 - val_mse: 8.6839e-04
Epoch 36/200
26/26          0s 159ms/step -
loss: 0.0145 - mse: 5.4064e-04
Epoch 36: val_loss did not improve from 0.01871
26/26          4s 167ms/step -
loss: 0.0145 - mse: 5.4117e-04 - val_loss: 0.0189 - val_mse: 8.2392e-04
Epoch 37/200
26/26          0s 158ms/step -
loss: 0.0144 - mse: 5.2420e-04
Epoch 37: val_loss did not improve from 0.01871
26/26          5s 172ms/step -
loss: 0.0144 - mse: 5.2552e-04 - val_loss: 0.0193 - val_mse: 8.8107e-04
Epoch 38/200
26/26          0s 159ms/step -
loss: 0.0153 - mse: 5.7882e-04
Epoch 38: val_loss improved from 0.01871 to 0.01871, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/seq_lab_model.keras
26/26          6s 202ms/step -
loss: 0.0153 - mse: 5.7904e-04 - val_loss: 0.0187 - val_mse: 8.1800e-04
Epoch 39/200
26/26          0s 159ms/step -
loss: 0.0146 - mse: 5.3458e-04
Epoch 39: val_loss did not improve from 0.01871
26/26          4s 167ms/step -
loss: 0.0146 - mse: 5.3475e-04 - val_loss: 0.0188 - val_mse: 8.0004e-04

```

Epoch 40/200
 26/26 0s 158ms/step -
 loss: 0.0135 - mse: 4.6318e-04
 Epoch 40: val_loss improved from 0.01871 to 0.01825, saving model to
 /content/drive/MyDrive/BA865/BA865 Group Project/Best Models
 MSBA/seq_lab_model.keras
 26/26 6s 194ms/step -
 loss: 0.0135 - mse: 4.6358e-04 - val_loss: 0.0182 - val_mse: 7.8849e-04
 Epoch 41/200
 26/26 0s 158ms/step -
 loss: 0.0134 - mse: 4.6182e-04
 Epoch 41: val_loss did not improve from 0.01825
 26/26 4s 167ms/step -
 loss: 0.0134 - mse: 4.6215e-04 - val_loss: 0.0184 - val_mse: 7.9387e-04
 Epoch 42/200
 26/26 0s 158ms/step -
 loss: 0.0129 - mse: 4.2608e-04
 Epoch 42: val_loss did not improve from 0.01825
 26/26 4s 167ms/step -
 loss: 0.0129 - mse: 4.2681e-04 - val_loss: 0.0186 - val_mse: 7.9218e-04
 Epoch 43/200
 26/26 0s 157ms/step -
 loss: 0.0128 - mse: 4.0989e-04
 Epoch 43: val_loss did not improve from 0.01825
 26/26 5s 165ms/step -
 loss: 0.0128 - mse: 4.1067e-04 - val_loss: 0.0188 - val_mse: 8.1253e-04
 Epoch 44/200
 26/26 0s 158ms/step -
 loss: 0.0126 - mse: 4.0224e-04
 Epoch 44: val_loss improved from 0.01825 to 0.01802, saving model to
 /content/drive/MyDrive/BA865/BA865 Group Project/Best Models
 MSBA/seq_lab_model.keras
 26/26 5s 194ms/step -
 loss: 0.0126 - mse: 4.0229e-04 - val_loss: 0.0180 - val_mse: 7.6627e-04
 Epoch 45/200
 26/26 0s 157ms/step -
 loss: 0.0120 - mse: 3.7496e-04
 Epoch 45: val_loss did not improve from 0.01802
 26/26 4s 166ms/step -
 loss: 0.0120 - mse: 3.7539e-04 - val_loss: 0.0185 - val_mse: 7.8110e-04
 Epoch 46/200
 26/26 0s 156ms/step -
 loss: 0.0121 - mse: 3.9227e-04
 Epoch 46: val_loss improved from 0.01802 to 0.01799, saving model to
 /content/drive/MyDrive/BA865/BA865 Group Project/Best Models
 MSBA/seq_lab_model.keras
 26/26 5s 197ms/step -
 loss: 0.0121 - mse: 3.9157e-04 - val_loss: 0.0180 - val_mse: 7.6722e-04

Epoch 47/200
26/26 0s 153ms/step -
loss: 0.0122 - mse: 3.7666e-04
Epoch 47: val_loss did not improve from 0.01799
26/26 9s 167ms/step -
loss: 0.0122 - mse: 3.7662e-04 - val_loss: 0.0183 - val_mse: 7.8072e-04
Epoch 48/200
26/26 0s 155ms/step -
loss: 0.0120 - mse: 3.5688e-04
Epoch 48: val_loss did not improve from 0.01799
26/26 4s 169ms/step -
loss: 0.0120 - mse: 3.5778e-04 - val_loss: 0.0180 - val_mse: 7.6540e-04
Epoch 49/200
26/26 0s 155ms/step -
loss: 0.0117 - mse: 3.5729e-04
Epoch 49: val_loss improved from 0.01799 to 0.01793, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/seq_lab_model.keras
26/26 6s 191ms/step -
loss: 0.0117 - mse: 3.5698e-04 - val_loss: 0.0179 - val_mse: 7.5692e-04
Epoch 50/200
26/26 0s 157ms/step -
loss: 0.0115 - mse: 3.3502e-04
Epoch 50: val_loss did not improve from 0.01793
26/26 4s 166ms/step -
loss: 0.0115 - mse: 3.3533e-04 - val_loss: 0.0180 - val_mse: 7.7111e-04
Epoch 51/200
26/26 0s 159ms/step -
loss: 0.0113 - mse: 3.2673e-04
Epoch 51: val_loss did not improve from 0.01793
26/26 4s 167ms/step -
loss: 0.0113 - mse: 3.2710e-04 - val_loss: 0.0189 - val_mse: 8.2373e-04
Epoch 52/200
26/26 0s 160ms/step -
loss: 0.0119 - mse: 3.5557e-04
Epoch 52: val_loss improved from 0.01793 to 0.01787, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/seq_lab_model.keras
26/26 6s 194ms/step -
loss: 0.0119 - mse: 3.5530e-04 - val_loss: 0.0179 - val_mse: 7.5557e-04
Epoch 53/200
26/26 0s 160ms/step -
loss: 0.0114 - mse: 3.2731e-04
Epoch 53: val_loss did not improve from 0.01787
26/26 4s 170ms/step -
loss: 0.0114 - mse: 3.2723e-04 - val_loss: 0.0182 - val_mse: 7.7940e-04
Epoch 54/200
26/26 0s 162ms/step -

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loss: 0.0118 - mse: 3.4367e-04
Epoch 54: val_loss did not improve from 0.01787
26/26          4s 170ms/step -
loss: 0.0118 - mse: 3.4305e-04 - val_loss: 0.0179 - val_mse: 7.5290e-04
Epoch 55/200
26/26          0s 160ms/step -
loss: 0.0108 - mse: 2.9069e-04
Epoch 55: val_loss did not improve from 0.01787
26/26          5s 170ms/step -
loss: 0.0108 - mse: 2.9078e-04 - val_loss: 0.0183 - val_mse: 7.8185e-04
Epoch 56/200
26/26          0s 159ms/step -
loss: 0.0114 - mse: 3.2132e-04
Epoch 56: val_loss did not improve from 0.01787
26/26          4s 168ms/step -
loss: 0.0114 - mse: 3.2141e-04 - val_loss: 0.0180 - val_mse: 7.6093e-04
Epoch 57/200
26/26          0s 157ms/step -
loss: 0.0112 - mse: 2.9825e-04
Epoch 57: val_loss improved from 0.01787 to 0.01776, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/seq_lab_model.keras
26/26          6s 191ms/step -
loss: 0.0112 - mse: 2.9852e-04 - val_loss: 0.0178 - val_mse: 7.4056e-04
Epoch 58/200
26/26          0s 156ms/step -
loss: 0.0109 - mse: 3.0250e-04
Epoch 58: val_loss improved from 0.01776 to 0.01759, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/seq_lab_model.keras
26/26          5s 190ms/step -
loss: 0.0109 - mse: 3.0218e-04 - val_loss: 0.0176 - val_mse: 7.3692e-04
Epoch 59/200
26/26          0s 157ms/step -
loss: 0.0103 - mse: 2.6985e-04
Epoch 59: val_loss improved from 0.01759 to 0.01754, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/seq_lab_model.keras
26/26          5s 191ms/step -
loss: 0.0103 - mse: 2.7008e-04 - val_loss: 0.0175 - val_mse: 7.3418e-04
Epoch 60/200
26/26          0s 156ms/step -
loss: 0.0101 - mse: 2.6132e-04
Epoch 60: val_loss improved from 0.01754 to 0.01721, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/seq_lab_model.keras
26/26          5s 192ms/step -
loss: 0.0102 - mse: 2.6146e-04 - val_loss: 0.0172 - val_mse: 7.1893e-04

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Epoch 61/200
 26/26 0s 156ms/step -
 loss: 0.0099 - mse: 2.5229e-04
 Epoch 61: val_loss did not improve from 0.01721
 26/26 4s 166ms/step -
 loss: 0.0099 - mse: 2.5251e-04 - val_loss: 0.0174 - val_mse: 7.2791e-04
 Epoch 62/200
 26/26 0s 156ms/step -
 loss: 0.0103 - mse: 2.6264e-04
 Epoch 62: val_loss did not improve from 0.01721
 26/26 5s 165ms/step -
 loss: 0.0103 - mse: 2.6270e-04 - val_loss: 0.0176 - val_mse: 7.2518e-04
 Epoch 63/200
 26/26 0s 156ms/step -
 loss: 0.0100 - mse: 2.5113e-04
 Epoch 63: val_loss did not improve from 0.01721
 26/26 4s 165ms/step -
 loss: 0.0100 - mse: 2.5118e-04 - val_loss: 0.0182 - val_mse: 7.7707e-04
 Epoch 64/200
 26/26 0s 157ms/step -
 loss: 0.0101 - mse: 2.5866e-04
 Epoch 64: val_loss did not improve from 0.01721
 26/26 4s 171ms/step -
 loss: 0.0101 - mse: 2.5840e-04 - val_loss: 0.0179 - val_mse: 7.6042e-04
 Epoch 65/200
 26/26 0s 158ms/step -
 loss: 0.0099 - mse: 2.4432e-04
 Epoch 65: val_loss did not improve from 0.01721
 26/26 4s 167ms/step -
 loss: 0.0099 - mse: 2.4431e-04 - val_loss: 0.0175 - val_mse: 7.3807e-04
 Epoch 66/200
 26/26 0s 158ms/step -
 loss: 0.0098 - mse: 2.4840e-04
 Epoch 66: val_loss did not improve from 0.01721
 26/26 5s 172ms/step -
 loss: 0.0098 - mse: 2.4827e-04 - val_loss: 0.0179 - val_mse: 7.4318e-04
 Epoch 67/200
 26/26 0s 158ms/step -
 loss: 0.0099 - mse: 2.5306e-04
 Epoch 67: val_loss did not improve from 0.01721
 26/26 5s 174ms/step -
 loss: 0.0099 - mse: 2.5286e-04 - val_loss: 0.0176 - val_mse: 7.3357e-04
 Epoch 68/200
 26/26 0s 158ms/step -
 loss: 0.0098 - mse: 2.3980e-04
 Epoch 68: val_loss did not improve from 0.01721
 26/26 5s 172ms/step -
 loss: 0.0098 - mse: 2.3975e-04 - val_loss: 0.0172 - val_mse: 7.1715e-04

Epoch 69/200
26/26 0s 158ms/step -
loss: 0.0094 - mse: 2.2950e-04
Epoch 69: val_loss did not improve from 0.01721
26/26 4s 172ms/step -
loss: 0.0094 - mse: 2.2949e-04 - val_loss: 0.0174 - val_mse: 7.2035e-04
Epoch 70/200
26/26 0s 158ms/step -
loss: 0.0095 - mse: 2.3086e-04
Epoch 70: val_loss did not improve from 0.01721
26/26 4s 167ms/step -
loss: 0.0095 - mse: 2.3069e-04 - val_loss: 0.0174 - val_mse: 7.1761e-04
Epoch 71/200
26/26 0s 157ms/step -
loss: 0.0090 - mse: 2.0758e-04
Epoch 71: val_loss did not improve from 0.01721
26/26 4s 166ms/step -
loss: 0.0090 - mse: 2.0774e-04 - val_loss: 0.0173 - val_mse: 7.1857e-04
Epoch 72/200
26/26 0s 158ms/step -
loss: 0.0090 - mse: 2.0528e-04
Epoch 72: val_loss did not improve from 0.01721
26/26 4s 172ms/step -
loss: 0.0090 - mse: 2.0548e-04 - val_loss: 0.0174 - val_mse: 7.2182e-04
Epoch 73/200
26/26 0s 158ms/step -
loss: 0.0090 - mse: 2.0361e-04
Epoch 73: val_loss did not improve from 0.01721
26/26 5s 171ms/step -
loss: 0.0090 - mse: 2.0384e-04 - val_loss: 0.0173 - val_mse: 7.2049e-04
Epoch 74/200
26/26 0s 158ms/step -
loss: 0.0093 - mse: 2.1781e-04
Epoch 74: val_loss improved from 0.01721 to 0.01716, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/seq_lab_model.keras
26/26 5s 199ms/step -
loss: 0.0093 - mse: 2.1752e-04 - val_loss: 0.0172 - val_mse: 7.0955e-04
Epoch 75/200
26/26 0s 155ms/step -
loss: 0.0090 - mse: 2.0439e-04
Epoch 75: val_loss did not improve from 0.01716
26/26 9s 164ms/step -
loss: 0.0090 - mse: 2.0464e-04 - val_loss: 0.0175 - val_mse: 7.2070e-04
Epoch 76/200
26/26 0s 158ms/step -
loss: 0.0092 - mse: 2.1335e-04
Epoch 76: val_loss did not improve from 0.01716

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26/26          4s 168ms/step -
loss: 0.0092 - mse: 2.1330e-04 - val_loss: 0.0176 - val_mse: 7.2568e-04
Epoch 77/200
26/26          0s 159ms/step -
loss: 0.0091 - mse: 2.0389e-04
Epoch 77: val_loss did not improve from 0.01716
26/26          4s 168ms/step -
loss: 0.0091 - mse: 2.0395e-04 - val_loss: 0.0174 - val_mse: 7.1149e-04
Epoch 78/200
26/26          0s 158ms/step -
loss: 0.0088 - mse: 2.0445e-04
Epoch 78: val_loss did not improve from 0.01716
26/26          5s 172ms/step -
loss: 0.0089 - mse: 2.0425e-04 - val_loss: 0.0173 - val_mse: 7.1134e-04
Epoch 79/200
26/26          0s 159ms/step -
loss: 0.0095 - mse: 2.1843e-04
Epoch 79: val_loss did not improve from 0.01716
26/26          4s 168ms/step -
loss: 0.0095 - mse: 2.1885e-04 - val_loss: 0.0178 - val_mse: 7.4938e-04
Epoch 80/200
26/26          0s 159ms/step -
loss: 0.0098 - mse: 2.2948e-04
Epoch 80: val_loss did not improve from 0.01716
26/26          5s 173ms/step -
loss: 0.0097 - mse: 2.2904e-04 - val_loss: 0.0174 - val_mse: 7.1597e-04
Epoch 81/200
26/26          0s 158ms/step -
loss: 0.0089 - mse: 1.9474e-04
Epoch 81: val_loss improved from 0.01716 to 0.01714, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/seq_lab_model.keras
26/26          5s 192ms/step -
loss: 0.0089 - mse: 1.9493e-04 - val_loss: 0.0171 - val_mse: 7.0355e-04
Epoch 82/200
26/26          0s 158ms/step -
loss: 0.0085 - mse: 1.8474e-04
Epoch 82: val_loss did not improve from 0.01714
26/26          4s 168ms/step -
loss: 0.0086 - mse: 1.8489e-04 - val_loss: 0.0174 - val_mse: 7.1055e-04
Epoch 83/200
26/26          0s 159ms/step -
loss: 0.0084 - mse: 1.7491e-04
Epoch 83: val_loss did not improve from 0.01714
26/26          5s 174ms/step -
loss: 0.0084 - mse: 1.7520e-04 - val_loss: 0.0175 - val_mse: 7.2677e-04
Epoch 84/200
26/26          0s 160ms/step -

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loss: 0.0085 - mse: 1.8036e-04
Epoch 84: val_loss did not improve from 0.01714
26/26          5s 175ms/step -
loss: 0.0085 - mse: 1.8048e-04 - val_loss: 0.0172 - val_mse: 7.1017e-04
Epoch 85/200
26/26          0s 158ms/step -
loss: 0.0085 - mse: 1.8146e-04
Epoch 85: val_loss improved from 0.01714 to 0.01714, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/seq_lab_model.keras
26/26          6s 193ms/step -
loss: 0.0085 - mse: 1.8148e-04 - val_loss: 0.0171 - val_mse: 7.0362e-04
Epoch 86/200
26/26          0s 157ms/step -
loss: 0.0084 - mse: 1.7651e-04
Epoch 86: val_loss did not improve from 0.01714
26/26          5s 170ms/step -
loss: 0.0084 - mse: 1.7651e-04 - val_loss: 0.0173 - val_mse: 7.1829e-04
Epoch 87/200
26/26          0s 157ms/step -
loss: 0.0082 - mse: 1.7013e-04
Epoch 87: val_loss did not improve from 0.01714
26/26          5s 171ms/step -
loss: 0.0082 - mse: 1.7011e-04 - val_loss: 0.0172 - val_mse: 7.1130e-04
Epoch 88/200
26/26          0s 160ms/step -
loss: 0.0078 - mse: 1.5495e-04
Epoch 88: val_loss did not improve from 0.01714
26/26          5s 174ms/step -
loss: 0.0079 - mse: 1.5534e-04 - val_loss: 0.0172 - val_mse: 7.1301e-04
Epoch 89/200
26/26          0s 157ms/step -
loss: 0.0083 - mse: 1.7263e-04
Epoch 89: val_loss improved from 0.01714 to 0.01712, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/seq_lab_model.keras
26/26          6s 191ms/step -
loss: 0.0083 - mse: 1.7252e-04 - val_loss: 0.0171 - val_mse: 7.0002e-04
Epoch 90/200
26/26          0s 157ms/step -
loss: 0.0081 - mse: 1.6825e-04
Epoch 90: val_loss did not improve from 0.01712
26/26          4s 172ms/step -
loss: 0.0081 - mse: 1.6816e-04 - val_loss: 0.0172 - val_mse: 7.0922e-04
Epoch 91/200
26/26          0s 156ms/step -
loss: 0.0080 - mse: 1.6303e-04
Epoch 91: val_loss did not improve from 0.01712

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26/26          4s 170ms/step -
loss: 0.0080 - mse: 1.6295e-04 - val_loss: 0.0173 - val_mse: 7.1758e-04
Epoch 92/200
26/26          0s 159ms/step -
loss: 0.0082 - mse: 1.7010e-04
Epoch 92: val_loss did not improve from 0.01712
26/26          5s 169ms/step -
loss: 0.0082 - mse: 1.6986e-04 - val_loss: 0.0171 - val_mse: 7.0590e-04
Epoch 93/200
26/26          0s 158ms/step -
loss: 0.0078 - mse: 1.5448e-04
Epoch 93: val_loss did not improve from 0.01712
26/26          5s 167ms/step -
loss: 0.0078 - mse: 1.5454e-04 - val_loss: 0.0173 - val_mse: 7.1458e-04
Epoch 94/200
26/26          0s 157ms/step -
loss: 0.0078 - mse: 1.5442e-04
Epoch 94: val_loss did not improve from 0.01712
26/26          4s 171ms/step -
loss: 0.0078 - mse: 1.5445e-04 - val_loss: 0.0172 - val_mse: 7.1002e-04
Epoch 95/200
26/26          0s 157ms/step -
loss: 0.0079 - mse: 1.5875e-04
Epoch 95: val_loss did not improve from 0.01712
26/26          4s 166ms/step -
loss: 0.0079 - mse: 1.5869e-04 - val_loss: 0.0175 - val_mse: 7.1942e-04
Epoch 96/200
26/26          0s 158ms/step -
loss: 0.0080 - mse: 1.5748e-04
Epoch 96: val_loss improved from 0.01712 to 0.01705, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/seq_lab_model.keras
26/26          5s 198ms/step -
loss: 0.0080 - mse: 1.5746e-04 - val_loss: 0.0170 - val_mse: 6.9781e-04
Epoch 97/200
26/26          0s 155ms/step -
loss: 0.0078 - mse: 1.4832e-04
Epoch 97: val_loss did not improve from 0.01705
26/26         10s 169ms/step -
loss: 0.0078 - mse: 1.4851e-04 - val_loss: 0.0173 - val_mse: 7.2106e-04
Epoch 98/200
26/26          0s 159ms/step -
loss: 0.0084 - mse: 1.6989e-04
Epoch 98: val_loss did not improve from 0.01705
26/26          5s 173ms/step -
loss: 0.0084 - mse: 1.6975e-04 - val_loss: 0.0172 - val_mse: 7.1159e-04
Epoch 99/200
26/26          0s 158ms/step -

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loss: 0.0081 - mse: 1.5823e-04
Epoch 99: val_loss did not improve from 0.01705
26/26          5s 172ms/step -
loss: 0.0081 - mse: 1.5822e-04 - val_loss: 0.0173 - val_mse: 7.1025e-04
Epoch 100/200
26/26          0s 159ms/step -
loss: 0.0082 - mse: 1.6073e-04
Epoch 100: val_loss did not improve from 0.01705
26/26          5s 173ms/step -
loss: 0.0082 - mse: 1.6067e-04 - val_loss: 0.0176 - val_mse: 7.3105e-04
Epoch 101/200
26/26          0s 159ms/step -
loss: 0.0080 - mse: 1.5503e-04
Epoch 101: val_loss did not improve from 0.01705
26/26          4s 168ms/step -
loss: 0.0080 - mse: 1.5505e-04 - val_loss: 0.0171 - val_mse: 7.0784e-04
Epoch 102/200
26/26          0s 159ms/step -
loss: 0.0076 - mse: 1.4560e-04
Epoch 102: val_loss did not improve from 0.01705
26/26          4s 168ms/step -
loss: 0.0076 - mse: 1.4558e-04 - val_loss: 0.0173 - val_mse: 7.0927e-04
Epoch 103/200
26/26          0s 159ms/step -
loss: 0.0078 - mse: 1.4928e-04
Epoch 103: val_loss improved from 0.01705 to 0.01687, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/seq_lab_model.keras
26/26          5s 203ms/step -
loss: 0.0078 - mse: 1.4940e-04 - val_loss: 0.0169 - val_mse: 6.8706e-04
Epoch 104/200
26/26          0s 155ms/step -
loss: 0.0077 - mse: 1.4740e-04
Epoch 104: val_loss did not improve from 0.01687
26/26          9s 163ms/step -
loss: 0.0077 - mse: 1.4729e-04 - val_loss: 0.0170 - val_mse: 6.9485e-04
Epoch 105/200
26/26          0s 159ms/step -
loss: 0.0075 - mse: 1.4200e-04
Epoch 105: val_loss did not improve from 0.01687
26/26          5s 175ms/step -
loss: 0.0075 - mse: 1.4213e-04 - val_loss: 0.0170 - val_mse: 6.9590e-04
Epoch 106/200
26/26          0s 159ms/step -
loss: 0.0077 - mse: 1.4851e-04
Epoch 106: val_loss did not improve from 0.01687
26/26          5s 172ms/step -
loss: 0.0077 - mse: 1.4843e-04 - val_loss: 0.0172 - val_mse: 7.0757e-04

```

Epoch 107/200
26/26 0s 157ms/step -
loss: 0.0077 - mse: 1.4532e-04
Epoch 107: val_loss did not improve from 0.01687
26/26 4s 171ms/step -
loss: 0.0077 - mse: 1.4540e-04 - val_loss: 0.0175 - val_mse: 7.2117e-04
Epoch 108/200
26/26 0s 158ms/step -
loss: 0.0081 - mse: 1.5468e-04
Epoch 108: val_loss improved from 0.01687 to 0.01686, saving model to
/content/drive/MyDrive/BA865/BA865 Group Project/Best Models
MSBA/seq_lab_model.keras
26/26 6s 196ms/step -
loss: 0.0080 - mse: 1.5443e-04 - val_loss: 0.0169 - val_mse: 6.8935e-04
Epoch 109/200
26/26 0s 157ms/step -
loss: 0.0076 - mse: 1.4165e-04
Epoch 109: val_loss did not improve from 0.01686
26/26 4s 166ms/step -
loss: 0.0076 - mse: 1.4176e-04 - val_loss: 0.0170 - val_mse: 7.0019e-04
Epoch 110/200
26/26 0s 158ms/step -
loss: 0.0078 - mse: 1.4984e-04
Epoch 110: val_loss did not improve from 0.01686
26/26 5s 167ms/step -
loss: 0.0078 - mse: 1.4976e-04 - val_loss: 0.0170 - val_mse: 6.9529e-04
Epoch 111/200
26/26 0s 160ms/step -
loss: 0.0077 - mse: 1.4617e-04
Epoch 111: val_loss did not improve from 0.01686
26/26 5s 175ms/step -
loss: 0.0077 - mse: 1.4602e-04 - val_loss: 0.0170 - val_mse: 6.9655e-04
Epoch 112/200
26/26 0s 158ms/step -
loss: 0.0076 - mse: 1.4166e-04
Epoch 112: val_loss did not improve from 0.01686
26/26 5s 166ms/step -
loss: 0.0076 - mse: 1.4163e-04 - val_loss: 0.0169 - val_mse: 6.8581e-04
Epoch 113/200
26/26 0s 157ms/step -
loss: 0.0074 - mse: 1.3606e-04
Epoch 113: val_loss did not improve from 0.01686
26/26 5s 171ms/step -
loss: 0.0074 - mse: 1.3596e-04 - val_loss: 0.0170 - val_mse: 6.8984e-04
Epoch 114/200
26/26 0s 157ms/step -
loss: 0.0071 - mse: 1.2723e-04
Epoch 114: val_loss did not improve from 0.01686

```

26/26          5s 171ms/step -
loss: 0.0071 - mse: 1.2725e-04 - val_loss: 0.0169 - val_mse: 6.9434e-04
Epoch 115/200
26/26          0s 156ms/step -
loss: 0.0070 - mse: 1.2434e-04
Epoch 115: val_loss did not improve from 0.01686
26/26          5s 170ms/step -
loss: 0.0070 - mse: 1.2450e-04 - val_loss: 0.0173 - val_mse: 7.1672e-04
Epoch 116/200
26/26          0s 158ms/step -
loss: 0.0076 - mse: 1.4131e-04
Epoch 116: val_loss did not improve from 0.01686
26/26          5s 172ms/step -
loss: 0.0076 - mse: 1.4115e-04 - val_loss: 0.0171 - val_mse: 6.9616e-04
Epoch 117/200
26/26          0s 157ms/step -
loss: 0.0071 - mse: 1.2735e-04
Epoch 117: val_loss did not improve from 0.01686
26/26          5s 171ms/step -
loss: 0.0071 - mse: 1.2738e-04 - val_loss: 0.0169 - val_mse: 6.9411e-04
Epoch 118/200
26/26          0s 157ms/step -
loss: 0.0071 - mse: 1.2721e-04
Epoch 118: val_loss did not improve from 0.01686
26/26          5s 167ms/step -
loss: 0.0071 - mse: 1.2713e-04 - val_loss: 0.0170 - val_mse: 6.9772e-04
Epoch 119/200
26/26          0s 157ms/step -
loss: 0.0069 - mse: 1.2181e-04
Epoch 119: val_loss did not improve from 0.01686
26/26          4s 166ms/step -
loss: 0.0069 - mse: 1.2197e-04 - val_loss: 0.0170 - val_mse: 6.9431e-04
Epoch 120/200
26/26          0s 157ms/step -
loss: 0.0072 - mse: 1.2992e-04
Epoch 120: val_loss did not improve from 0.01686
26/26          5s 166ms/step -
loss: 0.0072 - mse: 1.2981e-04 - val_loss: 0.0170 - val_mse: 6.9936e-04
Epoch 121/200
26/26          0s 157ms/step -
loss: 0.0070 - mse: 1.2431e-04
Epoch 121: val_loss did not improve from 0.01686
26/26          5s 166ms/step -
loss: 0.0070 - mse: 1.2429e-04 - val_loss: 0.0170 - val_mse: 6.9604e-04
Epoch 122/200
26/26          0s 157ms/step -
loss: 0.0070 - mse: 1.2454e-04
Epoch 122: val_loss did not improve from 0.01686

```

```

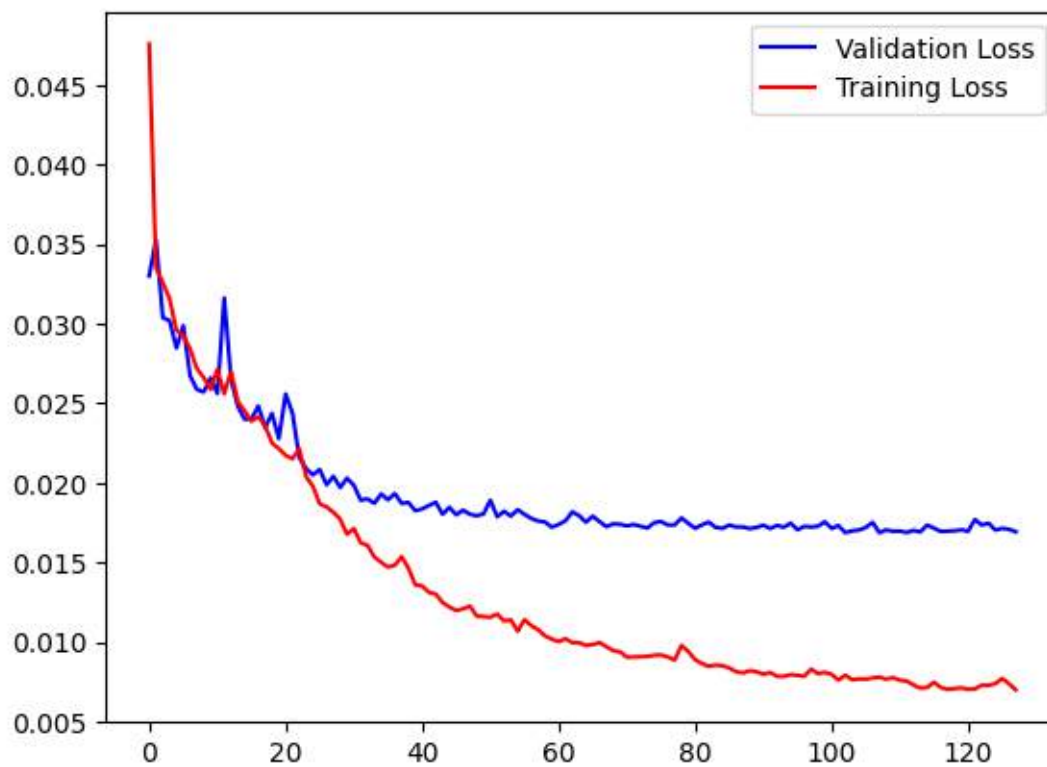
26/26          5s 165ms/step -
loss: 0.0070 - mse: 1.2455e-04 - val_loss: 0.0177 - val_mse: 7.2451e-04
Epoch 123/200
26/26          0s 157ms/step -
loss: 0.0074 - mse: 1.3575e-04
Epoch 123: val_loss did not improve from 0.01686
26/26          4s 165ms/step -
loss: 0.0074 - mse: 1.3559e-04 - val_loss: 0.0173 - val_mse: 7.1579e-04
Epoch 124/200
26/26          0s 157ms/step -
loss: 0.0073 - mse: 1.2780e-04
Epoch 124: val_loss did not improve from 0.01686
26/26          5s 166ms/step -
loss: 0.0073 - mse: 1.2787e-04 - val_loss: 0.0175 - val_mse: 7.2785e-04
Epoch 125/200
26/26          0s 156ms/step -
loss: 0.0074 - mse: 1.3351e-04
Epoch 125: val_loss did not improve from 0.01686
26/26          5s 164ms/step -
loss: 0.0074 - mse: 1.3347e-04 - val_loss: 0.0170 - val_mse: 7.0092e-04
Epoch 126/200
26/26          0s 156ms/step -
loss: 0.0078 - mse: 1.4286e-04
Epoch 126: val_loss did not improve from 0.01686
26/26          5s 171ms/step -
loss: 0.0078 - mse: 1.4280e-04 - val_loss: 0.0171 - val_mse: 7.0401e-04
Epoch 127/200
26/26          0s 158ms/step -
loss: 0.0075 - mse: 1.3678e-04
Epoch 127: val_loss did not improve from 0.01686
26/26          4s 171ms/step -
loss: 0.0075 - mse: 1.3661e-04 - val_loss: 0.0171 - val_mse: 6.9493e-04
Epoch 128/200
26/26          0s 157ms/step -
loss: 0.0070 - mse: 1.2019e-04
Epoch 128: val_loss did not improve from 0.01686
26/26          4s 165ms/step -
loss: 0.0070 - mse: 1.2026e-04 - val_loss: 0.0169 - val_mse: 6.9648e-04

```

```

[ ]: # Plotting the training and validation loss
plt.plot(history_seq_lab_model.history['val_loss'],c="b")
plt.plot(history_seq_lab_model.history['loss'],c="r")
plt.legend(['Validation Loss','Training Loss'])
plt.show()

```

Depiction of Model Performance on images from the Training Set

```
[ ]: show_predictions_lab(best_seq_lab_model, X_lab, Y_lab, num_samples=5)
```

1/1

1s 748ms/step



Depiction of Model Performance on images from the Test Set

```
[ ]: show_predictions_lab(best_seq_lab_model, X_lab_test, Y_lab_test,num_samples=5)
```

1/1 0s 102ms/step



The model successfully captures basic structure and object placement and applies skin tones and background hues that roughly match the ground truth. The colors are still muted and lack richness especially in the hair and clothing areas. The model struggles to recover finer color detail, leading to flat and slightly desaturated results. The output reflects the model's limited color diversity and its difficulty generalizing from grayscale to realistic tones.

6.7 Model 6: Transformer with VGG

```
[ ]: # Defining the custom loss functions that we experimented with
def perceptual_loss(y_true, y_pred):
    # Scale from [0, 1] to [0, 255] before using VGG
    y_true_proc = preprocess_input(y_true * 255.0)
    y_pred_proc = preprocess_input(y_pred * 255.0)

    y_true_features = feature_extractor(y_true_proc)
    y_pred_features = feature_extractor(y_pred_proc)

    return tf.reduce_mean(tf.square(y_true_features - y_pred_features))

def hybrid_loss(y_true, y_pred):
    mae_loss = tf.reduce_mean(tf.abs(y_true - y_pred))
    p_loss = perceptual_loss(y_true, y_pred)
    return 0.3 * p_loss + 0.7 * mae_loss
```

```
[ ]: # Defining the transformer encoder and decoder blocks
def transformer_encoder(inputs, num_heads=4, ff_dim=256):
    # Flatten spatial dims into sequence
    _, H, W, C = inputs.shape
    x = Reshape((H * W, C))(inputs)

    # LayerNorm + MHA
    x_norm = LayerNormalization(epsilon=1e-6)(x)
    attn_output = MultiHeadAttention(num_heads=num_heads, key_dim=C)(x_norm, x_norm)
    x = Add()([x, attn_output])

    # Feed-forward network
    x_norm = LayerNormalization(epsilon=1e-6)(x)
    ff_output = Dense(ff_dim, activation='relu')(x_norm)
    ff_output = Dense(C)(ff_output)
    x = Add()([x, ff_output])

    x = Reshape((H, W, C))(x)
    return x
```

```

def build_decoder(vgg_output):
    # 15x22x256 → 30x44x256
    x = UpSampling2D((2, 2))(vgg_output)
    x = Conv2D(256, (3, 3), padding='same')(x)
    x = BatchNormalization()(x)
    x = Activation('relu')(x)

    # 30x44x256 → 60x88x128
    x = UpSampling2D((2, 2))(x)
    x = Conv2D(128, (3, 3), padding='same')(x)
    x = BatchNormalization()(x)
    x = Activation('relu')(x)

    # 60x88x128 → 120x176x64
    x = UpSampling2D((2, 2))(x)
    x = Conv2D(64, (3, 3), padding='same')(x)
    x = BatchNormalization()(x)
    x = Activation('relu')(x)

    x = Conv2D(32, (3, 3), padding='same')(x)
    x = BatchNormalization()(x)
    x = Activation('relu')(x)

    output_rgb = Conv2D(3, (3, 3), activation='sigmoid', padding='same')(x)
    return output_rgb

```

```

[ ]: # Defining the transformer + vgg model architecture
# Input grayscale image
input_1 = Input(shape=(120, 176, 1), name='grayscale_input')
x_rgb = Lambda(lambda x: tf.image.grayscale_to_rgb(x))(input_1)

# VGG encoder
vgg_base = VGG16(include_top=False, weights='imagenet', input_tensor=x_rgb)
for layer in vgg_base.layers:
    layer.trainable = False

vgg_features = vgg_base.get_layer('block3_conv3').output # (15, 22, 256)

# Transformer block
x = transformer_encoder(vgg_features, num_heads=4, ff_dim=512)

# Decoder block
x = build_decoder(x)

output_rgb = Lambda(lambda x: tf.image.resize_with_crop_or_pad(x, 120, 176))(x)

trans_vgg_model = Model(inputs=input_1, outputs=output_rgb)

```

```
# Feature extractor for perceptual loss
vgg_feat_model = VGG16(include_top=False, weights='imagenet', input_shape=(120, 176, 3))
vgg_feat_model.trainable = False
feature_extractor = Model(inputs=vgg_feat_model.input, outputs=vgg_feat_model.get_layer('block3_conv3').output)

trans_vgg_model.compile(optimizer='adam', loss='mse', metrics=['mae'])
trans_vgg_model.summary()
```

Model: "functional_80"

Layer (type)	Output Shape	Param #	Connected to
grayscale_input (InputLayer)	(None, 120, 176, 1)	0	-
lambda_4 (Lambda)	(None, 120, 176, 3)	0	grayscale_input[...]
block1_conv1 (Conv2D)	(None, 120, 176, 64)	1,792	lambda_4[0][0]
block1_conv2 (Conv2D)	(None, 120, 176, 64)	36,928	block1_conv1[0][...]
block1_pool (MaxPooling2D)	(None, 60, 88, 64)	0	block1_conv2[0][...]
block2_conv1 (Conv2D)	(None, 60, 88, 128)	73,856	block1_pool[0][0]
block2_conv2 (Conv2D)	(None, 60, 88, 128)	147,584	block2_conv1[0][...]
block2_pool (MaxPooling2D)	(None, 30, 44, 128)	0	block2_conv2[0][...]
block3_conv1 (Conv2D)	(None, 30, 44, 256)	295,168	block2_pool[0][0]
block3_conv2 (Conv2D)	(None, 30, 44, 256)	590,080	block3_conv1[0][...]
block3_conv3	(None, 30, 44,	590,080	block3_conv2[0][...

(Conv2D)	256)		
reshape_2 (Reshape)	(None, 1320, 256)	0	block3_conv3[0][...]
layer_normalizatio... (LayerNormalizatio...	(None, 1320, 256)	512	reshape_2[0][0]
multi_head_attenti... (MultiHeadAttentio...	(None, 1320, 256)	1,051,904	layer_normalizati... layer_normalizati...
add_2 (Add)	(None, 1320, 256)	0	reshape_2[0][0], multi_head_atten...
layer_normalizatio... (LayerNormalizatio...	(None, 1320, 256)	512	add_2[0][0]
dense_2 (Dense)	(None, 1320, 512)	131,584	layer_normalizati...
dense_3 (Dense)	(None, 1320, 256)	131,328	dense_2[0][0]
add_3 (Add)	(None, 1320, 256)	0	add_2[0][0], dense_3[0][0]
reshape_3 (Reshape)	(None, 30, 44, 256)	0	add_3[0][0]
up_sampling2d_17 (UpSampling2D)	(None, 60, 88, 256)	0	reshape_3[0][0]
conv2d_83 (Conv2D)	(None, 60, 88, 256)	590,080	up_sampling2d_17...
batch_normalizatio... (BatchNormalizatio...	(None, 60, 88, 256)	1,024	conv2d_83[0][0]
activation_4 (Activation)	(None, 60, 88, 256)	0	batch_normalizati...
up_sampling2d_18 (UpSampling2D)	(None, 120, 176, 256)	0	activation_4[0][...]
conv2d_84 (Conv2D)	(None, 120, 176, 128)	295,040	up_sampling2d_18...
batch_normalizatio... (BatchNormalizatio...	(None, 120, 176, 128)	512	conv2d_84[0][0]
activation_5	(None, 120, 176,	0	batch_normalizati...

(Activation)	128)		
up_sampling2d_19 (UpSampling2D)	(None, 240, 352, 128)	0	activation_5[0][...
conv2d_85 (Conv2D)	(None, 240, 352, 64)	73,792	up_sampling2d_19...
batch_normalizatio... (BatchNormalizatio...	(None, 240, 352, 64)	256	conv2d_85[0][0]
activation_6 (Activation)	(None, 240, 352, 64)	0	batch_normalizat...
conv2d_86 (Conv2D)	(None, 240, 352, 32)	18,464	activation_6[0][...
batch_normalizatio... (BatchNormalizatio...	(None, 240, 352, 32)	128	conv2d_86[0][0]
activation_7 (Activation)	(None, 240, 352, 32)	0	batch_normalizat...
conv2d_87 (Conv2D)	(None, 240, 352, 3)	867	activation_7[0][...
lambda_5 (Lambda)	(None, 120, 176, 3)	0	conv2d_87[0][0]

Total params: 4,031,491 (15.38 MB)

Trainable params: 2,295,043 (8.75 MB)

Non-trainable params: 1,736,448 (6.62 MB)

```
[ ]: # Fitting the model to the training data
history_trans_vgg = trans_vgg_model.fit(X, Y, batch_size=16, epochs=200,
    ↪ validation_split=0.20, callbacks=[early_stop])
```

Epoch 1/200

23/23 82s 3s/step - loss:

0.0717 - mae: 0.2126 - val_loss: 0.3006 - val_mae: 0.4684

Epoch 2/200

23/23 15s 635ms/step -

```

loss: 0.0396 - mae: 0.1528 - val_loss: 0.2141 - val_mae: 0.3864
Epoch 3/200
23/23          21s 642ms/step -
loss: 0.0349 - mae: 0.1404 - val_loss: 0.1608 - val_mae: 0.3232
Epoch 4/200
23/23          20s 637ms/step -
loss: 0.0325 - mae: 0.1346 - val_loss: 0.1152 - val_mae: 0.2675
Epoch 5/200
23/23          20s 637ms/step -
loss: 0.0305 - mae: 0.1292 - val_loss: 0.0968 - val_mae: 0.2450
Epoch 6/200
23/23          21s 645ms/step -
loss: 0.0287 - mae: 0.1238 - val_loss: 0.0521 - val_mae: 0.1769
Epoch 7/200
23/23          15s 647ms/step -
loss: 0.0275 - mae: 0.1207 - val_loss: 0.0667 - val_mae: 0.2020
Epoch 8/200
23/23          20s 636ms/step -
loss: 0.0260 - mae: 0.1166 - val_loss: 0.0515 - val_mae: 0.1755
Epoch 9/200
23/23          21s 637ms/step -
loss: 0.0251 - mae: 0.1155 - val_loss: 0.0412 - val_mae: 0.1561
Epoch 10/200
23/23          16s 715ms/step -
loss: 0.0249 - mae: 0.1135 - val_loss: 0.0527 - val_mae: 0.1754
Epoch 11/200
23/23          19s 634ms/step -
loss: 0.0237 - mae: 0.1102 - val_loss: 0.0433 - val_mae: 0.1578
Epoch 12/200
23/23          20s 635ms/step -
loss: 0.0220 - mae: 0.1067 - val_loss: 0.0395 - val_mae: 0.1478
Epoch 13/200
23/23          22s 709ms/step -
loss: 0.0220 - mae: 0.1058 - val_loss: 0.0384 - val_mae: 0.1420
Epoch 14/200
23/23          19s 644ms/step -
loss: 0.0212 - mae: 0.1032 - val_loss: 0.0394 - val_mae: 0.1508
Epoch 15/200
23/23          20s 637ms/step -
loss: 0.0205 - mae: 0.1019 - val_loss: 0.0586 - val_mae: 0.1743
Epoch 16/200
23/23          20s 636ms/step -
loss: 0.0203 - mae: 0.1016 - val_loss: 0.0440 - val_mae: 0.1522
Epoch 17/200
23/23          21s 642ms/step -
loss: 0.0194 - mae: 0.0975 - val_loss: 0.0389 - val_mae: 0.1385
Epoch 18/200
23/23          15s 658ms/step -

```

```

loss: 0.0189 - mae: 0.0969 - val_loss: 0.0381 - val_mae: 0.1389
Epoch 19/200
23/23          20s 636ms/step -
loss: 0.0180 - mae: 0.0944 - val_loss: 0.0377 - val_mae: 0.1378
Epoch 20/200
23/23          20s 633ms/step -
loss: 0.0176 - mae: 0.0930 - val_loss: 0.0391 - val_mae: 0.1378
Epoch 21/200
23/23          15s 651ms/step -
loss: 0.0168 - mae: 0.0901 - val_loss: 0.0395 - val_mae: 0.1491
Epoch 22/200
23/23          20s 636ms/step -
loss: 0.0166 - mae: 0.0909 - val_loss: 0.0379 - val_mae: 0.1386
Epoch 23/200
23/23          21s 643ms/step -
loss: 0.0157 - mae: 0.0881 - val_loss: 0.0395 - val_mae: 0.1413
Epoch 24/200
23/23          22s 698ms/step -
loss: 0.0159 - mae: 0.0890 - val_loss: 0.0386 - val_mae: 0.1405
Epoch 25/200
23/23          19s 641ms/step -
loss: 0.0148 - mae: 0.0850 - val_loss: 0.0393 - val_mae: 0.1424
Epoch 26/200
23/23          20s 642ms/step -
loss: 0.0142 - mae: 0.0832 - val_loss: 0.0382 - val_mae: 0.1378
Epoch 27/200
23/23          20s 637ms/step -
loss: 0.0139 - mae: 0.0819 - val_loss: 0.0400 - val_mae: 0.1392
Epoch 28/200
23/23          15s 647ms/step -
loss: 0.0134 - mae: 0.0805 - val_loss: 0.0388 - val_mae: 0.1368
Epoch 29/200
23/23          20s 635ms/step -
loss: 0.0134 - mae: 0.0805 - val_loss: 0.0393 - val_mae: 0.1367
Epoch 30/200
23/23          20s 637ms/step -
loss: 0.0127 - mae: 0.0789 - val_loss: 0.0396 - val_mae: 0.1397
Epoch 31/200
23/23          15s 656ms/step -
loss: 0.0126 - mae: 0.0777 - val_loss: 0.0399 - val_mae: 0.1388
Epoch 32/200
23/23          15s 645ms/step -
loss: 0.0122 - mae: 0.0761 - val_loss: 0.0403 - val_mae: 0.1396
Epoch 33/200
23/23          20s 629ms/step -
loss: 0.0123 - mae: 0.0770 - val_loss: 0.0418 - val_mae: 0.1430
Epoch 34/200
23/23          15s 642ms/step -

```

```

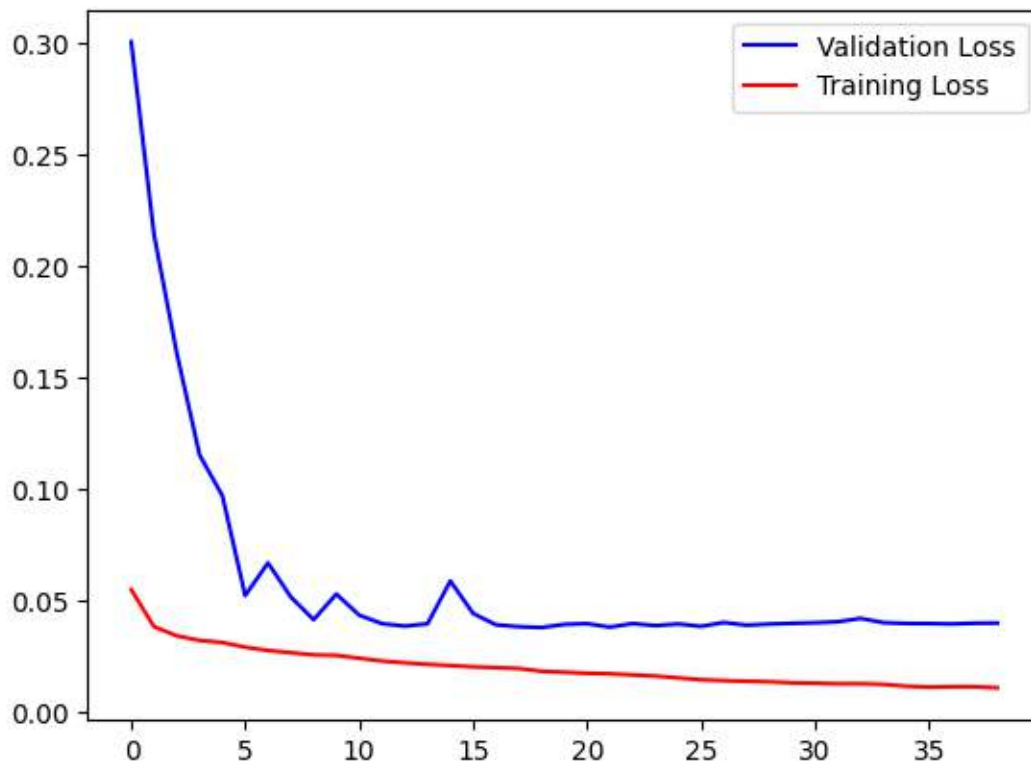
loss: 0.0121 - mae: 0.0765 - val_loss: 0.0399 - val_mae: 0.1395
Epoch 35/200
23/23          20s 637ms/step -
loss: 0.0113 - mae: 0.0733 - val_loss: 0.0395 - val_mae: 0.1368
Epoch 36/200
23/23          21s 641ms/step -
loss: 0.0109 - mae: 0.0716 - val_loss: 0.0395 - val_mae: 0.1389
Epoch 37/200
23/23          21s 642ms/step -
loss: 0.0109 - mae: 0.0727 - val_loss: 0.0393 - val_mae: 0.1384
Epoch 38/200
23/23          20s 641ms/step -
loss: 0.0112 - mae: 0.0730 - val_loss: 0.0396 - val_mae: 0.1408
Epoch 39/200
23/23          16s 707ms/step -
loss: 0.0107 - mae: 0.0715 - val_loss: 0.0397 - val_mae: 0.1375

```

```

[ ]: # Plotting the training and validation loss
plt.plot(history_trans_vgg.history['val_loss'],c="b")
plt.plot(history_trans_vgg.history['loss'],c="r")
plt.legend(['Validation Loss','Training Loss'])
plt.show()

```



Depiction of Model Performance on images from the Training Set

```
[ ]: show_predictions(trans_vgg_model, X, Y, num_samples=5)
```

1/1

1s 1s/step



Depiction of Model Performance on images from the Test Set

```
[ ]: show_predictions(trans_vgg_model, X_test, Y_test, num_samples=6)
```

1/1 2s 2s/step



As for the MSBA headshot dataset, this plot shows a steady decrease in both training and validation loss, with the validation loss stabilizing after around 10 epochs. The training loss continues to drop, while the validation loss stops decreasing at a slightly higher value, indicating good learning progress with minimal overfitting. The model shows decent generalization, though the small gap suggests there is still room to improve its performance on unseen data. However, as seen before, the images are not interpretable and completely lack detail even though the right colours are present.

7 Conclusion

Colorizing grayscale images is a fundamentally challenging task because it requires the model to infer missing color information without any semantic understanding. In grayscale, many distinct colors can appear visually identical. For example, green grass and red bricks may look the same in grayscale. Without contextual clues or prior knowledge, it's difficult for a model to make accurate predictions. This ambiguity becomes even more pronounced in full scene portraits, where multiple subjects, varied lighting, and complex backgrounds add further noise. Our relatively simple neural networks lacked the capacity to fully resolve this complexity. Without semantic context, grayscale is inherently ambiguous.

Additionally, there may have been unintentional bias in both our full scene portraits and MSBA headshots dataset. Since images were drawn from our personal camera rolls and cohort, most of the subjects were of Asian descent. The lack of tonal diversity in the training data limited the model's exposure to a range of skin tones and hair color diversity. As a result, the model may have learned representations that are not able to be broadly generalizable, reducing its performance on unrepresented demographics.

Interestingly, the MSBA headshots had the most consistent and visually coherent results. We believe this is due to the semi-standardized nature of the images. Each photo featured a different individual with varying skin tones, hair colors, clothing, and presence of accessories, but the overall framing, lighting, and background styles were relatively consistent across the dataset. The balance between controlled composition and subject level diversity may have provided the model with a more learnable structure, which allows it to generalize better within this subset compared to the chaotic full scene portraits.

8 Future Improvements

As we have seen, image colorization is ambiguous and highly dependent on context. There continues to be substantial room for improvement beyond the aforementioned models. Though our project produced encouraging results, particularly in the controlled portrait scenario, moving on to more complex and varied scenes is still a challenge. Future improvements will require more diverse data, increased model complexity, and potentially the integration of semantic understanding to guide color prediction accurately.

A key direction identified for future work is to expand the dataset by including a more diverse range of images that feature varied scenes, objects, skin tones, and lighting conditions to help the model generalize more effectively. This would increase the variability that would require more complex models and greater computational resources.

An option for increasing the model’s architectural complexity and along with necessary computational resources could enhance its ability to capture both global context and fine details. Additionally, the integration of descriptive text labels during training would provide semantic context about the objects in each image. This would help the model learn typical color associations and apply the knowledge to unfamiliar cases.

Beyond our primary models, we experimented with combining VGG and Transformer networks in parallel. The goal was to use VGG’s object recognition capabilities to inject semantic information into the colorization process while leveraging Transformers for detail retention. The results did not produce meaningful results. The predicted images from the model were a single color, regardless of input, and suggested something was inherently wrong in our implementation. Nonetheless, this direction remains promising conceptually and highlights a potential area for future work.

All together, these improvements may lead to a more powerful and color generalization network.